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Conference Theme

ACMC11 THEME: ORGANIC SOUNDS IN LIVE ELECTROACOUSTIC MUSIC

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For the sake of clarity, a dictionary definition of the term 'organic' is first offered.

organic
organic adj
1. of, produced by, or found in plants or animals, the rocks were carefully searched for organic remains,
2. not using, or grown without, artificial fertilizers or pesticides, organic vegetables, an organic farm,
3. (Chem) of or belonging to the class of chemical compounds that are formed from carbon,
4. (of change or development) gradual and natural rather than sudden or forced,
5. made up of many different parts which contribute to the way in which the whole society or structure works, an organic whole,

Collins English Dictionary Online - 10th Edition 2009 © William Collins Sons & Co. Ltd. 1979, 1986 © HarperCollins Publishers 1998, 2000, 2003, 2005, 2006, 2007, 2009. Retrieved December 2010 from http://www.collinslanguage.com

A contextual definition might include sounds made by plants and animals (including humans) as well as all sounds that exhibit the behaviour and/or form of living things (the surrogates of living organisms). In live electroacoustic music, this may include the use of real-world samples, abstract sounds produced by musical instruments, as well as those produced by generative systems.

Firstly, regarding the use of sound samples in live electroacoustic music: Australia and New Zealand are home to some of the most remarkable (and some of the most endangered) species of plants and animals in the natural world. Sounds from the Australasian biosphere such as wind, waves, insects, bird and whale song have, for centuries - if not millennia - provided musicians with a source of inspiration, and with settings in which to contemplate their individual and collective heritage. Body sounds and rhythms too, form a powerful subset of musical materials on which many claim human appreciation of musical phraseology and dance is based. Then there are the industrial and urban soundscapes - the sounds of every day life - that might that might be considered to be organic. Since the advent of recording, composers have made prolific use of real-world sounds in their music, and in both Australia and New Zealand there is an established culture of sample editing approaches to music making. Now, with the aid of multichannel technology, and faster computer processing, composers have the capacity to reproduce sophisticated real-world environments in 3-dimensions, and to use a number of once-non-real-time processes in live performance. Some interesting questions arise: What are the advantages of employing such technologies in this way? How might sound samples be used to greatest effect in live electroacoustic performance settings?

The term 'organic', within the context of music is not without precedent. In popular music it is often used to describe the quality of certain sounds as 'living', or to highlight a specific process of natural and/or evolutionary development – so it is with ease, that it migrates to the realm of abstract instrumental sounds. Traditional Maori and Aboriginal instruments have achieved a special status in their ability to evoke associations with living things present and past, while modern and ancient instruments made from wood, bone, seashells, clay, and stone appear to be being used more and more in live contemporary performance. Electroacoustic composers can now elect to transform acoustic instruments (both traditional and modern), and/or other objects of personal value into complex hyper-instruments that produce organic acoustic, and electroacoustic sounds. Very quiet, (and sometimes inaudible) sounds, such as those produced by some ethnic instruments, or sounds from the human body, can be amplified, and spectrally extended in real time, transforming our ordinary sense of spatial organisation. Within this sub-domain, some important questions arise: How might traditional Maori and Aboriginal instruments be respectfully employed within the context of electroacoustic music? How might technology be used to extend or change our own body boundaries?

The term may also be used to describes generative systems based on conversational or evolutionary models. For this special class, we might adopt the category of 'surrogates of living things'. These intelligent artificial agents are being used more and more (in various musical settings) as a means of extending the sonic capabilities of instruments, to manage information too complex for the ordinary performer to comprehend (such as multimodal gestural data), and as performers in their own right. How might this new technology be used to assist the ordinary electroacoustic performer/composer?

Scientific and socio-cultural approaches to the subject unearth several complex sub-topics each with its own set of detailed questions: What are the salient features (abstract sonic characteristics) that lead to the appreciation of sound as organic? Does psychological projection of 'the self' (or selves) play a part in the efficacy of such sounds? Does the use of organic sounds in electroacoustic music encourage a paradigm shift from *Music for Performance* to *Music for Personal Growth*?

Academic discussion on the theme *Organic Sounds In Live Electroacoustic Music* is a difficult proposition, as there are a number of remote, yet intersecting domains to consider – each with its own set of terms, and its own cluster of research priorities. Nevertheless, significant contributions to the topic (and sub-topics) have already been made. There is much to be gleaned from the music of Australasian composers and performers who work with the (often unique) sound sources specified; from their artist's talks and research publications; and from the expert domain literature associated with the fields of acoustic ecology, Maori and Aboriginal music, socio-cultural aspects of electroacoustic music, live electronic music, and psychology of human audition. The organisers of ACMC11 feel that the subject is important to Australasia, and worthy of further investigation.

Keynote Presentations

‘LIVING IN A PERFORMING WORLD – PERFORMING IN A LIVING WORLD’

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ABSTRACT

There has been a steady erosion of the ‘living/non-living’ dichotomy (of which ‘organic/inorganic’ is a version). This has given way to a more ecological model in which ‘living’ becomes an activity rather than a state [Ingold - 6]. Until a century or so ago the activity was limited to the cause/effect of human body action, but the sounding effect may now be displaced entirely from the original cause, both in time and space. This is ‘action over a distance’ rather than ‘dislocation’ – and technology might become the healer. Due to recording, the sounding world itself has become an ever wider source of musical material. More recently it has made a transition to becoming a fully fledged ‘musical instrument’. I will suggest that in this mediatised world ‘liveness’ is a network of activity in which subjects (performers) and objects (performed), causes and effects, can never unambiguously be ‘sorted out’ (our musical equivalent of the Turing test may not be very conclusive). Ingold’s point is that in many societies apparently inorganic (‘non-living’) objects *are* imbued with a spirit – an animistic world – which may be related to in the same way as to fellow creatures. Clearly the Schaefferian world of *reduced listening* has here transformed. How do we feel when we activate this world *as live music*? All kinds of causes may be evoked - breath, friction, percussion are inherited from the mechanical world, to which bio-activity and perhaps imagination itself may be harnessed to play our ‘animate net-work’.

1. EVOLUTION, PROGRESS AND ODYSSEY

This paper aims to be evolutionary. There is a clear narrative of journey – even an Odyssey (appropriate for my attendance at this conference). But I am not so sure there is in the end any goal – that’s one distinction between progress and evolution. I am seeking to understand a bit better where we are and what problems and opportunities we face. As the world evolves so must our ideas. So the first part of my talk brings together ideas I have thought and written about over many years but (I hope) in a way which sets the stage for another nudge along the way.

My own attempts to pin down the ‘live’ in ‘live electroacoustic music’ have changed over the years. When I was a student (in the late 1960s to early 1970s) it meant a live performer with a traditional acoustic instrument or voice, the sounds transformed in some (analogue) way and projected on loudspeakers. The French coined the term ‘mixed music’ for the combination of such a performer (transformed or simply amplified) with pre-recorded sound then on tape. In the 1980s I hated and resisted the slow and steady invasion of the term ‘real time’ which came from the earliest event-based (Midi) computer processes – I wrote about this as a binary opposition in the article ‘Live versus Real-Time’ [3].

As computers became powerful enough to combine event and signal processing, mixing and diffusion (as well as becoming more portable) in the 1990s, I stubbornly retained my views on the nature of ‘live’ against my more adventurous students and other practitioner colleagues – even when the term ‘live’ regained some standing once again in the last years of the century. DJs and laptop improvisers were called ‘live’ which was a step forward in the rehabilitation of the word – but inaccurately applied I thought. With a lot of what they were doing (no conventional instrument or voice in sight) I remarked ‘that’s not live!’ to which they steadfastly replied ‘yes it is!’. So to celebrate the new century I reconsidered my position and wrote in *Living Electronic Music* [4: 90] -

Working Definition of ‘Live’ (for the Time Being)

We need a working definition, not to define boundaries for all time, but to set the limits of the stage we will occupy for the time being.

‘Live’ will here mean:

- The presence of a human performer:

who takes decisions and/or makes actions during a performance which change the real sounding nature of the music;

- this embraces the historically accepted view of the ‘live’ as involving a human:

who produces sound mechanically; or who produces sounds on electronic substitutes for mechanical instruments using similar physical gestural input;

- but it also includes one:

  who does not mechanically cause the sound, yet who may cause, form or influence it through electronically mediated interfaces under their immediate control.

Today I want to shift territory once again – from observer to participant, from musicologist to performer. This temporary working definition just cited still focused on the world of the performer on stage, being listened to and observed by me – by us. This time around we live in a world as contradictory as ever - one more connected up, yet more distributed, in touch when far apart, yet still wanting human warmth and presence, creating 'virtual realities' while still seeking the noises and smells of the rain forest and the feel of warm wind.

I welcome the combination of 'organic' and 'live' in the theme of the conference. There has been a steady erosion of the 'living/non-living' dichotomy (of which 'organic/inorganic' is a version) which mirrors the later 20thC rejection of 'mind/body', 'natural/cultural' and other binary thinking. This has given way to a more ecological (holistic) model which seeks to place *interdependencies* more clearly to the fore.

> For life is not a principle that is separately installed inside individual organisms, and sets them in motion upon the stage of the inanimate. To the contrary […] life is 'a name for what is going on in the generative field within which organic forms are located and 'held in place' [6: 200]

In this (Ingold's) view 'living' becomes an activity rather than a state. Music is a very special subset of this activity – one which seems to have 'added meaning' as well as 'added value'. Furthermore technology has extended its range and ubiquity. Until a century or so ago the activity was limited to the cause/effect of human body action, but the sounding effect may now be displaced entirely from the original cause, both in time and space. This is a kind of musical or sound prosthesis allowed by technology and has progressively become global.

2. DISLOCATION AS *ENTFERNUNG* NOT *VERRENKUNG*!

Let us go back to what I originally wrote. In 1994 I wanted to look again at the Schaefferian idea of the 'acousmatic'. I was a bit unhappy with the simple Pythagorean idea of a 'venerated lecturer behind a curtain'. In fact Pierre Schaeffer's discussion seemed to revolve around a group of consequences of the development of 'technologies of sound'. I called these the 'acousmatic dislocations':

> Dislocation 1: Time (recording)
> Dislocation 2: Space (telecommunications (telephone, radio), recording)
> Dislocation 3: Mechanical causality (electronic synthesis, telecommunications, recording) [4: 95]

We can rewrite these in terms of the extraordinary final quarter of the 19th century -

> Telephone: Alexander Graham Bell (1876) - separating source and receiver in space.
> Sound recording: Thomas Edison (1877) - separating source and receiver in time and space.
> Radio: Marconi (1890s) - radio waves - across the Atlantic (1901).
> Synthesis: Elisha Gray – Thaddeus Carhill (1870s-90s) – separating sound production (as perceived) from immediate mechanical causality[1].

In 1996 I was invited to the *Digitale* conference and festival in Cologne. I speak basic German and risked presenting my paper *auf Deutsch* – I spent quite some time with dictionary in the preparation and enjoyed the experience. In fact it was a simple development of the 1994 paper, so of course contained the phrase 'acousmatic dislocation', and I translated this as *akusmatische Verrenkung*. Now *Verrenkung* is fine for dislocation if you mean you need a doctor to get your shoulder back into place, and I was met with at best humour and at worst incomprehension in using this phrase. I was told that speaking figuratively the German language would switch to something like the much more poetic and abstract *Entfernen* or *Entfernung* – which I would retranslate as *removal* or *distancing*. Now this led me while thinking about this paper (here) to an interesting insight – that we can make 'lost in translation' into something gained by correction and comparison.

The 19th century saw the creation of a EuroAmerican-centric order based in large part upon that very idea – a *distancing* not a *dislocation* - not so much 'something that needs correcting' as something that need traversing. The technological discoveries just referred to can be re-read as not so much the *causes* of any dislocation but quite the opposite – an answer to them. Technology reaching out across these gaps – oceans of space and time that our imagination had longed for centuries could be crossed.[2] We always did travel to far off lands and so the *dislocations* were always there - but maybe we wanted to see and hear those we had left behind. Such technology has the potential to bridge the gap – to heal the rift – to link together. Thus my list above may be turned on its head and seen as the start of a *creative response* to dislocation – not its *cause*. Sending letters was a great start after the invention of the alphabet – but to preserve and hear the absent voice (in both distance and time) was a dream that Edison specifically addressed at the time of his invention.

[1] That is we perceive (say) what we believe to be a human voice from the vibrations of a loudspeaker cone which remains mechanical.

[2] See the oft quoted Francis Bacon, writing in 'New Atlantis' (1624/27) "We have sound houses, where we practise and demonstrate all sounds and their generation. … We have all means to convey sounds in trunks and pipes, in strange lines and distances."

3. LISTENING, IMITATING, RECORDING AND CAPTURE

In the course of the twentieth century we have seen a steady transformation from the world (that is our environment) as the source of musical sound to the world itself as potential performance instrument. For centuries (millennia we might suppose) music has imitated the sounds around us – usually the sounds of living creatures but not always. But through recording, the sounding world itself has become an ever wider source of musical material well beyond what we can mechanically imitate with our voices or objects around us, projected into a variety of acoustics. The early 20th century futurists ensured that we consider all sounds including the most noisy and industrial – but before them one wonderful precursor:

> At a sufficient distance over the woods this sound [bells] acquires a certain vibratory hum, as if the pine needles in the horizon were the strings of a harp which it swept. All sound heard at the greatest possible distance produces one and the same effect, a vibration of the universal lyre. (Thoreau) [13: 168]
>
> To convince ourselves of the amazing variety of noises, it is enough to think of the rumble of thunder, the whistle of the wind, the roar of the waterfall, the gurgling of a brook, the rustling of leaves, the clatter of a trotting horse [...], and of the generous, solemn white breathing of a nocturnal city. (Russolo) [10: 85]
>
> […] We will sing of the vibrant nightly fervour of arsenals and shipyards blazing with violent electric moons; […] deep-chested locomotives whose wheels paw the tracks like the hooves of enormous steel horses bridled by tubing; and the sleek flight of planes whose propellers chatter in the wind like banners and seem to cheer like an enthusiastic crowd. (Marinetti) [7: 22]

Thoreau's ideas fed through of course to Cage and beyond - while the futurists' *intonarumori* had more bark in the telling than bite in their realization (judging by the few extant recordings and recreations). Then there was Arseny Avraamov's *Symphony of sirens* most famously performed in the city of Baku in 1922, to celebrate the fifth anniversary of the Russian October Revolution in 1917 (also performed a year later in Moscow) which has recently been thoroughly documented and recreated by a group of Spanish artists directed by Miguel Molina Alarcón [8][3]:

> […] Avraamov worked with choirs thousands strong, foghorns from the entire Caspian flotilla, two artillery batteries, several full infantry regiments, hydro-airplanes, twenty five steam locomotives and whistles and all the factory sirens in the city. He also invented a number of portable devices, which he called "Steam Whistle Machines" for this event, consisting of an ensemble of 20 to 25 sirens tuned to the notes of the *Internationale*. […] Avraamov did not want spectators, but intended the active participation of everybody […][4]. ([8: 19]

Avraamov himself wrote –

> The social problems of music can be solved in one of two ways: the active engagement of the broad masses with musical creativity (the socialization of music) or the modernization of its hitherto archaic material technology (the electrification of music). [He goes on to refer to the dream of] the creation of new, powerful, instruments […] that could sound simultaneously for millions of listeners. (Arseny Avraamov (orig. 1924 translated in [8: 70])

In 1986 I wrote about *mimesis* in electroacoustic music contrasting *timbral mimesis* in which the sound quality of something was imitated with *syntactic mimesis* in which a *pattern of relationships*[5] was modeled in sonic terms [2]. Of course the two are combined in raw environmental phenomena – but to a surprising extent they can be separated – as in the famous experiments on 'bouncing' and 'smashing' behaviours [15].

A very large part of algorithmic composition might be seen as a kind of *behaviour mimesis*. From environmental measurements of cosmic events to weather maps, through animal and human behaviours, from deepest rural to most complex urban phenomena – anything that may be described in mathematical symbolic form can 'drive' a synthesis environment. Iannis Xenakis is often cited as one of the first to harness the power of the *world's behaviours* to these ends - in Xenakis's case not (literally anyway) the *world's sounds*.

Thus a divided world of mimesis emerged in the last half of the 20th century. Behaviours and sounds became strangely separated when abstracted from the world's near infinite resources now available through recording and analysis. In many ways this exemplified the perennial argument, recurring over many generations but clearly witnessed in the contemporary music battles of the 1950s and 1960s, between those who composed drawing out relationships primarily from sound quality ('abstracted syntax') and those who relied to an extent on *a priori* symbolic/numerical 'systems' ('abstract syntax') [2]. The promise that empirical research might come to the rescue of this split way of thinking has not been delivered to any great extent. In this view research in psychology of perception (including sound behaviour)

[3] Molina Alarcón's book (with CDs) covers a lot more than this – it is subtitled 'Original documents and reconstructions of 72 key works of music, poetry and agitprop from the Russian Avantgardes (1908-1942)'.

[4] These are Molina's words but followed by the complete instructions for the performance written by Avraamov himself [8: 20-21).

[5] The word *behaviour* can be used here – although I did not use it when originally conceived. It is also the term used by Denis Smalley for one of his nine indicative fields [12].

could help us devise 'rules' – not idealized ones but strictly empirical, derived from observed examples – which might drive artificial intelligence applications finding 'suitable' sounds – that is sounds that 'suit' a particular behaviour.[6] Much installation art seems to focus on such syntactic (generative) concerns at the expense of the sound quality.

What I am suggesting here is the possibility of bringing two sometimes distinct areas of sonic endeavour together – the installation and the performance. I will suggest that in this mediatised world 'liveness' is a network of activity in which subjects (performers) and objects (performed), causes and effects, can never unambiguously be 'sorted out'. Liveness becomes a property of relationship and response, not something which resides solely in decoding 'the sound itself'.

4. LIVE – A LISTENER'S OR A PERFORMER'S PERSPECTIVE

In a high proportion of what I have written on liveness in electroacoustic music I have focused on 'decoding the sound stream' – that is trying to set the questions on 'living presence' almost like a guessing game – what are the cues and clues we have available to make an intelligent assessment of the agent at work producing the musical sound [4]. As I discussed above this tends to reflect an observer's perspective rather than that of an active and influential participant.

Thus in a more recent chapter [5] I looked again at liveness from the specific point of view of performance and 'bodily expression'. For this I revisited Alan Turing's famous test for machine intelligence (he did not use the term 'artificial'). This has been much simplified in almost every description of it[7]. I will not re-run that argument here except to say that it involves a complex game of playacting. It is also strongly *behaviourist* – that is if the machine behaves in a certain way it might be described as 'intelligent' – that is not to say it has reached decisions in the same way as you or I!

There is no direct and easy mapping of this to music performance – but some of the principles hold good. Let us imagine an internet mediated jam session. Dislocated in our old sense but now re-located by technology.

'H' is our real human and 'H?' may be human but then again may be a machine. In the first version 'I' (originally 'I for Interrogator' but it can be I myself), am observing the session (but cannot see the 'H'/'H?' participants through 'the curtain') and can begin to ask questions –

- firstly and most importantly is the music any good? Are the participants responding appropriately in the musical context?

And then I might risk the more subtle question –

- are the participants human or machine?

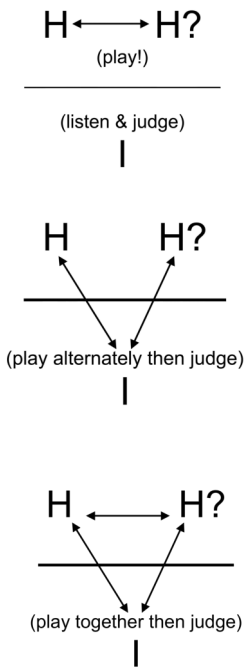


Figure 1: Rethinking Turing's Game for Music (three versions)

In the second set-up I join in, playing with each of the other participants in turn (I am clearly myself a participant now) and then judge likewise, and then finally as a trio (version 3) and ask the same questions. But in versions 2 I have not just observed 'response between others' but felt it as a 'response to me', combining the two questions in version 3.

This is a fundamental shift – one I was not so aware of when I wrote a first version of this paper in 2005 for the Live Algorithms for Music Conference in London. We will all be familiar with the recent ideas that listeners are clearly participants in music making. And ethnomusicologists have been aware for many decades that there is no such thing as an observer who is totally independent of (and so does not influence) the observed. That said we have a difference of degree which is important for musicians – a participant who actively seeks to create or to manipulate sounds to contribute to a performance is clearly involved in a more direct way than a participant listener.

When we join in *fully as performers* we shift the centre of gravity in an ensemble – we give and (usually) expect to receive – that is we expect a *response on a personal level*. We have clearly moved away from judging the sound as an observer – 'is that live?' – to judging as a participant performer – 'am I getting a response?'.

In Emmerson 2007 [4: 3] I divided the world of 'living presence' in sound into three 'worlds' – the first

[6] There is some evidence that such sophisticated research is being applied steadily in the computer games industry [1].

[7] It is much more than testing to see if a machine behaves 'indistinguishably' from a human – the experiment asks the machine to deceive us and we measure if it has succeeded as well as or better than a 'real' human! [14].

physical and mechanical (perceiving action and agency), the second psychological (working out ideas of will, choice and intention); both these enwrapped by a third 'personal and social presence (what you mean to me, where I am and who I am with)'[8]. Here too, my model tended to be one of observation rather than participation. 'I am at a concert – do I believe what they are performing on stage is 'live'?'.

Let us now extend it to include ideas of *response* just examined. Much more along the lines 'I am participating – how am I relating to others and how are they relating to me?'. This is an added dimension to the previous observations and does not displace them in any way. The first world becomes that of haptics (and all other physical input to our senses) – in a world with gravity and atmosphere there is no movement without some form of resistance. The second world moves us from observing a game or reading a narrative to playing the game and constructing the narrative. At a stroke our musicology has entered the realm of the computer game like environment in which the user is clearly a performer – whose will, choice and intention directly influence the outcome.

The third world which wraps these two immersively is the personal and the social. In my original article I bracketed these two words together – but now I want to separate them again. Personal – that is individual – sound and music space has emerged and become almost dominant since the launch by Sony of the Walkman in 1979 through to the iPod of recent years. Many computer games retain a solo emphasis – bettering your previous score rather than a simultaneous opponent – a trend broken notably by a small number of multi-user interfaces such as the Wii[9].

The *social* – community – aspects of media music making remain problematic in several senses. There are laptop orchestras. These face the problem that the laptop *is* already an orchestra – we need a kind of niche hypothesis[10] here such that each animal in the menagerie does not simply drown the others out, or ignore them completely. Then in internet collaborative music making in general, how do I gain a sense of 'others' performing with me. We might start with a kind of extended 'skype' – but this is totally inadequate in its present form. As a performer, visual and aural cues from other performers are tightly synchronized – it is only our analysis which has taken the two apart. And the intangible ability of some performers to anticipate a downbeat with apparently little by way of visual cue is well known – but only if they are in the same space together[11].

[8] Taking from Christopher Small the idea that *relationships* create the 'meaning' of musical performance [11: 93].

[9] With a maximum of 16 Wii remotes to a console.

[10] Evolution has tended to encourage a sharing of the available bandwidth in noisy environments (see the writings of Bernard L. Krause).

[11] This is not to say that tight synchronicity is a necessity for live music making – only that there will be consequences in its loss – as no network will have latency less than that determined by the speed of light.

But high quality versions of current media may function when our task is to link 'real humans' together in internet mediated traditional performance (whether on traditional instruments or electronic substitutes). But I wish to extend the idea of performing 'agent' for which this remains problematic.

5. THE ANIMATE NET-WORK

Many years ago I used to refer to one function of live electronic music as 'humanising the technology'. This shifted to thinking of it more as 'humanising through technology' – now I have arrived at describing this process as 'reanimation'. This is a small but important shift, embracing a much wider view of 'living' in the sense of activity as suggested by Ingold and in the terms discussed above.

Yet don't we seek out other 'organic' activity? – it is after all maybe more likely to respond to us. Take care here! Ingold's point (and that of many other anthropologists) is that in many societies apparently inorganic ('non-living') objects *are* imbued with a spirit – an animistic world – which may be related to in a similar way as to fellow creatures. It is this relationship activity which is 'living'. Clearly the Schaefferian world of *reduced listening* has here transformed – sound is certainly not only 'for sounds sake' in this case - it indicates far more than a cause when it indicates an object's spirit.

The final part of this paper tries to tie together the nudges forward of the several ideas I have presented over the years into one object. I shall call this the 'animate net-work'. The 'animate net-work' is a *Gedankexperiment* – that is it's a kind of imaginary beast which might exist and whose possible existence might get us exploring alternative possibilities – a *thought experiment* with (I hope) practical consequences.

Figure 2: The 'animate net-work' - an interconnecting web of agencies: human, environmental, computer-generated – this is impossible to draw but somehow we can visualize it.

Figure 2 is intended to combine some simple agencies in an imaginary group entity – it's not simply a 'work', an 'ensemble' or a 'performance' – but a kind of 'growth' which might combine elements of all three. Technology has given us the opportunity to reinstate 'animacy' to a range of agencies and activities on the network. A performance element used to be 'organic' but it is now 'animate'; everything has the potential to be thought of as animate – it need not be confined to carbon-based life forms. In this model liveness becomes responsiveness – in all the senses so far discussed. I stress – we are not discussing whether any *individual* element (agency) is alive or not. We can add the idea of *growth*, the way that nodes, that is active agents, on the network may be added or substracted (I really should use the terms 'are born' or 'die'!) – and then we have a fully fledged (strange turn of phrase!) ecosystem.

How do we (composers and performers) feel when we activate this world *as music*? The world sound bank is 'touched' and responds – *how* it responds is up to us, or another agent in the net-work.

6. PERFORMING THE ANIMATE NET-WORK

We have seen an increasingly sophisticated restoration of body senses in recent interface development – all the kinds of causes may be evoked and remodeled to give us back our sense of corporeal involvement in sound production and transformation. We can return to some fundamentals: from our first world (physical presence) we have the primary methods by which humans have provoked the environment to sound - these involve the use of muscles and the transfer of energy in the mechanical and physical domain – breath, friction and percussion.

But wait – this has until now applied to individuals. But our net-work now 'relocates' this individualism within the living world as a whole from whence it came and to which it shall return. The world is now a potential performer. So of course there are environmental examples everywhere we look and hear of these three basic energy transfers – a rainforest is jammed packed full of wind, stridulation, creaks and groans, gravity driven percussion – from animal, vegetable and mineral origins. Although there is one sound provocation which is not commonly found in traditional music production – the 'breaking' of substances in a way which leaves them changed – the crack of a branch, the sound of an earthquake, the pop of a balloon, explosions (elemental and human created - the sounds of fireworks etc.) – of course these have become common sources in electroacoustic composition but can now be restored to our repertoire of performance techniques[12].

Then from our second world (psychological presence) we can harness for our performance bio-activity – which in this context means 'the body electric', that is without perceptible muscle movement. This has been with us since the 1960s [9] but clearly steps are now being made at more sophisticated and deeper levels of neural activity – the foothills of harnessing imagination itself[13].

How might this be extended in my flexible network model – beyond the individually human? Clearly group mental activity has been the subject of much interest over recent decades (whether Jungian in origin or more pragmatic). But the modeling of real-world environmental (sound in landscape) behaviour touches on this, too. I know I am dangerously close to animism here – the idea of non-human organic and inorganic systems possessing 'will, choice and intention' may not be my intention to discuss here – but they certainly possess performative behaviour which we can 'play with' and respond to – and properly configured and addressed will respond to us.

My final world of 'presence' (the social) is paradoxically perhaps the one that does not seem to change much in the words of the retelling. 'What you mean to me, where I am and who I am with' will not change as a set of indicators of significance and meaning. But of course the 'you', the 'where' and the 'who' may be radically different in this model.

I may not know when performing if 'you' are human, computer, environmental, organic or inorganic – providing you are responsive! My sense of exactly where you are may also be no more than 'present for me, and responsive to me' – I have left open the third question 'who I am with' which unites these two - as one I do find intensely problematic with contemporary network technology. It is one which undoubtedly causes us some anxiety …

But anxiety is not a word to finish on. I find the possibilities – multiple and open – of a new musical world where technology connects up a wide variety of different sound makers in a responsive and animate network to be exciting and challenging - indeed I might have called my paper - 'Live Sounds in Organic Electroacoustic Music'.

London, April-June 2011

[12] The voice is a wonderful exception where we can create and control micro explosions, breakings, pops and clicks and yet *restore* the original materials – as Joan LaBarbara has said 'voice is the original instrument' with Trevor Wishart's paraphrase 'voice is the original synthesiser' [16].

[13] Or at least mental imagery – see http://www.emotiv.com/ - and their recent demo videos.

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NEGOTIATING N.Z. SONIC ARTS IN REAL-TIME & NETSPACE

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ABSTRACT

The contention is that it is worthwhile to combine some mainstream historical genres found in New Zealand sonic art – electroacoustic music, traditional Mäori instruments, found/invented instruments, and Asian influenced approaches; and that newer production techniques provide a means of negotiating between different sonic art perspectives in both real-time and netspace. Some early experiments by the author toward this are outlined, particularly in combining tradition Mäori instruments and electroacoustic music. The work *Mittsu no Yugo* (Whalley 2010), a telemusic event between Beijing, Hamilton, and Calgary for MUSICACOUSTICA10, is then discussed as a recent working example of the proposition.

1. INTRODUCTION

The exploration of timbre and gesture in music/sound art to reflect and explore identity has historically been a central concern of musicians/composers. It is then useful to first briefly discuss a theoretical approach to music, genre and identity, to frame why combining some mainstream New Zealand sonic art genres is worthwhile.

In discussing identity generally, Hall ((1996b: 115) notes that individual and group identity is 'formed at the unstable point where the unspeakable stories of subjectivity meet the narratives of history and culture', and it is maintained and updated once established. In this sense, while identity can be fixed, allowing discussion of it, identity is also a narrative used for experimentation or reaffirmation, rather than an attribute of character.

Both Hall (1991: 44-45) and Back (1996: 8) when discussing culture generally, note that people may mix older notions that were once used to define cultural divisions, and that once might have been in opposition, to form new identities. And they also argue that new ethnicities can evolve at a local level as the result of productive tension between global and local influences (Hall 1991: 55, Back 1996).

Music has always played a part in these ongoing processes. A useful bridge as to how this might work sonically is provided by Shepherd and Wicke (1997), who point out that music is a *medium* that allows a dialectic between sounds and people: that music is a process rather than an object. This people/sound dialectic is both structured and structuring in interaction with the human body. Music may then both restrict and facilitate simultaneously the range of meanings that can be constructed by recipients.

As narrative, music as a medium then allows for the creation and construction of experience. Frith (1996: 109) points out that 'music produces people since talking about identity is talking about experience'. Frith (1996: 111) expands on the link between music and identity when he notes that groups do not agree on values and then express these in their cultural activities (a homology model) but that they only get to know themselves *as groups*, *through* cultural activity, through aesthetic judgment. In this sense, making music is not simply a way of representing ideas, but a way of living them.

Given the music/identity link outlined, it is of little surprise that New Zealand sound based artists might want to explore an authentic voice based in their immediate environment (virtual or real) and/or take on aspects of their surrounding cultural influences. Apart from popular music and intermedia forms, four distinct historical New Zealand sonic art approaches might include the following.

The first stems from Douglas Lilburn's (1915-2001) pioneering work. Lilburn abandoned acoustic composition after completing his Third Symphony in 1961, to search for a more distinct New Zealand voice based in the natural environment. Using audio samples fused with electronic sounds made it possible for him to create a unique New Zealand soundscape: part local, part global. Early examples, quoting directly from the environment, are found in his later works like *Three Inscapes* (1972) and *Soundscape with Lake and River* (1979) (see Norris & Young 2001).

The search for authenticity in environment and local culture is also reflected in the emergence of an experimental 'performance' based 'soundculture', resulting from the use of found and invented instruments. Composers such as Phil Dadson, for example, pioneered this approach in New Zealand - his work with *From Scratch*, also incorporated Polynesian rhythms played on tuned percussion instruments made of

plastic pipes. The invented instrument approach extends to sounds sculptures, in works such as Chris Cree Brown's *Aeolian Harp*. (Dart, Elmsly, Whalley, 2001).

As part of an interest in Asian cultures and near neighbours, white New Zealand composers such as Jack Body, for example, took a documentary approach to Indonesian soundscapes in his electroacoustic CD *Suara* (1993) Continuing on from this, the interest in including Asia instruments and perspectives in New Zealand composition is increasingly common across a number musical styles.

But perhaps the most significant recent sonic art development has been the revival of traditional Mäori instruments, using an improvisational approach to creating content, and providing an alternative to Lilburn's sense of New Zealand authenticity. The approach is illustrated in the pioneering work of Hirini Melbourne and Richard Nunns. The CDs *Te Ku Te Whe* (1993) and later *Te Hekenga-ä-rangi* (2003) where a tradition of landscape and language referenced music drawing from a New Zealand pre-European tradition is referenced (see Whalley 2005).

A snapshot of the range of these approaches can be found in *New Zealand Sonic Art III* (Whalley 2002), where the call for works sought material generated from found/invented instruments, traditional instruments, and environmental sounds. The aim was to capture some of the range of cultural approaches and experimental music taking place in New Zealand at the time, to find commonalities and differences between diverse practitioners; and to explore issues of access and elitism, regionalism and internationalism, tools and output. The collection then reflects something of the established but also then emerging and evolving New Zealand identity. A summary of the outcome is given in *New Zealand Electroacoustic Music: Nexus, Flux, Negotiation* (Whalley, 2004a).

2. THINGS TOGETHER, THINGS APART

While each of these distinct sonic art traditions has found ways of expanding their historical base, such as through electroacoustic music's tape and instrument genre, or interaction using programs such as MAX/MSP, the possibilities of bringing together some of these sonic art streams *within works* are yet to be further explored.

There are of course practitioners that have, and continue to, negotiate the boundaries between at least two of the established New Zealand Sonic Art genres, as some of the works on *New Zealand Sonic Art III* illustrate. Further, the crossover between traditional Mäori instrumental approaches and electroacoustic music (see overview: Whalley 2005) is part of a wider interest in combining electroacoustic music with regional instruments/aesthetics internationally (Whalley 2005b, *Organised Sound* issue).

While then appreciating that stylistic diversity is something to be celebrated and expanded; the argument here is that it is often in the synthesis of technical and stylistic approaches that unique hybrids can be found: that music/sound can be used as a way of negotiating between differing perspectives, and that such negotiation is a healthy part of evolving cultural identity.

As part of this process, the development of New Zealand Sonic Art forms is likely to be influenced by the changing environment that we inhabit, one that is increasingly virtual and global through the application of digital technology and is likely to alter our sense of place, space and time. In addition, the increasing use of 'intelligent' agent based machines through automation and decision-making, is likely to influence us, forcing a consideration of the machine role, and 'conscious' machine participation.

Accordingly, concerns with making sound art in real-time, with involving more 'intelligent' machines in the process (Whalley 2009), and with exploring the use of netspace as a performance medium (Weinberg 2005), are areas of interest for computer-based music researchers internationally.

A question when writing the work *Mittsu no Yugo* (Whalley 2010), was then how we might bring at least some of the historical New Zealand Sonic Art traditions together in real-time; and explore new netspace and intelligent machine technology that is now available.

3. THREE TECHNICAL LIMITATIONS AND THREE EARLY QUESTIONS

On the surface, established New Zealand Sonic Art traditions are not immediately suited to our current new real-time digital environment.

Much New Zealand electroacoustic music, while continuing through the application of digital technology and extended through multi-channel diffusion techniques, is still largely based on an editing approach to production. While this is partly mitigated by the 'instrument and tape genre', 'on the fly' digital music generation remains somewhat neglected here. Many of the New Zealand works that use MAX/MSP (see Rowe 1994) as a performance medium in combination with a live instrument at least addresses part of the real-time medium, but without addressing the reactive role of the machine. This is in contrast to Paine's (2002) *conversational model* of interaction as an alternative to machine participation. The question to address here was then how to expand electroacoustic music as a performance idiom to involve the use of machine

intelligence that might improvise in a realistically conversational manner.

The adoption of extensive digital editing techniques to create virtual performances also typified the later work of Nunns and Melbourne. Despite the improvised origins of the material for the second album *Te Hekenga-ä-rangi* made in 2003 for example, digital technology was often used to edit and manipulate sound *after* performances, rather than being integral to performance. As such, the work increasingly drew on electroacoustic music approaches where the result is a hybrid of traditional music and new technology that could not be recreated in live performance. In this sense, *Te Hekenga-ä-rangi* (2003) departs significantly from *Te Ku Te Whe* (1994) in that the production and compositional approach used multiple instrumental parts, and extensive multi-track layering, mixing and effects. The second technical question was then one of how this new hybrid genre might also be realised in real-time.

Finally, the New Zealand 'soundculture' tradition is at least one that has a strong history of real-time composition through improvisation; but in this country, it has not been one that has extensively adopted digital technology as a means of extending sounds, or using effects as a means of manipulating sound after the fact in real-time - particularly in work based on 'instruments' taken from the natural environment. The final question here was then one of how one might extend the idiom through real-time effects processing beyond using simple reactive MAX/MSP patches.

4. SOME EARLY GENRE EXPERIMENTS WITH MÄORI INSTRUMENTS AND EA MUSIC

The central notion of combining New Zealand Sonic Art traditions, and expanding real-time performance with increasing machine intelligence, involved a few byways before arriving at a real-time work in netspace.

My interest in blending different electroacoustic music and Mäori instruments came about after hearing *Te Ku Te Whe,* and then subsequently persuading Melbourne/Nunns to put together a longer and more layered work than appeared on their first album, for *New Zealand Sonic Art III* (2002). Combining improvisation and digital editing, the result was *Te Hau Kuri (Dog's Breath),* the first track on the CD. The short audio sequences for this work were recorded onto a stereo track, with both players improvising using similar instruments panned L/R for most of the sound bites. This provided the source material. Melbourne composed the structure based on a legend he chose from the area he came from. I subsequently produced the track, suggesting further layering to create the sense of a larger ensemble. Apart from pan, EQ, compression and volume alterations that were added to get the ensemble to fit together, the recorded segments are left intact.

Subsequent to this, I took an editing and layering approach to create the work *Kasumi* (Whalley 2002) based on using similar production techniques of gathering short improvised materials to begin. *Kasumi* used samples from seven New Zealand Mäori instruments played by Richard Nunns. However, the source material was also manipulated extensively to create new 'instruments' and sonic textures. In addition, the recorded Mäori instrumental fragments were also loaded into a sampler. Extensive pitch and real-time sound transformation techniques were then added, and the altered sounds blended and contrasted with the original material to create the structure of the work. The sampler also allowed the instruments to be performed in ways that were not physically possible, such as two octave transpositions and extended pitch bending. This overcame the melodic limitations of many of the instruments that are often confined to four notes in one octave. Together with effects manipulations, the result was types of hyper-instruments. Further, the work explored an Asia/Pacific connection integrating Japanese aesthetics. The manipulated Japanese text for this piece is from The Manyoshu, number 4290 by Otomo no Yakamochi (718-785) written on 23 Feburary, 753.

From these first steps, the focus changed to ways in which one might recreate the sense of layering a real-time performance with Mäori instruments, extended through electroacoustic generation and manipulation techniques.

PiWeCS: a Public Space Interactive Web-based Composition System was one of the first outcomes (Whalley 2004b). For this, Richard Nunns provided a range of short samples from a number of instruments. The technical/compositional infrastructure involved creating a web-based system server/client to produce the work. Since access to leading players of the instruments was limited, and many of the instruments unique, the intention was to find ways to overcome barriers of time and scarcity to allow lay people to discover and explore some of the instrumental sounds in a public space, independent of passive reception through live performance and the CD medium; to allow them to explore the combination of some of these instruments in a compositional and interactive and real-time context; and to allow the possibility of extending the instruments in an electroacoustic context as a way of exploring a new hybrid sonic landscape unique to New Zealand.

A further experimental work was the HIEMPA project (Whalley 2008) - Hybrid Instruments from Electroacoustic Manipulation and Models of Pütorino and Aquascape. This project combined a team of people with technical, artistic, environmental and cultural expertise toward an artistic outcome of aiming to extend the New Zealand sonic art tradition. It involved collecting audio samples from the aquascape of the Ruakuri Caves and Nature Reserve in Waitomo, South Waikato, New Zealand, and samples of a variety of

Pütorino – a New Zealand Mäori wind instrument. Following a machine learning analysis of the audio material and an analysis of the performance material, hybrid digital instruments were built and mapped to suitable hardware triggers. The new instruments were playable in real-time, along with the electroacoustic manipulation of Pütorino performances. The project took into account the environmental and cultural significance of the source material, with the results to be released in a series of compositions. Sadly, we were funded for the background research for the project's first stage, but not the production/performance second stage, and so the full CD output remains to be completed. One of the outcomes covering part of the project can be found in *For Morehu Te Whare* (Whalley 2010). In this, two short samples of Mäori musical instruments, the pütorino iti (flute) and the ku (mouth bow) heard in the opening, were played by Richard Nunns. Apart from a short water sample taken from the cave, most of the material is electroacoustically derived and developed.

5. TECHNICAL: 'INTELLIGENT' MACHINES

Perhaps the most useful technical development made as part of the PIWeCS project addressed the need to extend machine intelligence in performance to make something more interactive, in line with Paine's (2002) *conversational model* of interaction. The machine implementation of PIWeCS was achieved through integrating multi-agent programming with MAX/MSP on a server. Human input was through web interface using then FlashMX to build an augmented mixing board. The system developed allowed a limited conversation to be initiated and continued by participants through arranging samples, and building larger compositional structures over time (Whalley 2004b). It also allowed the extension of a composition through the electroacoustic manipulation of source material. The Java based multi-agent system, scripted with compositional knowledge as an 'expert system', acted as a guide and mentor for participants. In addition, the system accumulated user preferences and patterns, allowing the nature of machine/human interaction to be unique to each session. It also allowed the exchange and sharing of ideas and the relationship between users and machine to deepen over time, in line with Paine's model.

6. GENERATIVE IMPROVISATION

The use of intelligent agent systems in music production and performance is of course now widespread and implemented through a variety of aesthetic approaches: a field recently surveyed in *Software agents in music and sound art research/ creative work: current state and a possible direction* (Whalley 2009). In this, I point to how composers, musicians and computer scientists have begun to use agent-based systems to create music and sound art in both linear and non-linear (non-predetermined form and/or content) idioms, with some robust approaches now drawing on various disciplines and industries apart from music. Further, the paper puts forward a theoretical framework for the use of intelligent agent technology in creating music/sound art works, and an overview of common methods of implementation is given.

Software based intelligent agent systems can exhibit varying degrees of persistence, independence, communication and collaboration with other software agents or people. Intelligent agents have the capability to make decisions, the capacity to learn within an environment, and the ability to be mobile over computer networks. More advanced intelligent agents monitor environments, can glean information, make decisions to react or not, and modify their behaviour according to the results received (see Whalley 2009). Software platforms such as *Grasshopper*, *Jade* and *Zeus* for the development of such system, then make the technology well suited to real-time decision making in multi-causal and dynamic situations. A Multi-Agent Systems (MAS) is one where a computational system allows agents to cooperate or compete with others to achieve some individual or collective task(s). A Distributed Artificial Intelligence (DAI) approach is a co-operative system where agents act together to solve a problem, and can include agents being used in decision support systems where the outcome is based on interactions between human and agent input.

Of the four common methods in applying agent based technology to creating form and/or content for sonic art rather than notation-based music, the most prevalent one is perhaps a generative approach reflecting process-based rather than goal-based paradigms of musical creation (Chadabe 1996). A *generative approach* often involves applying biological or evolutionary concepts (also see Dorin 2001), allowing the machine technology to create content and even structure based on the dynamic interplay of parts. This is a non-linear perspective where the composer creates a range of software-agent behaviours, and there will be many possible outcomes and durations of works that result. Process then becomes the artistic focus, rather than a set product. These systems are generally technically self-contained once set in motion, and the language used either self-referential or allowed to evolve (Whalley, 2009). Some recent deployments of intelligent agent technology can develop musical styles, through evolving a machine personality based on internal communication of agents.

But the most significant intelligent agent implementation for human/machine interaction, extending the generative method, lies in a *generative/improvisation approach* to machine agency (Whalley 2009). This allows for real-time improvised human input, but rather than the machine agent simply reacting, there is a human *and* machine adaptation (autonomy and learning) to the generative input of the machine as well as the input of human agency. In theoretical terms, if the adaptive

relationship between human and machine demonstrates balanced listening/dialogue, and if little of the final output is prescribed, a full *conversational model* of interaction results – one where intelligent machines become an equal partner in the creative process.

However, most implementations of the *conversational model* implemented through generative improvisation prescribe some of the machine sonic language in the first instance – an aesthetic judgement to start the creative process. This approach was adopted in the PiWECS system (Whalley 2004b).

Following PiWECS, a further early trial of the generative improvisation method using agent technology was developed with better sensor inputs to capture human gestures, and then applied to make several variations of a soundscape combining Mäori and electroacoustic manipulations. Aspects of the variations were used in the work *Pukera* (Nunns & Whalley 2004). Here, short audio samples were record by Nunns. The sampled fragments appear as recorded at times in the structure, but most are also processed and developed electronically and then blended to make the structure of the work, providing a dramatic dynamic between processed and natural sounds. While the soundscape work is self-contained, for one of the live performances, Nunns used one instrument to improvise a part along with the score. The small Pütorino (wind instrument), played with a variety of techniques, enhanced the texture and melodic shape of the work in real-time (Nunns & Whalley 2004).

Apart from the use of agent-based technology in combination with Mäori instruments, a recent attempt at a generative improvisation approach to making a conversational model of interaction was also explored toward making a real-time model of aspects of Smalley's (1997) view of gesture and spectra. GIIMP - Generative Improvisation and Interactive Music Project (Whalley 2010d) was the outcome. This was realized in MAX/MSP alone, coupled with an additive synthesiser. GIIMP implemented a machine/human generative improvisation system using human gesture input, machine gesture capture, and a gesture mutation module in conjunction with a flocking patch, mapped through microtonal/spectral techniques to sound. The intention of this system was partly to technically simplify some of the previous more complicated agent-based music systems developed, meld some established and current practices, and combine aspects of symbolic and sub-symbolic approaches toward musical outcomes.

7. MITTSU NO YUGO

High-speed Internet2 is being explored as a composition/performance space by an increasing number of electroacoustic music practitioners, and the subject of an upcoming edition of *Organised Sound* (17:1) that I am co-editing. Conceptually, Internet2 technology allows for electroacoustic studios and live performance venues to link across long distances internationally to create new art forms, communicating through data streams as well as high quality audio (Whalley 2010c).

Through the byways of various projects presented in earlier sections, netspace was subsequently explored in the recent international work *Mittsu no Yugo* (12'00) (Whalley 2010b). This was played and 'recomposed' in real-time over high-speed research networks using the new IPv6 format.

The composition was one of a number of ongoing projects with our international partners toward technically exploring sound-based composition/ performance and machine agent approaches suited to Internet2 music making; while also considering and exploring the juxtaposition and synthesis of regional aesthetics in the creation of new compositions.

Mittsu no Yugo as a performance rendition was made in real-time between Hamilton, Beijing and Calgary, and performed live for an audience at the MUSICACOUSTICA10 festival in Beijing.

The Hamilton node had the greatest telepresence in the project - the footprint of the biggest action,. In Hamilton, performers were Ian Whalley (Max/MSP patches, wind synthesizer, pedals), Lara Hall (violin and looper), Hannah Gilmour (spectral beds, rhythm and effects). Richard Nunns' samples were again drawn on and manipulated in the performance. In Canada, David Larsen played Buffalo drum; and in Beijing, Bruce Gremo played shakuhachi.

Aesthetically, the idea was to first explore different locations reflected in distinct instrument timbres and gestures. But the work also attempts to meld aspects of the sonic input from the three different cultural nodes through aspects of real-time conversation and negotiation. As a result, one can sometimes identify individual parts in the work, but sometimes it is difficult to discern who contributed what in the texture. This is partly because many of the real instruments acted as both pitch/duration *and* textural/spectral sound makers; but also because some of the instrumental sonic sources could either be digitally manipulated directly, or manipulated through adding real-time effects.

Net-wise, the technical process involved linking multiple bi-directional audio channels between the three countries, streaming high-quality audio. Independent of this, we linked three video channels using digital video conferencing units. This allowed performers to both see and hear each other live. Apart from the video conferencing units at each node, multiple Skype channels allowed technicians at each station to also type/chat 'off-stage', in case any technical problem arose during the performance.

Small delays with audio latency, shorter than video latency, were compensated for through the way the score was written as a performance guide, allowing for the entry of different parts to be rhythmically imprecise, so that musical timing was not absolutely critical. Accordingly, the work partly adopts a non-realistic paradigm of a Recording Studio/Remixer approach (Carot, Rebelo Renaud: 2007) where latency is accepted and the composition built around its limitations.

In approaching netspace as a composition decision arena from a production perspective, it seems at first blush to be similar to other interactive music decision spaces involving machine technology (Whalley 2009). One still has to make choices between a Process or Goal approach to composition, select some sort of starting sonic language and associated musical/sonic knowledge paradigm, and decide on what the role of the machine is to be in terms of autonomy. But the decision space is complicated by the need to also accommodate some new things that netspace presents.

In theoretically discussing net-based music, Follmer (2005) outlines a three-dimensional schema for NMP (networked music performance). This matrix includes the parameters of Interplay with Network Characteristics, Interactivity and Openness, Complexity and Flexibility. In his subsequent analysis of diverse music practices, Follmer identifies hypermusic (music from hybrid instruments), real-virtual/space sound installations, and algorithmic/generative installations as being best suited to telemusic space. This analysis seemed to resonate well with some of the prior experiences that I have outlined above in developing machine-based real-time electroacoustic music performance systems.

Limiting Follmer's analysis however is its basis in a compositional and machine perspective. Apart from this, a consideration of netspace needs to include communicative issues such as the phenomenology of sound making, music/sound as a means to explore real/virtual space, and social interaction in music making. Complicating this further, players in netspace might consider the phenomenology of wider audience reception, given a worldwide audience through internet broadcast, and they might also consider the reaction of an audience at any real node that takes part in a telemusic event.

To navigate the increasingly multi-dimensional decision space from a compositional perspective, the work on *Mittsu no Yugo* proceeded on the communicative principle of moving from the known to the unknown in sonic terms. This involved combining sonic streams where some parts were familiar to some participants and audience, and some aspects were unfamiliar. The framework for this approach (see Whalley 2010e) considers gesture, sound, language and player semiotics. In the work, it was achieved through layering and balancing three distinct sonic art practices: traditions distinctly associated with the body and 'instrumental'/extensions; environmental and organic sounds and gestures; and abstract gestures and sounds. Combining these approaches resulted in a work that was texturally diverse, yet would have some familiar points of reference for the instrumentalists to start a conversation from, and points of reference for subsequent audiences.

Both Follmer's analysis and the accommodation of a wider communicative dimension lend themselves well to a generative improvisational approach to making aspects of form/content, involving dialogue between players and players/machines. The social aspect of music making between players, for example, involved negotiation and relationship building in real-time that resulted in better sonic outputs as the rehearsals of the work progressed.

From a technical compositional perspective, I began from a goal-based approach in the first instance, because the work had to be played as part of a concert, and this dictated a time length. Accordingly, I wrote a graphic score for the work that loosely prescribed the macro structure, gestures within sections, and broad motives for the work. In addition, I prescribed the sectional tunings and parts of the spectra (see Sethares 2005) that each larger section might adhere to. The key here however, is the flexibility of the structure and the order of any material with each. Participants could make their own decisions within it and the structure could also be changed in real-time in terms of stretching and compressing sections.

Content was then largely made through a process-based approach, and while performers could prepare some of the content they might use, the work was generated interactively in real-time in the dialogue between machine and human participants.



Figure 1. Hall, Gilmour and Whalley at the Waikato node of *Mittsu no Yugo*

As a consequence, the work could arrive at different outcomes with each performance. Moreover, the compositional approach was flexible enough allow for additional performers and instruments to take part and negotiate new outcomes. It is then worthwhile noting that the traditional role of the electroacoustic music composer in netspace starts to shift to one of concept designer, director and participant.

Machine participation, in line with prior work, was first implemented through the real-time manipulation of some of the pre-prepared sample material, allowing for the development of motives using electroacoustic manipulation techniques.

Also adopting previous techniques that had been developed, intelligent machine participation involved a MAX/MSP agent-based system that I built for the Hamilton node. This could generate data and respond to human input, allowing human/machine dialogue in a gesture conversational manner. The output was sonically realised through a combination of resyntheisis and additive synthesis. This sound-making method was used to maintain an organic sensibility that would sometimes integrate with the acoustic instruments being playing in the work, but also contrast with them electroacoustically at times. An area to explore in the future is implementing the system at different nodes, allowing both people/machine dialogue at each node, and machine/machine dialogue between nodes.

Paine's (2002) notion of a conversational model of interaction was then adopted in *Mittsu no Yugo* to include human/machine dialogue. In addition to this, the machine/human relationship also includes reactive machines that some players might use to extend manipulate sounds in real-time or add effects after the fact. But the best implementation of the conversational model however was perhaps implemented in the real-time dialogue between performers as they made the work through netspace.

Work in the medium is ongoing, and an enhanced performance system will be used in an upcoming work between Hamilton, Singapore and Beijing for the upcoming MUSICACOUSTICAII festival.

8. CONCLUSION

The increasing influence of applying digital and Internet2 technology allows composers and musicians to rethink older sonic art forms. As more countries begin to explore the combination of these technologies, we are likely to partly preserve and value the distinct sonic nature of different nations' nodal points on the netspace grid – the sounds and gestures that identify us.

At the same time, as our sense of identity is confronted by others in this new digital space, we are likely to use our established regional sound arts as a means to take part in global conversations, through extending older organic music practices. In tandem, decisions may also need to be made about how to accommodate increasingly intelligent machine technology through intelligent agent technology.

Some interesting research questions then arise. Are we likely to develop broader regionalized aesthetics and undermine national ones? Should machines be subservient to regional aesthetics? And what sort of unique identity might be developed by just machine/machine conversations, or machine/human conversations? Many of these issues are of course far from new, and are likely to have as many answers as there are individual stories that engage with broader narratives of culture and history.

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Presentations of Research

SONIC ART: SHIFTING PARADIGMS IN ECOLOGICAL CRISIS

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ABSTRACT

Today's global ecological crisis challenges creative practitioners to devise new processes that can contribute to cultural change. There is an urgent need to listen to our environment and reconnect remnants of self, culture and place. 'Sound' is undoubtedly one of the most powerful means to stimulate this shift in consciousness.

The dramatic advancement of digital media and information technology has truly cultivated a paradigm shift in how we collaborate as artists today. These changes have evolved and expanded the tools of expression but most importantly they have opened the ability to communicate at a higher level in an interdisciplinary context. Sonic art is integral to this process and has a profound ability to ignite an awareness and connection to the environment.

This research seeks to identify the role of sonic art in the current ecological emergency through a series of case studies and experiments. Sonic art, particularly acousmatic music, is a powerful medium to inspire change, but can this genre extend beyond purely expression? What are the implications of organic soundscapes designed to stimulate growth and converse with ecosystems? What issues arise when ritual sounds of the pasts are recontextualised and augmented with technology?

This paper presents case studies outlining the impact of three contrasting projects recently implemented in regional Australia and New Zealand. These include; *Sonic Babylon*, a global project planting interactive gardens of sound; *Blue Gold*, a hybrid performance installation by Ros Bandt exploring water as the commodity of the 21st Century, and the sonic art initiatives of the Remnant/Emergency Artlab, a team of artists forging new processes targeting the roots of our collective ecological crisis.

_DERIVATIONS: AN AUTONOMOUS SYSTEM FOR RESPONSIVE LIVE MUSIC IMPROVISATION

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ABSTRACT

Much recent research in the area of interactive computer music is concerned with the creation of computer systems that respond dynamically to musical input provided by improvising instrumentalists. These systems have the capacity to participate both responsively and autonomously in a live performance context, acting as virtual improvisers imbued with an awareness of the current and past musical context provided by one or more human improvisers. Various systems of this type strive for novelty and spontaneity in their responses, but also aim to coherently integrate computer-produced sounds with those of the musician. This research examines practical and aesthetic motivations behind systems that derive their sonic and algorithmic responses directly from an instrumentalist's live performance. The current iteration of the author's *_derivations* system is used as a case study to examine the design approach and contextualize this work in relation to earlier works by the author and relevant recent work of others in this field, particularly works that focus on *timbre* and sonority derived from the live instrument, rather than pattern or rule-based melodic invention.

Interactive systems have been described in terms of their ability to meet the musical aesthetics of their system designers (Drummond 2009). The design of sonic responses within performative systems is also dependant upon the musical and sonic context envisaged for the live interaction. In the *Omax-Ofon* system (Assayag, Bloch and Chemillier 2006), interaction is achieved by presenting the instrumentalist with a clone of himself, with the system referencing a continuously updated repository time-stamped and descriptor-analysed recordings of the performance using pattern matching techniques. In Eldridge's *Fond Punctions* (Eldridge 2005, 2008), performer triggered live sampling and generatively (by Neural Oscillator Networks) controlled granular synthesis techniques create 'timbral reflections' of the sampled improvisations. In Hsu's work with improvisatory systems, the focus is on the analysis and storage of timbral characteristics of the sound produced by a human improviser, and how this data can control and influence the behaviour modes of the software agents interacting with the performer (Hsu 2005, 2006, 2008, 2009).

In designing system response in the current iteration of the author's work, *_derivations*, a central aesthetic concern was to create a unified timbral palette that blends computationally generated sound with the live instrumental sound, i.e. the human and machine must cohabit in the same, unified sound world. The system response is designed to augment and transform the sonic environment in which it is operating, organically feeding off the environment provided by the performer's musical gestures. To achieve this, the system has access to all that has previously been played by the instrumentalist to derive its sonic responses. *_derivations* employs a combination of continuous live sampling of the instrumental performance, as well the analysis and storage of timbral information captured from the live signal. In the current design, three synthesis modules – a granulator, a bank of four phase vocoders and an additive re-synthesis module - reference phrase-analysed segments of recorded audio and spectral data to generate their sonic material. Current transformation techniques are based on quasi-aleatoric processes devised by the author. It is suggested that the unique combination of analysis, storage and transformation techniques present in the current design provides a rich and complex environment for human-machine interaction, and fertile ground for ongoing research.

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CULTURE VS. NATURE?

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ABSTRACT

We are living in a world reaching a critical point where the equilibrium between a healthy climate environment, the energy our society needs to maintain or improve this lifestyle and the interconnected economies could pass more quickly than expected from the current complex balance to a complete new reality where unbalance would be the rule and human beings would need to be as creative as never before to survive.

Have the arts a role in all this? Have artists a responsibility in this context?

While listening to scientists, politicians, engineers, economists, philosophers, sociologists, technology specialists and policy-makers reflecting and debating about the fast changes of our environment seems to be something almost natural today, artists voices are increasingly taking a prominent role around these matters too.

It is remarkable how different societies react to this problem. Similar situations are seen under a different perspective in different countries, regions or cities. While people have mid- and long-term confidence on "the system" in some places, there is a general disbelief in others. The same could be observed in the artists' concerns regarding environmental issues. These are not the same in all latitudes.

The arts could play a major part in helping the global society to understand the magnitude of the crisis we are facing, and in promoting the awareness around environmental matters. And it could also be a very good vehicle to disseminate proposals that could produce changes in our behavior and decisions, influencing our chances for the future. Artists could promote inter and transdisciplinary actions focusing on the global environmental crisis and our responsibility regarding the turning point we are living in defining the future of life on Earth.

Electroacoustic music/Sound art and associated approaches could be a key point in the network of creative actions to help each of us find reasonable and peaceful strategies of survival (and empowerment). Biomusic, *musique concrète*, soundwalk practices, soundscapes, sound field recordings, sound ecology and soundmaps are just a few examples; the possibilities are endless if we think about music and sound-based artistic works related to other art forms as well as to a multitude of scientific fields.

Before we used to talk about the four basic elements of nature: water, earth, air and fire but now we are talking more about matters such as: climate change, biodiversity, water supply, air quality, food chain and even geo-engineering, among others.

The world is changing faster than we can - individually- perceive it.
Music and musicians have a lot to do with the surrounding world but we need to build the bridges, to open the communication channels and to develop the dialog strategies.
Electroacoustic music has many faces, and one could be to become a powerful tool of awareness and transformation, to the benefit of us all.

Culture distinguishes from nature but we enjoy the aesthetics of our wild environment. Do we need to confront culture vs. nature? Is that the only way we have learnt in our civilized world? Many of us have been living in an almost fully artificial environment most of our lives, forgetting that we only have one place, even if global, just one. We can move around but still we are contained inside the same big space.

The arts can be a catalyst to explore transdisciplinary intersections between nature, science, technology and society. As we move into an era of unprecedented ecological threats, the arts could be one of the keys to engender a deeper awareness of our critical situation and could also help to develop intellectual partnerships to work in the solving of our global environmental crisis.

Today, while writing this text, I was listening in the news about the ecological devastation, and the loss of species in the world's ocean, comparable to those great mass extinctions of

prehistory. Overfishing, pollution, acidification and more: what this has to do with music? What this has to do with us, musicians, artists/researchers, composers? If organic means something related or derived from living matter, from living organisms, produced by or found in plants or animals: What do we expect of our future? What do we do for our future? And, Are we, musicians, part of that future?

Life and live music. The beauty of nature and the beauty of culture. The physical forces and the intellectual mind. We need to search for the equilibrium between culture and nature if we want to have a future for the humankind, if we aim to find the organic in harmony with sophisticated sounds, if we hope to have organic sounds available to still be able to create live electroacoustic music!

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GIVING VOICE: FOUR EXPRESSIONS OF EXPERIENCE INTO MUSIC

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ABSTRACT

This paper presents details about the design of specific sound cues from four compositions, illustrating the artists' efforts in exteriorising into sound aspects of human experience that correlate with concepts of *the organic*. I approach my enquiry with awareness of the value that those experiences entail, and by responding to my listening with a reflective attitude and empathy for the expressions rendered in these pieces.

1. INTRODUCTION

The diverse manners in creating, performing and listening to electroacoustic music has been accompanied by a wealth of discourses from practitioners and audiences seeking to address the complex experiences that are provoked. These experiences range from the technical, logistical and architectural to the psychological, bodily, social or communal, to mention a few. The four cases presented here serve to distinguish the ways in which concepts of the organic are addressed in each composer's preferred discourse. The compositions are: *Speak Memory* (2010) by New Zealand composer John Cousins, *Abide with Me* (2009) by New Zealand composer John Coulter, *The Breaking of the Scream* (1999) by Argentinean composer José Halac, and my *Pomegranate Friends*, a recent cycle of works for mixed choir, saxophone and live electronics. These composers' concepts of "the numinous", "the essence", "syncretism", and "enfolded experience" are shown to correspond to attributes of "the organic" by means of a listening process that attends to sounds both in regard to their measurable physical properties and their psychologically implied realities.

2. THE BREAKING OF THE SCREAM

This composition was released on a CD from INNOVA in 1999 under the collective title of *Selections From Sonic Circuits VII –International Festival of Electroacoustic Music*. The piece is an expansion and elaboration of Halac's *Todo lo que nuestros ojos amaron ('All our eyes ever loved')*. My first encounter with the materials of this composition was from a few years before this release when I presented it during a music conference at Indiana University. Although that version was shorter, featuring a more direct sonic language, the most striking factor in the piece has remained, namely the tribulations of a human scream as it seeks to transform its manifestation of suffering into an expression of this through poetry. The tortuous arrival of the scream into the poetic expression is achieved through moments of confrontation between the composer's sustained scream against two other sound sources: the composer's singing of a *baguala* chant,[1] and the composer's reading of the source poem.

Figure 1 illustrates, against a grid representing the duration of the composition, how the element of the sustained scream goes from clear sound to increased distortion. As the scream element unfolds its organic, gradually paced transformations it moves to the background of the texture as the other two elements emerge. These two new elements follow a different design in regard to distortion. The chant is first heard with relative clarity, and reappears towards the end, first very distorted and subdued, and concluding with a sudden increase in dynamic level. The element of the poem reading follows a path somewhat as the reverse of the scream, from quite distorted to enhanced presence and proximity.

Whereas the first version of the piece was a brutal, raw outcry, the revisited version or over-composed work became more clearly an act of atonement through the syncretic acts that the composer had used. The now distant original work had a recognisable moment of initial shock caused by the opening scream, followed by an increase of tension by the distortion applied to the sound of the scream. After a relatively short middle section of harmonic and spectral expansion that seemed to attenuate the energy, the scream came back, leading into the final fast paced reading of the poem. The new *The Breaking of the Scream* achieves its function of atonement by deliberately extending the transitions from scream into poetry recitation by means of intersecting the scream with three other elements, where each of these intersections could be perceived as delayed seizing of the poetry. We first hear how the scream meets with its own spectral components. It then moves to a first encounter with the poetry (or a false entrance of the poetry), before we encounter the baguala chanting. All these encounters are nonetheless integral, self reflective of the sound of the voice of the composer, who is the organism that renders the scream, the chant, and the poem. These all are, at the most essential level, sounds that have been *lived*. Towards the end the piece is a clear silence that allows composer and listener some distance

[1] *Baguala* is a surviving archaic lyrical genre from the Argentinean Northwest region, disseminated and urbanized throughout the country during the 20th century.

from the experience. There is an interruption in the drama, and gradually from the silence something hints at emerging, we are uncertain whether this is spectra, scream, chant or a metamorphosis. The last phrase of the chant breaks in with a *subito fortissimo* carrying the scream on its shoulders, and as the chant comes to a close, the scream blends with, or becomes the voice that now holds the words of the poem.

In my experiencing of this work, it exemplifies the organic transformation that Halac's perspective on syncretism can achieve. In Halac's works, we can follow a process of amalgamation that acknowledges more than one point of origin, and at the same time gives him the leading role in integrating those origins. In the case of *The Breaking of the Scream*, we can perceive a choreography of gradual transformations applied to each distinct element, as embodied by the composer through learning and performing, to reach and expression of change that at the end affects all and each constituent part, including the composer himself. The magical transformations in the creative space become available to the composer to reconfigure the past experience and, with time, to lessen the impact of the conditions that led to the real screaming.

3. ABIDE WITH ME

To audiences of Anglo or British Commonwealth background, the title of this piece will bring immediate associations. Abide with Me is a popular hymn composed by William Henry Monk (1823-1889). Scottish Anglican priest Henry Francis Lyte (1793-1847) wrote the lyrics to this hymn a few weeks before he passed away. The text is a prayer asking God to dispel the fears affecting someone who sees death approaching. This hymn has become one of the best-known pieces played in memorial services, private and institutional. It is also a popular item during Evensong, an Anglican ecclesiastical service that takes place at the end of the day.

The music of this hymn is simple and calming, expressed in quadruple meter in Eb major, organised in two balanced periods over sixteen bars of homophonic texture. The opening phrases in each period memorably trace scales degrees 3-2-1, while the consequent phrases move higher in the scale to gradually wave downward seeking resolution, or reaching-over the initial pitches of the next antecedent phrase. The highest reaching point is on scale degree 6. Although these leaps (as the tune is repeated several times) give us some sense of expansion in regard to psychological association with space (high and low) or a possible shift of vantage point of an embodied persona (i.e. *this is me now* and *that was me then*), we should acknowledge that the melodic range does not reach an octave, perhaps because it cannot, or it does not want to, or does not need to. The inevitability of the organic processes set in motion in Lyte's body, as he laid down very ill, were taking due course as much as a tonal melody that will invariably reach the tonic, appropriately in its obligatory register.

I describe the music and text of this hymn because I feel they offer much correlation to how I experienced John Coulter's audiovisual work of same name. In various personal communications with the composer, we discussed the circumstances and inspiration behind this work. John, at that time, was coming to terms with understanding the untimely death of a close relative. It goes without saying that the acts of creation in such circumstances are potent affirmations of existence at the most essential level. On the two occasions that I listened to this composition, the "story" of the *rambling tone* remained with me. The most potent expression relates to a continuous tone of similar qualities as the human voice, warm and rich, a tone with continuous pitch fluctuation that appears to ramble around the space lost, seeking without aim, and gradually gaining vector, marked by an increase in perceived *mass* (my perception here constructed this feeling on the grounds of increased volume and acceleration in the fluctuations, as well as the narrowing of a frequency band that made for an apparent "zeroing on target" effect–the equivalent of a *Fortspinnung*. This gradual process is integral to the video, featuring in this section a rambling dot of yellow/red colour, with the screen increasingly filling up with other dots, reaching to a moment where all dots converge (Figure 2). I remember that, while listening to this moment in the music while watching at the video, I was somehow reassured in my role of as a spectator, not invited nor drawn into assuming any role within the sonic world presented to me. If anything, my imaginary role was to observe the imagined attempts of this rambling sound in the hope that it will stabilise itself into a direction, move towards its target, and when this would happen, this would then give both the rambling tone and myself a feeling of satisfaction. The quality of the experience was similar to the empathic, good-willing stance that the hymn required from me, as on one hand I wished for the tune to reach the octave, and on the other I had to accept the itinerary of sounds as they were.

Several months later, the audiovisual work *Abide with Me* was presented in concert at the 2010 Beijing Modern Music Festival (BMMF). The reactions of the 500-strong audience there were akin to mine, but multiplied by their exteriorisation. According to a recollection by the composer, members of the audience spontaneously started to cheer every so-called *attempt* of the rambling tone, time after time with increased enthusiasm in a remarkable display of empathy for the embodied tone and its predicament. Metaphorically, if the Scottish priest had been incapacitated to sing a hymn, he could have expressed the sentiments contained in his verses by whistling in contours like those which occurred in John's composition. I felt that the significant effort of the composer in maintaining a conceptual frame around essential attributes of existence (evident through the relative span of the composed section, and the

abundance and refinement of organised anamorphic associations) was leaving me at the end of the piece with this question: 'If the dots on the screen can do it, and my listening too can believe such outcome; will our souls also reach their aim?'

4. TWO HANDS OR WINGS, POMEGRANATE FRIENDS

I have adapted David Bohm's concept of "enfoldment" in the sense that experience can be viewed as fully contained in a movement of energy or waves (2005). Other processes at a different times, can make what is contained to become manifest, or to move into what he terms as "unfolded" or "explicate order". This notion is beautifully exemplified in his metaphor of a hologram in which 'the entire object is contained in each region of the hologram, enfolded as a pattern of waves, which can then be unfolded by shining light through it.' (Bohm 2005, 129). The 2005 reprint of Bohm's writings and interviews also contain a stimulating preface by Leroy Little Bear, a member of the Blackfoot Confederacy who also served as Director of the Harvard University Native American Program. He recognised an alignment between Bohm's ideas and the Blackfoot view of their language as process and action in synchronous correspondences with movement in flux, which in turn reflects their understanding of reality as a point of reference in the flux. I have referenced to the concepts of "enfoldment" and "unfoldment" before when describing consciousness and cognitive processes in my work *Moonless Etude* (Dirié 2009).

In the choral project discussed here, I was moderately aware of the elements that were condensing or enfolding into the music. These included being inspired by the writings of the Renaissance Jesuit priest Matteo Ricci (1552-1610), my involvement in creating a number of works for the Fusion vocal ensemble in Brisbane, and my desire to use a small-scale multichannel set up to contribute and perhaps influence the performance practices of chamber choirs in the region. Ricci's writings about memory and mnemonics had already influenced my other compositions, namely *The Dream of José Arcadio*, and *Hwatu, Decks & Steps*. The current project, however, developed from my encounter with Ricci's translations of popular sayings about friendship into Chinese (Ricci 2009). The overall composition took shape as a set of three moderately short works for chamber choir and 4-channel live electronics, two of which have been completed and premiered. I was experimenting at the time with various tunings, testing my physical and psychological responses to different situations using simple configurations of sinewaves. As the exploratory quality of listening to different tunings became very important as a situated practice, I decided to include a live performer with alto saxophone to mediate between the sinewaves and the choir, while the choir mediates between the text, the performative situation, and the architectural space. The shifting and clashing of different tunings are, for most audiences, imperceptible–although I hope they still impact as sensory stimulants. I believe that different tunings do affect our bodies and emotions in ways that our perception does not regard as deserving of detailed processing.

Despite these considerations about tunings, voices, instruments, text, and the gradual and organic integration of creative impulses and conditions, there is no *singular sound* in these pieces that condense an experience in the same manner as seen in the compositions by Halac, Cousins, or Coulter. In these instances, we see that a major effort went into shaping a sound organically and in reference to a personal experience. Retrospectively, I see that in my two choral works, the effort was applied instead to an overall musical event that presents itself under the guise of alternating simple, friendly situations. It became increasingly apparent that a metaphor of belonging and not belonging was achieved by means of prompting the performers and audiences to transiting through the given changes, hopefully weighing the possibilities of personal transformations as a result of this process. This was done, however, without a rigid imposition of method to the music, but rather in a more Cage-*esque* attitude of simulated detachment from outcomes.

The improvising saxophone takes a role of attempting to highlight the tuning peculiarities of each context, sometimes serving to integrate sections, yet always risking alienation. Conversely, the singers' music, with its feel of normalcy, order, and gentle bearing, remains rather foreign to the other environment of natural tuning and to the more organic or barely intentional sounding of pitches from the saxophonist. These situations contributed to an enhanced experience of personal and social space that allow for different and coexisting tunings.

Comparatively, the process of living organically through this sonic metaphor also corresponds to the concept of "sounding", through which we probe an environment by means of inflicting a stimulus and evaluating its response. My role during the performance of *Two Hands or Wings* and *Pomegranate Friends* was as laptop musician controlling the patches, yet I was mentally and emotionally shifting vantage points as chorister, saxophonist, audience, venue, city, and imagined country. The creative project became a subtle mechanism to condense, update my experiences, and to calibrate my sense of presence in an environment whose ways of resonating I still have to comprehend, and hopefully enhance or affect. All the elements in the performance remain in state of flux, while the peculiar characteristics of the work as *composed*, refer to a single moment or reality as a point in the flux.

5. SPEAK MEMORY

This composition was completed in the year 2010 at John Cousins' private 8-channel studio in Christchurch, and presented to audiences during the 2010 New Zealand Electroacoustic Symposium (NZEMS) in Auckland. The version I listened to consisted of a multichannel mix, diffused through a 24-speakers acousmonium at the Kenneth Myers Centre of the University of Auckland. *Speak Memory,* like other works by Cousins, generously and justifiably spans almost one hour in length at 57' 16".

During this span of time, the composer presents us with the voice of his mother reminiscing and recounting significant episodes in her life. The sonic language in the composition is of broad register, meaning that the approaches range from documentary style to richly transformed sonic scenes. The main narrating voice is combined with other sounds of various origin used to make the narrative more palpable, as well as to expand the boundaries of referenced realities. This may refer to what Mrs. Cousins narrates, or to what the composer adds as running commentary, among others.

Very soon into the composition, we feel certain that we are not listening about a factual sequence of events, but rather we are invited, or thrown, into an enhanced virtual world that is more effective for explaining, or for sharing how it was to live through those stories. In the process of recording the voice of his mother, prompting her into relating various stories and details, and then transforming and expanding the expressivity of the sounds, the composer included himself as another narrating voice and persona. As Mrs. Cousins memory speaks, John's memories resonate with the recreated stories. The audience is given the double bill of following at least two narratives. There is a particular segment in the composition that stayed with me long after the first listening. This is a section I informally call the "conception cue". It occurs at about 18'10" and lasts approximately up to duration 30'. Including the fireworks sounds that initiate the conception cue, the segment can be seen as occupying almost the entire middle third of the composition (Figure 4).

In the narration, we hear about Mrs. Cousins' date with Mr. Cousins senior as a soldier during a night of public outdoors celebration, and the ensuing multichannel fireworks puts the audience into a state of gasping awe about the amorous encounter. Besides the references to the encounter and the celebratory scenery, we are also presented with a new and very quiet sound, its presence emerges very high in the frequency bands amidst the exciting banging of the fireworks. The fireworks, I think, operate both as an obvious metaphor of the experience of the amorous encounter, and also as a cue for us, the respectful listeners who are graciously prompted to look the other way to avoid any indiscretion or impropriety, so to speak. As the story then moves on and out of the fireworks, we can still hear that high pitch hovering, very innocently and cosy above all other textural elements.

In my two experiences of listening to this piece, first siting in the centre of the circle in the acousmonium, and then standing on the side of ring and observing the audience, I quite vividly felt that I was witnessing the presence of life emerging, condensed in that high pitch. I felt that all the elements in the texture and the narrations that continued were done as an effective representation of the unawareness of the life that was suggested during the fireworks episode. The high pitch, smoothly sustained, seemed not to hold any particular interest in what was occurring in other parts of the composition, and it seemed quite content in not been heard. I felt for a moment that I might have spotted something that I was not supposed to have noticed. My short conversation with other audience members afterwards more or less confirmed that I might have simply heard things in my head. A few days after these comments had been written, I returned to the "conception cue" and produced a spectrogram, noticing that where I thought there was one high tone, there were actually two, with the cleverness of effective organic forms, parenting a soft tone through a harmonious relationship of F^7-C^9 (Figure 5). Each of these pitches, in turn, were not single discrete frequencies but very dense bands, roughly covering all frequencies from the lowest to the highest within the frequency bins for each tone (more or less spanning 5400-5600Hz, and 16400-16600Hz). One could either zoom into each of these bins and find more frequencies, or imagine that the frequencies in each bin were not originally sounded in the visible bin, but were the resonant outcomes of encounters of other frequencies taking place beyond the grasp of the spectrogram. The voice of the composer appears lower in the texture when his narration questions how emotions and intentions factor in the encounter and generation of life. The textural overlap thus comprises several surrogates of John Cousins, to the point where the notion of surrogacy itself becomes insufficient to address such imbricated encapsulations, and a wonderful pondering about the self and the emergence of life from a distant big bang. My conversations with the composer, later on, made me aware of his interest and effort in creating sounds as "recorded experience" and as manifestations of spiritual and supernatural forces; or creating to enhance the "numinosity" of sound, as he prefers to call it (Cousins 2010).

In his reflections about the benefits of organising energy over time, Henri Lefebvre stated that 'it brings compensation for the miseries of everydayness, for its deficiencies and failures' (Lefebvre 2004). In a more positive tone, the manners in which we integrate human experience into the design of sound, their rhythmic dispositions of the situations in which they unfold, help us to restore a sense of balance between conflicting perceptions and concepts, perhaps achieving a behaviour that feels organic.

6. FIGURES

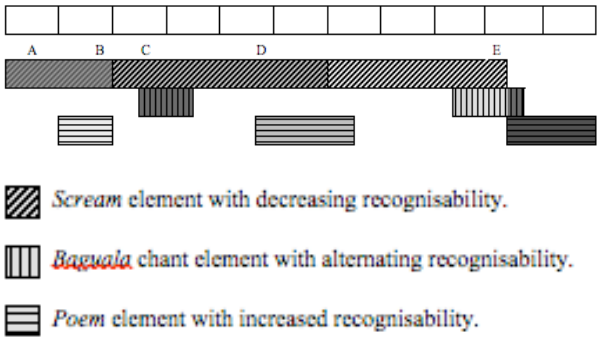


Figure 1. *The Breaking of The Scream* –relative distribution of syncretic elements throughout the 11 minutes of duration of the composition.

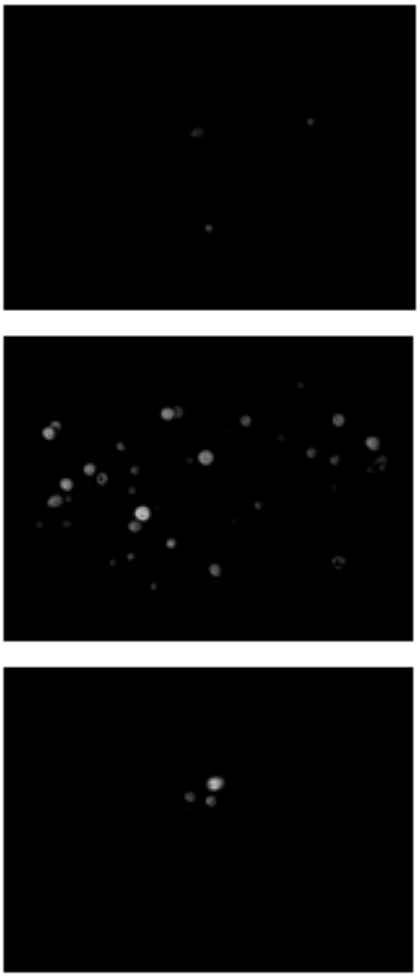


Figure 2. *Abide with Me* –three moments in the gradual eventfulness in the screen and gradual convergence of dots. (Screenshots from video selection provided by the composer).



Figure 3. The premiere of *Pomegranate Friends* – layered photos illustrating the notion of enhanced experience of personal and social space. (Photo by Katherine Chan).

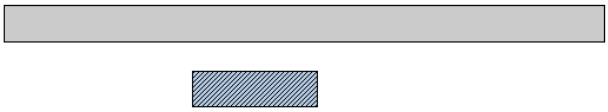


Figure 4. *Speak Memory* –position and span of the "conception cue" relative to the 57'16" of total duration.

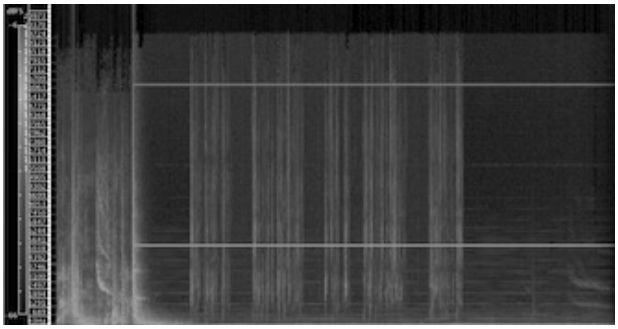


Figure 5. *Speak Memory* –spectrogram of the "conception cue" revealing the relative positioning of its component frequencies. (Produced through Sonic Visualiser).

7. ACKNOWLEDGEMENTS

I would like to thank the composers who kindly contributed with their works for this article, and to composer and ethnomusicologist Dr. Nicholas Ng for his attentive reading and helpful comments.

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E-SULING: EXTENDED TECHNIQUES FOR INDONESIAN PERFORMANCE

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ABSTRACT

The paper describes the E-Suling, an Indonesian hyper-instrument modified with sensors and a microchip. We describe the software and hardware designs in a variety of different languages and frameworks. We also discuss how the E-Suling is used in composition and performance to extend new techniques to traditional Indonesian music.

1. INTRODUCTION

The proliferation and reduced cost of data transmitting electronic sensors along with the free and shared nature of an open source programming environment has fuelled a healthy and creative culture of user specific projects, in which musical applications are just one of a number of areas where innovative design is taking place. Within this area, the emergent field of computational ethnomusicology (Tzanetakis, Kapur, et. al., 2007) provides the opportunity to capture raw numerical data through instrument-specific interface design, representing performance information in the form of a highly detailed fieldwork transcription for a variety of cultures around the world.

Indonesian gamelan percussion ensembles provide rich testing grounds for computational ethnomusicology, in particular the early 20th century Balinese *gong kebyar* style that proliferated and still dominates on the island to a great extent. It is explosively fast and frenetic, yet tightly organised. This strictly composed and largely improvisation-free art form consists of many new and/or variant compositions that have no written notation. Taught orally and through demonstration to the group many of these compositions are never played again and are quickly forgotten, I Wayan Gde Yudane explains, *'Most recent pieces disappear, because they are played by non-permanent groups that are formed solely for a competition (Body, 2002).* Even in Balinese gamelan works containing a western notated score one classically trained musician will dictate the musical information orally to the group and in some instances instrument specific melodic passages will be fleshed out from a prescribed note sequence in a way unique to the instrument and player/s, such as the sharing of idiophonic melodic sequences on the reyong. These patterns may not be played the same way by a different group, and if the reyong players are of a high calibre the composer may find it difficult or impolite to directly infer his performance technique opinions. With the use of custom technology geared towards Indonesian music, we can start addressing some of these issues with modern data capture techniques.



Figure 1. Gangsa players performing a Legong in Ubud, Gianyar. Feb' 2011.

This paper presents the E-Suling, a traditional Indonesian instrument that has been modified with sensors and a microchip. Section 2 describes related work on computational ethnomusicology and hyper-instrument design. Section 3 describes the designing of the E-Suling, with software, hardware and infrastructure. Section 4 describes how the E-Suling was used in composition and performance environments. We conclude with future work and how our team plans to contribute to the field of computational ethnomusicology.

2. RELATED WORK

This section describes a non-exhaustive history on artists who have combined electronics with Indonesian music, as well as a brief discussion on hyper instruments.

The Indonesian gamelan has influenced numerous music scholars around the planet, leading to varied and innovative directions using these unique instruments as a creative foundry. The integration of electronics and contemporary art music aesthetics into gamelan related composition is a natural progression for these ensembles, especially for those of us who are not intrinsically connected to the use of gamelan as part of our own socio-cultural and religious heritage.

Columbian composer Alba Fernanda Triana completed the creation of the Gamelan Electronico in 2007[1]. A Max/MSP programmed table with six light bars that react to the shadows of the interactive participant's hand motion, providing continuous control of parameters assigned to important elements of the compositional structure. The performance of the work is endlessly variable, based on the participant who takes on a dual role as performer and audience simultaneously. The outcome hinging on the balance between expectation and investment from said participant.

The Brooklyn based LEMUR gave birth to the Gamelatron[2] in 2008. Using robotics from other instruments and incorporating them into a mix of Javanese and Balinese gamelan, compositions performed by a fully robotic MIDI controlled orchestra were created and concerts are regularly performed. New versions of the Gamelatron are in development and this research will undoubtedly be of great value in those interested in reversing the information flow process to capture data from virtuosic and uniquely human performances, rendering a descriptive score that can be stored for future analysis.

The KarmetiK Machine Orchestra (Kapur, et. al., 2010) operates, a group of composers with specialisations in custom interface and robotic design. Works by the group include hyper-extension of instruments from Indian music as well as pieces where a human gamelan ensemble performs with a robotic *reyong*[3] playing traditional Indonesian music.

The advent of the hyper-instrument began with Tod Machover's Hyper Cello (Machover, 1992). With an invaluable performative collaboration with Yo Yo Ma at MIT the practise of adding sensors to acoustic instruments was pioneered.

At the 2003 New Musical Interfaces for Musical Expression (NIME) conference Cleo Palacio-Quintin presented her Hyper-Flute, a construction conceptually akin to my own E-Suling, 'A standard Boehm flute (the model used is a Powell 2100, made in Boston) extended via electronic sensors that link it to a computer, enabling control of digital sound processing parameters while performing (Palacio-Quintin, 2003).'

These examples have shown differing yet related ways that people and instruments can interface with new technology, audience/participant; prescribed MIDI performance; and hyper-extension of human performance.

The Gamelan Electronico's requirement of interactivity from anybody who wishes to participate rewards swiftly by transferring simple hand gesture into changes in process, a creative 'meeting in the middle' between composer and listener/interactor.

The robot percussion ensemble's automaton-like performances show promise, however pre-programmed MIDI score control alone through velocity insensitive hammer action lacks a musical expressiveness that is still easily surpassed by human control with even rudimentary training. Perhaps robotic actuators will soon become more sensitively designed at an accessible cost to allow for more expressive development in this area.

Hyper-instruments are most relevant in both topical areas of this paper due to the concept of data capture and output to interfaces and software, allowing for mapping and control from real-time human expression in live effects processing, sound manipulation, or meaningful data collection.

3. ETHICS AND CULTURAL CONSIDERATIONS

The music of Indonesia is intrinsically connected to its cultural and religious beliefs and practices, the depth of which probably cannot be fully understood by non-indigenous people. However, the good-natured spirit of community and the eagerness to share this music internationally has seen gamelan ensembles spread throughout the world during the 20th century and today, Budi Putra lecturer in Central Javanese gamelan at Victoria University, Wellington, New Zealand explains, '*...gamelan does not belong to the people of Indonesia any more, gamelan is now anywhere, spread anywhere (Putra 2002).'*

With regard to traditional Balinese *gong kebyar* form it is highly unusual, possibly even unheard of to encounter non-indigenous ensembles that can match the power, speed and virtuosity of performance that takes place on the island itself, therefore, after a certain amount of time acquainting oneself performatively with works from the traditional Balinese canon the first author opines that, imitation is the sincerest form of flattery, but where cultures and technologies collide and cross pollinate, the creation of new hybrid forms offer the potential for genuinely fruitful results. These offshoots should run in parallel to conventional forms and are not intended as a replacement of the traditional socio-cultural context in which they are situated. In *The Tempest in Bali* David George describes the use of Balinese *wayang*[4] theatrical style in a fresh representation of the Shakespeare play, firstly for

[1] www.albatriana.com
[2] http://gamelatron.com
[3] Balinese, a long row of metal gong 'pots'.
[4] In Indonesian *wayang* is used to describe theatrical performance, it literally means *shadow.*

western audiences and then presented back to Balinese audiences, *'...for instead of those performances saying: "look how another culture might present one of our plays," the offer now became: "look what we have learned from you and made of it (George 1989).'*

4. THE E-SULING

4.1. The Flute

The *suling* is a traditional Indonesian end-blown flute that varies in size and scale/mode across the archipelago, the regional style version 1 of my self-constructed flute originates from is central Java and is tuned to the seven note pelog scale. The flute was built from scratch due the unavailability of *suling* instruments in New Zealand and the unwillingness to potentially damage existing flutes during the modification process. Two *suling* bought in Java made from PVC pipe and owned by the Wellington based Gamelan Padhang Moncar[1] group were used as a measurement and construction template.

This custom electronic flute is an attempt to extend the traditional techniques of the instrument into the realms of live audio capture and / or effects processing for the accomplished player looking to experiment. The intent is primarily for the virtuosic player to have a flexible controller that can be easily assigned to the parameters of favoured digital audio workstations and effects, keeping the hands on the instrument and focusing on fluid performability and spontaneous sound manipulation. Attached sensors such as the movement tracking triple axis accelerometer and a vibration measuring piezo that can be applied to other gamelan instruments in a manner useful for fieldwork data capture for transcription purposes.

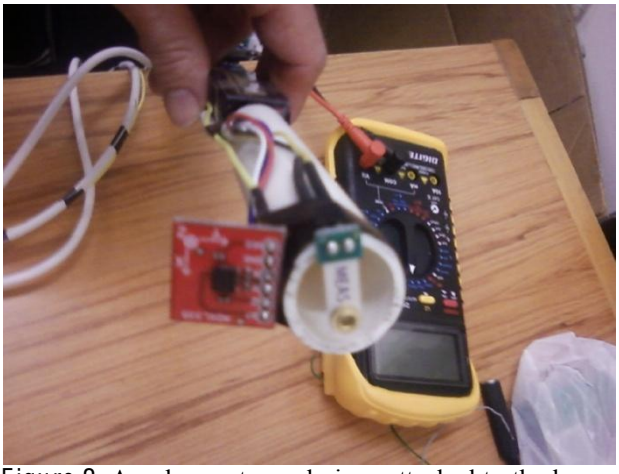


Figure 2. Accelerometer and piezo attached to the lower end of version 1 of the flute.

3.2 Hyper-extension

Modifications on the flute include an electret microphone with a stereo 3.5mm TRS jack; a small force resistant sensor; three assignable buttons; a 50mm softpot strip sensor; a piezo vibration sensor; and a triple axis accelerometer for tracking physical movement. All sensor data is captured by the Arduino Mega microcontroller[2].

Composition specific coding was used in a number of performances and some of the outcomes were:

1. Button triggering of pre-stored samples.
2. Variable speed playback of samples, forward and reverse using a twisting action with the accelerometer.
3. Microphone input gain continuous control via thumb pressure on the small FSR.
4. Return to sample 0 via pressure on the softpot sensor strip.
5. Sample playback volume via small FSR.
6. Pitch shifting value via piezo vibration sensor.
7. Changes in effects chain via button control.
8. Switching of composition movement specific coding via simultaneous dual-button presses, allowing program kill and start-up without touching the computer keyboard.

Using Processing,[3] a customised graphic user interface or GUI was created, in which to visually display incoming serial data from the arduino in the form of metered data displaying continuous control signals, and button position states. Processing then sent these messages via OSC (Wright et. al, 1997) to ChucK (Wang & Cook, 2003) , where it could be understood and interpreted as control change data.

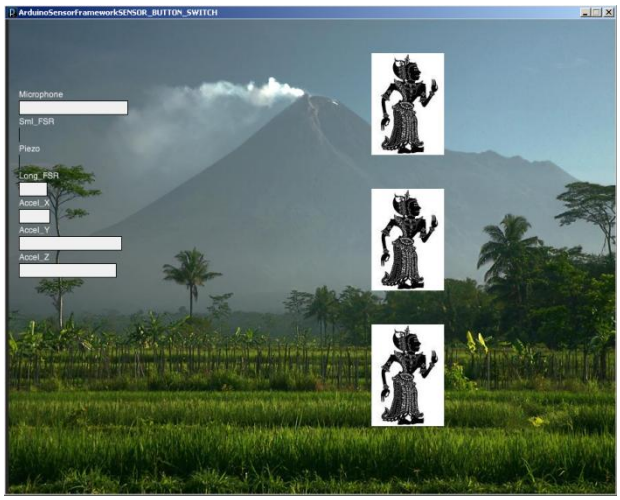


Figure 3. Graphic user interface rendered in Processing software to visualise incoming data.

3.3 Version 2.0 of the E-Suling

The first author was presented with a gift of a *gambuh suling*[4] during a trip to Bali in February of 2011 from his

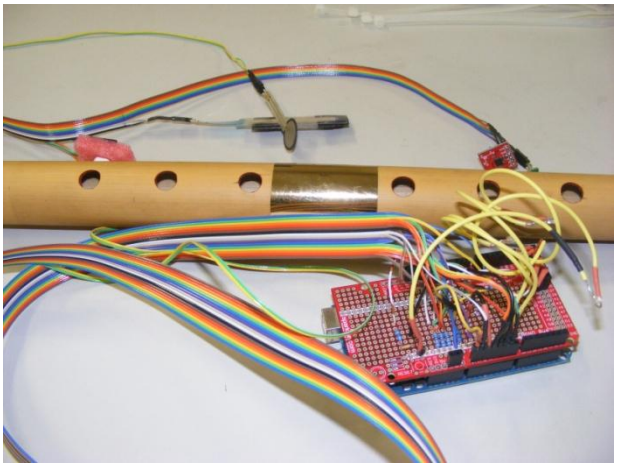
[1] www.gamelan.org.nz

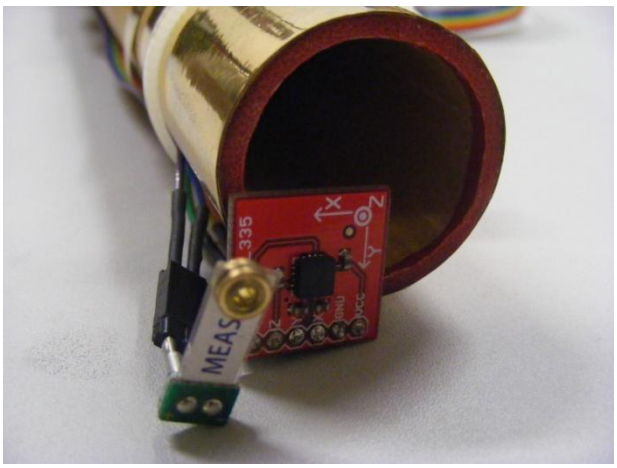
[2] www.arduino.cc

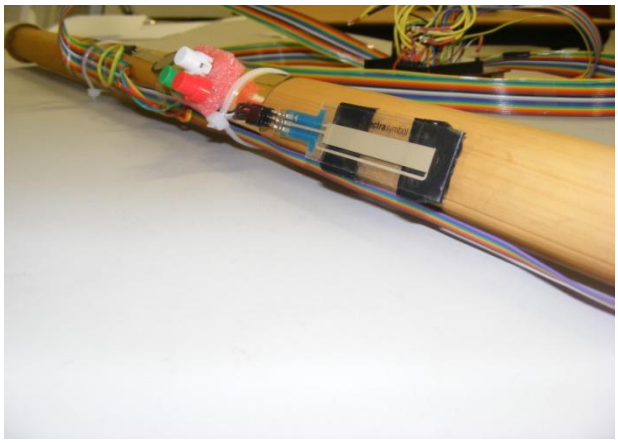
[3] http://processing.org

[4] Large Balinese bamboo flute nearly a meter in length

kendang[1] teacher, a larger and much more skilfully constructed flute than the first self-constructed Javanese prototype. A much wider tonal and dynamic range is accessible as a result of this to the skilled player, although some time must be spent by the first author on acquiring the circular breathing technique required to play the instrument with traditional authenticity. A full rewiring is currently taking place of all sensor and button components, along with a system of attachment that will cause no damage to this organic, handcrafted instrument. Improvements include a correction in the small force sensitive resistor circuit to provide a wider range of continuous control data, a correction to the 50mm membrane strip soft-pot circuit design and arduino code (Murphy et. al., 2010), and a rescaling of components to fit along the larger sized flute body.







Figures 4, 5, and 6. Version 2.0 of the E-Suling, new flute and wiring.

5. COMPOSITION / PERFORMANCE

5.1. Gendru Wo / Yamato Damashii

During mid-2010 the first author co-composed the work *Gendru Wo / Yamato Damashii* with fellow student Bridget Johnson for custom built interfaces, the E-Suling, and Johnson's Magnetic Tapes[2]. The work was a discourse between two idiosyncratic interfaces, influenced by the musical cultures of Indonesia and Japan. The work was structured through three set movements allowing room for improvisation in musical phrasing and gestural exchange.

The first movement involved triggering and manipulating sampled field recordings of organic material such as shaken tree branches and the shuffling of stones. By twisting the flute the playback rate was increased, decreased, and scrubbed back and forth. At slow rates branches sounded gigantic and small stones became boulders, creating the sound world that one could imagine the *Gendru wo*, a Javanese stone throwing, tree dwelling spirit would inhabit.

The second movement relied heavily on the incorporation of LiSa, and it's integration into the live sampling granular synthesis engine developed by Dan Truman, with some minor modifications of the first author's own such as randomisation of panning position in the stereo field of returning sample slices, and altering of the grain pitch shifting array to return tones suitable for the composition. Staccato flute notes divided and scattered, long blown, bended notes harmonised and spread out around the room, finger snaps fed back into the microphone via a nearby monitor speaker and unexpectedly took on the texture of flowing water.

The final movement utilised button presses to trigger subtly effected samples of the *demung*, a lower pitched foundational melody instrument in the Javanese gamelan. Improvised soloing on the flute was alternately completely dry and unamplified, and processed at will via a thumb pressure pad rounded out the work.

[1] Traditional Balinese drum.

[2] http://www.youtube.com/watch?v=SdNBaBlShDA

6. CONCLUSION AND FUTURE WORK

Now that two versions of the E-Suling have been designed and software has been implemented to drive compositions and performance that borrows from and expands Indonesian forms into an extended realm of sound, new compositions for the flute will be developed requiring fresh sound worlds. A Balinese gamelan composition currently underway that incorporates the E-Suling, a *jegogan* interface and a *kempli* beater interface will be performed in October this year. This composition also includes a process element using numbers extracted from the Balinese *pawukon* or permutational calendar. A tour to Indonesia is highly likely to occur next year and will include a trip to Bali where the first author will have the opportunity to present the work in the home culture of these extended instruments. Feedback from the Balinese will be collected while there and reported upon return.

The next step will look at how to address issues of the lack of the preservation of performances from an oral tradition outlined in the introduction and how computational ethnomusicology could assist with this. We plan to build new data capturing interfaces for the instruments for performance and transcription purposes. These interfaces will also allow for interaction with robotic gamelan systems such as the previously mentioned Machine Orchestra.

Controller connection to other digital audio workstations, such as Ableton Live will also help in electronic explorations of the flute's organic sound source.

7. ACKNOWLEDGEMENTS

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SYNCRETISM: ORGANIC FLOW OF ARTICULATED MORPHOLOGIES

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ABSTRACT

This paper presents some ideas on my compositional approach to a series of pieces I called "syncretisms". I will offer a view of the main operations involved and the conceptual particulars derived from the resulting sonority.

1. INTRODUCTION

To define "syncretism" I will use the definition used by composer Horacio Vaggione for the liner notes of my 1994 CD "Illegal Edge":

Syncretism: to merge everything together with the morphology of a (articulated) continuity. [i]

This paper focuses on electroacoustic compositions that include folk music or folk-indigenous music referential material particularly from NorthWest Argentina. However, the concept of fusion, hybridization and syncretism should not be restricted to referential materials in a particular medium. Hybridization is a cultural phenomenon present in many countries and without a doubt it generates discussions on the transformation of the inner texture of society and its inevitable effect on the art world of the 21st Century.

2. RELIGIOUS BACKGROUND

My first encounter with the word "syncretism" was in 1993 when a "santero" priest showed me a crucifix with Christ on one side and an image of Obatalá (Santeria deity) embedded on its back. He explained to me that the cross was the only way to survive the Church's imposition of faith in 16th Century Latin America. The cross was a syncretic object. It served both faiths and it bore an ambiguous yet pragmatic identity that satisfied the Church and the Santeros as well. The tension created by the ambiguity did not need to be solved (could not) and the symmetry of one priest on each side looking at what each one considered the "real" faith, gives the impression that the syncretic object, however complicated it may seem, is not complicated at all in that context. Actually, it makes a lot of sense. The constant flow of information (identities) around the cross, a loop of meanings, the pragmatic use of a symbol, an integration of alleged contradictions and the un-resolved tension between multiple discourses, made me realize that I was thinking in very similar terms when composing my pieces.

3. SYNCRETIC OBJECTS

The syncretic object is therefore a sonic entity that projects the view of the observer from his or her point of listening. When two or more elements are inseparable but identifiable parts of a context, their respective constitutive natures are re-signified when they enter in contact with the other elements of the group that they have come to relate to (in composition, by the decision of the composer). It is the composer who creates the context of belongingness and chooses which elements will interact and how they will behave. The composer must design methods of interaction between the elements that will be

understood by the listener as articulative markers, formal meaningful creators of sonic sense.
Any computer process such as filtering, phase vocoding, convolution, granulation affect a sound at different levels changing its nature, producing an output which represents its originality, its capacity to represent itself in its original existence, or its loss of its originality becoming convolved to an unrecognizable state, in which case it simply becomes something else.

A syncretic process involves therefore the recognition of a sound as an entity but also as a bearer of multiple identities some which may change while others remain the same. The result will be a structure whose balance principles are based on the composer's control of the nature of each element. What I call the "nature" of a sound may become its "organic quality" which the composer must define before entering in the structural game of construction.

The composer then chooses the level of intervention in the organic quality or nature of the sound which is the same as to say that the composer controls the level or referentiality that a sound will have in the composition. And that focus on control is what will determine that a composition is unique in its own syncretic existence.

When two sounds are actively operating in a context, energy is generated at many levels from their innate operative behaviour producing tension in the context. This tension is constant and doesn't resolve (as it does in a traditional tonal context) but is replaced by other dualities. This tensional dualism is created by the heterogeneous quality of the sources, which move in time and space with undeterred resolve. A syncretic composition therefore doesn't attempt to solve dualisms but to highlight them projecting the articulation as well as the sources. There is a sense of ambiguity in almost all the extension of these pieces which leaves the listener with a sense of continuous dis-continuity, a formal imbalance saved by the articulation proposed on each particular moment.

4. HYBRIDATION

This "hybrid" sensation is well described and explained by Argentine anthropologist Nestor Garcia Canclini who describes the "cultural hybridisation process" that Latin American countries go through today and how this may be a way out of the powerful mandates of modernist thought.

He uses the word "hybrid" to describe a culture that no longer can be considered as pure (following strict rules of Modernism) and that is defined by the level of fusion present at any given time and context.

"When one tries to understand transferences between borders and frontiers of countries, in the fluid networks that communicate peoples, ethnic groups and classes, what's popular, cultured, national and foreign appear not as entities but as stages. A stage….where a particular narrative is staged. It is important to include in that re-conversion, the re-staging, processes of hybridation through which the representations of society are elaborated with a dramatic sense.

To further clarify these distinctions, and to add yet another difficulty to the compositional process, I will cite a reference from this same book from Brazilian musicologist Jose Jorge de Carvalho: (referring to television, video-games, video clips, etc):

"*…they promise happiness coming from the cultural industry…they are basically the experience of the transitory: they help people, who live a fast and changing life, to get rid of the weight and responsibility of memory.*

Whereas in the popular and cultured arts they work within a tradition, where self-referentiality and constant commentary are essential in the establishment of a hermeneutic that's basic to their existence, contributing to the building of a collective memory."

In a syncretic composition, the clash of heterogeneities is confronted with yet another type of sound event: sounds with no history or attachment to traditions. Sounds that create a transient void that allows the piece to become temporarily "from nowhere". This void is essencial to provide a sense unlimited possibility. It embodies an abstract sonic self that neutralizes the imposed intentions of cultural attachments.

5. ARTICULATION

When confronted, two or more sound objects

share a musical space but connections are not yet implemented. The composer chooses paths of relations and these are called "articulations". The distance between sounds in any context is then bridged by these articulatory decisions which will give the piece its meaning, coherence and expressiveness. Articulations therefore are trajectories between sound objects that connect or transform them at multiple levels. Many articulations can be applied to a single sound event connecting it with one or more sound objects or with an electronic transformation process which will create yet another type of articulation with its new and old selves. In fig. 1, the chart shows a typical flow of a nodal syncretic context. A node may represent operations, meanings, sound typology, causality, sound parameters. Each connection represents a distinct articulation. One node may be articulated with many other nodes in different ways. Fig. 2 shows two sound parameters from different sound objects related by two articulations: high speed speech and expansion with phase vocoder x 2. The range of articulative choices is unlimited. From convolution and phase vocoding to rhythmic counterpoint, sound objects are subject to all sort of manipulations that will change or reinforce its identity by comparison to other sounds in the context. Sounds are "victims" of an "assault" (see later under "Patrimony") on its integrity, and its "nature" is challenged by the composer's action on it. An articulation becomes a game of dualities and the new "syncretic" context is the arena where the challenge takes place.

6. AMBIGUITY

In my first pieces which dealt with issues of a materiality centered on concrete folk sources, the issue of abstraction became immediately fundamental. The discussion of the Presence-Absence dynamic of the materials was for me challenging due to my own aesthetic belief regarding referentiality. My decisions, then, were that I would establish a mechanism which would control the dynamic of the Presence-Absence dualism of the folk material, and would relate it with the other synthesized, modified, instrumental sounds to build this "syncretic context" of relations in many different temporal and structural levels. When we deal with dualisms the issue of ambiguity is key. In composing syncretic pieces, this notion became more and more important especially due to the type of criticism I was receiving: Lack of directionality, gratuitous use of referential materials, inappropriate handling of sacred texts, indifference towards meaningful ancestral cultural heritage and on more positive reviews, a sense of conviviality of the heterogeneous, dissimilar sources which in a surprising manner were actually creating meaningful syntax, organically.

Ambiguity then was a force of organization since some sound events had a strong sense of directionality but others had not. Therefore, the temporality of the whole was affected profoundly since no single sense of real direction was present but instead there were many different senses of direction, including a non-directional, static sense of flow. Then a syncretic composition deals with the issue of ambiguity in the positive sense of the word and not the negative one. Instead of "it is neither this nor that" we go to "it is a bit of this and a bit of that".

7. ORGANIC FLOW

In a hybrid context, the sense of organic flow becomes fundamental to understand when materials stop delivering a "communal" meaningful discourse to replace it with an iconic representation of itself or, in words of Brian Ferneyhough " a tendentially absolutist iconic pretension…" or "strangely visible black holes, at the still center of their own burnt-out identity".[ii]

Form is an illusion of continuity created by many participants. It is an illusion because there is no such thing as a continuous indivisible texture that is heard as just one thing by everyone. There is always a detail in a micro, meso or macro level that will be detected and thus will cause an articulated break up in the whole. Time is continuous as long as there are single parts with their own temporality. This type of reception creates a sense of continuity of articulated actions. Organic flow does occur within the boundaries of articulation. Continuity is a creation both on the part of the composer and the listener.

8. INTEGRATION

I propose through this idea to contemplate the integration as a form of understanding each member of a group individually but at the same

time in its contribution to its group, however transient the group might be. An identity (that may last a few seconds or milliseconds) is projected just by being in contact with other sound impulses, spectra, voices, or any meaningful sound in the context. In a network-oriented syncretic structure, any sound can become functional and determinant of form and thus any meaning may be used to change the macro structure and the overall perception of the piece.

9. PATRIMONY

When materials belong to cultural contexts (most of them do), the use and transformation of these materials can be taken as an "appropriation" or "assault" to the patrimonial property. We may include here texts, repertoire of songs or chants, religious repertoire, speech recordings, traditional instruments, narrations of sacred texts etc. We should understand the sound in the context of the culture and its tradition and what it means to those who protect it, cultivate it or even adore it.

It may be useful to explain this word and concept to better understand the implications of dealing with musical sources that are created outside the borders of our own. Patrimony or cultural heritage is used here as a post-modernist construct. According to it, patrimony evolves: we must recognize the past so that present and future generations may re-interpret it. The subject looks at patrimony that evolves from the point of view of his/her historical context, this way the new perspectives enrich it. The post-modern notion then accepts a patrimony that evolves through interpretation and therefore from significance and symbolism, because when it is re-signified, it is also re-symbolized, that is, it acquires a different or new symbolic character. The post-modern view is the historical perspective of the subject that observes, filtered by his/her psychological and affective impression of the phenomenon.

10. SYNCRETISM: ASSAULTING IDENTITY

We begin with a profound understanding of the identity or multiple identities of a sound. Sound events have operative features that may be common to other sounds and thus we tend to group them analyzing and interpreting based on those features. Then, we work at the perceptual level and the sound becomes part of a complex process of creation and reception whereby the composer creates the symbolic sonic form.

The constant reflexion and reaction to the sound, transforming it, re-connecting it to other sounds can be called a creative "assault" to the identity of a source. This may mean to contradict the original purpose of it in its belonging to a different cultural musical or sonic context. We negotiate in a "neutral" ground, where artistic freedom is key, and changes occur. The consequences are these new syncretic contexts where new tensions and significances are attributed to the sounds and the old patrimonial ownership, has given way to a new "sonic habitat" for sounds to behave and operate differently. A case for lost identity and purpose may be raised but the new composition is justified on its own aesthetic purpose. Every sound is now a participant and actively creates articulative contexts where complexity, multiplicity, diversity and a sense of dynamic individuality complete an imbalanced and ambiguous yet highly articulated composition.

11. EXAMPLES

"INDIA VIEJA, sincretismo #1" Acousmatic (1991)*

I will now give an example of an electroacoustic syncretic piece and the reaction to it from a highly recognized Argentinean ethno-musicologist specialized in folk musics from the NorthWest of Argentina, Leda Valladares. In her very honest and generous letter from 1995, she comments on my first syncretic piece: "Old Indian Woman". She writes:

"I liked your work and the fact that my compilations on bagualas and vidalas served you as inspiration to deal with something cosmic and secret. I would prefer that the sonic textures were less dense and more incisive with respect to the voice, which in such nakedness and mystery has a lot to discover and study. The contrasts of sonorities distract from the magnetism and geneticism of the ancestral chanting, an endless reserve. To study it and to show its depths will be richer without mixing it with electronics, which sometimes invades you with pre-natal fear and paralyzes you. The ancestral chanting of the Third World is the largest reserve of music.

There you can find the mysteries to decipher, to de-frost…"

Leda Valladares is protecting the transcendental quality of the music I used, defending it to the point of denying its existence in contexts that may hurt the image she has of it, or the psychological energy she has vested in it. She believes in a pure and distinguishable status of existence of that music and anything that may undervalue it, will be deemed as impure and disposable disturbances.

"India Vieja, sincretismo #1" (Old Indian Woman) was composed at the Computer Music Center at Brooklyn College under the direction of Charles Dodge. In this piece I worked with a "baguala", a folk song, from the Coya Indian oral tradition from NorthWest Argentina. I processed the material generating textures descriptive of the song's images. One of them, the woman's desire to become young again by drinking condor's blood, was achieved slowing down her singing and drumming to depict old age and the condor's flight. This computer image intervenes in the sense of the poem and the force of the original singing style, creating new sound images absent in the original song but present now in the computer composition. The original "magic and mystery" that Valladares attributes to the style is replaced by a new hybrid sense due to my own intervention on the object which is no longer the same object but a new, syncretized one.

"UITOTOS, Sincretismo #2" (1992) 10 minutes. Acousmatic and string quartet*

In this composition my materials were bell-like synthesized sounds using pure sine tones filtered with comb / reson filters in the C-Sound software environment (using some of Jean Claude Risset's designs for bells), a recording of a cello sustained note and a recording of two native Uitoto priests reciting their version of the genesis in a magic/secret language in a hocket-type of counterpoint. One priest recited and the other replied but using the last word uttered by the first. The building of the piece centered in that rhythmic and textural qualities and the circularity of the overall sensation of the speech. The construction was about those qualities: circularity in the recurrence of events, small gestures against big ones, presence and absence of the voices, and slow speed gradually increasing at a similar pace with the eventual presence of the voices. Different trajectories (directional and non-directional), different speeds, rhythms, sonic qualities in an atmosphere of dissolution created by the reson filter quality, that would "give away" the voices very slowly. No interpretation or reference to the meaning of the text took place in this piece other than trying to assimilate the priests' ritualistic formulation. The parameters highlighted were others and the syncretic context was ruled by a different set of interpretative tools. The tape piece had a new life later as a tape and string quartet and later as a tape and violin piece.

"THE BREAKING OF THE SCREAM" (1999) 10'50". Acousmatic*

A folk song, from the oral tradition of the music of NorthWest Argentina, the "baguala" (Coya Indian song), talks about the desire of a man to compare himself with a tiger, making reference to the sexual power of the animal. The poem, by Argentine contemporary poet Pablo Anadón, is about a loved one leaving our life, and the existential sense of loneliness and abandonment that inundates our soul after the loss. The scream is a reference to both the sexuality of the folk song and the pain of abandonment of the poem, but it also connects to un-conscious and unknown fears and meanings.

In this "syncretic" environment, the elements reflect and exchange musical, acoustical and spectral qualities between one another, constantly adding new meanings to the multiple fusions that are formed in the different levels of the structure (music, poetry, rhythm, meanings, emotions, speech etc) . The words of the poem become screams, the main scream is granulated to create rhythmic textures, while the electronic sounds, derived from the voices and the reading of the poem, help create the environments where the elements will relate in multiple ways as sounds, as carriers of meaning, as creators of morphologies. In "The Breaking of the Scream" (which begins with a climactic and extreme human scream) there is an intentional exploration of the idea of excess whereby I constantly expand the dynamic range to give space to new peaks and climaxes.

CONCLUSION

In electroacoustic music, we may consider that

sounds carry a hybrid materiality that is abstracted and fused in our creative imagination to convey meaning many times in heterogeneous contexts of multiple sound events. Then we look into the sonic qualities of each sound we want to bring to our composing table, as a sort of musical anthropologist. We consider all the aspects of the sound, including the historical and traditional dimensions, as compositional.

Transformational operations (articulations) change the structures, and entities (traces) become fleeting (discreet) identities. Music becomes then a flexible structure that changes in time. We open its form to eventuality and ambiguity so that modulations can freely take place in the continuum between (as expressed by Vaggione) "autoreferential time embedded in referential sounds." A syncretic composer is liminal by nature, moving freely between borders, erasing and re-tracing them for purely aesthetic reasons.

ACKNOWLEDGEMENTS

I want to thank composer Horacio Vaggione, museologist Celina Hafford and Gerardo Dirié for their invaluable contribution to this presentation.

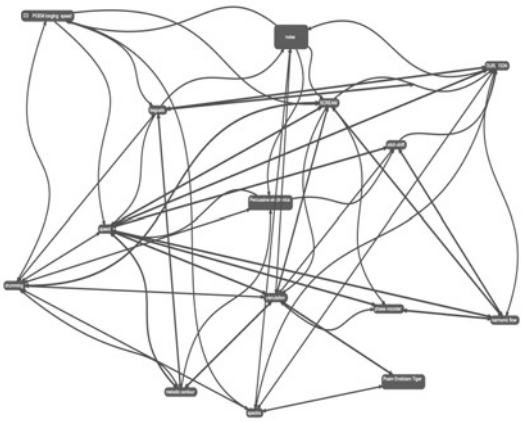


Fig. 1. *The Breaking of the Scream*. Nodal map of articulations between nodes: sounds, parameters, references, operations and meanings.

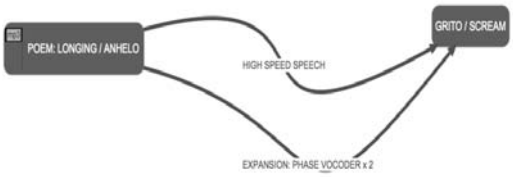


Fig. 2. *The Breaking of the Scream*. Dual articulation between two nodes: longing in the poem and the recorded scream. Two articulations are present: high speed speech contrasting with a time expansion process on the screaming voice. Emphasis on composer's own voice screaming and reciting at different speeds to provoke a dual sensation of longing and rage.

*** Pieces can be found at this link:**

https://www.myspace.com/halacjose

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[i] HALAC, José, *Illegal Edge*, Centaur Records, Liner notes, USA, 1994

[ii] FERNEYHOUGH, Brian, *Collected writings*, Chapter: "Form – Figure – Style: An intermediate assessment (1982)", Routledge, London, 1996, page 24.

MUSICAL PARAMETER MANIPULATION POSSIBILITIES OF A HOMEMADE REACTABLE

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ABSTRACT

Musical parameter control is an important part of live interactive electronic computer music. Due to the increasing availability and affordability of music technology, including powerful computer software, advances in this area are being made to enable easier and more effective parameter control.

The purpose of this paper is to investigate and discuss the musical parameter manipulation possibilities of a homemade instrument with a tangible tabletop interface based on the technology of the reacTable. The design and construction of the instrument is documented, including the physical build as well as the software component of the system, which incorporates the computer software *ReacTIVision*, *Max/MSP* and *Reason*. The core of the paper discusses parameter manipulation abilities by way of a comparison between two controllers: the homemade instrument and the *Korg nanoKONTROL*. Mapping strategies – in an interactive music sense – are explored in detail, while the execution and capabilities of parameter control by use of the physical interface devices of the two controllers are assessed.

1. HOMEMADE REACTABLE

Using instructional information found primarily on the *reacTIVision* (*reacTIVision 1.4* nd*)* website and in the paper entitled *ReacTIVision: A computer-vision frameworks for table-based tangible interaction* (Kaltenbrunner et al. 2007), and in conjunction with original creative ideas, a homemade reacTable was designed and constructed by Masters by Research student James Herrington (see Figure 1). The instrument, with a tabletop tangible user interface, incorporates multi-touch technology, and is based on the technology of the original reacTable (Jorda et al. 2005). It can be played by a single performer, or by multiple performers.

Like the *reacTable*, this instrument incorporates a clear tabletop with a camera placed beneath, which constantly examines the table surface, tracking the nature, position and orientation of the tangibles, or objects, that are placed, and moved around, on it. The tangibles display visual symbols, called *fiducials* (see Figure 2), which are recognised by the software. Each tangible is dedicated a function for generating or manipulating/controlling a sound.

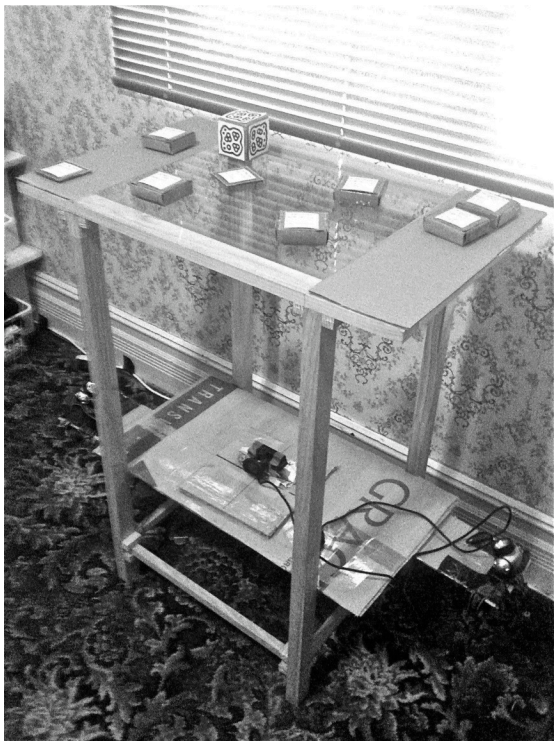


Figure 1: Homemade reacTable

Users interact by moving them around the tabletop, changing their position, their orientations, or their faces (in the case of, say, a cube object) (Jorda et al. 2005; Jorda et al. 2006).

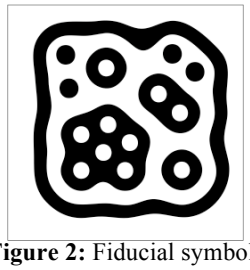


Figure 2: Fiducial symbol

This is where this instrument differs from the original *reacTable*. The vision captured by the camera is sent to the open source software *ReacTIVision*, and then to *Max/MSP*, which allows the instrument to work as a MIDI controller. This information is then sent to *Proppellerhead*'s *Reason*, where the final mapping is completed to allow note on/off events (determined by a tangible being placed and displaced in the camera's vision), along with the x-position, y-position, and orientation of each tangible assigned to manipulate different parameters of music[1].

In recent years, the availability of the previously mentioned software, and growing information on the subject, have resulted in a number of computer musicians and artists creating their own reacTables (or adaptations of them)[2]. At a tertiary level, researchers, such as the Computational Arts Research Group at the Queensland University of Technology, who developed the "Morph Table"[3] (Brown et al. 2007), are also working with the technology. While this use of the technology is utilized to develop an instrument with a set purpose of expanding certain performance and compositional techniques (i.e., morphing between musical patterns), my own homemade reacTable is set up to act as a basic MIDI controller, where the placement/displacement of any object on the table can be assigned any available note on/off function and the movement assigned to control any available parameter.

1.1 Basic physical design and build

The wooden frame structure of the homemade reacTable is based (as the name might suggest) on the shape and design of a table. The table stands 92cm high, at perfect mid-waist height. As it is intended to be performed while standing up, this gives the performer a "birds-eye" view of the tabletop, while relieving them from having to bend or sit down to move the objects around. The dimensions of the tabletop interface – clear Perspex – are 46cm (length) x 37.6cm (width). This provides the performer with quite a large area (1729.6cm^2) to move the objects around. As part of the design, on either side of the interface are two shelves (15cm x 46cm) intended for unused, or "standby", objects to rest on.

A *Sony PlayStation 3 Eye* webcam (*PlayStation Eye*. 2011) – with approx. dimensions of 84 x 67 x 57mm, and a video capture of 640 x 480 pixel – is placed 61cm directly beneath the tabletop, facing upwards in order to capture the vision of the objects being moved around. Two LED torches are placed on either side of the table, on the same level plane as the camera, but roughly 25cm to the left and right respectively of being directly underneath the tabletop interface. They are then angled to shine on the bottom side of the Perspex. This was necessary, as the visual fiducial symbols needed ample light in order to be properly recognised and sufficiently tracked when placed on the table surface. The camera is then able to constantly examine the interface, without any distracting light reflection – as the torches were strategically placed to the sides, rather than directly underneath the clear Perspex.

When it comes to the tangible objects of the homemade reacTable, they can be sorted into four categories of size. Objects can be of all shapes and sizes, and will work as long as they have an attached, and recognisable, fiducial. However, the four size groups used are as follows:

- Cube object – 7cm x 7cm x 7cm (Figure 3)
- Rectangular prism object – 7cm x 7cm x 2cm (Figure 4)
- Large flat object - 7cm x 7cm (Figure 5)
- Small flat object – 5cm x 5cm (Figure 6)

The first three listed objects – the larger objects – when placed on the tabletop interface, take up the same amount of surface area (49cm^2 – on which the fiducial is presented). The small flat object, however, takes up a lesser amount of surface area, with a smaller fiducial attached to a surface area of 25cm^2.



Figure 3: Cube object



Figure 4: Rectangular prism object



Figure 5: Large flat object



Figure 6: Small flat object

1.2 Computer software: *ReacTIVision*, *Max/MSP* and *Reason*

Regarding the computer aspect of the instrument, three software programs are used in conjunction with each other in order for vision to be captured, analysed and then interpreted into sound, or in other words, for the instrument to function. The three computer software programs, which act as the "engine room" of the instrument, are *ReacTIVision* (*reacTIVision 1.4* nd), *Max/MSP* (Cycling '74 2011) and *Reason* (Propellerhead 2011).

ReacTIVision, developed by Martin Kaltenbrunner and Ross Bencina, is the fundamental sensor

[1] Alternative software – and software combinations – can be used as opposed to *Max/MSP* and *Reason* in the set up of a homemade reacTable. This information can be found on the *reacTIVision* website.

[2] Several of these projects are documented anonymously online, including (http://www.youtube.com/watch?v=JuQo25KYELg) and (http://www.youtube.com/watch?v=15IE6d4zfME&feature=related).

[3] See (http://www.youtube.com/watch?v=nKXhfApKCms)

component of the homemade reacTable. The software is open source and can be found at the *ReacTIVision* website (*reacTIVision 1.4* nd), along with information about the internal structures and workings of the software, and instructions on how to set it up. As its function is the analysing of visual information captured by the camera placed beneath the tabletop, *ReacTIVision* does not contain any sound components. Instead, Tangible User Interface Object (TUIO) messages are sent to a TUIO-enabled client application: in the case of my instrument, this is *Max/MSP* (*reacTIVision 1.4* nd).

Max/MSP (version 5) acts as the client application in the instrument. Here, the fiducials' recognition, centre point and orientation information is processed and organised into four groups of numbers: note on/off (0 – 1), x-position (0 – 640), y-position (0 – 480) and angle (0 – 360) [The fiducials' recognition/derecognition relating to note on/off; centre point relating to x and y position; and orientation relating to angle, or rotation]. Using various techniques in *Max/MSP*, this information was organised in such a way that the zero point was located at the bottom, left hand corner of the table. For example, moving an object from left to right raises the value of the x-axis number, while moving an object from bottom to top raises the value of the y-axis number. The processing of information was also organised so that the value of the angle, or orientation, number rises when an object is rotated clockwise. These sets of numbers are then scaled to 0 – 127 in order to be sent as MIDI information to the computer software program *Reason*.

Reason (version 4) completes the process of interpreting object recognition and movement into sound generation and control. To sum up, *ReacTIVision* has analysed vision of objects and their placements, and sent this information to *Max/MSP* where it has been organised into sets of note on/off, x-position, y-position and orientation values and finally sent to *Reason*. *Reason* is where the mapping of these values to parameters of music occurs. An example would be if the y-position value of an object were assigned to the pitch shift parameter, therefore enabling the movement of this object from bottom to top of the table interface to raise the pitch of the sound produced.

2. MUSICAL PARAMETER MANIPULATION

One of the main advantages of the homemade reacTable is the number of musical parameter manipulation possibilities that can be achieved through the use of various mapping strategies. Mapping, in terms of interactive music systems, is the connection between the outputs of a gestural controller and the inputs of a sound generator. The method is typically used to link performer actions to the generation and control of musical sounds and parameters (Drummond 2009; Wanderley 2001; Winkler 1998).

2.1 Homemade reacTable mapping strategies

When it comes to the homemade reacTable, the mapping is the relationship between the movement of the tangible objects and the sounds produced. The relationships can be set up in an obvious, or not so obvious way, and ideally a balance between the two makes for a more interesting instrument. As discussed above, parameters can be mapped (or assigned) to the x-axis, y-axis and rotation of each object, while note on/off functions can be mapped to the recognition/ derecognition of objects on the table interface.

Throughout this paper, specific parameter mapping assignments of the homemade reacTable have not been discussed, other than in examples, because, discussing them in detail here (i.e., each object's function) is not important. It is only important to know that the placement of objects on and off the table surface can generate any accessible sound and activate any accessible effect, and that the x-axis, y-axis and angle movement of the object can manipulate any accessible parameter of the music.

There are four main mapping strategies that can be used in interactive music systems: *one-to-one*, which is the direct connection of an output to an input; *one-to-many*, which is the connection of a single output to multiple inputs; *many-to-one*, which is the connection of two or more outputs to control one input; and *many-to-many*, which is a combination of the different mapping types (Drummond 2009; Miranda and Wanderley 2006). This is certainly apparent in the homemade reacTable, where the output relates to one type of an object's movement (i.e., x-axis or y-axis or angle movement), and the input relates to any desired parameter of music to be manipulated.

A noteworthy feature of the objects is that, because of the way the instrument is set up – specifically the *Max/MSP* element – objects can continually rotate. That is so that the value assigned to the orientation of the object resets to 0 after a full rotation. This enables control of the parameter so that the value can switch straight from MIDI CC 127 to MIDI CC 0, or vice versa. The x-position, y-position and orientation of each object can be used to manipulate parameters of the same "parent" effect (e.g., x = reverb dry/wet amount, y = reverb decay, and angle = HF damp), or alternatively be used to manipulate completely different parameters (e.g., x = reverb dry/wet amount, y = pitch shift, and angle = panning).

2.2 Comparison

The parameter manipulation abilities of the homemade reacTable can be contrasted with the manipulation abilities of a commercial MIDI control device, the *Korg nanoKONTROL* (*Korg* 2011). The choice in comparing the *nanoKontrol* is because, as a general MIDI controller that sends MIDI information to MIDI-enabled devices, it incorporates the basic and universal note on/off functions, and also interface-controls that can access every MIDI CC value. Although new and

experimental MIDI interfaces are being developed all the time, the most common physical MIDI controllers contain pads, or keys, for note on/off functions, and knobs/faders for continuous signal control. This USB bus powered device offers nine faders, nine knobs and 18 switches, with four programmable scenes, along with a full transport section (controlling functions such as *start, stop, loop,* or *record* on the DAW software). It is a small controller with dimensions 320(W) x 82(D) x 29.5(H) mm, and a weight of 29g.

It is important to note that the assessment will only take into account the knobs and faders of the *nanoKONTROL*, which can access every MIDI CC value (0 – 127). Although they can transmit MIDI CC messages, the 18 switches on the device can only access, or trigger, two values. Attack and decay times can be assigned to the switches, and therefore access every value if set up correctly; however, this is a set function and the MIDI CC values cannot be continually controlled. When it comes to the homemade reacTable, the x, y and orientation position of each object (or more specifically, fiducial) can access every MIDI CC value.

In the comparison below, the parameter manipulating devices will be referred to as Physical Interface Controllers (PICs). In the case of the homemade reacTable, this term will refer to the tangible objects with attached fiducial symbols. In the case of the *nanoKONTROL*, this will refer to the knobs and faders of the device. The devices are compared below on four dimensions, in the form of questions relating to the execution and capabilities of parameter control. The two controllers are assessed as if performed as a solo instrument by a solo performer.

How many potential PICs can be used/assigned to manipulate parameters of the music?

Homemade reacTable:
In the setup of the homemade reacTable, the default 'amoeba' fiducial set[4] is used, which includes 216 distinct fiducials. Fiducial IDs 0 – 107 being a black image on white, while fiducial IDs 108 – 216 are the inverse, with the same images reversed (i.e., now a white image on black). Each of these fiducials can be tracked and therefore can be assigned to manipulate different parameters of music. This means that 216 potential PICs can be set up for use in the one instrument (*reacTIVision 1.4* nd).

nanoKONTROL:
The *nanoKONTROL*'s interface consists of nine knobs and nine faders with four programmable scenes. As each scenes memory allows the same settings to be retained, this results in 36 knobs and 36 faders that can be assigned to manipulate different parameters of music. Therefore, 72 potential PICs can be set up for use with the controller.

[4]Alternative sets of fiducials are also available to be used and are available from the *reacTIVision* website.

How many PICs can be used/played/controlled simultaneously?

This question requires two distinct responses. The first relates to the idea that multiple PICs can be accessed and easily moved between when interacting with the instrument, and also without the parameter assignment being changed or replaced mid-performance. The second response relates to the amount of PICs that can be controlled simultaneously under the physical limitation of the human performer.

Homemade reacTable:
Although 216 fiducials can be assigned to manipulate different parameters of music, the objects they are attached to cannot possibly fit on the tabletop interface of the homemade reacTable all at the one time.

In a practical experiment, multiple performances of the instrument using a different amount of objects were trialled, in ways utilizing alternative mapping strategies for each set of objects. Only larger objects were used, those with a surface area of 49cm^2 – which is placed on the interface. The kind of music being performed is not specified, nor the different parameters of music used in the mapping to the objects' movement, for this information is not relevant, as the experiment is to determine the *number* of objects that can be can be interacted with on the interface in a comfortable manner.

Firstly, parameters were only assigned to the objects' x-axis and orientation (i.e., no parameters were assigned to the y-axis of each object). Set up this way, the objects were lined up from top to bottom so that the full x-axis range of values could be realised (i.e., each object could be moved from the left most point of the table [MIDI CC 0] to the right most point of the table [MIDI CC 127]) without clashing with each other. Using this mapping strategy, four objects appeared to be an adequate number. Taking into account the size of the table and the size of the objects, there was comfortable room between the objects so they could all be rotated without interference while on the same x-axis point.

Secondly, the parameter-to-object mapping was set in the same way as mentioned above; however, only assigning parameters to the y-axes and orientations of the objects (rather than the x-axes and orientations). Once again, four objects was found to also be an adequate number. That is to say, when lining the objects from left to right, each object could access every y-axis value without clashing with other objects, and in fact more room – or empty space – was available between the objects. This is because the width of the tabletop interface is longer than the height.

Next, parameters were only assigned to the objects' orientation. Set up this way, the objects were lined up from bottom to top as well as from left to right in a grid fashion, with enough space between them so that the full rotation range of values could be accessed without clashing with each other. Using this strategy, 16

objects were found to be a satisfactory amount of objects on the interface. That is to say, the interface accommodated four rows of four objects.

Finally, parameters of music were assigned to the objects' x-axis, y-axis and orientation. With this mapping strategy employed, it is hard to give an exact number of how many objects allows for satisfactory performance. In a way, it depends on what type of piece is being performed. For example, an experimental free improvisation piece, where objects are moved around at free will, would accommodate more objects than a piece where one or two objects have to be moved at certain times to certain positions without interrupting the placement of other objects. If the piece is in fact structured in this way, the performer would need to pre-determine which objects need to move along entire x and y-axes, and arrange the objects accordingly when first placing them on the table. In saying this, nine objects (three rows of three) were found to be an adequate number where objects can move around freely enough.

When it comes to the number of PICs that can be used simultaneously (i.e., how many on the interface at the one time), it really depends on a variety of factors, such as the mapping strategies employed. Another factor, not discussed thus far, is the size of the objects. The examples above utilized objects with a surface area of 49cm^2. As mentioned, smaller sized objects can be used, which would enable more simultaneous PICs on the interface at the one time. Combinations of larger and smaller objects can also be used. It is also worth noting that when the performer needs to change or replace certain PICs, and thus the parameters to be manipulated, one object at a time can be replaced on the interface.

The second response to the original question involves the human performer's limitations in the physical controlling of the objects. Traditionally, or rather ideally, two objects can be controlled simultaneously by a solo human performer, that is, one in each hand[5]. We say 'ideally' because more than two objects *can* physically be controlled, however, when doing so, restrictions arise. For example, the performer could use his or her nose or teeth, like a modern day Jimi Hendrix, to control a third object, however – when compared to controlling objects with one's hands – this can hardly be achieved efficiently, as it would be awkward. A second example occurs when the performer moves two or more objects with the one hand. Once again, to avoid being extremely awkward, this can only take place if the two or more objects are to be executing the same control, that is, moving the multiple objects up, down, left or right simultaneously. Rotating the objects with the one hand would be difficult without also altering the x and y positions of the object.

[5]Multiple performers result in more objects being able to be controlled simultaneously. For example, two performers on the one instrument can control four objects, three can control six objects, and four can control eight objects, with each performer controlling the standard two objects. More performers can be added, but because of the size of the instrument, however, the space would become more and more cramped when consisting of more than four players. The square design of the instrument also neatly accommodates four players, with one performer on each side.

nanoKONTROL:

As realised in the previous question, 72 PICs can be set up from nine knobs and nine faders on the physical interface of the *nanoKONTROL*, however, only these nine knobs and nine faders (i.e., 18 PICs) can be used without altering the scene, and thus replacing the assigned parameter set of one scene with a completely different set of another scene. The act of switching between scenes directly opposes the idea of being able to easily access and move between parameters to manipulate. For example, it would not be possible to manipulate the parameter assigned to *knob 1* of the first scene and the parameter assigned to *knob 2* of the second scene simultaneously. This applies to any two separate parameters assigned to PICs on contradicting scenes. Therefore, only the 18 (physical) PICs can be controlled "simultaneously" when relating to the first response of the original question. Unlike the homemade reacTable, where one PIC can be changed or replaced at a time, the *nanoKONTROL* can only move between scenes, and therefore 18 PICs (or rather the parameters assigned to them) can only be changed or replaced by 18 PICs at a time.

As considered with the homemade reacTable, to answer the second part of the question involves discussing the human performance confinements in the physical controlling of the – in this case – knobs and faders. Like the homemade reacTable, the ideal number of faders and/or knobs to be controlled simultaneously is two – one with each hand[6]. The same "slight exceptions" examples also relate. That is to say, a performer *could* use his or her nose or teeth (or any other part of the body) to control a PIC, however, it would be awkward; while two or more faders (not so much knobs) can be controlled simultaneously, with enough amount of efficiency, with the one hand, although, only when performing the same control – in this case, being moved up or down.

How many parameters of music can be independently manipulated using the one PIC?

Homemade reacTable:

When it comes to the homemade reacTable, three parameters can be independently controlled using the one PIC. That is to say, a parameter of music each can

[6]Once again, multiple performers result in more faders and knobs being able to be controlled simultaneously, with each performer controlling two PICs. However, as opposed to the homemade reacTable, at a much smaller physical size, it would be extremely cramped with any more than two performers.

be assigned to the movement of the object on the x-axis of the tabletop interface, the movement on the y-axis, and the angle – or orientation – of the object. Each parameter can be manipulated separately. For example, an object can be moved from left to right on the table, controlling the parameter assigned to the x-axis, while maintaining the values, or settings, of the parameters assigned to the y-axis and angle of the object.

Alternatively, the parameters can be manipulated simultaneously at an independent rate. For example, moving the object in an oval-shaped manner, while continuously rotating the object. Each parameter is being manipulated at a different rate, while changes in the shape of the movement add to the independent parameter manipulation possibilities. Furthermore, changing the rotating speed of the object will affect the manipulation rate of the parameter assigned to the angle of the object independently of (i.e., without affecting) the parameters assigned to the x and y-axes[7].

nanoKONTROL:

Only one parameter of music can be independently controlled using only one PIC of the *nanoKONTROL*. Multiple parameters can be assigned to the one fader (or knob alternatively), for example, however, when the fader is controlled (i.e., raised and lowered), the value of each parameter is manipulated at the same rate.

Using various external software, one can alter the nature of how each of the parameters assigned to the one PIC is manipulated. Using a fader once again as an example, the minimum and maximum values can be reversed, so as the fader is physically raised, the value of the parameter is lowered. The minimum and maximum vales can also be restricted to a certain range, so as a fader is raised from the minimum position to maximum position on the physical instrument, the MIDI CC value of the parameter would raise, for example, from 10 – 80 (or whatever range the user has assigned) as opposed to 0 – 127[8]. However, even if both of these examples were allocated to two different parameters assigned to the one PIC, along with another parameter being manipulated in the traditional sense (minimum position to maximum position on the physical interface equates to 0 – 127 parameter value), the three parameters would all be manipulated at the same exponential rate when controlled by the one fader, that is, not independently of one another[9].

Are there any placement restrictions of the PICs?

Homemade reacTable:

Objects are placed and moved around the tabletop interface of the instrument in order to achieve their assigned parameter manipulation functions. In saying this, it is not physically possible for two objects to be in the same xy position on the table surface. Depending on the parameter assignments of each object, this means that certain combinations of audio effects are unachievable. Because of this limitation, a good mapping strategy technique would be to assign music parameters to the movement of the objects' orientation and only *one* of their axes (x or y). This is because an object can be rotated on the table surface without affecting its xy location, and therefore, without clashing with other objects.

Experimenting with the larger objects (49cm^2) on the homemade reacTable, it was found that two objects with the same y-position, and side-by-side as close as possible to the same x-position, could not access any values within MIDI CC 20 of each other on the x-axis. Two objects with the same x-position, as close as possible to the same y-position could not access values within MIDI CC 28 on the y-axis. Once again, this is due to the fact that the width of the interface is longer than the height. Using the smaller objects (25cm^2), two with the same y-position, could not access within MIDI CC 15 on the x-axis, while two with the same x-position could not access within MIDI CC 20 on the y-axis[10].

nanoKONTROL:

Unlike the homemade reacTable, where parameters are manipulated by moving each object around the one surface plane, the knobs and faders of the *nanoKONTROL* are allocated their own space. Because of this, there are no placement limitations of the PICs.

An example of the difference between the two controllers under assessment (the homemade reacTable and the *nanoKONTROL*) would be as follows:

[Using only x and y possibilities, and ignoring the rotation parameter control function for now] *Object 1* of the homemade reacTable controls the reverberation of the entire sound through the use of a reverb unit in *Reason*. The parameter assigned to the movement of the x-axis is the amount of decay, and the parameter

[7]Multiple fiducials can be placed on the one object, such as a larger object with four fiducials displayed on the one face, and played in a way where one hand is controlling the object. This raises the amount of multiple parameters controlled by one PIC, however, in doing so disregards the idea of independent parameter manipulation between the parameters assigned to the two or more fiducials on the one object. That is to say, however the object is moved around the tabletop interface, all fiducials on the object move in the same way.

[8]Using the *Korg Kontrol Editor* (*nanoKONTROL* 2011), one can edit settings on the *nanoKONTROL* itself (i.e., without using external software) to modify how each PIC manipulates its assigned parameters (as discussed above), although, three alternative ways cannot be designated to the one PIC using this method. That method can only be achieved by using external software, such as *Ableton Live* (*Ableton* 2010).

[9]*Korg* has released a controller, different to the *nanoKONTROL*, called the *nanoPAD* (*Korg* 2011) with an xy pad (where x and y parameters can be manipulated independently) controlled by finger touch. In this case, the user's finger would act as the PIC.

[10]The larger the interface surface area is, the smaller these number values would become

	Homemade reacTable	Korg nanoKONTROL
No. of potential PICs	216	72
No. of PICs that can be controlled simultaneously	*In the same space without being changed/replaced*: Larger Objects (49cm^2 surface area): X and ANG controlling parameters: 4 Y and ANG controlling parameters: 4 ANG controlling parameters: 16 X, Y and ANG controlling parameters: 9 *Under human limitations*: 2	*In the same space without being changed/replaced*: 18: 9 faders, 9 knobs on 1 scene *Under human limitations*: 2
No. of parameters that can be independently manipulated using one PIC	3	1
Placement restrictions	2 objects can not be in the same xy position on the interface	No restrictions

Table 1: Comparison of homemade reacTable and *Korg nanoKONTROL* across 4 dimensions

assigned to the movement of the y-axis is the dry/wet amount. Meanwhile, *object 2* controls the equalisation of the entire sound through the use of a Parametric EQ unit. The parameter assigned to the movement of the x-axis is the frequency value, and the parameter assigned to the movement of the y-axis is the gain value. If the performer wishes to achieve the audio effect of a reverberation with a decay amount of MIDI CC 50 and a dry/wet amount of MIDI CC 30, simultaneously with the equalisation emphasising the frequency at MIDI CC 50 at a gain of MIDI CC 30, this unfortunately would not be possible. This is because, in the case of both object placements, x = 50 and y = 30, and two objects cannot be in the same xy position. To overcome this problem, various mapping strategies can be employed, as discussed previously. One approach would be to switch the assigned x and y parameters of one of the objects, resulting in one object requiring x = 50 and y = 30, and the second object requiring x = 30 and y = 50 to achieve the desired audio combination effect. This method would not work, however, if all four parameters (the parameters assigned to x and y movement of both objects) required the same MIDI CC value, that is, if both objects required the values x = 50 and y = 50.

In the case of the *nanoKONTROL*, using the same parameters and intended values, the audio effect combination *can* be achieved. Using four faders (or knobs alternatively) *fader 1*, assigned to the reverb decay amount, can achieve the MIDI CC value of 50; *fader 2*, assigned to the reverb dry/wet amount can achieve the MIDI CC value of 30; *fader 3*, assigned to the frequency of the EQ, can achieve the MIDI CC value of 50; while *fader 4*, assigned to the gain of the EQ, can achieve the MIDI CC value of 30. These values can be achieved simultaneously, unlike on the homemade reacTable, however, two additional PICs must be used.

The potential of each controller in relation to the four questions is summarised in Table 1 above.

2.2.1 Other things to consider

The above assessment has compared the parameter manipulation abilities of two controllers – the homemade reacTable and the *Korg nanoKONTROL*. Although, if the two were to be compared as the better overall, or more useful, controller, various other aspects would need to be considered. This may include the lag or delay between the physical movement of a PIC and the assigned parameter value. The CPU usage of each would also need to be assessed, as would the restrictions due to size, and the ease of portability.

3. FUTURE WORK

Future work on parameter manipulation utilizing this type of technology could include setting up the homemade reacTable instrument in a way so that the distance between two objects acts as another value that can be calculated and assigned to an additional musical parameter to be controlled. This way, both objects would have to be present on the interface in order for the parameter to be altered. If three parameters each were additionally mapped to the x-axis, y-axis and orientation of the two objects, this would enable the performer to control seven parameters of music independently of each other using only two objects. For many pieces of music, this would be all the control the performer needs.

Although the lag and delay due to the quality of the webcam was not discussed throughout the paper, future work may also involve the use of a High Definition camera to track the fiducials more effectively.

For his own artistic endeavors, James Herrington intends to further this research to develop alternative ways to perform and compose contemporary electronic music, and use these extensive manipulation

possibilities to advance the homemade reacTable as a DJ instrument. He is also currently working with the instrument as a component in an integrated Dubstep Performance environment for his Masters by Research project.

4. CONCLUSION

Advances in music technology in conjunction with increasing accessibility and affordability have contributed to progress in the area of musical parameter manipulation in live electronic computer music. This is made apparent by comparing a homemade electronic instrument with a commercial controller, and showing that in many ways the homemade instrument possesses superior parameter manipulation abilities. Apart from the computer itself, the sum of components of the homemade instrument – including open source, and relatively inexpensive software – come at a reasonable price. The PICs of the homemade reacTable as tangible, freely moving objects generate the novel characteristics, and enable the excellent range of parameter manipulation, of the instrument. A greater scope of versatility and control of the musical output is produced, that, by comparison, can be restrictive when it comes to other electronic devices and controllers. By no means have all possibilities in parameter manipulation utilizing this technology been explored, which leaves the door open for further investigation and exciting advances.

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THE COMPOSER AND THE MACHINE: ORGANIC PROCESSES AND MUSICALITY IN COMPUTER PROGRAMMING FOR MUSIC.

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ABSTRACT

This paper contemplates different processes and results developed by 'programming composers' as compared to composers who use programmers to facilitate or realise compositional components of their works. Different models for the relationship between music composition and computer programming are examined, as are the outcomes for composers and performers.

For music programmers the compositional process varies according to the composer and the work they wish to create. Complex musical configurations involving sound synthesis, processing, aleatoric and improvisational approaches may be guided by conceptual ideas that do not always originate with programming skills, and can be outsourced within differing levels of collaboration. Gerald Strang's seminal 1970 essay 'Ethics and Esthetics of Computer Composition' asks if it is possible for a 'programming composer to apply similar kinds of aesthetic and analytical judgments as a composer who does not program [Strang 39]. This paper contends that things have changed, and if the act of music programming were thought of as 'musical' by all composers, it could be employed to further the timbral and structural palettes of music composition for all music. Using works of her own and her peers, and a discussion with a 'programming composer,' the author discusses some different ways to recognise musicality in computer programming.

1. INTRODUCTION

Since the introduction of electronic music, artists have been fascinated with the timbral possibilities the new 'instruments' electronics offered up. Composers have enjoyed the organisational and decision making possibilities of computers interactivity, sound synthesis, algorithmic and generative composition systems. Despite the number of ways computers can be used to make music, computers in music composition are usually employed as part of one of two processes: algorithmic composition or timbral design [Di Scipio 1]. This binary oversimplifies the role of computers in composition, as they may be employed in different levels of these processes. Sometimes the algorithmic and timbral decisions in a composition may be made by someone who does not in fact program these parameters, but rather 'outsources' any programming that may be required. This develops a kind of "within" and "without" approach to the processes of musical organization where computers are used.

I am one of those 'without' composers, I don't program, and have never had a desire to. But I use computer processing in my compositions almost all the time. This led me to wonder about any perceptible difference that may be apparent in my works, attributed to the fact that I outsource all the generative sound and score programming. I began a discussion with my collaborator in new music ensemble Decibel, Lindsay Vickery about his approaches and thoughts on having been a 'programming composer' for many years. It seems very likely, forty years after Strang's essay, that music programmer's know their tools well enough to make intuitive and artistic judgments during the composition processes they engage with. Rather than deciding which decisions 'ought' to be delegated to a computer according to some human esthetic value [Strang 41], the process has become much more intuitive and open. It begs the question, then, is there a musicality to programming?

2. COMPOSITIONAL APPROACHES

James Harley posits two main facets to compositional approaches. One is psychological, studying creativity as it is applied to music, and the other is cognitive, developing models of the mental procedures and structures used in music-making [Harley 221]. Music has often been referred to as the art closest to science [Ball 23 and others], and the involvement of computers may have well amplified that relationship, especially in regards to formalism. Musical processes can be produced using formal tools, such as algorithms or other formulae, as generative and transformative devices, yet other

compositional instances call for strategies relying on interaction in order to control and qualify results and choices [Vaggione 58]. Current computer functionality deal with both of these instances easily and instantaneously, and there are plenty of programs that enable straightforward manipulations of these parameters, using graphical interfaces. This permits computer programmers to drive sophisticted musical activity that gets close to fusing Harley's psychological and cognitive mindsets. Of course, the formal rigor of a generative function provided by a computer does not guarantee by itself the musical coherence of a result [Vaggione 54], but it can be argued that as computer programming has become more commonplace, complex and refined in its application to music. It is more likely to be imbued with a musical, rather than scientific or calculated, sensibility manifest in the degree of intuition, abstraction, improvisation and responsiveness seen in music programming.

Interactive programming environments such as Quartz Composer, MaxMSP and Pure Data have proven very attractive to composers and laptop performers, as they offer an environment to compose, control and parametise music using a variety of different processes [Puckette 31]. Yet to date, these programs are usually used by composers in their own works, and rarely asked for as 'instruments' by other composers. Some 'composer programmers' use the functionality these programs bring to their compositions without being able to operate them, and outsource the programming to others. Examples include my own work, but also compositions by Pierre Boulez and Anthony Pateras, as well as many others.

2.1. Ideas out of the air

A composer may come up with an idea, and look for the best way to realise it. The skills or craft to perform the realisation of every element in the idea need not lie with the idea maker. 'Non programming composers' have handed musical interpretation of their music to instrumentalists for hundreds of years, yet have been slower to delegate to computing, unless it refers to audio engineering processes, such as mixing and mastering. The prevalence of electronic music and the adoption of interactive programming in popular music such as that by Aphex Twin, Amon Tobin, Radiohead and others has meant that the contributions of these programs are more visible. So outsourcing programming to achieve certain aims is a reasonable solution for many composers. Even better if they can outsource to 'composer programmers', who are more likely to be able to negotiate the concepts, terms and potential involved in compositional processes. By bringing a conceptual idea to a programmer, they are offering a possibility or ambition that the programmer may have never had considered for that program, thus extending their own ability on and perceptions of the software tool. In a discussion with Lindsay Vickery, he noted that

> *it's pretty easy to keep using the same tools and therefore generating the same kind of piece. It makes sense of course to keep using things that work. A fresh concept and pair of ears can definitely drive things in interesting directions though* [Vickery & Hope par 16].

SO not only is this approach beneficial to the 'non programming composer', it may offer insights to the 'composer programmer' as well.

2.2. Finding the idea in the program

Some composers 'find' an idea within a computational process. The idea and way to realise it could come from knowledge of a software capability, but could also be a conceptual construct that they bring to the program. As Vickery notes,

> *The process involves a sort of to and fro between drawings, spreadsheets (to work out the maths), the score and the software. There is often a period of tightening up where there are small changes made to all of the elements of the piece.* [Vickery & Hope par 2]

There is an important difference in these approaches. The computer becomes a tool that grows with the composer, and like the performer of an acoustic instrument – a sound or approach might be discovered, developed, cultivated and refined through practice, performances and compositions. As John Bischoff describes the creation of his work *Audio Wave* (1979-1980):

> *As I worked, I tried to peer into the behavior of the machine to see from where the next musical angle would come. My strategy was to accept the medium 'as is', using its confines as a possible avenue of discovery, rather than allowing myself to be distracted by the wish for a more perfectly plastic material* [Bischoff 79].

Working with the software in-depth offers up possibilities for its use that might not occur to another. This can create new ideas and possibilities, but can also create a cyclic expectation that is limited by the possibilities of the program itself. As Vickery notes;

> *Perhaps people who program themselves are more aware of the limitations. This can be a good and a bad thing of course – perhaps that inhibits programmers from exploring areas that are difficult or unstable or unreliable. Sometimes the solutions for*

interesting new problems take a lot of thinking about, I know I've toyed with some ideas for weeks before the eureka moment arrives. I mean in an indirect, niggling kind of thinking that goes on intermittently night and day... I think that sort of obsession is pretty hard to inspire in someone else [Vickery & Hope par 18].

3. CONCEPTUAL STARTING POINTS

Lindsay Vickery's *Antibody* (2009) employs the biological principles of mutation as a structural framework for the work. Five musical cells, stated at the beginning of the piece, are subjected to increasing 'mutation', using processes such as deletion, duplication, inversion, insertion and translocation. This mutation takes place in two layers; within the generation of the score and through a process of live audio effect, both using MaxMSP, a program Vickery has employed in his compositions for many years. The score 'parts' are written on Finale and then adjusted in real time on a laptop for the performer to read. Each of the five cells is transformed into different arrangements using the processes outlined above. In addition, the performances of the musicians reading these real time arrangements are transformed electronically



Figure 1: and excerpt showing two of the performer interfaces in Vickery's *Antibody* (2009), showing segments of music as they have 'appeared'.

using the same transformative principles for audio processing [Vickery]. These processes are ones that the program chosen to perform them, MaxMSP, does very well. Vickery's experience means that he is able to control the programs potenitial and possibilities, using them to fulfill his compositional concepts. MaxMSP enables real time processing of instruments playing in real-time and is also very good at generating aleatoric choices within parameters set by the composer. '*Antibody*' is an example of a composition where the understanding of the processes of MaxMSP offers is integrated in the very structure, scoring and sonic outcomes of the work. The plurality of layers in which the programming is involved in the work makes it very much part of its musical fabric.

In the authors work, *The Possible Stories of Harry Power* (2010), the computer has no audio synthesis or output, but is used as a kind of score generator, operating according to parameters set by the composer. Here the role of the computer programming is mostly 'back end' and the musicality of the programmer is not as vital. Yet a 'composer programmer' doing this work would be more like to understand the concepts described to them during the process of 'outsourcing'. In this work, three performers play a score written by the composer, and the computer 'listens' to their performance, using that data to write the next part of the score. The computer listens again to the performance of this new score, and creates another. The final part is provided by the composer, influenced by the kinds of scores created by the computer. These have been informed by the initial testing done during the construction of the MaxMSP/Jitter patch designed to run the score generator. Here is a concept developed by the composer, based on ideas about story telling and the importance given on the written histories (written notation) over oral ones (improvisations). Like *Antibody,* it takes its idea from an external notion and realizes with a piece of music, facilitated by MaxMSP/Jitter. Yet this work was conceived differently outside of or 'without' the program. The program was employed when it was decided the best mechanism to realise this generative procedure, some time after the concept was developed. It was not an idea that came from years of working with MaxMSP.

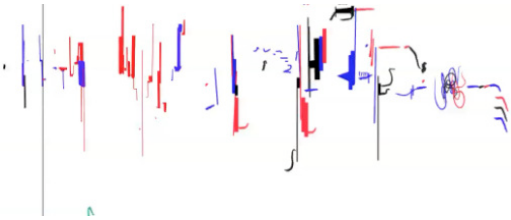


Figure 2: an excerpt of Hope's *The Possible Stories of Harry Power* (2010) showing the computer writing on the left, and the composers 'modified' writing on the right.

A 'composer programmer' is likely to try things, listening as they go – working inside the programming, creating actions, then coming out of that programming 'space' and working as the listener, perceiving their actions. This is a process common to all composers, the making/writing and playing/listening/workshopping. Improvisation offers a different system: a simultaneous making and listening on the fly. Computer programmers have become very good at this too, further bolstering a claim for 'musicianship'. Whilst a composer

performs a critical act with relationships and their representations [Vaggione 60], 'programming composers' do this on many more levels, as a program is another kind of representational system that can be used within and alongside more traditional music representations. As Vickery points out;

> *A composition is made up of many decisions. I felt somehow constrained by having to fix certain variables in a linear score; so many possibilities have to be ignored. What I have tried to do is to leave some pathways open, allowing each performance to explore a different trajectory. There is scored materials and the ways that performers interpret them and then audio processing of their performance and how that is distributed back into the space– my aim was to allow some of these components to unfold independently of one another* [Vickery & Hope par 10].

4. COMPOSING FOR, NOT FROM, COMPUTERS

Another way the 'composer programmer' may contribute to a music composition is when the computer is given a role on the score as an instrument within an ensemble. Rather than employing a programmer to help write the mechanics for a piece, the composer offers up something for the computer programmer/performer to read and interpret in a live situation. Graphic notation for electronics has become commonplace, but more often as an illustration of a prerecorded sound,

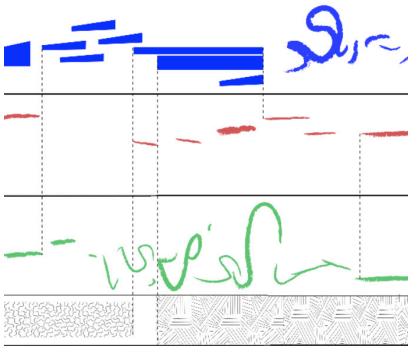


Figure 3: An excerpt from Hope's *Kuklinski's Dream* (2010) the computer part being the bottom most part.

followed for cues, volume and other control factors, rather than as a part to be 'interpreted'. This has made an important part of my own composition style, and in the illustrations below, two examples are provided. Figure 3 shows an excerpt the score for *Kuklinksi's Dream* (2010) where the computer part is notated along the bottom. The key explains three states for the computer: recording (sampling, to the left of this excerpt), playing back (to the right of the excerpt), and effected playback (not shown here). No instructions on how to effect the playback are provided; therein lay the artistry of the performer, deliberately permitted by composer.

In Figure 4, there is less freedom of interpretation involved, as the score does provide an instruction for a computer operator to transpose and extend a given moment in the strings, and route the sound into a bass amplifier. The skills here are more practical and operational, yet the quality of sound (in particular in the 'hashed' flags, denoting a distorted tone as opposed to black flags, which denote a clear tone) is very much the domain of the programmer, a timbral element the composer has given to the computer programmer to make their own. A range for the pitch is given, leaving the programmer to decide if the computer will make the choice of pitch randomly, or the program operator will decide in the live situation, or even decide beforehand.

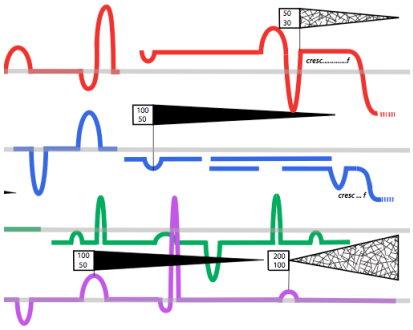


Figure 4: An excerpt of the score for Hope's *Cruel and Usual* (2011) showing the computer parts as flags (the shape describing the dynamic) with frequency ranges.

Both these samples provide an open invitation for musicality from the programmer; with their own part to perform in the score, as part of the ensemble. In a way, their artistry is amplified into the foreground of musicianship in the group, rather than the 'machine room' of the composition. Rather than programming used to set conditions for musical action, as is the case in Vickery's *Antibody,* here the musical action sets the conditions for programming. The ideas come from 'without' – outside of the programmers experience.

5. CONCLUSION

With the high level of musicality in the different situations where computer programming is featured in music, computers have been thought of as instruments in their own right for some years now. This has been propagated more by live performance (by such skilled practitioners as Robin

Fox, Kim Cascone and others) than by back end programming. However, it is less commonplace to find computer programmers as part of mixed ensembles, or programmers working to another composers brief, than you would expect. Computer music still seems separated from much contemporary classical music, sitting within its own 'electronic music' niche, and whilst there are examples of electroacoustic work (where electronics and acoustic instruments are combined in a group) beyond the ones discussed above, this area is a fertile one for development.

The computer in music is "a mechanism with which we interact, not a mathematical abstraction which can be fully characterized in terms of its results" [Winogard 391]. If 'non programming composers' can see the richness of musical possibilities that sophisticated programs offer, and treat music programmers as the musicians they really are, a huge range of possibilities becomes open to them, and computer assisted composition will move out of 'electronic music' and into the full realm of musical possibility.

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MULTIDIMENSIONAL DATA SETS: TRAVERSING SOUND SYNTHESIS, SOUND SCULPTURE, AND SCORED COMPOSITION

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ABSTRACT

This article documents some of the conceptual developments of some various approaches to using multidimensional data sets as a means of propagating sound, manipulating and sculpting sound, and generating compositional scores. This is not only achieved through a methodology that is reminiscent of some of the systematic matrix procedures employed by composer Peter Maxwell Davies, but also through a generative signal path method conventionally termed *Wave Terrain Synthesis*. Both methodologies follow in essence the same kind of paradigm - the notion of extracting information through a process of traversing multi-dimensional topography. In this article we look at four documented examples. The first example is concerned with the organic morphology of modulation synthesis. The second example documents a dynamical *Wave Terrain Synthesis* model that responds and adapts in realtime to live audio input. The third example addresses the use of *Wave Terrain Synthesis* as a method of controlling another signal processing technique - in this case the independent spatial distribution of 1024 different spectral bands over a multichannel speaker array. The fourth example reflects on the use of matrices in some of the systematic compositional processes of Peter Maxwell Davies, and briefly shows how pitch, rhythm, and articulation matrices can be extended into higher-dimensional structures, and proposes how gesture can be used to create realtime generative scores. The underlying intent here is to find an effective and unified methodology for simultaneously controlling the complex parameter sets of synthesis, spatialisation, and scored composition in live realtime laptop performance.

1. INTRODUCTION

The idea of traversing data sets is well established in the realms of sound synthesis and computer music. One only has to consider the concept of *wavetable lookup*, and we find this method is the basis of generating digital oscillators, audio sampling, *Waveshaping Synthesis*[1], audio analysis via *FFT* or *wavelet* methodologies and the subsequent resynthesis of this data, *Graphical Synthesis*, *Convolution Reverb*, *Wave Terrain Synthesis*[2], *Head-Related Transfer Functions* (HRTF's) for the binaural treatment of existing sounds, and *Multi-Impulse Response* techniques found in the *MIR-Project*[3]. Table lookup procedures for sound synthesis developed not long after the birth of the microcomputer in the mid-1970's, and it was in 1978 that Rich Gold first coined the term *Wave Terrain Synthesis*. There have been many advantages to the table lookup procedure - it is considered to be computationally efficient as processing requirements are consistently low, and the procedure allows for storing and reloading tables of data. These data sets might consist of either arithmetically generated values, statistical information, and measurement data, provided the data range is adequately sufficient. For example we may use "samples" of real world data, be it audio, video, or any other collection. Extending the dimensionality of the table lookup procedure allows for describing the complex non-linear behavior of electronic components and other complex phenomenology such as the localized perception of sound in a binaural setting using HRTF's.

Investigation into the wide variety of multi-dimensional data sets for use with *Wave Terrain Synthesis* presented many possibilities.[4] One of the main attractions to *Wave Terrain Synthesis* for the purposes of this research is that the system of traversing these data sets can be intuitive and effective to control, especially when mapped to gestural controller input.[5] The fact that the system also is so adaptable for varied applications as the examples will demonstrate, is also leading toward a hope that this research may find an organic, efficient, and unified methodology for the simultaneous control of the parameter sets associated with sound synthesis,

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1 Le Brun, M. 1979. "Digital Waveshaping Synthesis." *Journal of the Audio Engineering Society* 27(4): 250-264.

2 Gold, R. 1979. "A Terrain Reader." In C. P. Morgan, ed. *The BYTE Book of Computer Music*. Byte Publications, Petersborough, NH.

3 The MIR-Project http://www.vsl.co.at/en/65/73/500/320.vsl

4 James, S. 2005. Developing a Flexible and Expressive Realtime Polyphonic Wave Terrain Synthesis Instrument based on a Visual and multidimensional methodology. *West Australian Academy of Performing Arts, Western Australia. 2005.*

5 Mills, A. and R. C. De Souza. 1999. "Gestural Sounds by Means of Wave Terrain Synthesis." *Congresso Nacional da Sociedade Brasileira de Computação* XIX. http://gsd.ime.usp.br/sbcm/1999/papers/Anderson_Mills.html

sound sculpture and manipulation, and generative scored composition.

As a means of clarification, the term sound sculpture is used here to describe the physical act of sculpting a terrain surface with physical input much like Dan Overholt explored with the *MATRIX* interface.[6] Since the terrain largely affects the resulting sound, the perceived effect here is such that the performer is "sculpting" the sound gesturally via this process.

2. SOUND SYNTHESIS MORPHOLOGY

The concept of morphing sound synthesis is a concept that arose during the course of my Masters thesis. Initially I was more concerned in the thesis to observe the characteristic traits and topographies of conventional sound synthesis processes as it could inform and develop a further understanding of the many other kinds of topographies commonly used in *Wave Terrain Synthesis*. It was only later it occurred to me that this would open the possibility for both an organic and uninhibited way to efficiently morph between different forms of sound synthesis. In this example we take three forms of modulation synthesis and map the associated parameter spaces, keeping in mind that the audio input source will be sinusoidal in most cases. These three synthesis methods discussed are *Ring Modulation*, *Frequency Modulation*, and *Waveshaping Synthesis*.

It is immediately apparent why *Frequency Modulation Synthesis* creates more sidebands and spectral complexity when one observes the number of undulations found in its associated parameter space in Figure 3. This is significantly more than that of the *Ring Modulation* (Figure 2) and the *Waveshaping* parameter space (Figure 4). The terrain structures are generated in MaxMSP utilizing the Jitter matrix processing library. This library allows for traversing not only 2-dimensional matrices, but higher dimensional structures which may be created using the *jit.expr* object (Figure 1).

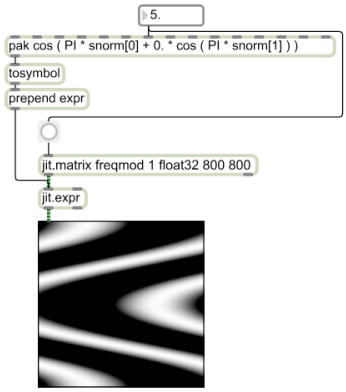


Figure 1. *Frequency Modulation* parameter space curve as generated in MaxMSP

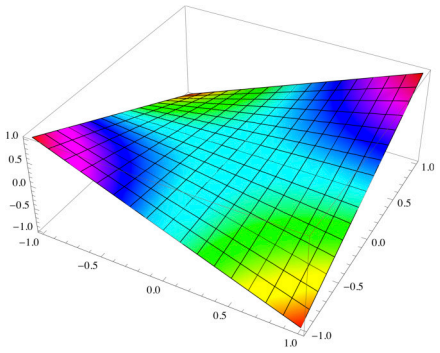


Figure 2. *Ring Modulation Synthesis* parameter space determined by the equation $f(x,y) = x_t . y_t$

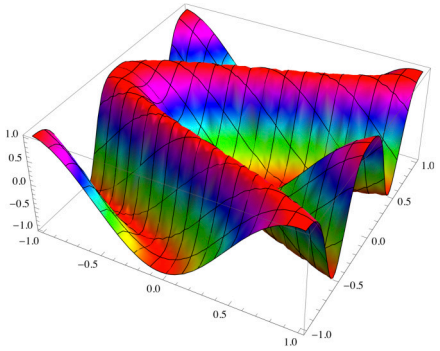


Figure 3. *Frequency Modulation Synthesis* parameter space determined by the two dimensional equation $f(x,y) = \cos\left(\pi x_t + I\cos\left(\pi y_t\right)\right)$ where $I = 3$

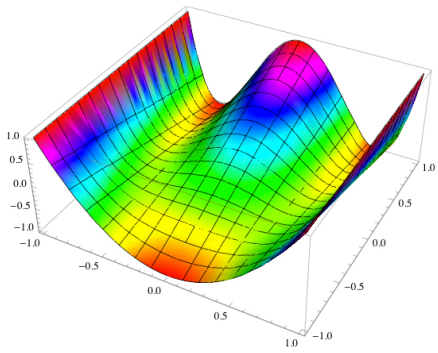


Figure 4. *Waveshaping Synthesis* parameter space determined by the two dimensional equation $f(x,y) = 0.5\left(y_t + 1\right)G_x - 0.5\left(y_t - 1\right)H_x$ where *G(x)* and *H(x)* are two Chebychev Polynomials

6 Overholt, D. 2002. "New Musical Mappings for the MATRIX Interface." *Proceedings of the 2002 International Computer Music Conference*. http://www.create.ucsb.edu/~dano/matrix/ICMC2002.pdf

Previous research has shown that many of the curves used will result in greater levels of spectral complexity. James has suggested that depending on the way the methodology is perceived, *Wave Terrain Synthesis* could also be described as multi-dimensional *Waveshaping Synthesis*.[7] The potential for using terrain surfaces that exhibit random and "noisy" topographies translate to a resulting waveform contour that exhibit these same sorts of characteristics. Previous research has investigated many different terrain curves ranging from finite solutions to constrained Algebraic, Trigonometric, Logarithmic/Exponential, Complex, and composite mathematical functions, to data extracted from analyses of global seafloor and land topography[8], digital image files, video[9], perlin noise functions, elliptic functions, recurrence plots, OpenGL NURBS surfaces, fractal and iterative structures, and video feedback.

The concept of morphology in matrix terms involves the interpolating between two or more different pre-defined states. In terms of signal processing, this can be achieved most simply by a linear crossfade. The *jit.xfade* object in MaxMSP is used to perform this (Figure 5).

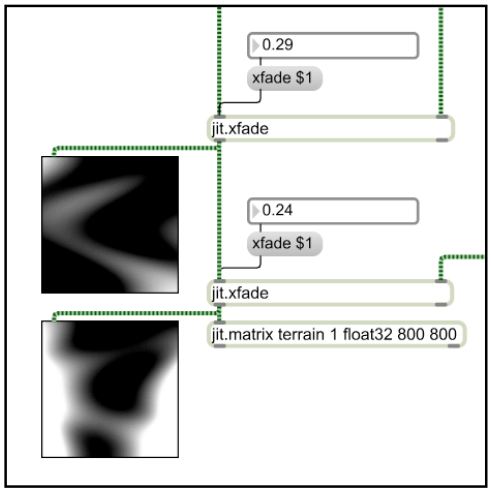


Figure 5. The use of *jit.xfade* to morph between terrain structures

Another useful modulation to explore in this process is the spatial distortion and contortion of a terrain structure through the process of 2D wavetable lookup; this is sometimes referred to as *spatial remapping*.

As one would find in any *Wave Terrain Synthesis* model, the multidimensional data set needs to be traversed with a trajectory signal that is often comprised of two independent audio signals. Previous research into *Wave Terrain Synthesis* has largely focussed on highly periodic trajectory structures as a means of traversing a terrain. Although other kinds of structures have been used including quasi-periodic orbits such as pseudo-phase space plots, chaotic orbits and random orbits.

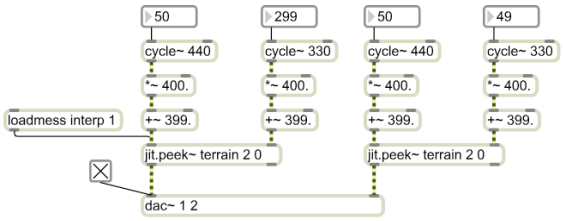


Figure 6. Two periodic trajectories created using the combinations of sine tones of varied frequency

In this example, and in keeping with the generative methodology used for generating terrain curves, the trajectories used followed periodic structures such as elliptical curves and orbits attributable to the *Harmonograph*. These have been described by the parametric equations:

$$x(t) = A_1 \sin(tf_1 + p_1)e^{-d_1 t} + A_2 \sin(tf_2 + p_2)e^{-d_2 t} \quad (1)$$

$$y(t) = A_3 \sin(tf_3 + p_3)e^{-d_3 t} + A_4 \sin(tf_4 + p_4)e^{-d_4 t} \quad (2)$$

Just as one can spatially distort a terrain structure, geometric transformations can be applied to the trajectory resulting in further modulation possibilities - some of these transformations include phase distortion or pulse width modulation and affine transformation which can involve the translation, scaling, and rotation of a trajectory.

An effective physical controller for these kinds of transformations requires a multi-sensory device. Hsu has explored the use of the Wacom Tablet for the purposes of drawing trajectory structures[10], but new technologies are emerging that promise new directions through customisable multi-sensory and tactile control such as the Arduino, as well as mobile and programmable multi-sensory devices such as the Apple iPhone and iPad. Finding effective ways and means of gesture mapping for trajectory motion and how to effectively generate and/or manipulate a terrain surface strikes right at the heart of effectively performing using this method. The choice of physical controller or sensor is critical in being

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[7] James, S. 2005. Developing a Flexible and Expressive Realtime Polyphonic Wave Terrain Synthesis Instrument based on a Visual and multidimensional methodology. *West Australian Academy of Performing Arts, Western Australia. 2005.*

[8] Thibault, B., and S. Gresham-Lancaster. 1992. "Terrain Reader." http://www.mcs.csuhayward.edu/~tebo/TerrainReader.html and Thibault, B., and S. Gresham-Lancaster. 1992. "Songlines.DEM." *Proceedings of the 1992 International Computer Music Conference*. San Jose: 465-466. http://www.mcs.csuhayward.edu/~tebo/Songlines.txt

[9] Dannenberg, R. B., and T. Neuendorffer. 2003. "Sound Synthesis from Real-Time Video Images." *Proceedings of the 2003 International Computer Music Conference,* San Francisco: International Computer Music Association: 385-388. http://www-2.cs.cmu.edu/~rbd/papers/videosound-icmc2003.pdf and Dannenberg, R. B., B. Bernstein, G. Zeglin, and T. Neuendorffer. 2003. "Sound Synthesis from Video, Wearable Lights, and 'The Watercourse Way'." *Proceedings of the Ninth Biennial Symposium on Arts and Technology*. Connecticut College: 38-44. http://www- 2.cs.cmu.edu/~rbd/papers/conncoll2003.pdf; http://www.cs.cmu.edu/~music/examples.html

[10] Hsu, W. 2002. "A Flexible Interface for Wave Terrain Synthesis." *PERformance & NETworking Colloquia*, San Fransisco State University, Department of Computer Science. http://cs.sfsu.edu/news/pernet/02/04-24-02.html and http://userwww.sfsu.edu/~whsu/TERRAIN/

able to effectively map gestural information from a performer. A *Polhemus Stylus* or similar device provides position and orientation information for a single point in space via a stylus tip, whilst a gesture interface can input many positions since the system tracks multiple features simultaneously. Further investigation will involve experimentation with two dimensional and three dimensional motion cameras such as the *Vicon* and *Kinect* interfaces to test their complementarity with the traversal of multi-dimensionality. This is a topic for further discussion at a later stage.

One can also easily see how the concept of morphing synthesis can be extended to three dimensions or more. By extending the matrix dimensionality, it opens up further parameters to be modulated at audio rate by using additional trajectory curves.

3. LIVE INTERACTIVE, DYNAMICAL, AND FEEDBACK-BASED METHODOLOGY

Performing with the *Decibel* ensemble at *Createworld* 2009 in Brisbane we performed one of my own pieces that involved live alto sax and a MaxMSP patch that reprocessed this live acoustic instrument in various ways. The reprocessing involved a *Wave Terrain Synthesis* patch that would generate dynamical terrain surfaces. In this model the rate of dynamic terrain evolution can be adjusted; effective rates can be achieved of up to 30 frames a second.

One of the attractions to this kind of model was a fascination with evolutionary and dynamical systems.[11] As opposed to the purely generative system found in example one, this second example explores more of an organic system where the structure of the contours are determined by real-world acoustical events, thus creating a relationship between the acoustic instrumentalist and the computer. Rather than the laptop performer determining the structure of all elements, terrain and trajectory, they are not in complete control of the terrain surfaces derived from the live audio input stream.

This model was inspired by Di Scipio's research,[12] who has used iterative structures for generating terrain curves. Di Scipio describes some of the sounds produced to be reminiscent of boiling water, cracking of rocks and icebanks, the sound of wind, various kinds of sonorous powders, burning materials, certain kinds of insects, thunder, electrical noises, sulphureous or volcanic events, and the sound of wind flapping against thin but large plastic plates. With higher order iterates the audio would be reminiscent of a large fire, rain or hail, or frying oil.

In this example, instead of using iterative based structures, this model draws a terrain based on the audio input. A low-pass filter is applied to the audio input in order to control the topographical complexity of the terrain surface itself, allowing the laptop performer to control the resulting spectral complexity of the synthesized sounds generated. As a means of introducing further complexity, sounds produced by the instrument are fed back via software and blended with the live input in order to introduce further complexity in the evolution of the terrain dynamics.

The terrain surfaces are recurrence plots generated in realtime based on the audio input. Recurrence plots are normally used to reveal non-stationarity of a series as well as to indicate the degree of aperiodicity in a signal.

$$\left\| f(t) - f(\tau) \right\| < r \tag{3}$$

For the purposes of creating smooth contours for *Wave Terrain Synthesis*, we look at the use of the recurrence plots without the Boolean stage. In other words, we are interested in a Recurrence Plot of a function f (t) that is described:

$$f(t,\tau) = \left| f(t) - f(\tau) \right| \tag{4}$$

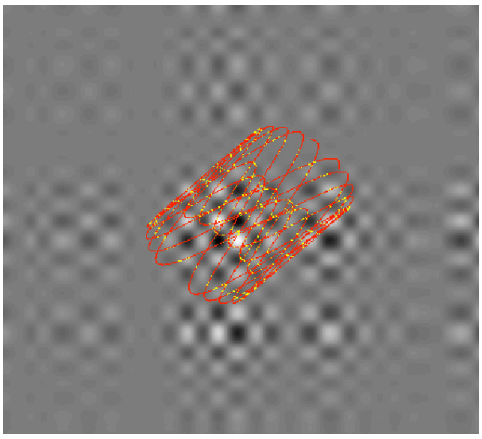


Figure 7a. A recurrence plot terrain generated using live audio input with the trajectory superimposed

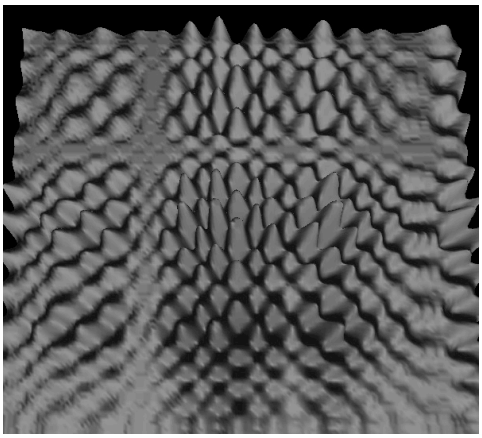


Figure 7b. The same recurrence plot in Figure 7a plotted in 3D

11 James, S. 2003. "Possibilities for Dynamical Wave Terrain Synthesis." *Converging Technologies, Proceedings of the Australasian Computer Music Conference*: 58-67.

12 Di Scipio, A. 2002. "The Synthesis of Environmental Sound Textures by Iterated Nonlinear Functions, and its Ecological Relevance to Perceptual Modeling." *Journal of New Music Research*. 31(2): 109-117.

The trajectories for this instrument were generated by "collecting" fragments of musical phrases by the instrumentalist. By using an attack and release detection mechanism, the phrases are catalogued and replayed in a different order to which they have been originally performed. There is also a pitch detection algorithm used to determine a pitch set used to generate long drone-like passages with the *Wave Terrain Synthesis* instrument.

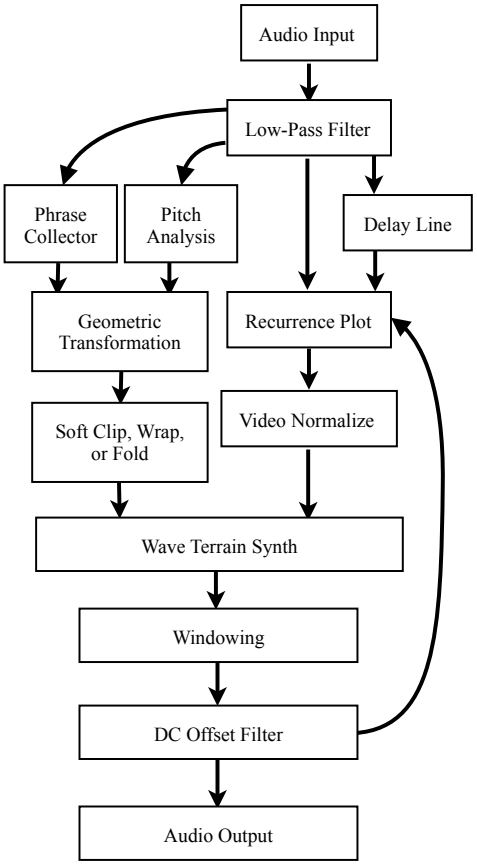


Figure 8. A schematic describing the instrument design

4. SPATIAL SCULPTURE

In the concluding paragraphs of my masters thesis I concluded that *Wave Terrain Synthesis* methodology could be used for the control of another process. In this example we will be looking at the use of *Wave Terrain Synthesis* methodology to control the discrete independent spatialisation of 1024 spectral bands and has been described *timbral spatialisation*.[13] This methodology may be adapted to any multi-channel speaker configuration, for example 4, 8 or 16 channels. In this example the performer uses *Wave Terrain Synthesis* as a bridging system for controlling the complex parameter sets associated with timbral spatialisation, allowing the performer to sculpt the localisation of many individual frequency bands with a comparatively small number of control parameters.

The process is achieved using a windowing technique that scales the relative amplitudes of each spectral bands, which are themselves dependant on the topographical nature of the terrain curve. A flat terrain contour translates to spectral uniformity in all loudpeakers. However as the terrain tips around on one axis the spectral distribution starts to separate so that higher frequencies moves across to one side of the spatial field, and low frequencies to the other.

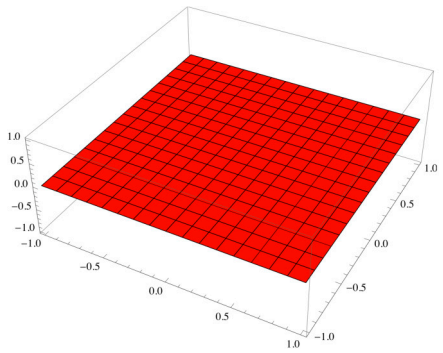


Figure 8a. Signal equally spectrally distributed out to all loudspeakers

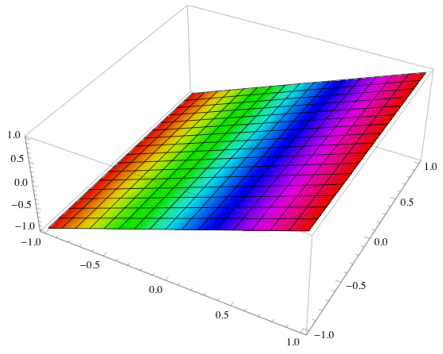


Figure 8b. Signal is distributed out to the loudspeakers in such a way that the sound spectrally shifts from low to high frequency across the room

More interesting results can be achieved with non-linear and curved terrain surfaces. The introduction of undulating terrain surfaces create very effective evolutional characteristics, and a sense of emersion in terms of the spatial listening experience., even when the source sound is a mono static oscillator.

One can see that there is an obvious benefit in this case to keeping the trajectory elliptical and to map each corner of the terrain with its relative spectral distribution. In this way the trajectory can be rotated about the polar plane causing a shift in azimuth for all spectral components and their associated localisation.

Spatialisation can also be influenced at audio rates by geometrically influencing the trajectory signal by rotation, scale or translation, or by spatially contorting

[13] Normandeau, R. "Timbre Spatialisation: The Medium is the Space." *Organised Sound, 2009. Vol. 14. Issue 3.*

the terrain surface using the *jit.repos* object. Chaos and randomisation can also be introduced in this way.

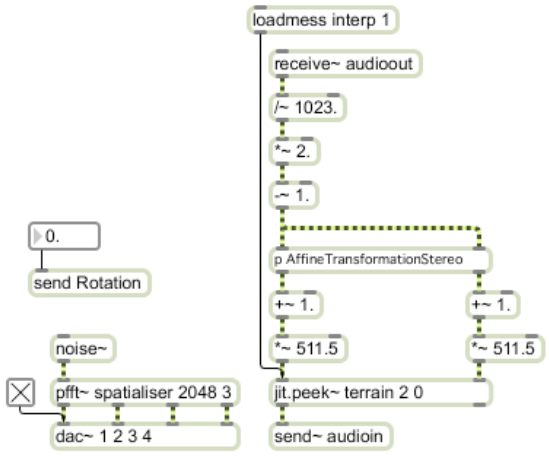


Figure 9a. The main patcher design for the spatialiser

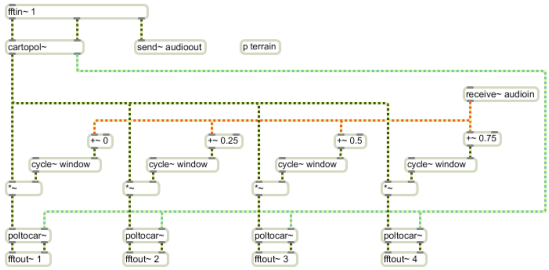


Figure 9b. The sub-patcher showing the spectral manipulation applied

5. SCORED COMPOSITION

Peter Maxwell Davies documents a use of pitch matrices and rhythm matrices in the composition of many of his works. in this way he is able to take a source melody and manipulate this with the contour of another. The different permutations that occur through this process create complexities in terms of harmonic and melodic material, thus also influencing the overall tonality.

Davies is known for his use of magic squares as a source of musical materials and as a structural determinant in his works. This is evident in Davies' compositional workings for *Ave Maris Stella* (1975), Symphony no. 3 (1984) and Symphony no. 6 (1996) to name but a few. In Davies' *A Mirror of Whitening Light* (1977) he took 8 notes from the plainsong *Veni Sancte Spiritus* which is arranged into an 8x8 square and then re-arranged according to the magic square of Mercury.
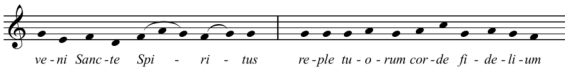


Figure 10a. Plainsong melody *Veni Sancte Spiritus*
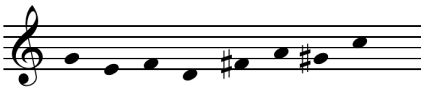


Figure 10b. Davies' pitch series from *Figure 10a*

G 1	E 2	F 3	D 4	F♯ 5	A 6	G♯ 7	C 8
E 9	C♯ 10	D 11	B 12	D♯ 13	F♯ 14	F 15	A 16
F 17	D 18	E♭ 19	C 20	E 21	G 22	F♯ 23	A♯ 24
D 25	B 26	C 27	A 28	C♯ 29	E 30	D♯ 31	G 32
F♯ 33	D♯ 34	E 35	D♭ 36	F 37	A♭ 38	G 39	B 40
A 41	F♯ 42	G 43	E 44	G♯ 45	B 46	B♭ 47	D 48
G♯ 49	F 50	F♯ 51	E♭ 52	G♯ 53	B♭ 54	A 55	C♯ 56
C 57	A 58	B♭ 59	G 60	B 61	D 62	C♯ 63	F 64

Figure 10c. Pitch matrix derived from *Figure 10b*

Peter Maxwell Davies has then generated melodic and harmonic material by traversing the matrix using zig-zag formations, diagonal lines, and spiral formations. One can see how the fundamental process here can be attributed to *Wave Terrain Synthesis*, but instead of extracting audio signals from a matrix, we are retrieving pitch, rhythm, or articulation data. Alternatively, the matrix data could be used for determining higher level constructs in generative composition such as Markov Chains and formal grammars.

Implementing the traversal of pitch, rhythm, and articulation matrices requires a slightly different approach in procedure, as the traversal should refer to specific cells rather than smooth gradations between cells as is found in *Wave Terrain Synthesis*. These constructs can be triggered and manipulated through the use of gesture, determining not only the pitch, rhythm, dynamic, and articulation sets used, but also the context of this material, whether it be harmonic, melodic, or largely textural. The question here is how does one effectively categorize these gestures, and distinguish between these many varied kinds of materials - chords, melody, fast-fluid material, and various kinds of layers and densities.

As is found in many of the systematic compositional procedures documented by Milton Babbitt, here we focus on other permutational and matrix transformational procedures such as matrix inversion, matrix multiplication, and other procedures involved in multi-signal processing. Rhythm and articulation maps are either kept synchronised with their original respective pitch, or can be processed independently through their own concurrent procedure.

C 8	A 58	B♭ 59	F♯ 5	D 4	D 62	C♯ 63	G 1
G♯ 49	F 15	F♯ 14	E♭ 52	G♯ 53	F 17	C♯ 10	C♯ 56
B♭ 47	F♯ 23	G 22	E 44	G♯ 45	E♭ 19	D 18	D 48
G 32	D♯ 34	E 35	C♯ 29	A 28	A♭ 38	G 39	D 25
B 40	B 26	C 27	F 37	D♭ 36	E 30	D♯ 31	F♯ 33
F 17	B♭ 47	B 46	C 20	E 21	G 43	F♯ 42	A♯ 24
E 9	A 55	B♭ 54	B 12	D♯ 13	F♯ 51	F 50	A 16
F 64	E 2	F 3	B 61	G 60	A 6	G♯ 7	C 57

Figure 10d. The pitch matrix found in *Figure 10c* re-ordered according to the magic square of Mercury

Matrices in Jitter can also reflect structures greater than two dimensions. So for transpositional pitch matrices this allows for pitch transformation over two or more melodic contours. This research will investigate how effective this might be in terms of modulating tonality.

In terms of scoring these musical structures in realtime, there are currently two music scoring systems developed for MaxMSP: *MaxScore* and *Lilypad* embedded in MaxMSP. This research project has been using *MaxScore* due to its support for complex metric structures and rhythmic tuplets.

6. CONCLUSIONS

The focus of this paper has been to observe four different applications of a similar methodological paradigm, and how such a model can be used for significantly different outcomes. The purpose is to encourage interest in this area, and to throw further questions. The underlying intent here is to find an effective and unified methodology for simultaneously controlling the complex parameter sets of synthesis, spatialisation, and scored composition in live realtime laptop performance.

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AN ORGANIC SYSTEM FOR TONAL COMPOSITION

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ABSTRACT

This paper aims to discuss and question the separation between computational musicology research involving tonal musics, and the creative composition and generation of tonal musics. It considers how style modelling and synthesis relate to creative pursuits, and the problems in applying principles from computational musicology to creative purposes. Potential hurdles in this relationship include a pervasive reluctance to recognise computer-created works as genuinely creative, as well as the musical specificity required by analytical style-modelling. I argue that there is a teleological aesthetic difference between *computational creativity* and *compositional creativity*, which affects the perception of a composed work as being 'creative'. The crucial difference between these paradigms is the agency of the (human) composer within the creative process. The author's own composing system, *Deviate,* is described briefly and evaluated in regard to these issues.

1. INTRODUCTION

Computational musicology, analytical musicology, and music psychology offer unique insights to how musics are composed, performed, and interpreted. Research in these areas includes such areas as sound (the perception of pitch and timbre), general musical rules relating to pitch and rhythm, as well as style-specific analyses. This exploration of the formations of tonal musics can be traced back to the development of rules for counterpoint, serialism, Meyer's theories of expectation (1956), Schenkerian analysis, and subsequent analytical models (Narmour, 1977).

1.1. Music analysis, psychology, and computation

A substantial portion key research in these areas has occurred over two decades ago, and many current studies still rely on these seminal texts. These include tonal hierarchies (Krumhansl 1979, 1990), grammatical hierarchies of structure and rhythm (Lerdahl and Jackendoff, 1983), improvisational performance analysis (Pressing, 1987) and consonance and musical style (Cope, 1991). More recent studies include music, expectation and emotion (Huron, 2006) and musical probability (Temperley, 2007). Style-specific analyses and modelling have been undertaken in a variety of areas, from Kippen and Bel's work with tabla drumming (1995) to Collins' generation of breakbeat patterns (2002).

1.2. Creative generation of tonal musics

These analyses are, however, seldom applied for practical or creative purposes. Meredith (1996: 5) writes, "Only a small number of programs have been developed for the purpose of automatically generating *tonal* music" (italics in original). He cites Ebcioglu's assertion (1992: 327) that musically trained composers typically do not consider tonal music a priority, and "typically the musically trained avant-garde composers who use computers for algorithmic composition do not consider tonal music important, and those computer enthusiasts who do have an interest in algorithmic tonal music composition are not musically trained" (cited in Meredith, 1996). Today, fifteen years after Ebcioglu's statement, it is not apparent (to the author) that this tendency has changed significantly. It should however be recognised that the definition of 'algorithmic tonal music' needs some clarification; for the purpose of this paper, tonal music will be taken to mean note-level generation (and analysis) using the Western twelve-note or diatonic scale. Musics that are not primarily tonal in focus employ found sound, timbral manipulation, granular synthesis, and microsonic processing, as well as atonal and microtonal composition.

The relationship between computers and music is well explored in a practical sense, whether for synthesis, information retrieval, sound recording, or music production. Meanwhile, *creative* generation of tonal musics is still relatively rare, as stated above, despite the substantial research into this area for musicological purposes. In this paper, I will consider the relationship of these fields—computational musicology, analytical musicology, and music psychology (henceforth referred to under the umbrella term of *computational musicology* for brevity)—to composition and creative musical applications.

1.3. Broad, modular, and organic systems for tonal composition

This paper discusses the development of "organic" music systems, where general musical rules can be recombined with different parameters to generate creative works with variety and customisability, rather than pastiches produced by empirical style modelling. It is proposed that combining broad, rather than specific, musical rules permits more creative applications for generating tonal musics. This will be discussed with regard to the author's own experiments.

2. MUSICOLOGY, COMPUTATIONAL MUSICOLOGY, AND COMPOSITION

2.1. Musicology and computation

The relationship between musical analysis and computation is complicated. From a musicology perspective, Cook (2005) argues that musicologists have yet to engage fully with the potential that computation

affords, in part because traditional musicology is oriented primarily towards the study of scores (ibid: 3) which have well-established (non-computational) analysis methods. Crawford (2010) likewise states, "For many years I have been an optimist about what computers can do for the musicologist, but, to be honest, little of what I hoped for when I started has yet come about". These statements identify the difficulties inherent in quantitative analysis of musics, and express concern that many areas are under-explored. Nonetheless, some useful discoveries, cited in the Introduction, have been made, which have been used computationally in empirical musicology, style synthesis, and composition.

2.2. Computation and composition

The relationship of computational music analysis to creative composition is likewise unclear. Describing the four major types of automated composition, Pearce et al (2002) identify the following: algorithmic composition, the design of compositional tools, the computational modelling of musical styles and the computational modelling of music cognition (ibid: 119). This indicates a persisting divide between the first two categories as creative endeavours, and the second two as research-oriented pursuits.

Although algorithmic and generative composition are recurring in music ranging from serialism to granular synthesis, the use of analysis-derived processes in tonal composition still appears to be relatively uncommon (see Ebcioglu, 1992: 327). Some potential reasons for the rift between creative computer-based composition and computational musicology are proposed below.

3. SPECIFICITY VERSUS BROADNESS

3.1. Style modelling and creativity

Robust and complex generative grammars, such as Cope's Experiments in Musical Intelligence (1996) have demonstrated that convincing works can be created based on an existing style and corpus of material. Although such corpus-based systems can reliably generate musical material of a particular style, the output is necessarily confined by the limits of the that style. Furthermore, musics such as Western classical and jazz are often bound by prescriptive rules relating to melodic contour, cadence, and voice leading. This makes *creative* generation (rather than style modelling) of these musics more difficult. *Strasheela* by Anders (2007) is one example of a "generalised" music generation/ representation system, which permits recombination of constraint parameters to generate music including Fuxian and florid counterpoint; this application, however, requires fairly extensive rule specification and processing (ibid: 164).

3.2. Broad approaches to creative generation

A broader, non-style-specific approach to musical generation means that many prescriptive rules are approximated or disregarded in favour of more general rules for rhythm, pitch spaces, and so on. By focusing on what might be generally called "popular" music styles, the composer (or user) can generate more diverse musical results. One problem with this 'broad' approach is that it is much more difficult to assess computational or musical success, as firstly there is no single style similarity as a goal. Processes and parameters are more difficult to derive through quantitative analysis of a musical corpus. Because the output of a broad system is, by design, variable, qualitative and quantitative analysis of the system's output is similarly problematic. Furthermore, attempting to create original, new compositions involves intentional *overgeneration* (after Meredith, 1996: 10), that is, producing material beyond the foreseen capabilities of the system. This results in (aesthetic) variability of musical output.

4. THE CREATIVITY PROBLEM

Another issue with the creative application of computational musicology relates to creativity. Style synthesis of tonal musics can firstly contain elements of pastiche which are contrary to the modernist and avant-garde prerogatives to innovate. This is relevant to the broadness issue discussed above, but is also defined in "moral" terms; can a tonal work created by a computer be considered creative?

4.1. Computer creativity and composer creativity

To answer this, it is necessary to distinguish computer-creativity from composer-creativity. Jordanous' (2011) research defines computational creativity as the ability of a system to produce quality output that is also varied and novel (ibid: 1). However, Jordanous also recognises that "a creative practitioner's primary aim [...] is to produce creative work, rather than to critically investigate creativity". This positions computational creativity as a process of critical investigation, rather than one focused on the act of creative composition. In other words, while a musical system may create, this does not necessarily imply a creative act on the part of the composer; and consequently, if composer-creativity is not present, the artistic merit of the work is compromised.

The friction between these two types of creativity is highlighted by critical reactions to David Cope's Emily Howell project. Although Howell's work was occasionally well received, there is a reluctance to recognise heavily machine-mediated composition (and the exaggerated depiction of Emily Howell as an independent composing entity) as genuinely creative, that is, as "original" or "artistic", rather than "clever". As music critic Mark Lawson states, "A computer, cleverly programmed, could probably produce the *Doubting Thomas Passion* by JS Bach or *More Snow on Kilimanjaro* by Ernest Hemingway. But the exercise would be worthless because the works from the software would not be informed by being a God-fearing kapelmeister in 18th-century Germany or a suicidal macho male in mid-20th century America." (Lawson, 2009). This antipathy towards recognising a computer as genuinely (rather than computationally) creative, in regard to Cope's research and composition, is extensively covered by Hofstadter and others (1996).

4.2. Creativity as a "human" pursuit

The Emily Howell paradigm represents an extreme example of how composers, as opposed to researchers, work with computers, in that it (admittedly somewhat inaccurately) positions the computer as the sole composer.[1] Although reactions to Howell insisting that the composer's intention and prerogative is crucial to the quality of the composed work may seem old-fashioned, the rationale of an artistic work is crucial even in seemingly progressive musical contexts, as evidenced by the importance of program notes. Even though a musical composition, if created by a computer program developed by a (human) composer, is logically the work of the composer, it is argued that a difference is perceived. This is specifically the case for generative and algorithmic composition of tonal musics, where style synthesis complicates the composition of original works.

5. AGENCY

The above example does not mean to suggest that automated composition of tonal musics is necessarily without artistic merit for the composer. However, *agency* is critical in determining how the aesthetic significance of the musical work is evaluated. This notion will be explored in regard to two primary creative fields within computational musicology of tonal musics; accompaniment/improvisation systems and live coding.

5.1. Accompaniment and improvisation systems

Score-based accompaniment systems need to engage with aspects of musical rhythm, pitch, and timing in order to recognise input from the primary musician; this involves pitch, beat, and pattern recognition to effectively follow a score (Dannenberg and Raphael, 2006). Accompaniment systems which improvise to a musician's input likewise monitor acoustic (or other musical) input, but generate new, rather than predetermined, output in response. The contexts of these improvisation systems are for the most part tonal, and commonly based around jazz. Such systems (including Rowe's *Cypher* (1993), Thom's *Band-OUT-of-a-Box* (2001) and Pachet's *Continuator* (2003)) employ musical grammars, recombination, and probability tables, techniques common to style synthesis.

5.2. Live coding

Live coding is more difficult to precisely define, as given the ephemeral nature of live coding, few examples of code exist (see Collins et al., 2003); furthermore, there is little commonality of musical style, aside from perhaps a general trend towards contemporary electronic music. Although many live coding performers engage with timbre, granular processes, and samples, a number also deal directly with tonal note generation as well (Brown and Sorensen, 2009). The situation of live coding as a primarily performance-based pursuit engenders the building of musical structures from scratch, and can involve probability routines, recursion, and defining pitch spaces.

In both these circumstances, the agency of a performer (as composer) is key. While style synthesis assumes a minimal amount of creativity (as opposed to research input) by the composer, both accompaniment/ improvisation systems and live coding require a real-time creative agent in order to operate. The condition of creative agency is thus fulfilled by the performing musician/composer or the real-time 'operator' of the system, rather than by the system itself or by its designer. The following section will discuss further applications of agency in creative computation of tonal musics.

6. 'ORGANIC' SYSTEMS FOR GENERATING TONAL MUSIC

6.1. Defining 'organic'

It appears that the creative application of generative tonal musical systems is limited by the specificity inherent in style synthesis, by a focus on computational rather than compositional creativity, and by the issue of compositional agency. Using the fifth definition of 'organic' in Collins' English Dictionary (online), "made up of many different parts which contribute to the way in which the whole society or structure works, an organic whole", it is proposed that an 'organic' musical system can be used for creatively generating tonal music. This involves interlinking different musical processes and parameters to create a variety of musical results, rather than to model one specific musical style.

6.2. Commercial precedents

SSEYO's *Koan* (now known as *NoaTikl*, *MixTikl*, and several other programs) (Intermorphic, 2011) is probably the most well-known system for broad, non-style-specific generation of tonal musics. Similarly, musical 'games' for generative tonal composition are plentiful; see Eno and Chilvers' *Bloom* application for iPhone (2008), or Wolfram Alpha's *WolframTones* music generator (2011), which uses cellular automata to generate music spanning several styles. These however appear to be mainly for gaming or recreational purposes, rather than for composition as such. The lack of use of systems such as these for composition may be attributed to several factors; oversimplicity, limited customisability for the user, lack of quality control, limited context-awareness, and restricted directability. As third-party products, these programs do not permit the necessary creative agency for the composer.

A recent review of Apple's *GarageBand* application for iPad clearly illustrates the problems (and also the advantages) of increasing computational creativity at the expense of composer creativity. This easy-to-use version of *GarageBand* incorporates some very basic generative features, including changeable note densities for drum loops, arpeggiation, and key-limited pitch spaces. The reviewer writes, "No musical talent?

[1] Cope remarks that composing with Emily Howell involves accepting or declining Howell's output, "much in the way he would if he was composing 'conventionally'" (cited in Adams, 2010).

No problem", and, "Even if you can't play a note, GarageBand for iPad will have you playing and even composing music soon after you first fire it up" (Silverman, 2011).

The approaches discussed so far—style synthesis, accompaniment/improvisation systems and live coding, and commercial applications—represent three different approaches to generating tonal musics, each with particular benefits and drawbacks. Characteristics of style synthesis include musical complexity, specific output, and use of higher-level structures, but limited creative agency. Live coding and accompaniment systems allow more agency, but are primarily performance-based endeavours. Third-party applications can permit generation of musical events across a range of styles, but limited customisability and agency for the end user. The following will consider an organic system for tonal composition based on the author's research.

7. DEVIATE (2006–)

Deviate was originally designed as a live improvisation system for contemporary popular electronic music and related loop-based styles as part of the author's PhD thesis (2010). Unlike improvisational accompaniment systems, it takes computer-based interaction as the only input, and does not perform any sort of signal- or MIDI-based machine listening. Built in Max/MSP, it generates MIDI notes which can be routed to a multi-track sequencer for recording and to VST/AU plug-ins for sound generation. Audio and video examples are accessible online at www.cetenbaath.com/cb/ and www.cetenbaath.com/cb/thesis-tracks/.

7.1. Components

Deviate's components aim to cover fundamental musical processes and features, such as tempo, time signature, rhythmic patterns and pitch spaces. Further controls include note transposition, variation from rhythmic patterns, and variable density and tonality of generated notes, while more mechanical features such as velocity and duration of note events can also be altered.

Drums (6 total):	**Pitched output (16 total)**	**Master-level control**
Rhythm template selection (10)	Pitch template selection (12)	Tempo
Variation from rhythm	Rhythm template selection (12)	Time signature
Density	Tonality	Swing
Automatic template variation	Density	Playback alteration
Automatic pattern variation	Tonic note	
Automatic density variation	Chord selection	
	Note duration	
	Order of playback	
	Melodic contour	

Table 1. List of controllable parameters in *Deviate*

A central aim in building this system was the ability to generate music in a range of styles, and as such *Deviate* is built in a modular fashion, where many generators are used to create note events. Generation parameters for each of these modules can be independently selected.

Deviate incorporates features including melodic contour, hypermetric organisation (Huron, 2006: 248), probability-based determination of pitch (Temperley, 2007: 3), reinforcement of tonal centre (Jones, 1993: 76) tonal hierarchies (Krumhansl and Kessler, 1982), and rhythmic grouping (Toussaint, 2005). Up to 22 modules (6 drums and 16 for pitched content) are used to generate musical output, each with parameters that can be independently controlled. A brief list of these parameters is detailed in Table 1.

8. EVALUATING AGENCY

Deviate's construction, involving modular note generators with individually alterable parameters, has proven to be capable of generating very varied musical output. Its capacity for customisation and user interaction with processes provides substantial room for compositional agency. Furthermore, its construction based on general musical rules, rather than analysis of a musical corpus, allows processes to be independently varied. These loosened constraints extend the applicability of the system.

8.1. Known drawbacks

There are, however, some issues with *Deviate*'s construction. Its modules and processes are not interdependent, and there is zero context-awareness between modules and in note generation, meaning that processes such as voice-leading are impossible. Because it was developed to be a live system, usable and comprehensible in real-time, it generally lacks complexity and higher-level structures. There is so far no facility for timbral awareness, as only MIDI note data is generated. Lastly, given the 'broad' nature of this system, it does not always generate musical output that is consonant or aesthetically pleasing, although it is certainly capable of doing so.

8.2. Advantages

It is possible that these drawbacks are, to an extent, a key part of its application as a creative composing system. A system capable of robustly generating musical output would suggest that it possesses sufficient 'computational creativity', thereby diminishing the agency of the composer. This would also imply that many musical rules and processes are in operation, as well as higher-level rules determining aspects such as aesthetics and musical form. This in turn may make meaningful and comprehensible compositional agency within such a system difficult.

8.3. Areas for future research

Still, many improvements are possible. Firstly, further testing of rhythmic and pitch templates could improve musical output. These templates could be designed via a genetic algorithm process. Secondly, interface and control elements could be improved through detailed recording and analysis of performance. A final area for

investigation is the incorporation of a limited amount of inter-module awareness for more consonant polyphonic output, although this may come at a trade-off for compositional agency.

9. CONCLUSION

The question of user agency is important in considering the application of findings and techniques from computational musicology in composition. A system for creative composition must permit sufficient compositional agency in order to be used creatively. Style synthesis, although it generates high quality musical results, often does not allow agency because it is tailored to specific ends and its rules are highly complex, often requiring analyses of an existing corpus. Live coding and accompaniment/improvisation systems necessarily allow creative agency as they incorporate a performative aspect. Finally, commercial applications are often aimed at a "game-like" rather than a creative use of music generation, and can suffer from lack of musical complexity, customisability and agency for the composer, where a lack of musical talent becomes irrelevant (Silverman, 2011).

Agency is ultimately dependent on the system's complexity and flexibility; the issues discussed support a modular, organic approach, which exponentially increases the range of musical results and permits a high level of engagement and agency for the composer. It is proposed that there is a trade-off between computational creativity and compositional creativity, and that a successful system for generative tonal composition must balance both these aspects.

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LIVING WAVES: GEOLOGICAL MUSIC

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ABSTRACT

This paper describes the creative and technical processes involved in producing the multi-channel fixed-media composition *Living Waves,* commissioned for the interdisciplinary Ruskin Rocks Project[1] led by Leeds University. The project brought together musicians, artists and scientists to create two new 21st Century stone percussion instruments (the Brantwood Musical Stones[2]) to be housed at Brantwood, the former home of artist, naturalist and social commentator John Ruskin, now a museum situated in the Lake District. The two tuned percussion instruments have been created from ringing rock extracted from several Cumbrian Quarries.

Living Waves has been inspired by the thoughts, writings and paintings of Ruskin. It is the sub-title he himself gave to *Deucalion*, his book on geology. It is also the name of a recent exhibition at Brantwood of some of his drawings and watercolours in which he explored the many repeated patterns which occur in nature. Mountains are viewed as fluid and dynamic forms, living waves created by the shifting of the earth's crust.

Through the compositional process I have tried to mirror Ruskin's visual explorations from a sonic perspective, using a range of transformational techniques to convey repeated patterns in different soundscapes at both the micro and macro level. On a conceptual level, Ruskin identified with the medieval scholars' concept of *Musica Mundana*, or 'music of the spheres'. A celestial harmony. The three branches of the medieval concept of music were laid down by Boethius as *Musica Mundana, Musica Humana* (the internal music of the human body), and *Musica Instrumentalis* (sounds made by singers and instrumentalists). In the context of this piece I have interpreted these as: sounds naturally occurring in nature; speech and the sound of manmade machinery; and instrumental sounds (in this case improvisations, on the new Brantwood Musical Stones, played by percussionist Dame Evelyn Glennie).

During the period from January 2010 – September 2011 ambisonic field recordings were taken of the natural environment that the rocks are taken from as well as location recordings at several quarries including the sounds of diamond saws, water-jet cutting, and rock blasting. These contrasting soundscapes are presented as sound paintings, weaved together by more abstract musical passages employing the sounds of the Brantwood Musical Stones and passages of text by Ruskin. Sound transformation and spatialisation techniques are employed to mimic some of the patterns occurring in the sound paintings, including the use of BATBoids[3] which uses an implementation of Craig Reynolds' Boids algorithm (mimicking the flocking and swarming of birds, bees, fish etc). Close collaboration with the performer during the recording of improvised material on the instruments ensured there were adequate opportunities for transformation and mimicking between the instrumental and soundscape material.

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[2] Designed by Kia Ng (Leeds University) and Marcus de Mowbrey (freelance musical instrument maker)

[3] 'BATBoids' v 1.00 © 2007 A P Lewis, Bangor University (Wales), part of the Bangor Audio Toolkit, www.bangor.ac.uk/music/studios/BAT

AWKWARD ECOLOGIES: THE 21ST CENTURY BAROQUE

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ABSTRACT

An examination of those forms of music that are genealogically related to musique concrète – identified variously as acousmatic music, electroacoustic music, and sound-based music – which makes the core argument that the mastery of sound matter, and by extension of nature, is now extensive enough that the modes of intention and reception underpinning these forms of music, are secondary to a particular mode: the technological. The talk unfolds this argument as a necessary preamble to my conclusion, which sketches a relationship between this argument and the notions of ecology and a 21st century baroque.

1. AWKWARD ECOLOGIES: THE 21ST CENTURY BAROQUE

My talk is part of a larger project that seeks to engage in an ecologically informed way with the practice of making sound-based music (Landy 2007) with technology. Clearly I am far from alone in doing this, as evidenced by many of topics being addressed at this conference and the recent EMS conference in New York, and the relatively recent rise in the arts and humanities of what is sometimes called ecocriticism, which has made its way into electroacoustic music studies via the most recent issue of *Organised Sound*.[1] What I think I am working towards is the replacement of the ethos of technocratic modernism – which underpinned musique concrète and perhaps still underpins electroacoustic music – with ways of thinking about and making music with technology that is appropriate to the large-scale challenges of our contemporary milieu, especially at an ecological level. Certainly this informs my own practice. Set within the specific context of this present conference, this might be defined as a re-auralising of the organic and our relationship to it. It is also predicated on a deliberately naïve approach, so as to enable me to get around the presuppositions that I'm most likely to bring to bear upon my listening and thinking.

Before launching the body of my subject, let me give some historical context by reprising the notional failure of musique concrète which is the background to my talk. This is summarized from an essay of mine recently published in *eContact!* (McKinnon 2011), and it should be pointed out that the gist of this argument is by no means entirely original (see Kahn 1999, and Harrison 2006).

Toward the end of his life, Pierre Schaeffer soundly dismissed musique concrète:

> Musique concrète in its work of assembling sound, produces sound-works, sound-structures, *but not music*... The whole problem of the sound-work is distancing oneself from the dramatic… Just as a painter or sculptor moves away from a model, stops representing a horse, or a wounded warrior, and arrives at the abstract… And *if you continue this abstracting movement… you get to music*. (Schaeffer and Hodgkinson 1987) [emphasis mine]

Schaeffer was at pains to find a means to eliminate the extrinsic from sound-based music as he – like his nemesis Pierre Boulez – maintained that the referential and semantic qualities of sound significantly diminished its formal, which is to say musical, potential. This he argued could only be developed at an intrinsic level, through the perception of similarities and differences in purely sonorous qualities which would "weaken the interest which could be taken in identifying objects which would otherwise present as a series of heterogenous events" (Schaeffer in Chion, 32)

This was spurred in part by Boulez's scathing critique of musique concrète in his short essay *Music (Concrete)* (1966), a title playing sardonically upon the meanings of the word concrete, but clearly favouring the noun:

> The word concrete reveals how [electroacoustic technology] was misled and the gross way in which the problem [of using this technology] was envisaged; it was aimed at defining a material manipulation of sonority, that sonority itself not responding to any definition or being subject to any restriction… That lack of directing thought resulted in total insolvency of the exploration… (290)

This effort to 'purify' sound was largely addressed through the practice of reduced listening [*ecouté reduite*], an aesthetic-cognitive practice derived by Schaeffer from the phenomenological reduction or bracketing [*epoché*] developed by Edmund Husserl. Reduced listening was conceived as the cognitive practice that would support Schaeffer's "Solfège of the Sonic Object [which] purports to take, from the practice of sound-producing bodies… a universal musicalness through a technique of hearing… [from this] new musics are expected to arise, based on reference structures that should play, for each of the seven morphological criteria, a role similar to that played by interval relations and the games of tonality and modality." (Palombini 1999).

[1] *Organised Sound*, 16(2)

Schaeffer's solfege was the core of his effort to establish a conceptual structure through which sound-based music could be understood to operate: reduced listening acting as the cognitive apparatus which maintains the solfege system as an agreed set of units within a hierarchically organised structure, such as already exists in instrumental music.

In short, Schaeffer sought a means to cleanse sound of its picaresque impurities so as to allow musique concrète to function in the same abstract and formal ways deemed essential by the Boulez and his fellow high modernists. It was this direction that Schaeffer followed, rather than embracing the vibrant heterogeneity of sound and the consequent problematisation or complexification of the notion of musical discourse. Concomitant with this theoretical endeavor to master sound was the drive towards the practical instrumentalisation of sound, a project integral to Schaeffer's leadership of the GRM: "Pierre Schaeffer would promote the construction of new machines that would simplify the operations on sound." (Terrugi 2007, 216). Machines such as the phonogène in various forms, the morphophone, and others, have played a key role in the development of electroacoustic technologies, as the ongoing popularity of GRM Tools attests. It is perhaps fair to say that electroacoustic technologies are now sufficiently advanced to be understood as musical instruments, through which virtuoso composers can largely master the sound world. As Boulez (2010) recently put it "With electronic media, one cannot only create new sounds and timbres, but one is free to fulfill all wishes, limited only by the imagination itself." The extent to which such a utopian vision is true is open to some debate, but it certainly points to the intentional will-to-power, manifest through technology, which is a strong aspect of the making of sound-based music, and also, a dominant aspect of its interpretation also. I'll get into this argument by looking at intentionality.

Philosopher of music Roger Scruton maintains that "The notes in [instrumental] music float free from their causes… What we understand, in understanding music, is not the material world, but the intentional object: the organisation that can be heard *in* the music" (Scruton in Hamilton, 2007). Needless to say the idea of music-as-intentionality is exactly what Schaeffer was trying to get at through the cognitive transformation of sound (reduced listening, the solfege) and its technological transformation also (see Schaeffer in Terrugi 2007, 215). In this, the concrete sonorous experience, the experience of sonic materiality, is to be set aside in favour of the abstract and the formal: we listen *past* sound, whether that of banal noises or rarefied instruments, to encounter music. As expressed in terms familiar to electroacoustic composers, "Musique concrète… starts from 'concrete' (i.e. observable) qualities in sound material itself, from which structural implications are extrapolated and links to other sound materials made. What finally emerges… is an 'abstract' musical discourse" (Harrison 2006, 4). Despite whatever differences there might be still be between instrumental and electroacoustic music, they are closely allied in this abstracting tendency, which derives from an ideology of transcendence in Western art music, which in turn is a more general and persistent mindset in Western cultures. Much more needs to be said about this, but in this context such my shorthand will have to suffice.

In both instrumental and electroacoustic music then, we listen through sound to arrive at music. But where does this leave Scruton's intentional object in regard to sound-based music? If we do not have an abstract system that somehow lies behind sound, as exists in note-based instrumental music, where do we arrive at apropos the "the organisation that can be heard *in* the music" that both Schaeffer and Boulez hold dear? If, following Harrison, in sound-based music the abstract emerges *out of* the observation of the concrete, and, following Smalley (1997, 14) "There is no permanent type of hierarchical organisation for all electroacoustic music, or even within a single work", then is there any intentional object to be found *beyond* sound in sound-based music? A response to this conundrum might follow three lines.

Firstly, following Schaefferian principles, there may well be music beyond sound, but it is most certainly not of the kind that Schaeffer hoped to find; that is, there is no structural equivalent in sound-based music to the structures of note-based music. Secondly, an answer obvious to electroacoustic composers: surely the intentional object of sound-based music is found in the qualities of sound itself, rather than in something which floats magically behind it? This is the logic and value of Smalley's spectromorphology project; intrinsic sonic qualities, extrinsic references, the threads between these, are concrete rather than abstract. Even here though, the principles of abstraction and formalisation remain in play. It is not that Smalley's taxononomy of texture, gesture, behavior and so forth cannot be discerned in sound-based music, but that these are *knowingly* discerned and this – like reduced listening – is a secondary level of awareness, a cognitive practice that has to be learned. Thirdly, is there a more immediate sense of the intentional object? The standard way to approach this follows Schaeffer's line: first we listen referentially, for sources and causes, before moving onto more symbolic modes of listening (semantic and reduced listening). I'd like to dwell a little longer in this initial causal level, because perhaps it is the case that the technologically mediated art of electroacoustic music doesn't necessarily lead us to the notional object dubbed the source-cause, even when what we hear is primarily phonographic, involving only an absolute minimum of technological intervention. Instead, it might be said that the most fundamental aspects of referential listening are on the one hand listening for *intentionality itself*, listening in the awareness that what we hear is intentional rather than accidental, and on the other hand, listening for clues as to how this intention is played out. In electroacoustics, this is analogous to saying of

instrumental music, "I'm listening to someone play the violin". There is a technology and someone using the technology. There is an interaction between someone and a something, or somethings, and one of those somethings is always a piece of technology. Perhaps this interaction is the most basic intentional object we can access, and technology is implicit in it. At one level this is an inane statement to make in the context of a computer music conference. But what I'm doing, following Heidegger rather than Husserl, is trying to draw awareness to something that is so close to us that we do not hear it anymore.

One significant consequence of such renewed awareness is that a primary aspect of referential listening is technology, or at least the interaction between an originating source-cause and a technological intervention with it that is a secondary source-cause. This is more or less to say that technological listening is unavoidable to some degree because it inheres in the electroacoustic experience itself. It is only the practiced listener who can detach sound from the electroacoustic apparatus, to hear it as sound rather than as technologically mediated sound, and this in itself might well be classed as an entry-level form of reduced listening. Were this *not* something that we all tend to do as technologist-artists then we might have witnessed the advent of an electroacoustic equivalent of *"Hörspielstreifen* [in film], the delicate atmosphere of recorded silence whose purpose is to imperceptibly confirm the presence of a reproduction underway and not frighten the audience into thinking there has been a technical malfunction (which would require a break in the silence of the audience itself)… (Kahn, 181).

Given all this, Smalley's concern for technological listening needs to be reframed somewhat. Smalley writes that "The composer, or other listeners conversant with technology and techniques, cannot easily brush aside a particular listening mode which I call technological listening. Technological listening occurs when a listener 'perceives' the technology or technique behind the music rather than the music itself, perhaps to such an extent that true musical meaning is blocked." (1997, 109) What I am suggesting is that technology *cannot* be brushed aside. This is because one important aspect of the intentional object of sound-based music is the listening for the technological manipulation and organization of sound. Intentionality is, at one level, a sonic concretion of technology; the intentional object emerges from the interaction of composer, technology, and sound. Where Smalley states that "Ideally the technology should be transparent, or at least the music needs to be composed in such a way that the qualities of its invention override any tendency to listen primarily in a technological manner" (109), I would rather say that the technology is never transparent and that what constitutes technological listening is instead an informed listener's awareness of the level of taste and skill with which technology has been deployed. For a naïve listener this is not the case and technological listening, as a mode of referential listening, tends to predominate, attaching itself along the way to whatever semantic reference points are suggested by technology, hence the reactions Andra McCartney (2002) encounters in her reader-response study "Alien Intimacies: Hearing science fiction narratives in Hildegard Westerkamp's *Cricket Voice*".

In terms of the ecological task of this talk, I am interested in the ways in which this version of technological listening alerts us to the fact that we are inside technology and that it changes, in some instances profoundly, the ways in which we hear and relate to the world through sound. There are many ways in which this idea might be developed but for now, given that time is pressing I would like, in closing, to offer one suggestion: that it offers a refigured understanding of our relationship with technology. Rather than approaching it as something other to ourselves, as tools that we somehow "just use", we must understand ourselves as part of an electroacoustic *assemblage* (see Deleuze and Guattari 1987, and Bennett 2010) comprising human and non-human, the latter both natural and artificial. This might be understood as an *awkward ecology*, not least of all given the modernist ethos of mastery that I mentioned earlier. As part of an assemblage, which is not an organic unity "in which the parts obediently serve the whole" (Bennett, 23), we both cede power to and draw power from other elements in the assemblage. Thus the composer is not detached from technology, or the soundfield, or the sound materials that yield so-called sound objects, but is intimately enmeshed in these. Not only this, but compositional outcomes are as much determined by the technology and the materials the composer is working with as they are by the composer. Sounds and technologies become primogenitors of music in collaboration with composers. Compositional consciousness is altered by the sounds and technologies it is engaged by and with. Our auralisation of the world is transformed by electroacoustic technologies, just as the visual technology of linear perspective transformed Western culture in the 16th century (see Romanyshyn 1989). This is baroque or grotesque, in that it challenges our sense of the proper order of things, and also in the sense that it is suggests a thing "roughly thrown together", in Jane Bennett's words an "ad hoc groupings of diverse elements… living, throbbing confederations that are able to function despite the persistent presence of energies that confound them from within" (2010, 23). This seems an excellent definition of the raw energy of early musique concrète, though it is certainly not the baroque Schaeffer talked of towards the end of his life, which steps backwards into the future: "It will be when our contemporary researchers… realise that there's no way out of traditional music, that we can get down to a baroque music for the 21st century" (Schaeffer and Hodgkinson 1987).

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THE GALLERY AS AN INSTRUMENT: USING REMOTE SENSING TECHNOLOGY TO INTERFACE WITH MUSICAL ROBOTICS

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ABSTRACT

This paper describes the study and use of proximity sensors as a method of control for musical robotic systems. Through the use of such sensors, installation viewers can interact with musical robotic systems in an organic, highly gestural manner. The non-synthetic nature of musical robot timbres is discussed, as are the highly specific options for sound spacialization in a musical robotic installation setting.

Throughout the paper, we describe the creation of a testbed-style installation called *Metal+Motors*. A variety of remote sensing options are presented, as well as reasons for our decision to use ultrasonic transducer rangefinders. Test results verifying the suitability of the ultrasonic rangefinders are shown, followed by an in-depth discussion of the spacialization and modes of interaction used in *Metal+Motors*. Finally, options for future work in the field of interaction with robotic musical systems in an installation context are presented.

1. INTRODUCTION

Modern widespread availability of remote sensing technologies allow for the creation of highly expressive musical environments. Using physical computing platforms and such devices as ultrasonic transducers, infrared sensors, and hall effect sensors, artists can explore the possibility of using audience and performer position to affect audiovisual elements of a work.

Remote sensing presents an interesting set of creative possibilities to artists who work in the domain of musical robotics. Through the use of the aforementioned sensing technologies, the artist can allow the installation viewer to directly interact with the mechatronic components of a robotic sculpture, opening up heretofore unexplored methods of human/robotic interaction. Viewers, for example, can experience their physical movements become translated in real time into timbres which morph as a result of the viewers' movements within a space. In this manner, then, the audience interacts with sensing technologies and musical robots to turn a gallery space into a single cohesive instrument: one played with the viewers' motions and projected throughout the space by the organic localization offered by mechatronic musical assemblies.

This paper explores a robotic sound art installation (entitled *Metal+Motors*) in which viewers interacted with musical robotics through physical motions which were collected via ultrasonic sensors. After a section focusing on the history of musical robotics and an examination of sound art as a potentially organic and immersive experience, the remote sensing technologies used in the aforementioned installation will be discussed, along with tests undertaken to determine a variety of sensors' suitability. Section 4 focuses on the localized and organic aspects of the soundscapes generated in Metal+Motors, as well as a detailed examination of the piece's methods of audience interaction. Finally, avenues for future work are discussed.

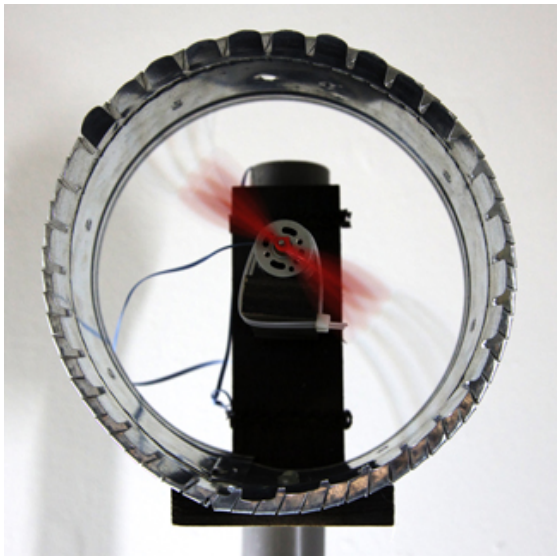


Figure 1. A Robotic Appendage from *Metal+Motors*

2. RELATED WORK

While there has been much work done on the musical potential of remote sensing technologies [1], the literature holds few examples of such technology being utilized in an installation environment involving musical robotics. The goal of the research undertaken in this paper was to combine the intense sonic localization possible with musical robotics with the spacial expressivity available through the use of remote sensors. The subsequent two subsections focus on the history of musical robotics and on remote sensing in interactive sound art.

2.1 A Very Brief History of Musical Robotics

Sound artist Trimpin is one of the early explorers in the field of both musical robotics and interactive kinetic art installation. Since the mid-1970's, Trimpin has created a diverse array of sonic artworks often based on found objects and surplus materials, often making use of novel forms of audience interaction [2]. Trimpin's work has provided an underlying current for much subsequent activity in the medium of musical robotics, and was a major inspiration for both the robotic and interactive components of *Metal+Motors*.

Other significant early workers in the field include Godfriend-Willem Raes [3], a builder of both installation-based and performance-oriented mechatronic sculptures, the California-based Survival Research Laboratories [4], and the group of robotics researchers at Waseda University [5]. Australia-born performance artist Stelarc [6] has explored noise-oriented kinetic sound art. Stelarc's installations and performances will be discussed in more detail in the next subsection. Recent workers in the field of musical robotics include the second author [7] and Gil Weinberg [8], both of who have engaged in extensive work on human-robot interaction and performance scenarios, New Zealand-based musician Greg Locke, and the New York-based installation and performance artist Eric Singer [9].

While all of the above have contributed greatly to the field of musical robotics, few have explicitly explored the potential for remote sensing technologies to be used as a means of controlling the spacialization of sounds generated by musical robotic assemblies.

2.2 Sound Art as an Immersive Experience

Kinetic installation art has a rich history of innovative systems of interaction, some of which were highly inspiring during the inception and execution of Metal+Motors. Performance and installation artist Stelarc [6], for example, makes extensive use of nontraditional methods of interaction with his artworks. In addition to electrodes and pressure sensors, he has explored computer vision technology to allow for remote audience interaction with his sculptures.

In a more traditional musical context, sculptor and instrument builder Patric Flanagan, creator of the JAZARI african robotic percussion ensemble, has made use of wireless accelerometers to enable highly gestural and unrestrictive control of his creations [10].

While many sound artists choose to focus on the use of loudspeakers, some eschew such techniques in favor of highly organic physical means of sound generation. Sound artist Zimoun, a key inspiration for Metal+Motors, makes use of highly dense clusters of actuators, solenoids, and other novel noisemaking devices [11]. Such arrays allow for unprecedented density in sound and specificity in spacialization, enabling audiences to become truly immersed in a soundfield.

The above are merely a few workers in the diverse field of interaction and spacialization, we have made an effort to draw upon their work to create a dynamic sculpture wherein the entire gallery space becomes a playable instrument featuring localized families of timbres.

3. REMOTE SENSING

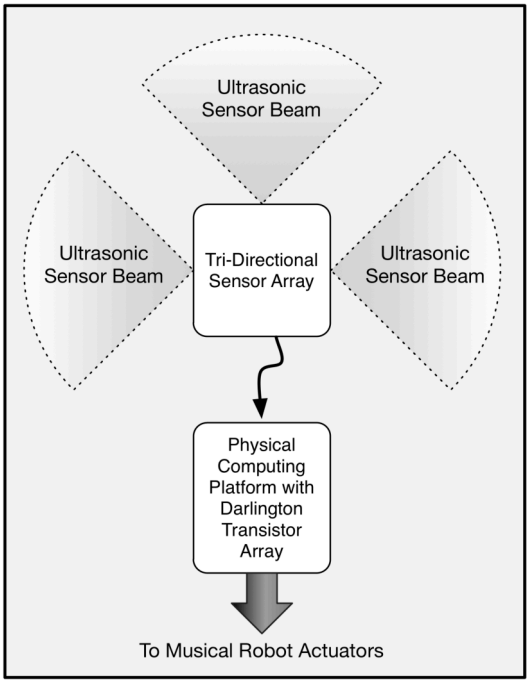


Figure 2. Remote Sensing Apparatus as used in *Metal+Motors*

3.1 Remote Sensing Technologies

In this context, the term "remote sensing technology" applies to non-contact sensors: sensors which don't involve a direct physical connection between user and sensor. We explored a variety of sensor technologies with a focus on photoresistors, infrared distance sensors, and ultrasonic transducer rangefinders. Photoresistors have a long history of use in musical interfaces: Geoff Rose's 1977 Laser Harp [12] and Szajner et Al's MIDI Laser Harp [13] date back to the early days of electronic musical interfaces and make use of photoresistors to detect the presence of absence of light.

More recently, Leila Hassan's Terminova [14] and Meason Wiley's MLGI [15] use modern analog to digital conversion technology to detect different levels of light reaching a photocell, allowing for variable control and proximity sensing. Photoresistors, though, were found to be highly susceptible to interference from ambient light sources and for this reason were determined to be unacceptable for use in a lighted gallery space.

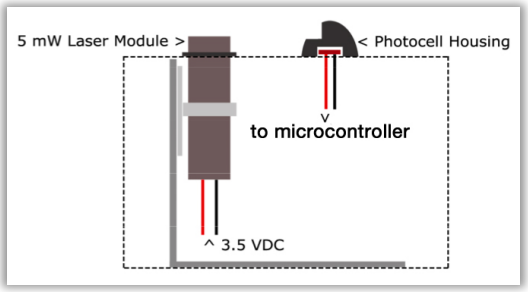


Figure 3. Laser Proximity Sensing using 5mW laser and photocell. Diagram courtesy of Meason Wiley.

While photoresistors are highly susceptible to visible light interference, infrared-based sensors behave much more predictably under normal lighting conditions. For this reason, we closely examined the possibility of using infrared sensors in the *Metal+Motors* installation (see section 3.3). Infrared sensors typically function by measuring beam angle as it is reflected off of an object.

A myriad of infrared sensors are available: many are manufactured by Sharp, including the popular GP line of sensors. Infrared sensors have found their way with much success into numerous mass-market musical instruments and interfaces including Roland's D-Beam. In academic circles, IR sensors have likewise found wide use: Ivan Franco's Airstick [16] and Chikashi Miyama's Peacock [17] both make use of arrays of IR sensors to provide non-contact interaction with musical systems.

Ultrasonic rangefinders were the third family of remote sensors closely examined for use in the *Metal+Motors* project. These sensors function by emitting a high frequency pulse and timing the duration between pulse emission and receipt of reflection signals. Like the aforementioned infrared rangefinders, these sensors are not susceptible to ambient light interference. Numerous models of ultrasonic rangefinders are available, including bare-bones transducers driven with user-built software from a parent microcontroller, surplus units salvaged from camera rangefinder equipment, and purpose-built modules complete with a dedicated microcontroller. Section 3.2 details the sensor test process and discusses our final sensor choices.

3.2 Sensor Tests

The use of non-contact sensors in a gallery environment presents a unique set of challenges compared to the above uses of such sensors in self-contained musical interfaces. Unlike the above interfaces, we were interested in receiving data over a much larger range: while Miyama's Peacock sensor, for example, is intended to pick up signals less than a meter from the emitters, we required much greater ranges on our sensors to permit gallery attendees to interact with the installation over ranges of more than three meters. In order to determine the suitability of sensors for use in the *Metal+Motors* installation, we began by examining manufacturer-provided datasheets to choose which sensors were worthy of closer study. Table 1 provides data for four such sensors.

Table 1. Exemplar Proximity Sensors

Sensor Name	Sensor Type	Range	Linear
GP2Y0A2YK0F	Infrared	1.52m	No
GP2Y0A21YK	Infrared	0.8m	No
Maxbotix LV-EZ0	Ultrasonic	6.45m	Yes
Maxbotix XL-EZ4	Ultrasonic	7.65m	Yes

The relatively short range and lack of linearity of infrared sensors quickly ruled them out: greater range and a more linear response curve were desired. While we experimented with applying a linearization transform function to the output values of an infrared sensor, the short range nonetheless resulted in our choosing to make use of ultrasonic rangefinder units in the *Metal+Motors* piece.

Price differences and differences in beam width led to our choice of the LV-EZ0 over Maxbotix's XL-EZ4. To verify our decision to use the LV-EZ0, we conducted a range and data linearity test of the sensor. Table 2 illustrates the test results.

Table 2. Ultrasonic Sensor Output Compared to Measured Distances

Measured Distance	Sensor Output Result
100cm	95cm
200cm	191cm
250cm	252cm

As shown in Table 2, the measured distance differs slightly from the actual sensor output. In practice, though, the differences were felt to be negligible: high levels of precision were not required for the installation. Additional precision could possibly be obtained by sending data via serial or pulsewidth modulation rather than raw analog data: the analog data likely contains additional noise and jitter which would not be present in a serial signal transmitted from the LV-EZ0's onboard microcontroller.

3.3 Ultrasonic Sensor Placement

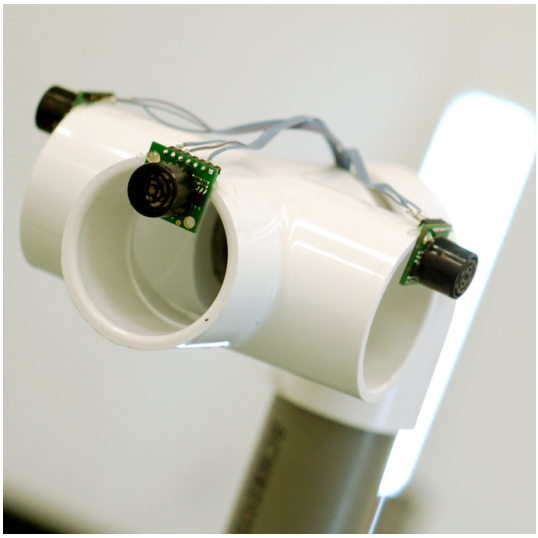


Figure 4. Tri-directional Ultrasonic Sensor placement

In order to allow viewers to more fully interact with the dimensionality of the gallery space, we decided to utilize a tri-directional sensor placement technique (see figure 2 and figure 3). Three sensors were placed at 90 degree angles to one another, each of which controlled a subset of robotic musical appendages. The wide-ranging area covered by the use of three sensors allowed for a deeply organic mode of interaction: once the three sensors were placed in the space, a large percentage of the gallery space became usable as a gestural performance space wherein audience members could convert their movements to real-world sounds created on the musical robot assemblies.

While we decided early on in the build process to utilize a three-sensor arrangement, we encountered technical issues specific to ultrasonic transducers: the use of multiple rangefinders resulted in interference as one rangefinder's pulse emission was picked up by another sensor. To overcome this, we made use of a feature on the LV-EZ series of sensors: multiple sensor units can be connected such that their pulses are emitted in a predictable manner and therefore do not produce interference with one another. Once the trio of sensors were connected in such a manner, the issue of interference vanished.

4. LOCALISED SOUND, INSTALLATION, AND VIEWER INTERACTION

4.1 Localised Sound Through Musical Robotics

The conceptual underpinning of the *Metal+Motors* installation was that of non-synthetic objects creating highly spacialised sound in response to audience interaction with non-tactile sensors. Robotic instruments by their nature allow for quite specific localization of sound: since the sounds are being physically produced in an organic, non-synthesized manner, each instrument's sounding body occupies a very real piece of physical space. Through creative placement within the gallery space, then, we were able to create a spatially diverse array of localized sounds.

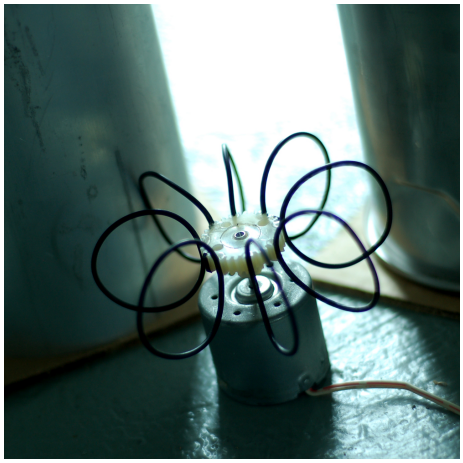


Figure 5. *Metal+Motors* consisted of a variety of DC motor actuators striking metallic objects in response to ultrasonic sensor data. Above is a DC motor bowing assembly shown striking two tuned metal pipes.

4.2 Installation and Interaction

Metal+Motors consisted of three zones of robotic instruments with each zone under the control of an ultrasonic rangefinder. The zones were spaced throughout the gallery, with each zone's corresponding sensor aligned in such a manner that a viewer interacting with a specific sensor would be affecting the zone's instruments in his or her field of vision. In this manner, then, three discrete zones of interaction were created: the gallery was not merely one instrument but rather three separate ones. Figure 2 illustrates the sensor arrangement and corresponding connection between the sensors, the microcontroller unit, and the robotic actuators.

The sensors were mapped into discrete zones: each zone was 100cm in depth and was connected to a variety of robotic arms. As audience members moved through the different zones on each sensor, the musical robots would produce differing sounds. Users were observed to quickly learn the different zones and become able to, in essence, play the sculpture by moving their bodies in and out of different detection zones. Through this mode of interaction, a wide variety of metallic and organic timbres were produced.

5. FUTURE WORK

While we were happy with the combination of the highly localized organic timbres and the deep level of audience interaction and expressivity obtained during the installation of *Metal+Motors*, much work remains to be done on the concept of using a gallery's space as an expressive instrument. While the use of non-contact sensors prevented any form of haptic feedback, we hope to explore such force feedback systems in future installations and performances. Haptics have the possibility of further immersing viewers and performers in musical robotic soundscapes and creating a deeper connection between performer and instrument.

Additionally, we plan on extending the concepts explored in *Metal+Motors* to the use of other non-synthetic objects such as wood, stone, and water as timbal sources. Finally, we intend to make use of the knowledge gained during the research and development of *Metal+Motors* to further enrich performance-oriented musical robotic scenarios, such as Kapur and Darling's Machine Orchestra.

6. CONCLUSION

Metal+Motors proved to be an excellent testing ground for proximity sensors in an installation context. We found ultrasonic transducers to be a highly satisfactory solution for detecting and tracking the position of installation viewers: their relatively long ranges and predictable response curves allowed us to map robotic programs to various points in their response zone. We feel that the non-synthetic sounds produced by physical objects being controlled in a deeply gestural manner led to the creation of interesting organic and metallic soundscapes which warrant much further exploration.

7. ACKNOWLEDGEMENTS

Thanks to Rodrigo Restrepo and to Trimpin.

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INTENTIONALITY, TEMPORALITY, CORPOREALITY AND ORGANICISM IN LIVE ELECTRONICS

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ABSTRACT

This paper responds to John Croft's critique of compositional approaches to works for instrument(s) and live electronics, as presented in his article "Theses on Liveness"[1]. Backgrounded by a discussion of the metaphor of "organicism" in music, and its extension from purely formal considerations to include the nature of the relationship between performer and computer, this paper teases out further limitations of the medium: namely, an inherent *intentionality imbalance*, an *arrow of causality* and an innate *temporal lag* between performer and computer. Furthermore, if we agree with Croft's diagnosis of much of live electronics' lack of intrinsic "difficulty"—in the sense of the kind of physical labour seen as integral to the experience of live music—then the only route left open is to accept the role of the computer as passive and insentient. Drawing the analogy to both models of passive instrumental design (e.g. the Aeolian harp) and spatial forms (e.g. the reverberant cathedral), this paper proposes a number of possible models that extend beyond Croft's preferred 'instrumental' paradigm, into areas that touch more upon sound installation praxis, to provide 'future work' for composers and sound artists who are concerned to uphold the ontological necessity for retaining the corporeality of the live experience, whilst avoiding the pitfalls of triviality.

1. INTRODUCTION

This paper concerns aesthetic issues that arise in the composition of music for solo instrument (or instrumental ensemble) and live electronics. For the purposes of this paper, I will limit myself purely to discussion of the context in which the live electronics are *not* being controlled by an on-stage musician with a physical, performative presence—which may, *potentially*, mitigate some of the objections rehearsed here—but is instead processed by a device (which I will call, for ease of reference, a "computer") with minimal, or at least unobtrusive and non-performative intervention from either a technician or the performers themselves.

Within this scope, I argue that the dominant paradigm of composition is, in essence, an extension of the late eighteenth-century musical metaphor of "organicism". In live electronics, this approach incorporates not just the formal processes of material proliferation and part/whole relationships, but also encompasses the sonic and interactive relationship between live sound and electronics. The electronics carry much of the work of proliferation and variation, and maintains close sonic ties to the performer's sound material, with allowances for sonic departures as a developmental strategy. It has become the dominant paradigm not just because of historical and cultural exigencies, but also, I argue, because of two inherent problems with the very concept of live electronics: firstly, there is an intrinsic *arrow of causality* from musician to computer; secondly, the two parts of the live electronics equation—human and computer—possess, and clearly display, an inherent *imbalance of intentionality*.

Live electronics systems have an inherently fixed linear temporality from human to computer precisely because *both* the computer is non-sentient *and* the performer is sentient. We accept, even before a single sound of the piece has been heard, that the computer is not "truly" responding, in any sense that might be called intelligence. But any response is intrinsically infallible and high fidelity. On the other hand, we accept immediately that the human is intelligent, whilst being highly fallible.

This imbalance of sentience and fallibility is perceived by the listener as an imbalance of *intentionality*. The performer is always perceived as the *initiator* of any discursive act, for an *intention* must be generated by them for any musical act to take place. Furthermore, there is always the chance that the performer's fallibility may render them mute (thus every musician's nightmare of picking up their instrument for a solo, and being unable to make a note). This fallibility further heightens the tensional aspect of the live experience.

Meanwhile, the computer is perceived as an insentient, passive *responder*. It is worth comparing this situation to the experience of listening to purely acousmatic music, in which we do not try to

[1] Croft (2007)

anthropomorphize the technological agency: we hear the music as a form of 'sonic document'. Without a corporeal presence, there is no imbalance to be resolved.

A good analogy for live electronics is that of the Aeolian harp, an instrument that responds unthinkingly, but not unbeautifully, to the wind passing through it. But a performer cannot engage in a meaningful and truly contrapuntal musical discourse with an Aeolian harp, not even by blowing *really* hard on it: its passivity and lack of intentionality prohibits this[1]. Its passivity, however, is also its beauty, and we can take some measure of aesthetic pleasure from the way it sonifies changing wind currents. I will return to this point later.

The imbalance of intentionality causes a *fixed linear temporality* and *arrow of causality*: the computer must always temporally *lag* the performer, or, at best, shadow the performer. It seems difficult to conceive of a situation in which this is *not* the case, in which we can say that the computer is truly *preceding* the performer. I will also return to a finer-grained discussion of this point later.

Organicism in live electronics, then, accepts as an *a priori* fact this fixed temporal disposition of the live electronics. In fact, it not only accepts it, it exploits it. In organic works, both performer and computer have well-defined roles: the performer acts as the *germinator* of material, which the computer *proliferates*, typically using processes involving variable temporal delays and variation techniques to create a web of material permutations, in the spirit of conventional developmental strategies.

In the article "Theses in Liveness", composer John Croft provides a substantive aesthetic critique of the field. He questions whether most of the dominant paradigms in live electronics are a symptom of "technology fetishism" rather than poetic considerations. I contend that while, in many ways, I believe Croft's objections to have a *prima facie* validity, I do not believe that organicism in live electronics is therefore dead. But it asks of us to maintain a critical approach to the relationship between performer and electronics, particularly in the conception of a live electronics practice that does not fall foul of "shifting the procedural into the foreground", thus becoming "a lamentable 'showcase' of the technology".[2]

2. THE METAPHORICAL ROOTS OF ORGANICISM

The metaphor of organicism arose in the late eighteenth century as concepts of growth and biological and physiological development in the physical sciences were adopted by music critics to replace the earlier rhetorical models of musical discourse. That is not to say, however, that a single, consistent metaphor was arrived at: indeed, there are competing metaphors of organicism that focus on different aspects of the musical work as simulacra of a biological organism. The most prominent and persistent metaphor, however, holds the image of the temporal unfolding of material as a *proliferation* and *variation* of autonomous musical units following accretive and degenerative processes. It could be seen to draw analogies with the processes of cell division, mutation, DNA encoding, natural replication errors and cell death that provide the underlying processes by which biological organisms grow, vary and decay[3]. The central image, however, is that of the concept of a germinating 'cell', which contains within it the recipe for its forthcoming formal organisation, as it sends out and accretes branches, rhizomes and inflorescences to the point where a singular, coherent whole is perceived. To quote nineteenth-century music critic Adolph Bernhard Marx, the motif is "the germinal vesicle, that membranous sac filled with some fluid element…the *Urgestalt* of everything organic."[4]

This metaphor owes its aptness to the cognitive faculties of the brain, particularly its ability to form, and maintain within memory stores, perceptual Gestalts. It can recognise cognitively pertinent feature-types, such as contour and interval of prominent motivic units, and track changes and accretions to these features over time. The metaphor of organicism, through the innate drive towards proliferation and variation extraneous of any sacred or secular function, arises almost contemporaneously with the notion of autonomy in art, which enters the music literature in about 1790, only 20 years after the concept of organicism first appears.

3. THE HISTORICAL AVANT-GARDE AND ITS DECONSTRUCTION OF ORGANICISM

The dismantling of the privileged position of musical organicism came most concertedly with the radical overhaul of materials, techniques and processes that accompanied high modernism. Although serialism was conceived as a fundamentally organic strategy, the limits of human Gestalt-forming abilities were, arguably, breached. It is ironic that Schoenberg ended up having to rely heavily on conventional non-pitch features, such as rhythmic character, texture, register and approximate contour, to convey a sense of familiarity, which the row itself—which was supposed to have been the 'germ'—was unable to provide. With the advent of total serialism in the early 1950s, music achieved the most unusual of situations: a piece in which virtually every notated parameter derives from a generative rule-set, not unlike the construction of a complex organism, but strangely ending up with what many commentators have dubbed as "inorganic".

[1] This partly explains why there are not many Aeolian harp concertos in existence.
[2] Croft (2007), 61

[3] Even though many of these processes were unknown to science at the time of the metaphor's genesis.
[4] Marx (1856), quoted in Spitzer (2004), 291.

Stockhausen's famous description of Messiaen's *Mode de valeurs et d'intensités* as "star music", is one example, in which the inherent stasis and lack of conventional gesturality requires reference to something extraterrestrial.

For a genre as technologically progressive as live electronics, it seems surprising that most works do not engage more wholeheartedly with inorganic formal constructions. But in the light of the inherent temporal and causal limitations of live electronics, not to mention the technological ease of certain modes of audio processing, perhaps it is not surprising. After all, the construction of a situation in which inorganicism dominates proceedings does, in fact, take on not just a conceptual disinclination but also a technological obstacle. The works of Karlheinz Stockhausen perhaps stand out as exemplars of an "inorganic" live electronics practice, though with the qualifier that his works predominantly rely on synthesis, which has the advantage of not being implicated so clearly in the one-way causality I mentioned earlier. I will return to questions of inorganicism later; for now I want to turn to a recent critique of the current dominant practice within live electronics.

4. CROFT'S THESES ON LIVENESS

John Croft's article "Theses on Liveness" is a provocative assessment of the current status of music for instrument and live electronics. It provides an intellectually robust critique of the common practice in live electronics, arguing for the necessity of maintaining a musical ontology that values corporeality as an essential part of the aesthetic experience.

If one agrees with the premises of the argument, then the conclusions are profound for a live electronics practice. If organicism in live electronics extends to the *relationship* between the acoustic and electroacoustic components, then Croft's taxonomy of relationships attempts to reduce these relationships to five basic types, of which at least three would be necessarily organic:

- *Backdrop*: the electronics serve primarily as a ground on which the instrumentalist provides a figure. There are no perceptually causal relationships between the background sound and the live sound.
- *Accompanimental:* the electronics server primarily to accompany the instrumentalist in a conventional sense, perhaps with the aid of triggering or score-following.
- *Responsorial/proliferating:* the electronics have an antiphonal relationship to the live sound.
- *Environmental:* the electronics simulate aspects of a virtual environment.
- *Instrumental:* the electronics combine with the live instrument sounds to create a "composite instrument".

Croft argues that not only is there an imbalance in intentionality between the two components, but also that we respond to a physical system differently through our empathising with physical labour—the ease or difficulty with which any given sound might be produced—and with the psychological tension of fallibility. Croft spells out this position clearly: "A loudspeaker can, in principle, produce any sound; on an instrument, almost all sounds are impossible, and of those that are possible, some are more difficult to produce than others, and this difficulty is patent in the act of performance". He continues: "This is surely why performance engages us in a way that cannot be accounted for in terms of the sound alone: the difficulty, the impossibilities, the encounter with limits, the *finitude* of the instrumental performance resonates with wider human experience."[1]

For Croft, the labour and precariousness of our corporeal reality is *stylised* through live instrumental performance. A listener's innate tendency to empathise with struggle, the overcoming of the odds, and to marvel at the mastery of such fallible systems is crucial to an aesthetically valorised experience.

The very real possibility of electroacoustic sound being just *too easy* and too divorced from physical labour strikes at the very understanding of music. As Croft points out, it suggests that there are two competing ontologies of music: that of music as irreducible from its sound, and that of music as inseparable from its physical conditions; as he puts it: "from the intersubjective and social act of performance, with the significance of effort, labour and expenditure, and indeed with the erotics of physical engagement."[2]

5. WHERE TO FROM HERE?

Croft's taxonomy seems, on the face of it, to be highly proscriptive to the composer. It might seem to shut the door on the majority of approaches to the genre, a position that might be construed as an *a priori* critique of the entire field.[3] But it is my contention that there are sufficient byways, tributaries, and fertile backwaters within live electronics that still adhere to Croft's preferred ontology for composers to continue to develop compelling and meaningful. What I propose below is therefore a sketch for further work: an outline of potential fields that are ripe for exploration and refinement, that hold true to the essential corporeality of the live experience, but in which the infallibility and fidelity of the electronics may be exploited in a non-trivial way.

[1] Croft (2007), 62
[2] loc. cit.
[3] Which, given he is an accomplished composer of music for instrument and live electronics, would be largely unfair.

5.1. Memory nets

I would first like to suggest that the ability of the computer to accurately record and reproduce live audio with great fidelity is the primary strength of live electronics, and any system that uses this to its strategic advantage in a thoughtful and non-trivial fashion may well go some way to overcoming Croft's objections. I suggest a new paradigm here, that of a "memory net". In a common live-electronics setup, sound is recorded and immediately *recycled* typically through some sort of time-domain brassage or delay process, which leads to a *texturisation* or *fragmentation* of the audio stream. In Croft's taxonomy, this is the *responsorial/proliferation* paradigm. As mentioned, this is the most commonplace approach to live electronics: it is relatively simple to implement such a system without much work. A more artistically potentiated approach, however, is to store incoming audio in a buffer and recall it at a much later point in the piece in such a way that it will fulfil a considerably different function than mere texturisation. Digital audio can compellingly, even shockingly, recall exact moments in time, and its recollection might constitute either a *summation* of past events or even the *disruption* of current events with a dispute from the past. A poetic example of this is provided by Karlheinz Essl's *Entsagung* for chamber ensemble and live electronics, in which live instrumental gestures from the opening of the piece are stored away and return at the end in an increasingly dense texture, whilst the live players drop out of the score completely. The psychological impact is compelling: we perceive the electronics as a kind of distorting mirror of the past, re-folding prior events into the ever-present now. The re-presentation of the past presents the listener with a situation we are rarely presented with in everyday life, for it has neither the role of an echo—which we take to be causally linked to a sound event that occurred in the very recent past, typically no more than 10 seconds ago—nor as archival footage, which we take to have occurred in the more distant past, anywhere from a few days ago to centuries ago. Instead, the replaying of audio from 5 or 10 minutes ago has the effect of *déjà vu*, a disorientation of the perception of now.

An objection is that this form of aural recollection has been around for centuries, in the form of recapitulatory structural processes in instrumental music. In this case, however, the re-appearance of past events is *allusive*, in the sense that two people talking about a past event *report* what happened, rather than literally replaying the occurrence verbatim.

With live electronics, however, the past incurs on the present with shocking fidelity, in a way that *cannot* be merely allusive, for its very literality acts as a sonic document.

5.2. Dominance/subordination

The formal process at the end of Essl's *Entsagung* suggests another paradigm in live electronics, that of the persuasive manipulation of the *hierarchy* between performer and computer, between that of dominance and subordination. In conventional works, the live performer is active for most of the piece, due to the composer's necessity to maintain the physical presence throughout. As such, the instrumental part often could be played without the electronics without significant aesthetic loss.

But we might conceive of an instrumental part less as a potentially standalone piece, and give space for the computer to emerge as a dominant discursive voice at times throughout the piece. It may very likely require the performer to drop out altogether: the difficulty here is the composer's aesthetic prerogative to convince us that this is a meaningful and musically enriching relationship: the computer must significantly vary and direct the discourse, without the crutch of the live sound.

5.3. Non-linear temporality

Aside from the *recollection* of events past, live electronics also has the potential to develop non-linear temporal structures beyond the conventional teleological "unfolding". Recent (and even not-so-recent) literary and filmic developments have questioned the necessity for linear narrative; but in conventional instrumental works, with the abstractness of its musical material, it is very difficult to achieve the illusion of non-linearity, as cultural conditioning and, potentially, cognitive constraints lead the listener to infer a linear organic unfolding even where one was not actually present in its construction. In other words, we can only rarely stop ourselves from hearing a beginning, middle and end, and strive valiantly to relate the sound material to each other.

In live electronics, however, the role of literal sonic recollection has the possibility to produce a much more non-linear presentation of material. Material that, in a linear presentation, would *precede* other material can be presented later in the work and contextualised with a literal representation of the later material. Again, the composer faces the task of convincing a listener of this non-temporality, and it remains an open project as to how true non-linearity can be achieved.[1]

5.4. True contrapuntality

If the implicit linear temporality of the relationship between performer and electronics—the "fixed imitative canon" I alluded to above—provides us with conceptual pause for thought, then a potential development is to give considerable weight to finding a technological and conceptual solution to the diversification of contrapuntal interrelationships. Implicated in this is the intractable notion that the composer must posit a situation in which the performer may truly be able to respond to or delay the electronic

[1] Significant work by composers such as Lindsay Vickery is progressing the possibility of non-linearity in these domains.

part. For this to be perceived as a true *response*, however, the audience must be able to perceive the *intentionality of response* on the performer's behalf. A conventionally notated score would seem to proscribe this possibility: even if the response and delay is written into the score, the perception is still that the performer is following the score, not the electronics. An objection might be that in a conventionally notated instrumental contrapuntality, performers do not *really* respond, at least in any improvisational way, but we must not forget that in this case there is no imbalance of sentience between members of, say, a string quartet, viola jokes notwithstanding. It might also be objected that it is easily possible for live electronics to generate material that precedes a response from the performer; this returns us to the *accompanimental* category of Croft's taxonomy, and we perhaps return to asking whether this would be better served as an instrument-and-tape paradigm. With an open score, or improvisational practices, it may be more conceivable to posit a situation of true interrelationship, but, even so, it may yet prove elusive, and it remains an open question, as to whether we can ever develop a *truly* two-way, interactive relationship between human and computer.

5.5. Microsound

Croft rightly cites Nono's pioneering works in live electronics, *Das atmende Klarsein* and *A Pierre* as exemplars of music that is predicated on the poetic implications of accentuating the transitory, fragile and vestigial in sound. It is, as Croft put its, "the grain, the 'imperfections', the unrepeatable" which constitute "the reason for the continued importance of performance". Live electronics has an almost unprecedented possibility of drawing to our perceptual attention those elements of sound that are typically of ancillary or even non-existent cognitive focus. These elements may be highlighted not only purely through the role of electronic amplification, but also through processual algorithms aimed at the analysis, decomposition and resynthesis of sound.

5.6. Sonification and space

As I mentioned at the outset of this paper, an Aeolian harp responds unthinkingly, but beautifully, to varying wind currents. It reifies the ephemeral fluctuations of air pressure, through a modal transmutation: it *sonifies* our environment, it gives voice to the ineffable. We do not, however, criticize an Aeolian harp for its inhumanity and infallibility. Rather, we might draw aesthetic pleasure from the way that different meteorological conditions give rise to very different 'voices' within the one physical system.

With live electronics, we have the potential to exploit the *sonifying* quality of a fixed system that the computer might provide. Already Croft hints at this, with regards to Nono's reinforcement of transitory aural phenomena, but I think we can take this further. We might think of other ways in which the performer might contribute 'input', other than purely sonic means. In Ruth Anderson's work *Centring* (1979), audience members are measured for their galvanic skin response which feeds the frequency of a bank of sine-wave oscillators. An on-stage dancer moves to the resulting audio, and, in theory, the audience members further respond to the dancer, creating a feedback loop between sound and stimulus. As a work of purely sonic art, it perhaps lacks the compelling sonic surface that would further transform it into a more sophisticated formal disposition—but it does provide a suggestion for further work that certainly asks us to interrogate the blurred boundary between mixed live electronics works and sound installations.

By expanding the concept of sonification beyond the dimensions of the object into the territory of *spaces*, however, we can utilise the insentience of the computer in a manner analogous to the acoustic response of a spatial environment. Not only can we simulate a virtual spatial environment, which is, perhaps, the least interesting version of this concept, but we can also interact with an existing physical space in ways that would be impossible with live performers. As a thought experiment, we might conceive of a work housed in a large resonant acoustic, such as a cathedral, in which the single performer generates subtle, almost inaudible sound which is, however, close-miked and proliferated out to multiple loudspeakers distributed as widely as possible throughout the space. As audience members pass through different parts of the cathedral, the changing resonances, changing processing and changing sonic impetus from the performer would provide a much more complex and engaging spatial encounter — most importantly to this concept would be the sense that it is not possible to hear all parts of the "piece". Like the stone garden at Ryoanji in Kyoto, in which you can never see all of the "rock islands" at once, in this piece there would always be sound occurring that would be unable to be heard by any one person.

5.7. Orchestral

Taking this idea of a spatially-oriented electronics further, we might conceive of a situation in which a massively multi-loudspeaker setup exists within a large acoustic, in which the organicist conception is taken to an *orchestral* extent. Through the development of sophisticated microvariation techniques, multiple microscopic delay offsets, and the storage of an intelligent recall of sound input, the composer could feasibly generate the kind of fine-grainedness that we appreciate in a live orchestral performance. Part of the joy of attending an orchestral concert is the possibility of attending to individual voices; in this case, massively multiple outputs would be required, each routed through a separate loudspeaker, to "contribute" in a heterophonic manner to an overall impression of global form. With sufficient computing and programming sophistication, the "orchestra" could

even be "sectionalised", with different loudspeaker subsets — "choirs" in orchestral parlance — taking on different material in a stratified disposition.

The potential within this idea for macrosructural formal processes to be articulated through spatial disposition is hinted at by Janet Cardiff's *Forty-part Motet*, in which Tallis's *Spem in Alium* is spatialised through 40 separated loudspeaker channels. In this work, the *tutti* sections take on a heightened perceptual significance, to the point where the sudden entry of all 40 voices surrounding the listener contributes to a heightened affect.

5.8. Inorganicism, textuality and extrinsic referentiality

Finally, we might conceive of a musical work that features inorganicism in its formal construction, emphasising disjuncture, fragmentation, chance operations, and the deliberate fracturing of linear growth and decay trajectories. *Or* we might focus our attention on the organic relationship between performer and computer. Yet a simplistic, deliberate mismatch between electronic sound and live performer probably only returns us to the same paradigm as instrument with tape: the *environment* category in Croft's taxonomy, with its attendant degenerate case of rainforest flute.

One solution is to conceive of a relationship in which, rather than the typically unquestioning acquiescence to and subsuming of the live sound by the computer, instead there is a *provocation* by the computer. The problem of one-way intentionality still limits the degree to which provocation can be heard as truly a performative act on behalf of the computer. But one area that might be worthwhile to explore is in the area of vocal textuality, in which the extrinsic referentiality of recorded text might fruitfully interplay with live text.

Indeed, extrinsic referentiality, especially in the form of historically-encoded sound sources or other sounds that resonate with wider sociocultural contexts, may well continue to present compelling artistic forms in which the relationships of live electronics are still

6. SUMMARY

If we accept the cautionary tone of Croft's writing, and I contend that there are indeed artistic and poetic reasons to attend to his injunctions, then I think we are faced with a choice: to either reject live electronics as, *a priori,* a dead end—which indeed many composers do—or to accept and indeed even exploit its inherent imbalances and limitations, in order to conceive of pathways for future work.

I argue that forms which accept and exploit the inherent insentience and passivity of the computer, in a manner perhaps more akin to sound installation, have a better chance of generating new musical situations than the conventional contrapuntal method.

Of course, in the end, we will just compose the music we want to compose, but attendant to the creative urge should always be the question of "why?" And if a search for answers energizes our creative urge, rather than diminishes it, then such discourse is not without its place.

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ECO-STRUCTURALISM IN PRACTICE

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1. ABSTRACT

The eco-structural composition technique uses natural and organic sound source materials at the beginning of a composition project with the intention of imbuing composed music with a natural or organic structure. This paper will describe a case study exploration of an eco-structural musical work and the process of composing it, from the source material to the finished work. The paper concludes by discussing the effectiveness of the process and how it could be improved for future musical compositions.

2. INTRODUCTION

Flow is a musical piece written by Tim Opie in 2009, shortly after the Black Saturday Bushfires in Victoria. For months the land had been getting very dry. On Black Saturday the temperature rose to 47°C in some areas. Reports were coming in on the radio of fires near the composer's home and a decision had to be made, either to evacuate or prepare to fight the fire. He and his family chose to stay.

Fortunately the fire didn't come too near the house. The plants around the house had all dried and withered, including the normally lush green fire retardant agapanthus. The waterfall and stream that ran through the national park near the house had also dried up. However about two weeks later it started to rain.

With the rain came relief. The waterfall again began to flow. Although it was not much more than a trickle, seeing the waterfall again in action was inspiration enough to get out recording equipment and begin work on a new composition.

This paper will briefly review eco-structuralism, the compositional technique used to create the musical work, and then examine the source materials, the eco-structural analysis performed on them, the processes applied to the resulting data, and, the compositional arrangement of the work. Finally the paper will discuss insights arising from eco-structuralism in practice, and explore ideas that may further the creative use of eco-structuralism.

3. ECO-STRUCTURALISM

Eco-structuralism is a composition technique whereby the composer uses the analysis of recordings of natural and organic sounds to dictate structure and other musical events within a composition (Opie & Brown 2006). This compositional technique is founded on the concepts of soundscape composition (Schafer 1993) and eco-composition (Keller 2006). It also, unlike these techniques, uses a formalised approach to achieve its outcome reminiscent of some twentieth century compositional approaches (Schoenberg 1975, Xenakis 1971).

When using eco-structuralism, the composer firstly chooses their source material, then using eco-structural analysis software they generate XML data files that detail various sonic attributes of the recording. Those attributes include amplitude, fundamental frequency, timbre and space. The data files are then used within a rule-set for composition. The rule-set defines strict use of the data, the intention of the strict rules being to maintain the integrity of the original sound recording structure, whilst still allowing for creative musical decisions.

The next section describes the how eco-structuralism was employed in the creation of the work, *Flow*.

4. RECORDING AND COMPOSING

At the time of the bush fire event that inspired the composition of *Flow*, only a standard definition video camera was available for audio capture. The audio was set to the highest quality available, which was a non-compressed 16bit, 44.1khz mono sound file, recorded through the built in microphone. About an hour of footage was captured whilst walking along the bank of the stream, and sitting up at the top of the waterfall.

A decision was made early on to only use audio from the excerpts of that video. After the filming process the footage was reviewed and sections selected. This was a purely aesthetic decision based primarily on the clarity and subject of the audio, but with some concern as to the clarity and subject of the video. Elements were selected that were interesting both sonically and visually. Seven clips were chosen, representing various parts of the waterfall and stream. The audio was extracted from these clips and the visual elements of the film were used to create a visual narrative, without being distracted by sonic considerations at that point.

Then began the audio analysis process. Having already extracted the audio from the clips, it was a simple matter of running these files through the eco-structural analysis software. An important feature of the eco-structural audio analysis process is the ability to add extra information to the resulting XML files as part of the analysis process. The extra information is identifier meta-data that allows anyone to find out exactly when the recording was made, the site of the recording, a geographical location for easy geotagging, and a description of the events around the recording. This meta-data proved to be extremely useful when reviewing eco-structure XML files, as it instantly reminds one of the recording, including events that may have been forgotten.

For example the file flow1-amp.xml stated:

> Date of Recording: Saturday 21 February 2009
>
> Description of Audio: Section 1 of the audio extracted from the waterfall near The Basin. The Basin waterfall is almost dried up due to the ongoing drought in Australia. Filmed two weeks after Black Saturday. There had been some ash falling in the area a few nights previously, but today has been quite a good, and substantially cooler day, having had a tiny amount of rain the night before, just enough to produce flow. There is an almost white noise tone to the piece, due to the speed the water is travelling, but there is a much louder trickle sound, which is fluctuating in frequency
>
> Location of Recording: Waterfall in the Dandenong Ranges National Park, near The Basin
>
> Geographical Location: 37d 51 58.80 S 145d 19 49.00 E

As an ecological endeavour, this descriptive information is useful not only as a contextual reminder or cataloguing system, but also as a historical marker of sound events of that location at that moment in time. The composer is strongly encouraged to include this information when working with eco-structuralism. R. Murray Schafer discussed the importance of the sound event, in which the place and time are critical in understanding the context of the sound from which it was taken, as no sound event will be identical (Schafer 1993).

The rest of the information in the resulting XML file pertains specifically to the audio, and is generated automatically.

When viewing the XML audio analysis data in a graphic form (see figure 1), and comparing it to the original sound (see figure 2), coinciding peaks can be easily identified, indicating that the data is correlated (Opie 2005).

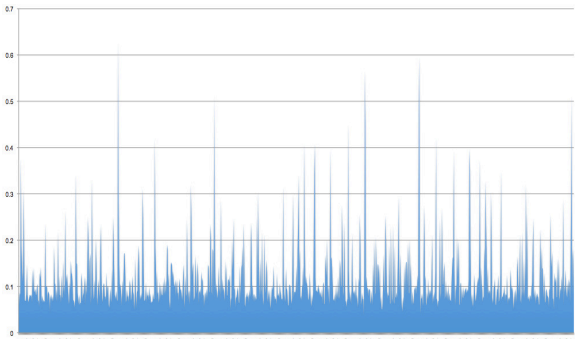


Figure 1: Eco-structural data presented as a graph.

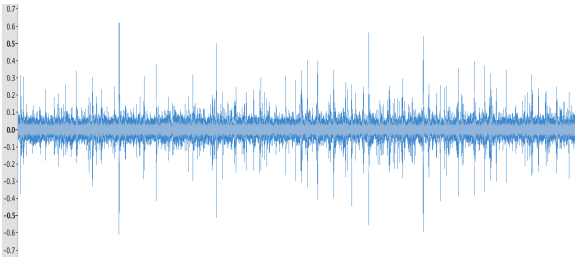


Figure 2: Original waveform of the sound recording that was graphed in figure 1.

Using the data so generated, some new sounds were created. The data was used for mapping the elongation of some of the original sounds, and to emphasize certain frequencies. It was eventually decided, after some experimentation, to use data from only four of the selected clips, as this already created a huge variety of permutations and interesting sounds. After creating numerous audio clips in this way, they were then arranged in a multitrack audio editing environment. The video was used at this point as a guide for which kinds of sounds to use at which point, rather than following a strict eco-structuralist process for the macro timescale.

5. ANALYSIS OF FLOW

In Schoenberg's (1967) Fundamentals of Musical Composition, he gave advice for self-criticism. Although this advice was written for more traditional music, it can be easily modified to critique other musical forms. We have chosen to use a modified version of this form of critique for the analysis of *Flow* in section 5.1, as well as formulating a series of questions that critique eco-structuralist intention in particular as outlined in section 5.2.

5.1 Musical Analysis and Critique

Listen for stylistic coherency: Is the music well constructed? Does it accomplish the composer's intention? Is there a teleological coherency throughout the work?

Whilst not having any traditional sense of melody, this piece does however have a strong motif that is integral to the piece. The water flowing motif keeps reappearing in various transformations. There is a constant sense of movement, as if the piece is trying to reach a particular goal, although there are certain points where the piece slows down to for a breath, which is a welcome break for the audience.

Analyse motivic development: How do parts in the music lead to the unfolding of new events?

The very first motif begins sounding very much like the original sound source, but undergoes a slow transformation, which seems at first to sound more metallic, but eventually becomes more bouncy. This idea follows a composition characteristic noted by Barry Truax (1994) in regard to stretching sound:

> By keeping the attack portion relatively intact and stretching only the body of the sound, each signal retained its recognizability, but allowed listening associations to be savoured, along with the inherent musicality of its constituent harmonics (Truax 1995).

By preserving the initial attack of the sound source before it is heavily affected, helps invite the listener into the music that is about to unfold.

During the middle section of *Flow* the sound undergoes much more radical transformations. This happens through a slower unfolding of transformations. The texture is also denser. It sounds as if there are many voices speaking from the water.

The end section involves a slowing down of events. The texture thins out and becomes less transformed. In some parts the sound almost slows to a silence. This is comparable to the

waterway in which this was filmed. Initially the stream is peaceful and placid, but as it gets closer to the rapids and waterfall it speeds up and gets louder, but after reaching the bottom of the fall is disperses much more slowly and begins to stagnate and divide into small slow moving bodies of water.

Eliminate non-essentials but avoid monotony: What is the balance of simplicity and complexity? Is there an overload of textures or tones? Is there enough interest in the material?

In the first draft of this piece more source materials were used, however it quickly became apparent that there was too much going on. It was becoming saturated to the extent that one sound could not be discerned from another. It was decided to strip the materials back to just four sound sources, as if it were an SATB choir. This simpler structure allowed more reflection on the intricacies of the composition and the works temporal structure.

Within this simpler framework, there was still sufficient timbral variety and variations in form to give the piece sufficient interest.

Countermelodic Construction: Does the interaction between parts add interest, vitality and direction to the musical work?

Special attention was paid to using a broad range of timbres and frequencies overall, but with particular focuses at certain times. There are low parts throughout the piece that give good balance and depth. During the latter part of the piece, the low parts take the thematic role for a short while, whilst the higher parts fade out. The "bass line" definitely adds variety and vitality to the piece, and helps move the piece forward, over the waterfall.

5.2 Eco-structural Analysis and Critique

Aesthetic Significance or Affordance of the Sound Source: What signifiers are present that can be used in a compositional manner?

Sound source 1 consists of two strong sounds. In the foreground is the sound of trickling water. It is very soft and fluttery, with constantly moving pitches. In the not too distant background is the sound of rapidly moving water, sounding more like white noise and almost drowning out the foreground sound. This background noise is actually the soundmark of the area whenever the waterfall is flowing. Locals use it to locate the waterfall whenever they go bushwalking in the area. To get home, you just need to find the river and follow the flow of the water until it reaches the back of the houses. During non-drought seasons the waterfall is so loud that you need to shout to communicate when you are nearby. This sound also provides two strong frequency bands as can be seen in its spectrum shown in figure 3.

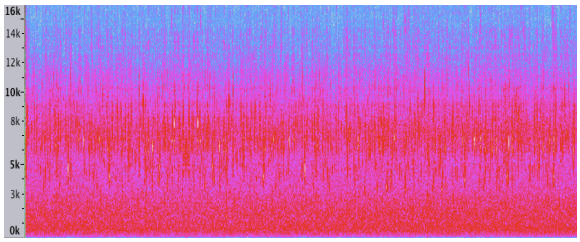


Figure 3: Spectrogram of source sound 1.

Sound source 2 contains the sound of water flowing through rocks into a deeper pool. The recording is made at a very intimate level to the point at which the water is hitting the pool. The background noise is much lower in amplitude. The spectrograph of this sound, shown in figure 4, indicates that there are some distinct harmonics present in this recording. These harmonic levels could be used in the timbre study (Opie 2008) to dictate some kind of harmonic ratio that could be implemented within the composition.

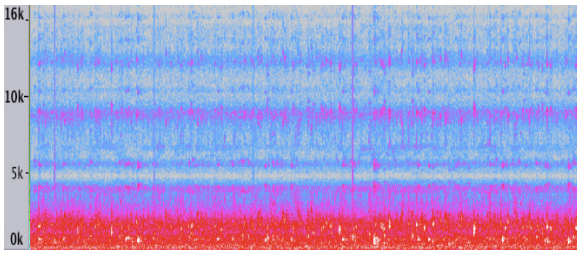


Figure 4: Spectrogram of source sound two.

Sound source 3 has similarities to a running tap. It is a small, but strong flow of water dropping into a small pool of water. Interestingly, the spectrogram indicates a wider spread in frequency than I expected. The

spectrum of this recording has more randomly distributed elements, as seen in figure 5, although when listening to the sound, it does not seem so unpredictable but sounds like a thematic return to a particular frequency. Such a part might make a good melody or perhaps countermelody.

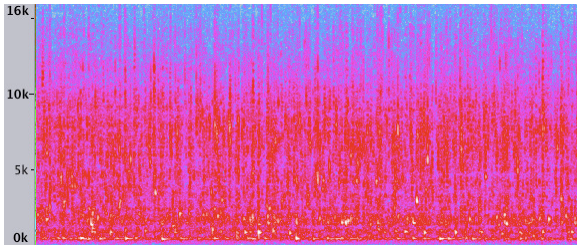


Figure 5: Spectrogram of source sound three.

Sound source 4 is a small trickling sound that was recorded further away from the waterfall, such that the background noise from the waterfall is barely audible. The simplicity of this audio recording allowed it to work well as a kind of melody line. In figure 6 we can see a distinct lower frequency range with a harmonic band above it.

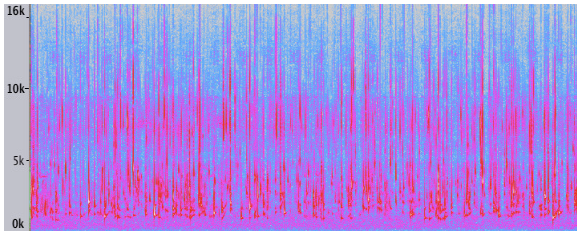


Figure 6: Spectrogram of source sound four.

Structural Integrity Post Transformation: How has the transformed material maintained or gained useful sonic organisation?

The structure of the first motif is quite evident. It begins by sounding identical to the source, however, it is undergoing a slow transformation that is stretching the structure. The stretch is so minute that at first it is not noticeable.

The middle section seems to be least like the original sound source, however, there are still many tell-tale signs that it is from the same source. We hear the transformation because it is created as an evolution, becoming more radical with each transformation.

The ending sounds more like the original sound sources, so the structural content here is more easily identified, as it was in the beginning.

When comparing the spectrograms of the initial sound sources to the final piece in figure 7, we can see where some of the sound sources have been implemented. For example sound source 4 is evident in the first section, and sound source three is used in the last third, with sound source one mixed in. The second part in the piece was influenced by sound source two.

Aesthetic Outcomes: What is the character of the transformed material?

The sounds of nature through the structures of filed recordings have been blended in a very tight manner in this work. When creating new sounds from the structural data, the original recordings were again reused as the sound source, so that events such as structural elongation would match in a very precise manner. Also events such as amplitude enveloping and frequency filtering were applied to the original sound in order to emphasise significant events within those recordings.

Although some sections have a very ‘digital’ sound, they are just highly transformed sound recordings from the same source. Blending the analyzed structure data back with the original sound source trough transformation mappings

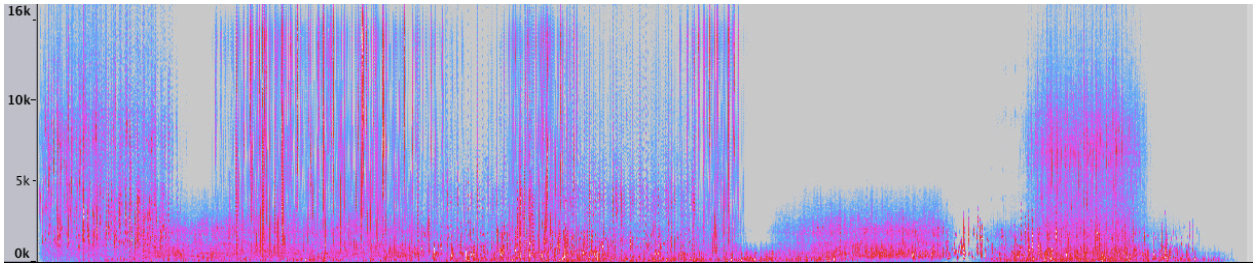


Figure 7: Spectrogram of Flow

highlights the sonic signifiers, as well as the cultural identifiers. This process also amplifies the motion and change inherent in the original sound source. Interestingly, the water sound becomes 'chatty' after it has undergone a more tumultuous transformation, yet there is no denying that it is water, because the transformation happen as we listen. From a political ecological perspective an audience member might describe the piece as trying to speak to us about the way we treat our waterways, however from an aesthetic perspective it sound as if the steam is singing as it merrily makes it way forward.

Evidence of Affordance: To what extent have the micro-structures of the source material permeated the final work?

The notion of moving water is quite evident within this piece. Also some of the transformations seem to mimic frog sounds. Although there were no frogs recorded, it adds to the sense that there is water about, and with it water species. There is also a low gurgling sound that sounds like water gurgling through pipes.

The structural forms used in this piece all had unique qualities, as mentioned previously, that supported analytical thinking about ways to compose this piece.

6. AREAS TO IMPROVE

The eco-structural approach was not strictly adhered to in this composition. Whilst all the sounds were made using this technique, the arrangement was not. A stronger commitment to the process would be required to discover the full potential of this compositional technique. Perhaps, like any 'serial' process, trust in it develops through experience, and perhaps blind adherence is not the path to satisfactory expressiveness.

One way to encourage a more strict approach would be to use a lot less data. One small piece of recorded sound has so many possibilities that in order to create a cohesive musical piece, it might be useful to limit the compositional explorations to just one sound source, in the same way that one tone-row in atonal serialism is used to give the piece a sense of internal coherency.

Another aspect of being more strict to the eco-structural approach would be to create a piece in which the source sound is used only as structural data and not used as audio material – even if transformed. Having the source sound playing, as in *Flow*, seems too easy a way of making the listener aware of what the sound source is.

7. CONCLUSION

We have described eco-structuralism and shown a detailed example of how it can be used in practice. Although eco-structuralism is a recent technique in composition, and although there are still many more avenues to explore, we feel this exploration indicates that there are many interesting possibilities. We have also suggested and demonstrated some techniques for analysis and critique of eco-structural music and we suggest that there is value in pursuing this technique further.

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ORGANICISM IN LIVE ELECTROACOUSTIC MUSIC

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ABSTRACT

This paper explores the potential for an organicist or relational holistic approach to experimental electroacoustic music composition that is indeterminate with respect to performance. It follows a phenomenological interpretation of the musical work as the product of dynamic, temporal or relational processes involving the performers, their instruments, the sounds themselves, the whole acoustic space and the audience. An analysis of an electroacoustic composition and Decibel ensemble performance is offered for which organic indeterminacy is described in terms of a performative openness towards the creation of experimental music.

1. INTRODUCTION

The performance of experimental electroacoustic music is here posited in terms of the compositional processes involved in the musical organization of sound the aesthetic outcome of which is essentially indeterminate. This performative indeterminacy can be more or less organized, even determinate in part, with respect to the materials used (such as the instruments dictating the timbre, frequency and amplitude of the sounds), the methodology involved in progressing from sound to sound (whether notated in part, graphically scored or using a set of instructions), and the structural division of the whole performance into individual parts[1].

The composition lays out the organizing principles to be put into play in the musical performance, and it is this organization of performative processes and the relations between them that gives a structure to the temporal continuity and aesthetic form of that performance. Compositions that are indeterminate with respect to performance cannot by definition determine the aesthetic outcome but rather must set up the performative processes within which performers work together to bring the musical work to its form unique to the space and time of its performance. Indeterminate compositions must allow the musical work to unfold organically: That is the aesthetic form of the musical work is dependent on the individual elements at play in the performance collectively opening up the space for the musical work to take its course. These processes and the relations that bind them together are set up by the composition and its musical organization of the sound. But what does it mean to musically organize sound in such an indeterminately organic sense?

2. MUSICAL WORK

The term ‘organization’ has its etymological root in the ancient Greek term ὄργανον (organon) which broadly speaking means ‘instrument’ or ‘tool’. Organization thus means the systematic ordering of the instruments supporting a complex whole, such as occurs with the organs that together constitute and support an organism. One instrumental sense of the ὄργανον in ancient Greek usage refers to both the material used in a work as well as a product of that ongoing work, such as the wood (material) used to create the timber (a product) that both go towards the construction of a house (the intended completion of the work). Organon also refers to the intellectual resources put into play in the production of that work. All of these individual organon, both the goal based (teleological) practical action, thinking and planning as well as the materials used, must be organized as a whole in working towards the production of the finished work.

Etymologically ὄργανον is related to ἔργειν (ergein) to work and *ἔργον* (ergon) the completed work itself that comes to a stand as the finished product. The organon (the intellectual and material instruments of work in the verbal sense) and the completed work itself (ergon) are bound together in ancient Greek usage in a way that is not readily evident in the English language. From a phenomenological perspective[2] however, the ergon is brought to a stand in presence by having been produced, and it endures as something present only by virtue of its relation to the instrumental processes of production (ergein and its organon) that have set that completed work free to be itself.

The work itself (as ergon) is not merely the static end result or sum of the past actions and materials used but rather should be understood in the active or verbal sense as constantly becoming present as work, as presencing, so long as that work stands. The relation of the produced work to its production in this sense is not that of an object standing free of its past work, where cause gives

[1] Following John Cage’s process oriented notions of indeterminate composition in “Composition as Process: Indeterminacy” (Cage, 2004, pp. 176-186).

[2] See for example Heidegger’s phenomenological interpretation of Aristotelian notions of ergon and energeia (Heidegger, 1973, p. 5).

way to the effect. Rather, the work is experienced as a whole, its past production being present as the constantly realised potentiality of what is produced.

This dynamically temporal sense of the ergon as work is rather subtle and can be difficult to grasp, most especially where one's own modern understanding is conditioned from a young age to think and perceive in terms of individual objects and discrete timelines of causes followed by effects. However, the dynamic potentiality inherent to all works is perhaps most readily evident in the production of live music, for the performative musical work itself is present, or rather presences, only so long as its individual organon continue to perform their own work in the ongoing temporal production of the whole: And once the individual organs cease to function, the musical work as a whole dissolves back into the silence from whence it came.

Furthermore, from this holistically relational perspective the musical work itself (as ergon) is more than merely the sum of its parts (organon); more than the musicians' actions, the musical instruments or the resonance within the performance space; more than the composer's teleological intent; and more than the aesthetic predispositions of the listening audience. Yet all these organon together form a dynamic relational whole, an organized temporal process of musical relations, without which the work itself could not become itself. It is this organicist notion[1] of relational holism that I should like to explore with respect to the Decibel ensemble's concert work *Variation on Electroacoustic Feedback for Multiband EQ Filter, Flute and Cello* (Riddoch, 2011).[2]

3. COMPOSITION

The motivation for this composition came about through my ongoing interest in developing a performance oriented research practice, melding phenomenological analysis with an electroacoustic music practice, and in line with the current ERA (Excellence in Research for Australia) guidelines for the creative arts sector in the Australian university system[3]. Phenomenologically descriptive analysis requires an investigation of the phenomenon of music itself as it occurs in listening and performance, rather than a more generalized musicological approach to theorizing about forms of music after the fact, so to speak. Music itself must first be disclosed in its lived experience and analysed from the perspective of that experience. The research problem here is therefore how to construct a composition that might bring the whole performative context of the work to presence in the musical work itself. Or in other words, how might a composition involve and highlight all the performative relations at work in the actual performance of that composition? From the performers and their instruments, to the reverberating sounds themselves, the compositional processes and the relation of this dynamic whole to the audience within the acoustic space.

The Decibel composition involved a simple instruction set for the performers and was performed at the Perth Institute of Contemporary Art, Western Australia in March 2011:

> **Variation on Electroacoustic Feedback for Multi-Band EQ Filter, Flute and Cello**
> Malcolm Riddoch (2011)
>
> Use the Larsen effect (microphone feedback) to explore the various natural resonant frequencies of the acoustic space. Acoustic performers should listen and play the nearest microtonal equivalent in pitch, loudness and duration.
>
> Performative attention should be paid to the resonance of one's acoustic instrument in the space as it interacts with that space's reinforcing resonant frequencies.
>
> Variations in the acoustic instruments' pitch and loudness may also drive new resonant electroacoustic feedback frequencies. In this case performative listening may alternate between passive (following the electroacoustic resonance) and active (playing the acoustic space itself).

As an experimental electroacoustic composition the work is indeterminate with respect to its performance. That is, the aesthetic form that a performance takes is dependent on the acoustic properties of the performance space and the dynamic interplay between the acoustic musicians' pitch recognition and the electroacoustician's manipulation of the resonant feedback across the various frequency bands.

4. TECHNICAL SETUP

The technical setup involved spatially separated acoustic musicians and two microphones output to a quadraphonic speaker array surrounding the audience. The two storey performance space (PICA main hall, see

[1] Organicism as a contemporary philosophical concern largely derives from a renewed interest in reinterpreting Plato's Timaeus (Plato, 1969). For a discussion of his notion of the organic universe see for example McDonough (2002) and Johansen (2004).

[2] Performed by Decibel artists Cat Hope (flute), Tristen Parr (cello) and Malcolm Riddoch (electroacoustics), at the Perth Institute of Contemporary Arts, Western Australia, March 28 2011 for the concert *A Voice from a Dark* Space (http://decibel.waapamusic.com). An audio recording and goniometer display are available at http://malcolmriddoch.com/?p=241/

[3] For an outline of this practice led research approach combining phenomenological analysis with an experimental electroacoustic practice see my paper for the 2010 ACMC (Riddoch, 2010).

figure 1) has a 4+ second reverberation although this was dampened significantly by the presence of 100 audience members, a very concrete effect that altered the electroacoustic feedback resonances and thus the performance itself. In this sense, at least from the performer's perspective, the audience has a direct physical relation to the musical work beyond their function as passive listeners. Different audiences in the same or different acoustic spaces, even different atmospheric conditions, all contribute to the indeterminacy of the resonant response of a performance space and thus of the performance itself.



Figure 1. PICA main gallery performance space. Note the side gallery alcoves, mezzanine and high roof of the main gallery which gave a very rich reverberance of at least 4 seconds duration (photography courtesy of Lisa Businovski).

The electroacoustic setup used two condenser microphones pointing upwards into the space rather than close miked to the acoustic instruments. The microphone audio outputs were input into a digital mixer for EQ filtering. In the mixer the microphone channels were muted then pre-fader bussed into six auxiliary channels each using a brick wall EQ filter to separate the audio output into six frequency bands: 20-100Hz, 100-200Hz, 200-500Hz, 500-1000Hz, 1-2kHz and 2-5kHz.

Heavy limiting was used on each auxiliary buss to control runaway feedback levels and the busses alternated between quadraphonic outputs 1 and 3 or 2 and 4 to create a surround audio effect. Additionally a six band high Q notch filter was used on the master output to both attenuate and accentuate resonances as they arose during the performance.

The electroacoustic setup allowed for a reasonably fine degree of control over the natural resonances of the acoustic space across the various frequency bands along with finer control within each band (using Apple Logic Pro environment mixer controlled via a Euphonix hardware mixer allowing for a high degree of tactile control, see Figure 2). For this performer controlling the electroacoustic feedback loop felt akin to playing a large mostly monophonic pipe organ but from inside the resonant instrument.

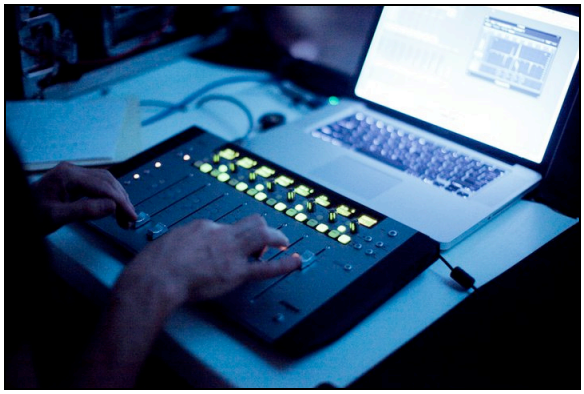


Figure 2. Digital mixing setup using a Euphonix mixer and Apple Logic Pro environment (Photograph courtesy of Lisa Businovski).

5. PERFORMANCE

Initial manipulation of the resonant frequencies within the acoustic space involved mixing one or more of the auxiliary busses until the Larsen effect was triggered and then waiting as the acoustic musicians tuned their instruments into the resonance thus reinforcing it. As the performance progressed, further complexity was introduced via the notch filters on the master output allowing finer control over the acoustic space's resonating frequencies as well as following and enhancing the acoustic instrument frequencies evident in the analyser display thus producing further resonant feedback.

The acoustic musicians, both aurally trained in pitch recognition, started by following the natural resonant feedback frequencies to the closest microtonal equivalent or its harmonic. Once accustomed to the tonal range and response of the acoustic space they began to push the resonance in unexpected directions with beat frequency effects and sudden higher harmonic feedback resonances adding to the complexity of the audio response. The performance then naturally evolved into six or more parts punctuated by the collapse of resonance and the beginning of a new resonant peak for the acoustic musicians to follow.

This simple interplay between the acoustic space, the acoustic instruments and electroacoustic feedback resulted in a dynamic performance centred around the natural resonant frequencies of the space. A dominant lower register resonance occurred at 196Hz or around a G♭ with higher harmonic resonances at 395Hz (G) and 1040Hz (G). Further resonance occurred at 485Hz (B) and 660Hz (E♭) amongst several others (see Figure 3). This natural resonant regularity thus produced a harmonic structure in the work dictated by the acoustic properties of the architectural space and its occupants.

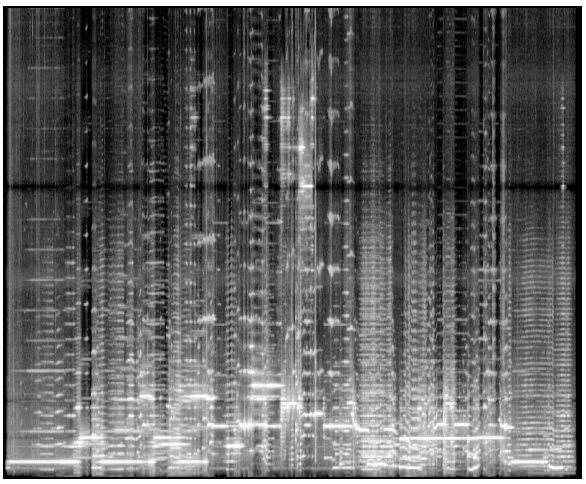


Figure 3. Spectrograph of 7min 40sec Decibel performance. Vertical axis shows frequency response from 0-6kHz. Note the regularity of resonant frequency peaks across the horizontal timeline as well as of the vertical harmonic intervals.

From a listener's perspective the performance was highly spatialized with the two acoustic instruments (flute and cello) sounding from two different alcoves off the main performance space, the electroacoustic quadraphonic output surrounding the audience, and the acoustic space resonance resulting in standing waves that appeared to resonate inside one's head with beat frequencies sliding in and around. In effect the entire enclosed acoustic space itself became a separate instrument played by both the acoustic and electroacoustic musicians, even as that space mediated the interaction between the acoustic and electroacoustic instruments and their performers.

6. CONCLUSIONS

The reciprocal play described above, between the acoustic and electroacoustic performers and their instruments along with the resonant space, is the main organizing principle for the composition. The organization of the interplay between these performative processes and relations (the organon) allows the musical work to emerge from that interplay in an open ended organic way. Rather than a top down or hierarchically deterministic structure imposed by a determinate composition on the performers and the resultant aesthetic form of the performance, an indeterminate musical organization of those processes sets the music free whilst still giving a structure to the temporal continuity and aesthetic form it takes. Organicism, in terms of a compositional focus on the relational processes inherent in music performance as a whole, allows the composition to musically organize sound in an indeterminate sense without relinquishing the composer's control over the musical work. The work has an organic unity that is a reflection of that compositional control.

This control, the composer's intent, is thus as inherent to the indeterminate musical work as it is in a deterministic fully notated score. The main difference between these two compositional methodologies, broadly speaking, would seem to be that the former (experimental) approach intentionally sets up the conditions for an open ended aesthetic outcome while the latter attempts to impose an aesthetically predetermined closure. The determinate composition makes use of the organon literally as a tool of the composer in the production of the predetermined work. In a peculiar sense the tools, or rather the performative processes, involved within a deterministic composition are not organically related to the creation of the musical work itself. The composer here is the creator.

Indeterminate compositions however, at least in the organicist sense, can only produce an open work if those organa are organized as an organic relational whole. Openness rather than closure, as a compositional theme, requires that each organon is fundamentally involved in the creation of the musical work itself. The composer in this instance creates the conditions within which the performers can become open to the musical possibilities experienced in the performance of the musical work. 'Openness', in this lived experiential and performative sense, is the creator.

The indeterminate musical work therefore must be experienced as a whole in the unique spatiotemporal context of its performance, for the organically open experience of that specific performance and thus the precise aesthetic form it takes on can never be repeated. A repeat performance here is limited to the repetition of the conditions within which the performers together become free for the possibility of openness towards the creation of the musical work. The composition provides those conditions as the constantly realized potentiality of what is produced yet the performative musical work itself is brought to presence only by virtue of the openness at play in the performance. This experiential openness makes the experimental electroacoustic musical work itself more than merely the sum of its parts, for the organa together form a dynamic relational whole as an organized temporal process of musical relations directed solely by that openness to the ongoing moment of creation.

Phenomenologically speaking, openness (die Lichtung) and presencing (enargeia)[1] are explicitly disclosed by the composition in the open coming to presence of the live experimental electroacoustic work. There is no overriding musical theme, no predetermined aesthetic, other than the openness within which the sounds themselves constantly arise and dissolve according to the structure provided by the materials and methods prescribed by the organically indeterminate

[1] Following the later Heidegger's notions of the relation between technology, openness and presencing in "The End of Philosophy and the Task of Thinking" (Heidegger, 1972, pp. 69-71).

composition. Organicism thus provides an abstract aesthetic telos for live experimental electroacoustic music and perhaps fulfills the promise Ferruccio Busoni saw in the beginnings of electronic music technology for which "music was born free; and to win freedom is its destiny".

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ELECTRO-ACOUSTIC CAN ENHANCE MĀORI MUSIC: A MODEL FOR COMPOSING.

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ABSTRACT

This article examines the possibility of composing electro-acoustic music and integrating traditional Māori waiata/ mōteatea[1] and taonga pūoro[2]. The focus will be on generating a model for composing new works and proving that electro-acoustic can enhance Māori music. The findings of a current PhD research will be highlighted along with 'work in progress' that will give support to this article.

1. INTRODUCTION

As a composer of Māori waiata and a new composer of electro-acoustic music, my principal concern was whether electro-acoustic can enhance Māori music and become an addition to its repertoire. Furthermore can a model for composing new works be generated as a guideline to encourage and develop new composers?

During my PhD research there has been a mixed`rl nana8800 reaction from composers of *waiata* and *taonga pūoro* musicians regarding the fusion of electro-acoustic with Māori music. Rangiiria Hedley (2010), a *taonga pūoro* musician prefers the organic pure sounds of these instruments as a 'stand-alone' genre and regards electro-acoustic as 'weird' due to its inorganic and manipulation of sound via the computer. Also many *taonga pūoro* musicians have a close connection to the instrument whilst playing due to the *'wairua'* the spiritual and emotional aspects of Māori music as explained in Chapter Two *Waiata* and *Tāonga Pūoro.*

On the other-hand Horomona Horo (2010) another taonga pūoro musician enjoys both the pure sounds of the taonga pūoro and experiments with new sounds from these instruments during the recording process. Dr Richard Nunn (2010) a well-known taonga pūoro expert and musician embrace both the organic indigenous sounds of the taonga pūoro but believes that taonga pūoro should be exposed to other 'art-forms'. Nunn has played taonga pūoro on many recordings including an 'electro-acoustic' recording '*Te Hau Kuri*' (2002, NZ Sonic Art III) composed by the late Dr Hirini Melbourne.

In reference to *waiata* and electro-acoustic there have been numerous recordings of *taonga pūoro* and contemporary new *waiata* such as *'Te Ku Te Whe'* (1994), *'Te Hekenga-Ā-Rangi'* (2003), *'Puhake Ki Te Rangi'* (2008) to name but a few. However these are not electro-acoustic recordings but organic sounds of vocal singing with *taonga pūoro* as an accompaniment. I have never heard an electro-acoustic composition that incorporates a traditional Māori *waiata/ mōteatea.* This will be investigated further in the composition of four original music projects in my PhD research. The composition will include two ancient chants of the Tainui and Waikato people, '*Waikato Te Awa'* and *'E Pā Tō Hau.'*

Although electro-acoustic is fairly new to the Māori the debate being, does it have a place in Māori music. This is best argued by Da Cunha (1995):

'Culture is production and not a product, we must be attentative in order to not be deceived; what we must guarantee for the future generations is not the preservation of cultural products, but the preservation of capacity for cultural production'.

Da Cunha argues that indigenous people should treasure their heritage and culture but keep evolving by producing new cultural production. In reference to this article, the suggestion is made that Māori music should adapt and be flexible if Māori as a whole is to survive in the future.

Furthermore, according to Dr Robert Joseph (2007) he expresses the importance of 'reflexivity and adaptability':

'As in the past, Māori have survived dramatic changes of colonisation, urbanisation and now globalisation, individually and collectively, by deploying their capacity for adaption; on the one hand modifying traditional forms to serve new functions and on the other creatively adapting introduced forms to their own ends, transforming both in the process'.

[1] **Waiata** is the generic term for all Māori songs. Mōteatea refers to laments and poetic prose of Māori composers.
[2] **Taonga pūoro** refers to all traditional Māori instruments.

The Māori have survived into the 21st century and they have brought forth their tribal heritage, genealogy, traditional belief systems, cosmology, customs and traditions, language, myths and legends, guardianship *kaitiakitanga*, and on the 'creative side' all the arts including *waiata, taonga pūoro* and the possibility of including *electro-acoustic*.

2. WAIATA AND TAONGA PŪORO

Waiata and *taonga pūoro* are part of the traditional music of the Māori people. Tiramorehu (1849) stated:

'Kei te Pō te timatatanga o te waiatatanga mai a te Atua. Ko te Ao, ko te Ao mārama, ko te Ao tūroa'

Translated, 'it was in the night that the Gods sang the world into existence. From the world of light into the world of music.

According to McLean (1996, 1999) the traditional Māori instruments consisted of idiophone (percussion instruments) and aerophones (wind instruments). Membrnophones (drums) were absent and there was only one chordophone (stringed instrument) represented by the *Kū* originating from the South Island of New Zealand.

Tiramorehu (1849) states that Māori music is connected with the Gods who in turn provide the sounds that are imitated by the musicians who sing *waiata/ mōteatea* and play *taonga pūoro*. Flintoff (2004) states that *taonga pūoro* instruments are part of the family of traditional Māori instruments that is categorized under Māori gods such as *Ranginui* the tunes; *Papatūānuku* the rhythms; *Hine-Pū-Te-Hue* the gourds; *Tānemahuta* father of song makers; *Tangaroa* the father of shell instruments; *Tāwhirimātea* the wind voices, and *Raukatauri* goddess of flutes.

Since the 1980s there has been a revival of both the *waiata/ mōteatea* and *taonga pūoro* traditions in recent times. Extensive collections by Ngata & Jones (2004, 2005, 2006, 2007) in the published series *Ngā Mōteatea 1, 2, 3, 4* and McLean Waiata Collection and published books, *Māori Chant:a study of ethnomusicology (1965)*, *Māori Music (1996), Traditional Songs of the Māori (1975)* has inspired the re-learning of ancient chants that are performed on the *marae*, tribal workshops and at regional and national kapa haka competitions.

Whalley (2005) mentions the cultural revival of Traditional New Zealand Māori instruments led by pioneer and performer/ composers Dr Hirini Melbourne and Dr Richard Nunn. Whalley also mentions live performance, improvisation, crossover work and collaboration between the composer and performer of *waiata* and *taonga pūoro* in recordings such as *Te Ku Te Whe* (1994) and *Te Hekenga-Ā-Rangi (2003)*. Both these recordings were a collaboration between Dr Hirini Melbourne and Dr Richard Nunn. The albums featured live recordings of *taonga pūoro* and *waiata.*

Other works featured Gillian Whitehead (1998), Helen Fisher (2000) and Martin Lodge (2004) who created a number of acoustic 'art music' thus giving rise to 'cross over' works. Whalley (2005) states;

'Although outside the digital domain, this work is touched on here to provide an insight into methods of working with composers both within and beyond the Māori music tradition.

To date the recording of *Te Hau Kuri (Dog's Breath)* composed by Dr Hirini Melbourne and featured on New Zealand Sonic Art III (2002) provides a fusion of electro-acoustic sound with *taonga pūoro* being the main instrument. The sounds of the nine traditional Māori instruments were deliberately chosen and arranged to retell the story of the Okiwa wind musically. This particular recording gives a true organic sound of *taonga pūoro* within an electro-acoustic setting.

3. A MODEL FOR COMPOSING

Analysing data from literature, personal communications with composers (of waiata, taonga pūoro and electro-acoustic), and examples of works gave rise to formulating a model for composing, thus the **Harakeke and Tūī** model. Although this is only one model for composing it takes into consideration important stages of composing original music; the pre-requisite stage, the production stage and finally the product stage.

3.1. The Harakeke and Tūī model

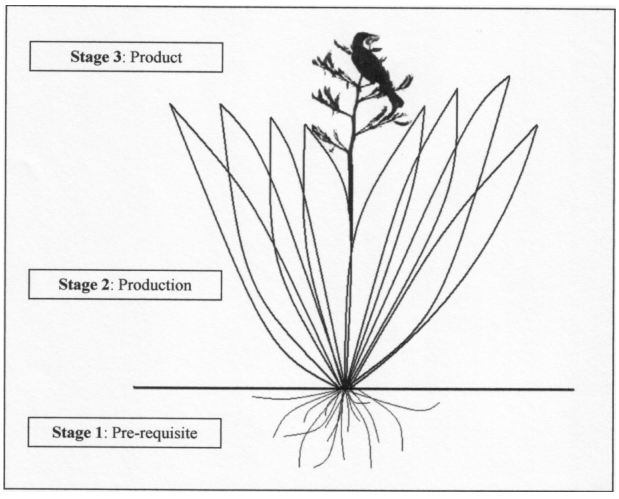


Figure 1. The Harakeke and Tūī model (Rollo, 2010)

Key

Stage One in the model is the pre-requisite for the composition of waiata, taonga pūoro and electro-acoustic. Each genre requires different knowledge and

skills in the composition process before progressing on to stage two and three.

Stage Two is 'the hands on' production of waiata, taonga pūoro and electro-acoustic composition. Each genre will be treated differently in the composition process and requires artistic creativity and an extensive knowledge of music production before progressing on to stage three.

Stage Three is the final product and digital recording of the waiata, taonga pūoro and electro-acoustic composition. The final outcome is the recording (music recording studio or mobile recording studio) and live performance of each composition to showcase the original compositions and to prove the validity of the new generated model for composing.

During stage three there will be an evaluation of each composition by invited participants (composers, performers and listeners) with the intention of improving the quality of the new works. Even after the final digital recording, there will be on going critical review which is eminent in creative arts.

3.2. The Philosophy of the Harakeke and Tūī model

Figure 1 represents a Māori perspective of the three main stages in the composition of waiata, taonga pūoro and electro-acoustic music.

Stage One is represented by the roots of the *harakeke*, Phorium tenax – New Zealand flax. The flax cannot survive without soil, water, the sun, its nutrients and roots that gives sustenance to the *harakeke* plant.

In regards to the composition process, the pre-requisites, the knowledge and skills of the composer are vital.

Stage Two is represented by the *rau* or flax leaves. Each leaf represents the important aspects of composition (the production) or 'hands on' approach. This includes the theme, organic sound samples, synthesized sound samples, live recordings, the arrangement, sections of the composition, the editing and so on.

Stage Three is represented by the *kōrari* (the flower stem of the flax) and the *tūī* bird (Prosthemadera-novaeseelandiae) a native New Zealand bird which eats the nectars of the flowers on the *kōrari* stem. In Māoridom the *tūī* is known for its sweet chirping singing voice and portrays the sound of the final composition hence the performance and final digital recording.

'Ka tangi te tūī, te kanakana te hae'
When the tūī sings, the jealous eyes in on the watch

4. MUSIC TECHNOLOGY AND COMPOSITION

Music technology has always played a big part in electro-acoustic music from the 'reel to reel' recorder, the phonograph recording, the 'tape cassette' recorder, sampling machines, Telharmonium (1897), Chamberlain (1946), Mellotron (1960s), Birotron (late 1970s), Optigan (1971), Optigan (1971), EMS Synthi AKS (1972), Orchestron (1975), Synclavier (1975), Fairlight CMI (1978), Kraakdoos, DX7 (1983), UPIC to the more sophisticated Macintosh e Mac, G4-700 combo, Roland PC-300 Keyboard, Steinburg Cubase AV, and supporting software programs such as Protools, Metasynth syntheziser and Kontakt sampler.

In reference to the history of electro-acoustic/ electronic music Collins and D'Escriván (2007) gives an overview and the importance of electrical technology:

'From musique concrete to audio-visual sampling, from eletronische music to minimal techno, from the Telharmonium to the laptop, electrical technology has facilitated more than a century of original music, spawning a multitude of new styles, instruments and methods'.

How can technology and the composition process support the fusion of electro-acoustic and Māori music focussing on the introduction of *waiata* and *taonga pūoro*? This will be further investigated during the composition of the four music works in the PhD research. The **Harakeke and Tūī** model provides a starting point in accumulating ideas and planning the new composition and, as technology advances into the 21st century and beyond will bring the music to reality.

The exciting part of the production stage is to experiment with live recordings of *waiata* and *taonga pūoro* and to decide whether to input the organic sounds as they are or to manipulate and invent a new sound, keeping in mind cultural considerations (Māori context). The composer must also give thought to where the *waiata* and *taonga pūoro* should be placed in the composition to highlight the motif.

5. WORK IN PROGRESS

As part of my PhD research, there will be four music works to test the models for composing *waiata, taonga pūoro,* and *electro-acoustic.* This is the exciting and creative part of the research. I will be working collaboratively with composers of the fore mentioned genres, musicians and engineers to produce the original music using the **Harakeke and Tūī** model. Although the work is still in progress I will give you some background information into the concept and process.

5.1. The Planning Stage

Figure 2 gives an outline of the overall theme of the four original music compositions which includes:

- One *taonga pūoro*[3] music composition.
- One *waiata tawhito*[4] music composition

[3] **Taonga pūoro** refers to all Māori traditional instruments. In reference to this article the composition refers to a live performance recorded.

[4] **Waiata tawhito** refers to ancient chants and song of the Māori.

- One *waiata hou* music[5] composition
- One *electro-acoustic*[6] music composition

The four music compositions will be integrated together to give a dramatic account of the origins of the Waikato river, the establishment of the *Kīngitanga* Māori king movement, a lament to the late Māori queen Te Arikinui Te Ātairangikaahu, historical account of the Waikato wars and the battle at Rangiriri *pā*[7] and finally the celebration of Kīngi Tuheitia, the sixth Māori king. *Waiata, taonga pūoro* and *electro-acoustic* will feature in this music project.

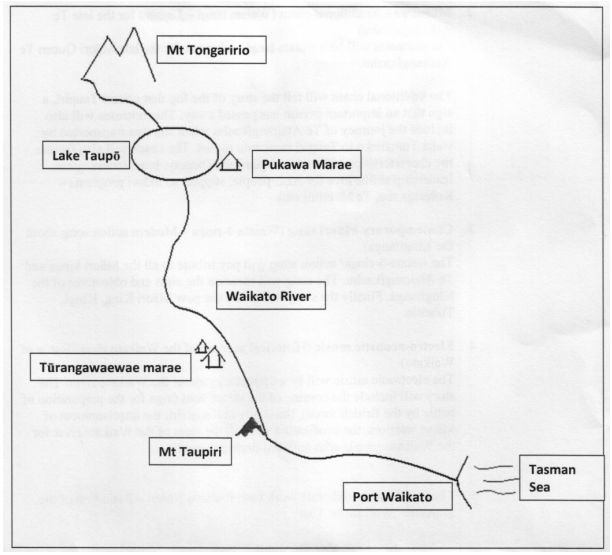


Figure 2. The Planning of the Four music compositions

5.2. The Four Music Composition

The reoccurring theme that will unite the music together is *Waikato Te Awa* a traditional Māori chant. This *waiata* celebrates the Waikato River the landscape that embraces the four music compositions as detailed in Figure 2. The other *waiata* that will feature in the composition is *E Pā Tō Hau* a lament.

Composition One is a *taonga pūoro* music that will tell the story of the origins of the Waikato River and the meeting at Pukawa marae that established the *Kīngitanga* Māori king movement. This will be a collaborative work with Horomona Horo, a *taonga pūoro* musician.

Composition Two is a *waiata/ mōteatea* a lament to the late Māori queen Te Arikinui Te Ātairangikaahu. The lament mourns her death and her travels on the *Tumanako* canoe to her final resting place at Taupiri mountain. Māori lyrics will be written and a melody line sorted. The form of the *waiata/ mōteatea* is a *waiata tangi*[8] and will follow traditional Māori music aesthetic.

Composition Three is an *electro-acoustic* giving a historical account of the Waikato war and the battle at Rangiriri pā. A period of unrest between the colonial army and the Waikato tribes.

Composition Four is a *waiata hou* a contemporary Māori action song that acknowledges all the Māori kings and Te Arikinui Te Ātairangikaahu. This song celebrates Kīngi Tuheitia, the sixth Māori king.

The purpose of the four music compositions is to test the new generated models for composing *waiata, taonga pūoro* and *electro-acoustic*, a guideline to develop new composers in the future. After the compositions are completed, invited composers, performers and listeners will evaluate the music and give feedback. In turn, the compositions will be edited and re-worked before a final digital recording is produced. There will also be a DVD recording of the *waiata* and *taonga pūoro* performances.

6. ELECTRO-ACOUSTIC AND MAORI MUSIC

Why not?

Composers and performers of electro-acoustic music have already indulged into creating new works individually or in collaboration with others. There has been new works that have incorporated other music forms and themes such as the recording *Kasumi* (2002, New Zealand Sonic Arts III) composed by Ian Whalley.

This particular composition includes a Japanese text from *Manyoshu* number 4290 by Otomo no Yakamochi (23 February 753) and the integration of *taonga pūoro.* The short instrumental samples used in this work are from seven New Zealand Māori instruments. In *Kasumi* (for instrument (s)/ voice and tape) the original sounds provided the basis for the electro-acoustic manipulation and integration of the work. Based on the text, the composition is constructed using parallel narratives like a folk tale, juxtaposing simple binary elements (2002, New Zealand Sonic Arts III).

The cultural/ environmental exploration of a New Zealand musical voice has seen the revival of both *waiata/ mōteatea* and *taonga pūoro.* This extended the Māori instrumental tradition through the application of digital music production techniques (Whalley, 2005), thus creating new works. The PhD research involves the composition of the four music compositions as detailed in 5.2. This will explore further organic and manipulated sounds of voice (*waiata)* and traditional instruments (*taonga pūoro).*

[5] **Waiata hou** refers to contemporary Māori songs with Western and other world influences.

[6] **Electro-acoustic** is electronic music.

[7] **Pā** is a fortified Māori village or settlement.

[8] **Waiata tangi** is a lament to the dead.

Māori music is evolving and Māori composers, performers and audience/ listeners have been exposed to many music genres including electro-acoustic. Although the revival of traditional *waiata* and the making and playing of *taonga pūoro* is eminent today, Māori music will explore new avenues, create new sounds and add to the vast repertoire of New Zealand music.

Electro-acoustic can enhance Māori music and the PhD research will prove that the models for composing is practical and the new compositions support the idea that *waiata* and *taonga pūoro* are important Māori art-forms that can be fused with Western music such as electro-acoustic, or any other forms of music for that matter.

7. CONCLUSION

This article examines the intellectual and creative aspects of composing new electro-acoustic music and combining *waiata/ mōteatea* and/ or *taonga pūoro* in to the soundscape/ abstract composition.

The ongoing current PhD research will generate a model for composing *waiata, taonga pūoro* and *electro-acoustic* with the aim of cross pollinating musical ideas in a cross-cultural environment. This model will be tested with four 'creative' music compositions not only to validate the model for composing but to support that electro-acoustic can enhance Māori music, and the production of new sounds and new works.

The Māori proverb *'He taurangi te hau, he hurihuri te ao'* supports the idea that *waiata, taonga pūoro* and *electro-acoustic* music is evolving and undergoing many changes, thus providing new concepts, theories, music practices, methodologies and technologies to enhance compositions for the future generation to employ and create new works of art.

8. ACKNOWLEDGEMENT

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THE CHRONOME: A CASE STUDY IN DESIGNING NEW CONTINUOUSLY EXPRESSIVE MUSICAL INSTRUMENTS

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ABSTRACT

While the Monome and its variants have created a large user community (and a variety of custom applications) the devices do not afford performers direct access to continuous control data from either the buttons or the LEDs. With the addition of continuous control, the Chronome provides a new open-source iteration of the Monome that is more expressive and organic to perform with. This increased control is provided through the addition of pressure control via the buttons, and multi-colour support for the LEDs. By increasing the level of control, but keeping a familiar instrument design, the authors aim to facilitate an increase in the virtuosic potential afforded by the Monome family of devices. With this in mind, the authors have developed several new applications that utilize the Chronome's new functionality, and provide increased levels of visual feedback, while taking advantage of the continuously variable "aftertouch" from the pressure buttons.

1. INTRODUCTION

In being a "digital lutherie" (Jorda 2005) how does one design new musical interfaces that provide the same expressive potential found in acoustic instruments? How do we learn from the "organic" interaction between musicians and their acoustic instruments, and how do we use these insights to design new digital instruments that push the potential of that connection even further? Exploring these questions reveals the many difficult design decisions that must be addressed in creating new interfaces for musical expression.

One approach to developing new expressive instruments has been to attempt to understand the principles that facilitate virtuosic performance. Dobrian describes "virtuosity" as "a person having complete mastery of an instrument, such that s/he can call upon all of the capabilities of that instrument at will with relative ease…" (Dobrian and Koppelman 2006). This implies that, instead of being distracted by the mechanics, a musician's mastery over an instrument allows them to focus instead on the musical ideas they wish to convey (Hunt and Kirk 2000). In this sense, virtuosity is the mastery of an instrument, which can then allow a performer to expressively explore a musical space.

A design solution to easily afford this level of mastery would be to create an instrument that was extremely simple, and which had a very limited set of discrete controls; however, this approach would decrease the overall virtuosic potential inherent in the instrument. This virtuosic potential and the difficulty in mastering an instrument are related to the level of controller complexity. This complexity can come from the implementation of continuous controls in the instrument design, or the mapping scheme used to link the controls to the sound they manipulate (Dobrian and Koppelman 2006). While it is possible to have too many of these continuous controls, having too few is far more dangerous (Dobrian and Bevilacqua 2003).

The above design principles governed the conceptualization and development of the Chronome controller. The Chronome came out of the open-source Monome/Arduinome interface, which is an 8x8 grid of discrete buttons paired with LEDs. For a more complete history of development and thorough historical context see (Vallis, et. al. 2010, Vallis and Kapur 2011). While the power of the Monome/Arduinome lies in its open-source configurability, the lack of continuous control means that there is great potential for expanding the expressiveness of the instrument. The authors' experiences performing with the Arduinome in The Machine Orchestra (Kapur, et. al. 2010), and their own practice has illustrated the need to increase the controller complexity in an effort to expand the virtuosic performance potential.

There are three ways in which the Chronome increases the potential of virtuosity, allowing it to act as an organic musical interface: each of the 64 discrete buttons is pressure sensitive. This allows parameters such as timbre to be modulated throughout the entire duration of the triggered event; each of the 64 LEDs allow for visual feedback using the entire visible colour spectrum. This provides "continuous" programmable feedback from the instrument, something that is not possible with acoustic instruments. Similarly to how the weight of piano keys inform the performance of a pianist, the "feel" of the silicon gel buttons creates a distinct "feel" to the Chronome.

The following sections discuss the evolution from the Monome to the Chronome, the interface construction, and the design choices made to realize the final device.

The authors will then discus the issue of mapping the new data to OSC and MIDI, while simultaneously allowing the device to behave as a traditional Monome/Arduinome. Lastly, we will look at new software applications that take advantage of the increased continuous control afforded by the Chronome.

2. EVOLUTION FROM MONOME TO CHRONOME

Since its creation in 2005 the Monome interface has developed a large community of users who have contributed a plethora of custom applications and hardware modifications, through open-source access to Monome schematics, firmware and software. A brief overview of the history of the Monome community modifications was discussed in (Vallis, Hochenbaum and Kapur 2010), and included the Arduinome project by the Authors that ported the code over to the Arduino microcontroller. Additionally Monome users have mutually exclusively explored both the use of RGB LEDs (multi-color light emitting diodes) for more dynamic visual feedback, and pressure sensitivity using conductive fabrics.

The Chronome is an attempt to not only incorporate these separate extensions of the Monome's functionality into a single device, but also to simplify the process of building the interface, integrate the interface more closely with the existing Monome/Arduinome serial protocols, and provide for easy expandability of additional analog sensors.

While existing Monome style interfaces are binary devices (LEDs and buttons only handle on/off messages), the Chronome affords continuous data from both the input and the output of the interface. This is achieved through 64 independent RGB LEDs and pressure sensitive buttons. In order to facilitate this new functionality, the use of the Arduino Mega is required to handle the large number of sensors.

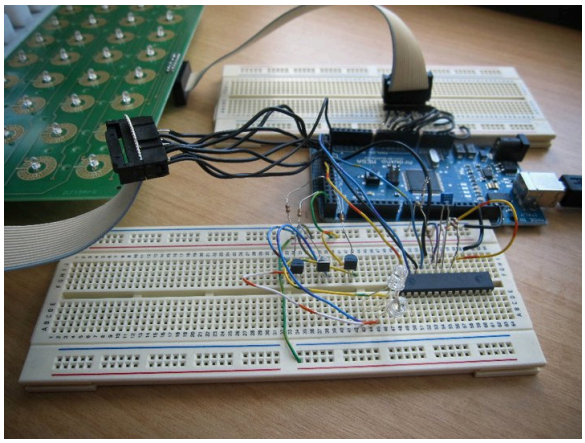


Figure 1 Chronome Prototype Using the TLC5940 and Arduino Mega

Similarly to the Arduinome, the following constraints were placed on the Chronomes design: ease of building, open-source access to the hardware, firmware, and schematics, and integration within the existing Monome serial protocols.

3. INTERFACE CONSTRUCTION

While both commercial and custom PCBs (printed circuit boards) for monome-like interfaces have been available since at least 2008, they only provided 4x4 button and LED grids that could be chained together to create larger configurations. This chaining of PCBs increases the difficulty of wiring, as many connections are made between the button pad/LED PCBs. For an electronics hobbyist, the difficulties brought on by the wiring may be enough to discourage them from taking on the project. Early versions of the Arduinome had a similar issue that was solved when Monome forum members took it upon themselves to design an 8x8 version of the button pad/LED PCB. This considerably simplified the Arduinome build process, and was a motivating factor to provide a similar 8x8 PCB for the Chronome. Designing a custom 8x8 PCB not only simplifies the build process, but also reduces the number of connections between the Arduino Mega and the PCB itself, and places all additional chips directly on the button PCB itself.

After researching existing RGB Monome clones, it was decided that the TLC5940[1] chips would be used to drive the 64 individual RGB LEDs; however, the existing RGB firmware only provided RGB support for less capable Arduinos (such as the Duemilanove), and would have made it difficult to simultaneously handle the 64 analog button inputs. With this in mind, additional research revealed an alternative approach to driving the RGB LEDs. The Atmel AVR chips are equipped with special hardware Output Compare Match pins that allow for a pulsed output of the Timer/Clock it is tied to. This means that the Arduinos can send out a clock pulse without requiring any instruction cycles from the main program loop. This function was used to drive the grayscale cycle of the TLC5940 chips, and as a consequence, left the bulk of instruction cycles open to other tasks such as reading the buttons, and handling serial data.

Research into existing attempts to add pressure sensitivity to buttons through conductive fabrics (Freed 2008) was used as the bases for the pressure sensitive buttons (see **Figure 2**). The increase in ports on the Arudino Mega meant that no additional multiplexing chips were required to read all 64 analog buttons. Eight digital pins were used to cycle 5V through the rows, while eight of the 16 ADC inputs were used to read the columns.

[1] http://www.arduino.cc/playground/Learning/TLC5940

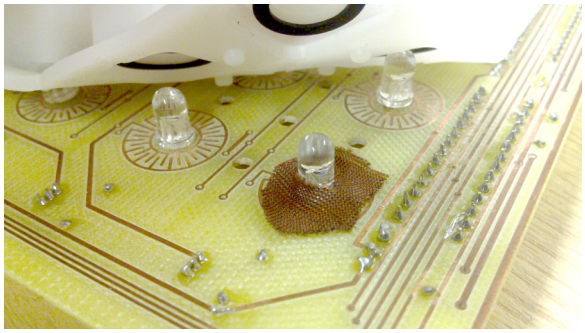


Figure 2 Conductive Fabrics for Pressure Buttons

Lastly, in a similar fashion to the Arduinome, the Chronome was flashed with a unique serial number for the Arduino Mega's FTDI USB chip[1]. This allowed ArduinomeSerial (see section 4), a custom version of the MonomeSerial application, to recognize Chronome devices when they are connected to a computer.

4. MAPPING

In section 4.1 we describe how real-world analog signals from the Chronome are converted into OSC and MIDI messages.

4.1 Communication and Configuration

Translating the Chronome's serial–data messages into musically relevant formats (OSC(Wright 2005)/MIDI) was a two-step process. Firstly, we developed an application (ArduinomeSerial) to convert the Chronome's serial-messages into OSC/MIDI. In writing this application, the authors ensured that all existing Monome/Arduinome messages are supported in order to allow the Chronome to be compatible with all existing musical applications developed in the Monome community. This was of extreme importance, as well as providing a way to set a global default colour for the Chronome, allowing a way for older Monome applications to change the LED colours (albeit globally and not for each LED individually). Secondly, we expanded the original MIDI implementation of ArduinomeSerial (which only chromatically assigned each button a MIDI note) to include pressure sensitivity and basic support to change the colour of the LEDs. More extensive MIDI customization however would be implemented using a secondary application, which is further discussed in the following paragraph. Ultimately, through developing the ArduinomeSerial application, we sought to allow computer musicians to connect the Chronome in such a way that it was fully compatible with its predecessors (Arduionme/Monome), while providing access to the extended expressive potential provided by the Chronome's unique sources of continuous data.

In tandem with work on the ArduinomeSerial application, we have developed a sister program called ChronomeState (see **Figure 3**) which provides users with an intuitive graphical way to easily customize the behaviour of the Chronome interface. Through the use of the ChronomeState application, users are able to independently customize the following parameters of each button:

- Buttom mode:
  - Toggle (on/off),
  - Trigger (momentary on/off)
  - Note (sends out assigned MIDI note number)
- Continuously variable colour for *each* button
- Pressure sensitivity can be enabled/disabled for each button

ChronomeState also allows each device mapping to be saved/loaded on the fly, allowing for users to quickly remap the controller during a live performance.

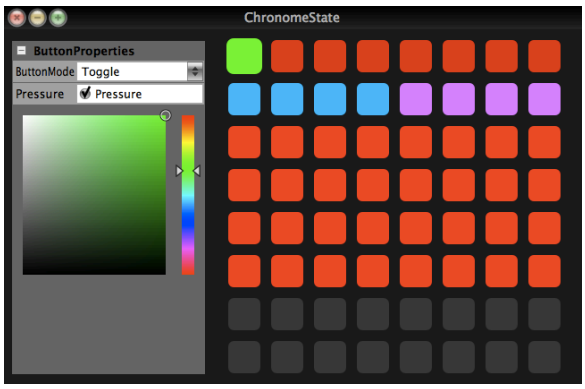


Figure 3 ChronomeState application for configuring Chronome behaviour

While it isn't necessary to use ChronomeState to interface with the Chronome, it is a convenience application that allows the Chronome to communicate with nearly any music software on the market via MIDI. Other exploratory methods of mapping the Chronome can be seen below in section 5.

5. MUSICAL CASE STUDIES

While in section 5.1 we discuss various musical case studies that use the additional control parameters provided by the Chronome to expand the expressive potential of the Monome/Arduinome family of instruments.

5.1 Chronome Applications

Several applications have been developed to take advantage of the new data afforded by the Chronome.

[1] The FTDI chip provides the USB link between a computer and the Arduino chip

These include: the navigation of a SOM[1] (Self Organizing Map), which maps the button location in the matrix to the nodes in the SOM, while simultaneously mapping pressure to navigate samples stacked inside each of the nodes; and a utility application written in ChucK(Wang 2008) that maps the buttons to various functions in Ableton Live, while simultaneously mapping the RGB LEDs to indicate relationships and states of various Ableton functions.

While existing applications for the Monome/Arduinome allow for similar control, the additional pressure data creates a more organic control input, and allows input event variability over time (analogous to "aftertouch" functionality). The fact that the Chronome also has more than one colour available provides a clear and immediate way to differentiate or group button behaviours and application states, as well as providing visual feedback of the newly variable controls.



Figure 4 RGB LED Run From MIDI in Ableton

6. CONCLUSIONS

As the expressive potential of an instrument can be related to the "complexity" of its controls, it is worth a brief look at how the level of control has evolved over the development of Monome centric devices. If we regard control as happening over a window of time, then the original Monome/Arduinome can be seen as having three dimensions of input: time, whether a button has been pressed, and which button number it was. This allows users to map the two input data streams (button state, and button number) to parameters such as pitch and envelope triggering. Devices such as Keyboards and Pianos can be thought of as four dimensional control systems in that they add velocity (a value sampled at the onset of the pressed key) as another control parameter. This inclusion of allowing dynamics with every triggered key was a major development between harpsichord style instruments and pianoforte based instrument designs. Finally, the Chronome brings a fifth dimension of expressive control by allowing the velocity parameter to be continuously variable over the duration of the triggered event. Through the addition of pressure sensitivity to the buttons, the Chronome now can facilitate much greater virtuosity then its predecessors. This input virtuosity is also coupled with a similar multi-dimensional visual feedback design, making for an extremely powerful interface.

The Monome and its variants have proven to be versatile and effective interfaces in their own right, with a broad array of applications, and a large user base; however, the lack of continuous control has often seen the interface paired with other devices, or modified to include sensors like accelerometers, and potentiometers. With the addition of continuous input and output, the authors are interested in exploring new applications of the device that will take advantage of nuanced control over the events triggered by the Chronome's buttons. The authors are particularly interested in the dual mappings afforded by each button—receiving a location in the button matrix, while also receiving variable pressure data, and tying this to the decoupled RGB output.

As with the Arduinome project, through open-source hardware, firmware, and schematics, the Chronome will be made freely available to anyone interested in building one. The authors feel that this will help to create new applications for the interface and continue to refine the hardware. In an effort to make building the project as simple as possible, we have provided a list of all the necessary parts, links to all required files, and complete build instructions. Additionally, we have made available files for a laser cut enclosure to house the finished instrument (see **Figure 5** below).

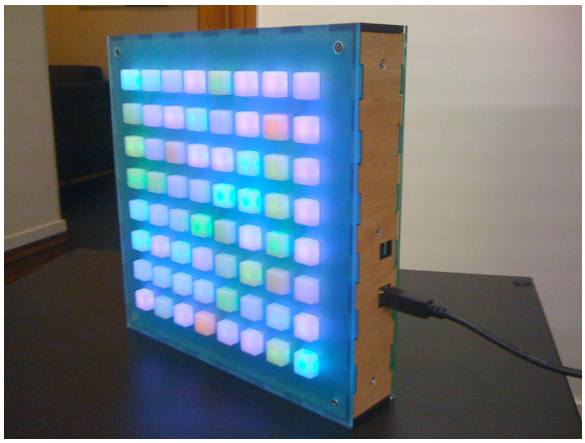


Figure 5 Chronome Laser Cut Enclosure

7. FUTURE WORK

The Chronome's increased controls greatly extend the versatility of the original Monome, but also add a level of complication to performing with it. The authors are interested in performing an evaluation of the new functionality, and comparing it against the ease of use of the original. Existing research shows that it is possible to take advantage of a musician's existing skills by designing an instrument based off of one they are already familiar with (Dobrian and Koppelman 2006). With this

[1] A SOM is form of unsupervised learning used to "cluster" a data set based on similarities between the individual data vectors. For more information refer to http://en.wikipedia.org/wiki/Self-organizing_map

in mind, it will be interesting to see whether the consistency between the Monome and the design of the new Chronome results in reducing the difficulty of mastery, and thereby accelerating the rate at which musicians can take advantage of the new control parameters and become virtuosic. Additionally, the open-source release of the Chronome provides an opportunity to ask potential builders some basic question regarding the interface. With this information, we will look at what the users motivations are for building a Chronome, and how the process could be improved or changed. Finally, the use of the Arduino Mega provides an increase of analog to Digital ports. These additional ports will be set up to allow users to plug in any analog sensor and enable it from the ArduinomeSerial application.

8. ACKNOWLEDGMENTS

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The possibilities of novel formal structures through computer controlled live performance

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ABSTRACT

Computer controlled performance opens a range of novel structural possibilities. This paper explores the mechanisms and ramifications of this approach, and its potential to expand the repertoire of formal structures available to the composer.

Traditional and computer coordinated performance models are compared. Modes of computer control, permutation, transformation and generation are discussed and their implications are evaluated.

The range of implications of this approach to the performance environment are given together with illustrations from the author's own work.

1. INTRODUCTION

The advent of computing provided a platform for controlling musical performance in a manner far more rapidly, seamlessly and interactively than previous media. James Hollan identifies two key features that distinguish computer interfaces from previous forms of representation:

> they provide the most plastic representational medium we have ever known, and they enable novel forms of communication. (…) This plasticity in combination with the dynamic character of computation makes possible new interactive representations and forms of communication that are impossible in other media [12].

This paper examines the scope of the structural possibilities that computer control of live performance implies and permits.

In a traditional acoustic performance model (Figure 1.), coordination of the performance is, in the first case, determined by the composer who provides materials which incorporate both events to be performed and a tempo/metric framework for their synchronization. Coordination of the actual performance is managed by the performers alone, through visual cues and auditory feedback.

This is not to diminish the crucial role played by performers who, in addition to their technical and interpretive skills, draw their own experience and judgments (still far more complex than any computer) into the equation. As John Zorn states in regard to the role of the performers of his work *Cobra* (1984): they "bring in their own private perceptions, past experiences, instrumental techniques, and interpersonal attitudes" [4].

Computer coordination of live musical performance (Figure 1.) allows for the control and synchronisation of

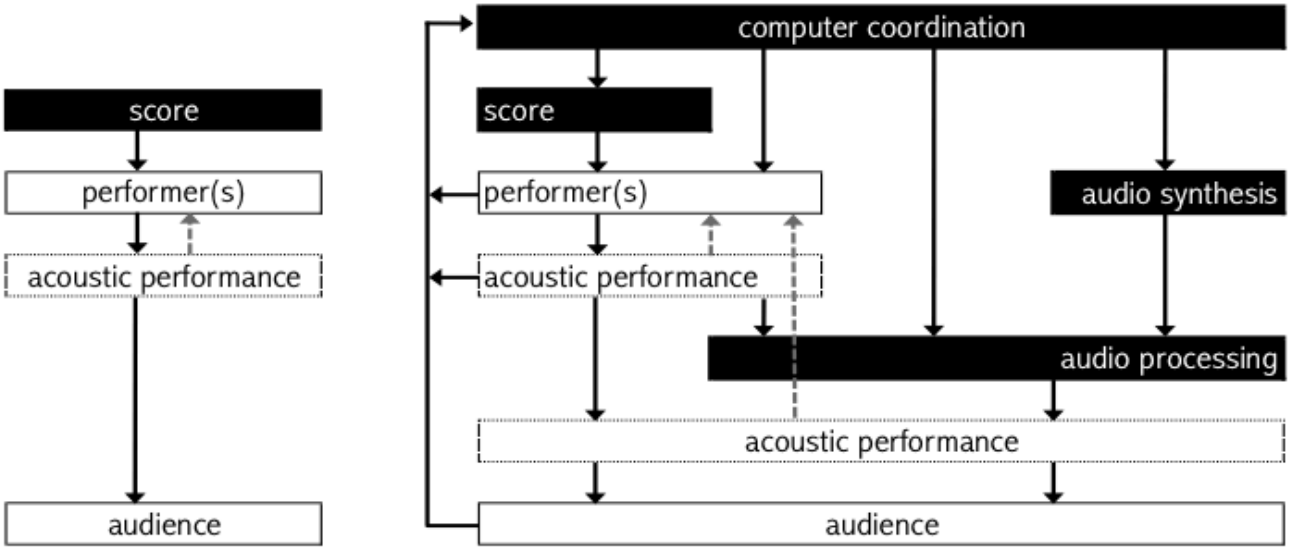


Figure 1. Traditional "Classical" acoustic performance model (left) and the Computer controlled performance model (right).

the score and the temporal framework, in addition to the generation of electronic sounds and electronic transformation of both the acoustically and electronically generated sounds. The computer-generated clicktrack creates the opportunity not only to independently control the tempi of multiple performers, but also to transmit formal (for example nonlinear selection of score materials) and performance (such as articulation, dynamics and so forth) parameters in real-time.

Computer coordination can control many actants in a performance in a manner analogous to the team of players necessary to bring symphony to life. Auditory and visual cues still play an important role in the coordination of the live performance, importantly however, in a computer controlled performance feedback into the system can also be achieved though other means:

- the performers may interact with the computer via hardware interface(s);
- the acoustic performance itself may be used as an interface through computer analysis; and
- the audience may interact with the computer, playing a role in defining the performance.

The performance of music requires extremely fine-grained coordination between events in the order of tens of milliseconds. Computer interoperability allows for the coordination and the requisite rapidity of distribution and interactive analysis that was not previously attainable with performers alone. As such, computer controlled performance potentially permits the conception of formal structures that were previously unrealizable and/or impractical.

2. STRUCTURAL IMPLICATIONS OF COMPUTER COORDINATED PERFORMANCE

The paradigm of the traditional score is that of a continuous scroll, albeit one chopped into segments and arranged sequentially on successive pages for convenience, with the resulting "systems" conventionally read sequentially from left-to-right, top-to-bottom. The rate at which the musical score is read is governed in performance by a synchronised tempo predetermined by the performers. Consequently, the structure of the performed music is, innately linear in character.

Although composers have developed methods to create novel formal structures by subverting the implications of the traditional score with approaches such as mobile musical modules and multiple conductors, they have done so by contravening the conventions of the traditional score. Such approaches are fettered by these conventions, for example: the minimum and maximum length of musical materials presented to the performers and the ability to sequence and coordinate the materials in real-time [18].

The performative, and potentially structural, implications of computer control derive from the nonlinear, hypertextual nature of computational capacities and are musically manifested in three principal organisational procedures: the permutative, generative and transformative.

2.1. Permutation

Computer coordination allows the permutation of musical materials that are presented to performers and the synchronisation of their performance. Permutation of scored materials may involve translocation, insertion, duplication and/or deletion of musical materials. The materials may vary in size from large structural blocks, to sub-structural cells or even individual parameters.

The permutation of large structural blocks of music may be found in traditional paper scores such as Stockhausen's *Momente* (1962-69) and *Mixtur* (1964), however synchronization issues rules out real-time permutation in these works [17].

Although short fragments of a few seconds length, are permutated in the performance of Feldman's *Intermission 6* (1953), the fragments remain isolated "sound objects" rather than functioning at any time as components of a continuous musical passage or discourse. In my works *Antibody* (2009) and *Improbable Games* (2010) sub-structural units are permutated to create novel passages of music in real-time.

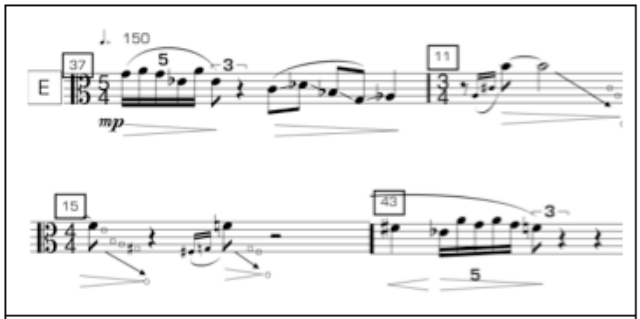


Figure 2. Detail from Vickery's *Antibody* (2009) showing permutated "cells".

The independent manipulation of even smaller units, the parameters that are combined to form musical events is also possible in digital media. This approach is exploited in Gerhard Winkler's *Hybrid* series (1991-) [20]. In *Hybrid II*, for example the glissando, string position, bow pressure and dynamics are graphically conveyed to the performer in real-time.

The structural implication of permutation of blocks, cells or parameters in "mobile" forms are the same as those identified by Boulez in Stravinsky's *The Rite of Spring* (1913) [2], namely that synchronous permutation of all parts simultaneously results in "vertical" changes in the performed materials, and asynchronous permutation of the parts, given that they are sufficiently distinct, results in "horizontal" or layered changes. (See Figure 4.)

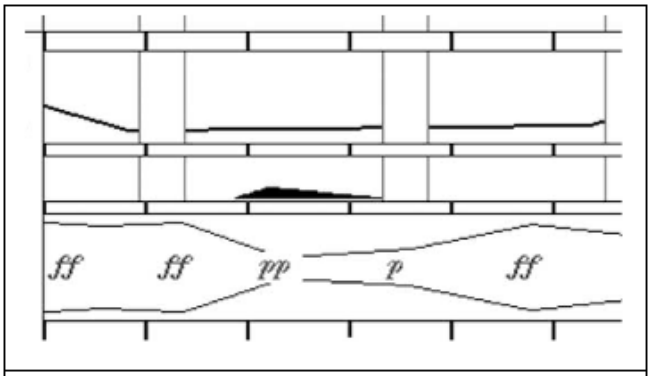


Figure 3. Detail from Winkler's *Hybrid II* for viola and interactive live electronics (1996/2001), showing graphs representing (from the top) string position, bow pressure and dynamics [19].

The vertical and horizontal structural implications of the permutation of audio that is digitally derived are identical to those pertaining to scored materials. However, the source of the sounds may strongly bear on the sense of formal structure that arises, especially in the case that the materials are referential to existing, familiar sources, or derived from processed acoustic sounds from the performance itself. In the first case, familiar sound sources carry their own signification: a ten second burst of a sample from Beethoven or Elvis will sound more isolated in the context of the average minute performance, for example, than a ten second burst of white noise.

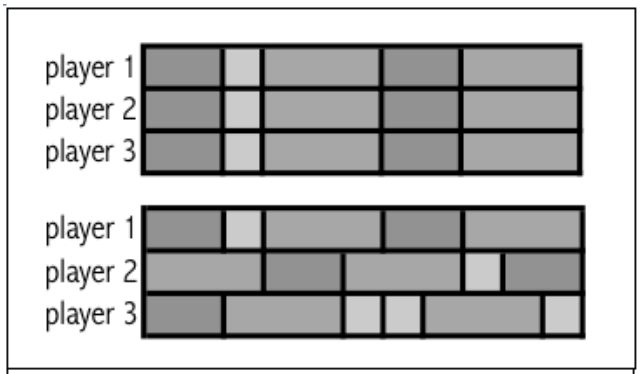


Figure 4. Synchronous "vertical' permutation (top), A-synchronous "horizontal" permutation (bottom).

The permutation of processed live acoustic sound, likewise may be closely bound in the mind of the listener to the instruments that produced them in the recent past. Structurally, live processing may act quite independently of the source sounds, following an opposing trajectory that reinforces, cancels or is in counterpoint with those of the live performer.

2.2. Transformative

Transformation differs from permutation in that it acts upon an "original" object to which alterations occur over time. In this sense transformation is related to the musical concept of development, as permutation is related to "concatenation" or "block" forms [7]. The notion of development is expanded by digital transformation in that the alterations need not be predetermined: they may act uniquely on the materials in each performance.

Transformations may be applied graphically to a digital score altering how it is to be performed. The transformative screen-score is the digital descendant of Stockhausen's *Refrain* (1959), a work in which the paper score is overlaid by a mobile clear plastic strip that modifies whatever the material is below it a structural approach he referred to as "Variable Form" [7].
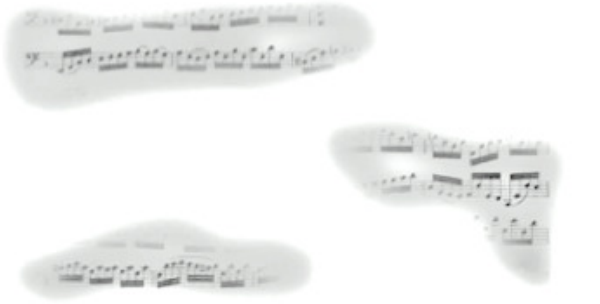


Figure 5: Excerpt from Kim-Boyle's *tunings* (2006), showing "real-time blurring and other distortion techniques" used to transform the score [14].

In David Kim-Boyle's *tunings* (2006) for cello and computer, "real-time blurring and other distortion techniques" are employed to reveal only portions of an underlying score. Boyle states that the work is modeled on "the idea of an old-fashioned radio tuning into different stations, sometimes pausing, often moving on" [14]. In this Open Work [9], Kim-Boyle refers to a range of musical materials, amongst them Bach's second Cello Suite. The reference to this work extends the "tuning" metaphor, drawing on the performer's own memory and familiarity with this core repertoire work.

My work *Transit of Venus* (2009), utilises a transformative real-time score, live sound processing and independent click tracks to control a guided-improvisation by the players. The score-player for *Transit of Venus* separates the functions of the traditional score, where performance indications are normally vertically unified.

This configuration allows temporal independence to be established between parameters such as texture, pitch, dynamics and articulation. The graphical-score component of the score-player displays a continuum of transformations from silence to free improvisation to be is followed by each performer.

Although transformation occurs over time and is therefore principally a "horizontal" technique, it may contribute important structural distinction according to how it is deployed through the distinction between "vertical" application to all actants in the work, or "horizontal" application to independent layers within the performance.

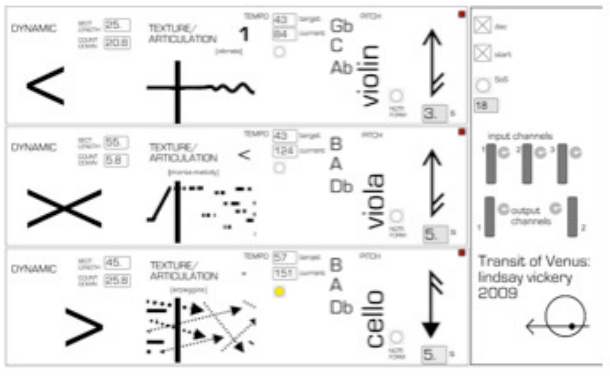


Figure 6: Screen-shot from Vickery's *Transit of Venus* (2009)

2.3. Generative

Algorithmic or interactive generative processes may be employed to construct components of a digital score in real-time. This approach opens broad range of structural possibilities often linked to a narrative or dramatic concept.

In the broad sense permutation and transformation may both be viewed as having generative characteristics. The distinction here is the complete absence of any "object" prior to the performance in generative works. Although algorithmic processes may be predetermine in a generative work, the outcome, in the form of a score or sonic product is completely undefined prior to the performance. For this reason, this form of "dynamic scoring" is sometimes euphemistically referred to as "extreme sight reading" [11].

For example in Polish composer Marek Chołoniewski's *Passage* (2001) a conductor directs a silent performance of hand gestures by the performers, which are measured by changes in luminosity measured by light sensitive resistors mounted on their music stands. The recorded gestural data in turn generates a scrolling score that is subsequently performed by the ensemble [5].

Similarly, Cat Hope's *Possible Stories of Harry Power* (2010) explores a concept not unlike the children's game "Chinese Whispers". Specifically in this case, the fabrication of mythologies around the life of the Australian folk hero, the bushranger Ned Kelly. Three instruments representing the Kelly, his mentor Harry Power, and Kelly's mother, read their pitch and volume from a projected graphical score. Each instrumental line is depicted in a different colour in what are essentially four a maps both of the pitch, duration and timbre of each line and the versions of the recorded movements of Power, Kelly and his mother in relation to each other in time.

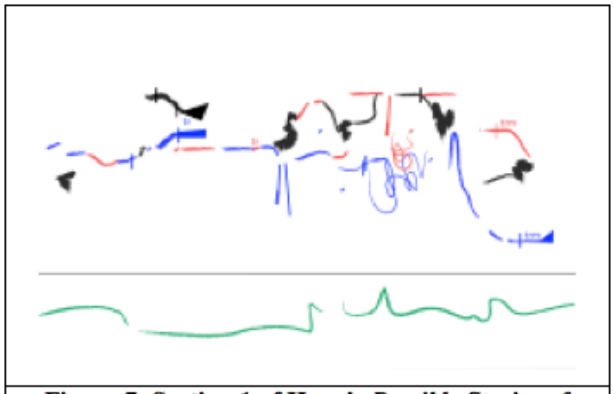


Figure 7: Section 1 of Hope's *Possible Stories of Harry Power*: the initial graphical notation score presented to the musicians.

In performance, the first "map" (Figure 8.) is played and simultaneously rendered by the computer into a second map incorporating the computer (and performer's) errors, this second map is then played and the process is repeated, recasting the score into a third map incorporating yet another layer of "Chinese whispers" style elaboration. The final section returns again to the composer's rendering of yet another version of the intertwining trajectories of the three characters (Figure 9.).

The computer patch, created in Max/MSP by Kynan Tan, "listens" and "retells" the performer's reading in an analogy to oral transmission. The three performers are recorded via contact microphones, the computer analyses the instruments' pitch, inverts them and then renders them proportionally in three contrasting colours as the next portion of the score. As such, *Harry Power,* casts the computer's generative potential in a narrative role, taking advantage of the inaccuracies in the

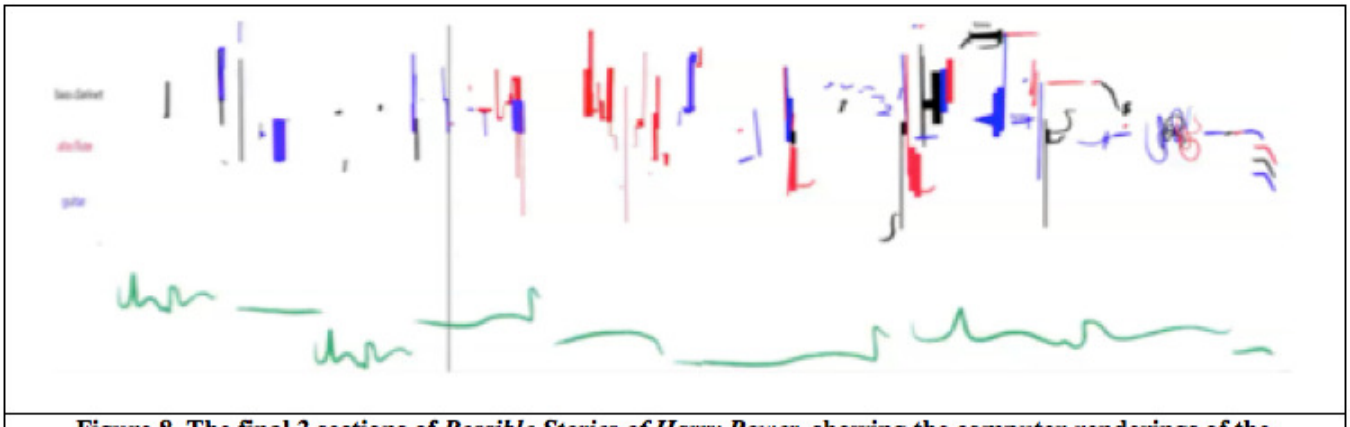


Figure 8. The final 3 sections of *Possible Stories of Harry Power,* showing the computer renderings of the performer's realization of section 1 and 2 and the final graphically notated section.

computer's transcription, to enact an analogy to the transformation of oral transmission of narrative.

Interaction with a generative model may also take place directly with the algorithmic processes themselves as is the case with "live coding", an approach that "involves writing and modifying computer programs that generate music in real-time. Often this music making activity occurs in a live performance situation with the code source projected for the audience" [3].

The structural implications of generative approaches are particularly open-ended: Choloniewski's work is gestural and performative, Hope's programmatic and the live coding approach is in some respects narrative and performative. It might be expected that many other paradigms will be explored as this practice (and technology) matures.

3. FORMAL IMPLICATIONS

A sense of structure is derived from changes in continuity and discontinuity in materials, processes and transformations evident in the sonic outcomes arising from a particular performance model. In the traditional classical model the sense of structure derives principally from the score, with a relatively minor contribution drawn from the performers' interpretation and interaction. Computer coordination allows a radical redistribution of the relationships between the performers, the score, the digital components and the audience. Structural decisions may arise from any actant in the performance model and may be the result of interaction and improvisation as well as predetermination. In Jason Freeman's *Glimmer* (2004) for chamber orchestra and audience participation, for example, the audience influences the unfolding composition "by waving four-inch battery-operated LED light sticks back and forth" in front of video cameras' [11].

Computer coordination allows for greater distinction between voices and layers in a musical work through expansion of timbral, dynamic, spatial and temporal qualities both "vertically" and "horizontally".

Sampling provides a pathway to unprecedented referentiality to sound objects outside the performance model. Networking and telepresence expand the potential of these possibilities beyond the specific environment of the performance model. Perhaps these possibilites will someday allow for the realization of Anthony Braxton's "orchestral musics" conceived "to be performed simultaneously in different cities, (if perhaps not "on different planets and even in different galaxies" [1].

3.1. Multiple versions

In *The Open Work*, Umberto Eco theorised the possibility of the "work in movement" permitting "numerous different personal interventions" [10]. The Computer coordinated performance provides just such a possibility, allowing for the existence a precise, unique but variable, multi-versioned work, in which each performance renders a new outcome.

3.2. Immancence

Computer coordination reduces the cognitive load on the performer. The manipulation of musical materials and the provision of coordination for their performance reduces non-musical decision-making, and potentially allows the performer give greater focus to their performance. It is also possible to apply structure to materials that are freely improvised, placing the performer(s) in an environment where the only consideration is the "performed moment". George Lewis the composer of the *Voyager* (1987), an "interactive musical environment that privileges

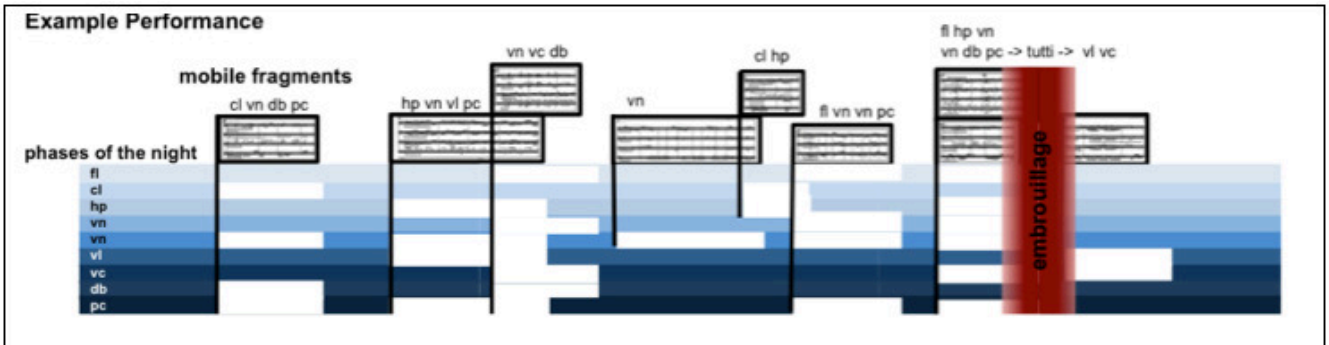


Figure 9. Representation of the formal structure of Vickery éraflage (2007)

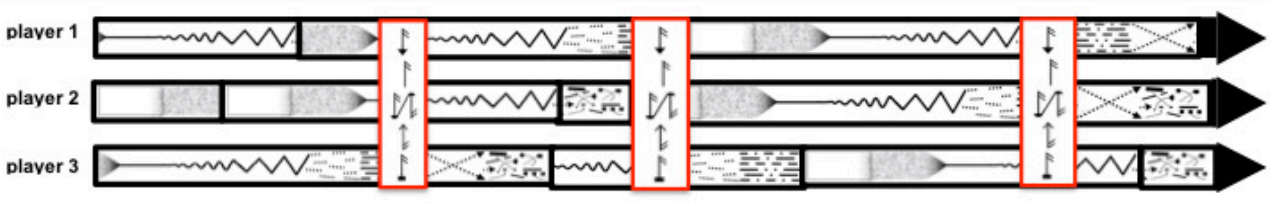


Figure 10. Representation of the formal structure of Vickery's Transit of Venus (2009)

improvisation" states "with no built-in hierarchy of human leader/computer follower—no "veto" buttons, footpedals or physical cues—all communication between the system and the improvisor takes place sonically" [16]. Other systems specifically designed for improvised performance include those of Lawrence Casserley and Evan Parker [4] and William Hsu and John Butcher [13].

In this environment the performer might be potentially capable of playing in an "immanent" state, what Deleuze defines as "a pure stream of a-subjective consciousness, a pre-reflexive impersonal consciousness, a qualitative duration of consciousness without a self" [8].

3.3. Poly-structure

Permutative, generative and transformative strategies can be independently employed a single work through computer coordination. The combination of formal structures in a single work leads to structural polyphony – poly-structure.

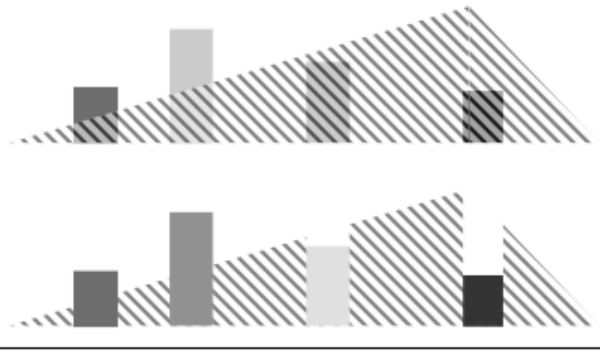


Figure 11. Representation of poly-structure (top) and interleaved poly-structure (bottom)

In my work *Éraflage* (2007) two formal structures co-exist: one a continuous loop of 27 bars at a constant tempo, that is performed throughout the work and the other a "mobile" collage of 12 musical fragments with five varied tempi. At indeterminate junctures the computer instructs between two and four players via headphones to disengage from the continuous texture, and to perform one of the fragments. The performers are coordinated both in their performance of the fragment and in their return to the continuous texture via click-track. This gives rise to a poly-structure comprising a dynamic permutative collage and a static looping continuum.

Poly-structures may also be interleaved by alternating sections that are based on distinct formal principals. My work *Transit of Venus* is an example of such a structure. The work alternates between two principal modes. The first presents a scrolling continuum of musical textures to the performers, the second is a improvisatory section during which the continuum and the metronomic click are suspended for all three performers. During these periods each performer follows the note-form indications that appear for short periods on the right of the screen (See Figure 6).

3.4. Subtractive structures

Poly-structures are additive in nature allowing the accretion of formally distinct material. The converse process – removing structural material – is also facilitated by computer coordination. Precise real-time excisions of material provide a novel structural approach.

An example of the potential of this tactic is my work *ghosts of departed quantities* (2010) that explores the paradox of "the devil's staircase", in which the proportional removal of material is repeated successively leading to the formation of an increasing number of fragments of decreasing size. The sonogram in Figure 13. shows the progressively increasing number and decreasing size of "holes" cut into the both the electronic and acoustic components of the performance.

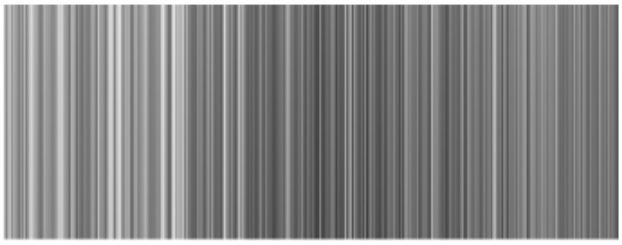


Figure 12: Condensed Sonogram of a 12 minute performance of Vickery's *Ghosts of Departed Quantities* (2010)

3.5. Substructural organisational procedures

Computer coordination may also act upon materials at the sub-structural level, allowing permutation, transformation and generation of material from within musical structures. In my work *Antibody,* for example a formal structure based on biological principles of mutation is explored. In this work, five contrasting blocks of musical material, each nine measures long with a distinct tempo and texture, are progressively "mutated" by translocating a greater number of cells between them to create increasingly hybrid arrangements. The order in which the five blocks appear is also permutated continuously.

Between the iteration of each block are interstitial spaces during which the players are instructed to improvise on the material they have just performed and link it to the next block. The length of these unscored sections is increased by one beat completion of each block.

Digital processing of the acoustic instruments provides a final level of transformation. Each player is separately recorded and processed and the degree and rate of processing grows throughout the work.

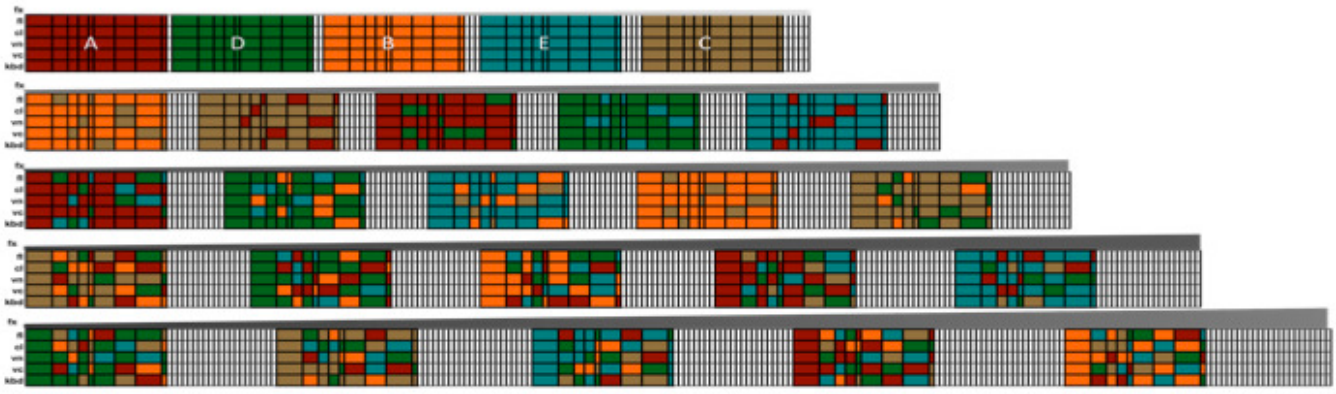


Figure 13. Representation of the "mutational" formal structure of Lindsay Vickery Antibody (2010)

4. CONCLUSION

The invention of the paper score provided composers with unprecedented control over the coordination of large musical forces and structures. However, in the last fifty years many composers have pushed the capabilities of the paper score to their logical limits.

Use of the computer as a source of coordination of musical forces provides a new step forward in furthering the development of musical organization and structure. Computer coordination allows for:

- the synchronisation and interaction of performance elements such as the score, performer(s), audio synthesis, acoustic performance, audio processing and audience;
- the permutation, transformation and generation of these elements;
- the instantiation of multiple versions of works;
- the formation of new formal paradigms such as poly-structure, subtractive structures, substructural organisational procedures.

Although the exploitation of this medium is in its early stages, the advantages in terms of exploring novel structural approaches are difficult to deny.

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PROBLEMS OF IMPLEMENTING MUSICAL COUNTERPOINT GENERATION SYSTEMS THROUGH THE PROLOG COMPUTER LANGUAGE

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ABSTRACT

This paper discusses typical problems associated with the construction of systems for the automatic generation of musical counterpoint, and how these problems can be overcome by using the PROLOG computer language and its unique features.

1. INTRODUCTION

The rules of counterpoint grow organically from the laws of psychoacoustics. For example, in the first species counterpoint only perfect and imperfect consonances can be used. In acoustical terms it means that the relations between notes can be expressed by *rational numbers* with small values used for both *numerator* and *denominator*, such as 2/1 (perfect octave), 3/2 (perfect fifth), 4/3 (perfect fourth, which is allowed in 3 or more voices counterpoint), 5/4 (major third) and 6/5 (minor third). Of course, on equally tempered instruments these values are only close approximations. Nevertheless, the fact remains that contrapuntal rules are in direct relation with mathematical proportions between frequencies. It is almost guaranteed that a composition based on these rules will sound natural to a listener who grew up in any of the western countries.

Today, in the era of computers, we tend to seek alternative methods for organising musical material, such as Markov chains, chaos theory, fractals, and many others, mostly based on mathematical models of what we perceive as natural. Such methods are quite natural for computers, which have no connection with the nature of sound. However, it is always a challenge and a good exercise to apply computational power to compose counterpoints.

On the surface, the construction of a system for the automatic generation of counterpoint seems to be quite a trivial task. Contrapuntal rules present themselves as easily programmable. However, after closer examination, several difficult problems arise. These problems are presented here and a method of overcoming them by using the PROLOG programming language is proposed.

This paper proposes using the PROLOG computer language to construct a contrapuntal generation system. The declarative nature of this language, and several another unique features, make it a suitable tool for constructing an expert system specialising in the rules of counterpoint. A system of this kind has been constructed at the University of Adelaide.

2. HISTORIC BACKGROUND

Counterpoint generation systems have already been successfully constructed before. The first such system was written by William Schottstaedt from Stanford University (Schottstaedt 1984). He used the SAIL programming language (*Stanford Artificial Intelligence Language*). Kemal Ebcioglu constructed his CHORAL system for automatic harmonisation of four-part chorales (Ebcioglu 1988). He acknowledged the potential usefulness of PROLOG. However, at the time, PROLOG implementation were simply too slow and not performing well enough. Therefore, he devised his own PROLOG-like language called BSL (*Backtracking Specification Language*). Ebcioglu's CHORAL system is based on roughly 350 rules and a backtracking mechanism. An interesting solution was proposed by David Cope. His GRADUS program, written in LISP, is based on a self-learning principle. His system learns from its own mistakes by maintaining a database of melodic dead ends (Cope 2005). The implementation of a counterpoint generation system as a logic constraint satisfaction problem was proposed by Ovans (Ovans 1990). Anders also recognises the usefulness of PROLOG in this context (Anders 2003).

3. PROLOG LANGUAGE

The PROLOG computer language was created in Marseille around 1972 by a group of researchers led by Alain Colmerauer. Initially it was intended to be just a tool for construction of a software system able to understand the French language in its written form (Colmerauer *et al.* 1992). The name PROLOG stands for PRO-gramming in LOG-ic.

Over the years several misconceptions about PROLOG have arisen. It is frequently perceived as some form of esoteric or ancient dialect, totally outdated with dubious usability. In reality, it is a general purpose language. It possesses several unique features, which can not be found in other mainstream languages available today (Bratko 2000).

It is perhaps necessary to list and briefly to describe several unique features of PROLOG, which are beneficial in the context of implementing a counterpoint generating system.

3.1. Declarative style

In general, a computer program consists of a series of instructions, which lead to solving a particular problem. A program written in PROLOG differs substantially from this model. In PROLOG, a program describes *what* has to be solved instead of *how* to solve it. Instead of writing a sequence of instructions on *what to do,* a programmer specifies the problem itself. The machine *reasons* upon such specification, and in consequence is able to devise the algorithm automatically. This is called Declarative Style, as opposed to Imperative Style; the later is predominant in computer programming today.

3.2. Symbolic calculations

Typically, computers work on numbers. PROLOG is one of the few languages able to process symbols. This feature frees the programmer from encoding everything into numerals, and makes computer code more human friendly.

3.3. In-built Inference Mechanism

Inference is the ability to make deductions, and to find solutions based on the facts and rules provided. In other words, the language itself has the ability to find ways of solving particular problems without the programmer telling the machine precisely how to do it. From a theoretical point of view, PROLOG's deductive system is based on so called *first order logic*. It is a formal method of processing logical implications (Bratko 2000).

3.4. Backtracking

Backtracking is one of the parts of the in-built inference mechanism. It is the ability to reverse the order of execution. In particular, when the program reaches a dead end (no solution), it retraces its steps, trying to find alternative solutions.

3.5. Meta-logic ability

This is the ability to write programs which operate on other programs. It is similar to lambda expressions, which can be found in LISP and several other languages. However, in PROLOG it is even possible to implement primitive control structures, such as IF..THEN..ELSE conditions, WHILE..DO loops and many others. It is also possible to modify the way in which PROLOG programs work, by defining alternative methods of processing facts and rules. It is even possible to implement different inference mechanisms.

All the aforementioned features make PROLOG a very powerful tool, especially for implementing expert systems. In practice, however, PROLOG requires some experience from the programmer to be able to unleash its entire potential. Nevertheless, with skilful programming, it is possible to push a lot of thinking activity onto the machine.

4. COMMON PROBLEMS

During the construction of a contrapuntal expert system, several problems have been identified, analysed and compared with examples available in the literature. They are all presented below, in brief, and effective solutions in PROLOG are proposed.

4.1. Declarative Nature of the Rules of Counterpoint

The rules of counterpoint are not written in a way which guarantees plausible outcomes. In particular, they do not provide any series of steps which could be translated into an algorithm. They do not specify whether the composer should start composing the melody from the beginning, from the end, or from the middle. Instead, they tell us what is good and what is bad. They are declarative in their nature. They do not tell us how to compose. It is entirely up to the composer how he or she comes up with melodies. This situation presents the quintessential problem to the computer programmer. Among all the contrapuntal rules there is only little written about how to start composing a composition.

Computer languages are mostly imperative in their nature. A computer program is usually nothing more than a series of instructions on how to achieve a solution step by step. This paradigm gap forces the computer programmer actually to devise a system for making compositions.

This particular approach takes advantage of the declarative nature of PROLOG, which is similar to the declarative nature of the contrapuntal rules. In particular, the rules can be translated from English to PROLOG without specifying any instructions on how to compose. PROLOG's internal inference mechanism takes care of generating melodies by itself.

4.2. Melodic Dead Ends

Typically, a composer makes a series of choices when writing a counterpoint. Sometimes the melody comes to a dead end, where contrapuntal rules prohibit any further continuation. In such cases the eraser is used; several previous choices are invalidated and alternative notes are chosen. Unfortunately, computer programs can only keep moving forward or crash. The majority of modern computer languages carry an in-built mechanism for exception handling. Nevertheless, the computer programmer needs to find a way of keeping track of already tried dead end solutions, cancelling part(s) of the current solution, and making alternative choices.

PROLOG's unique backtracking ability makes it particularly effective for dealing with melodic dead ends.

4.3. Music Notation versus Data Representation

To calculate counterpoint, the computer must be able to work on musical notes. In other words, musical knowledge somehow has to be represented inside the machine. A good contrapuntal generation system will also make distinctions between notes and intervals from their enharmonic equivalents. Within this context it is beneficial to use symbols and symbolic calculations instead of encoding everything into numbers.

The musical information that has to be encoded formally would include the following: notes with alterations and their respective octaves; pauses; rhythmic values; intervals (with both chromatic and diatonic parts included); melodic lines; and multi-staved compositions. In an object-orientated programming language, dedicated data types shall be defined for each of those entities. Additionally, a number of sub-routines (or methods) for accessing data within the structures would need to be constructed (Cope 2005) (Anders 2003).

In PROLOG, data structures do not need to be defined explicitly before use. It is enough to focus on preparing accessing methods only. Additionally, PROLOG can perform symbolic calculations. Therefore, information such as note names and alterations can be expressed directly as symbols, without encoding them as numbers.

Another interesting feature of PROLOG is the fact that methods designed for performing one particular transformation are frequently able to do the inverse transformation as well. A typical scenario would be to create a program to calculate musical intervals between two different notes. The reverse function of such program is to calculate notes based on the interval provided. In the majority of languages, this would require to create two different blocks of code, one for calculating intervals, another for calculating notes. In PROLOG, however, it is usually possible to use exactly the same code to do both.

4.4. The Contrapuntal Rule-Set

In a typical scenario, the contrapuntal rule-set is a moving target. The computer programmer faces a difficult question: “should I use J.J. Fux (Fux 1725/1965)? Or should I select Knud Jeppesen (Jeppesen 1939)? Should I do it in modal scales or major/minor?”. In practice, this can lead to a situation where a computer program specialises in one particular contrapuntal style. Therefore, it is necessary to design its architecture so that contrapuntal rule sets can be exchangeable. In particular, the set of rules responsible for keeping the counterpoint in a particular musical style should be designed as a replaceable module. This way, one and the same program can be used for generating both modal and tonal counterpoints. In such architecture, the only thing that has to be changed is a reference to a particular scale-assessing module.

In this respect, PROLOG, with its meta-logic capacity can be very useful.

4.5. The Choice

The process of finding counterpoint for a particular cantus firmus can be considered as finding a solution to a problem. Composing a counterpoint resembles to an extent solving a puzzle, such as Sudoku. The main difference is that Sudoku always has one possible solution, while there are multiple solutions for any given cantus firmus. As long as the composer strictly adheres to one particular set of contrapuntal rules, the composition process can be treated as choosing one possible solution which seems most pleasing to the composer. Typically, the number of first species counterpoints for a 12-note-long melody can reach up to several thousands. However, the number of second and third species counterpoints is so high that it can easily exceed million.

In this context the counterpoint generating program is nothing more than a mechanism for finding solutions to a set of logical constraints (Anders 2003). PROLOG with its inference engine based on first order logic is very well suited for this purpose. In particular, it is not necessary to code the entire searching mechanism. It is sufficient only to translate the rules of counterpoint into PROLOG's clauses. All possible counterpoints are calculated automatically by language's in-built mechanisms through successful application of the encoded rules.

The solutions found by PROLOG are not generated in random order. Rules are applied in the order in which they are written in the code. Therefore, the first solution is always the same, and reflects the ordering of the rules. In cases where only one, randomly selected solution is required, PROLOG's inference mechanism can be easily instructed to modify its own working and to apply rules randomly.

4.6. Aesthetics

The first counterpoint found may not necessarily be the most pleasing one. The rules of counterpoint offer several guidelines on what is better and what should be avoided. For example, they state that the composition should contain more imperfect consonances than perfect ones. These kinds of contrapuntal guidelines can be easily translated into a set of evaluation rules. These rules can be used to assess why some counterpoints are better than others. In this context, an automatic counterpoint generating system can be implemented as a search mechanism. It should apply contrapuntal rules in random order and keep looking for alternative solutions until it finds the one with a higher score than some predetermined threshold.

5. A CONTRAPUNTAL SYSTEM IN PROLOG

Typically a program written in PROLOG displays properties of either expert or knowledge based systems. In such systems, the role of a programmer is to encode a domain knowledge in formal notation. It could be said that the computer programmer assumes the role of a knowledge engineer. One of the unique features of

PROLOG is its ability to deduct automatically a solution based on this knowledge. In other words, PROLOG comes with an inbuilt inference mechanism, which can be considered as a form of machine intelligence, making the task of the programmer easier.

This particular contrapuntal system was constructed in a bottom-up fashion, starting from modules responsible for exporting calculated counterpoints to score documents and ending in the top-level contrapuntal solvers. The construction of the system is described briefly below.

The dialect of PROLOG used for the construction was HERCs Prolog, written by the author. It is loosely based on Micro-PROLOG dialect, with several MIDI and music related features.

5.1. Data Representation

The construction of the system was started by making several decisions on how musical data are represented and processed. The entire composition is denoted as a list of melodies. Each melody is a list of notes. Each note is a pair consisting of pitch and a number specifying duration. Each pitch is a pair consisting of a symbol specifying note and a numerical octave value. The symbols for notes are: C, C#, Cx, Cb, Cbb, D, D#, Dx, Db, Dbb... and so on. An empty pair ([]) denotes a pause. For the durations the following numbers were used: 12 – eight, 24 – crotchet, 48 – minim, 96 – longa. Hence, the following musical example
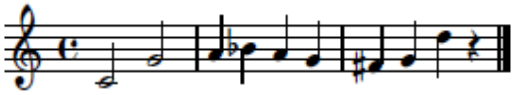


Figure 1. Sample melody.

can be formally notated as:

[[[C 1] 48] [[G 1] 48] [[A 1] 24] [[Bb 1] 24] [[A 1] 24] [[G 1] 24] [[F# 1] 24] [[G 1] 24] [[D 2] 24] [[] 24]]

A cantus firmus can also be represented with durations omitted. Hence the following cantus firmus:

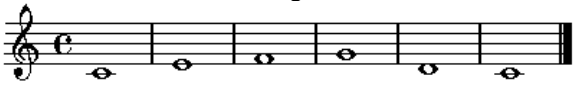


Figure 2. Sample cantus firmus.

can be denoted as:

[[C 1] [E 1] [F 1] [G 1] [D 1] [C 1]]

5.2. Input and Output of the System

Cantus firmus can be entered into the system directly from the terminal using the aforementioned notation. It can also be entered using external MIDI keyboard. The incoming MIDI transmission becomes translated directly into the internal PROLOG representation.

The calculated counterpoints, together with the original cantus firmus, are exported into PDF documents using Lilypond music engraving software. Large number of counterpoints can be exported in bulk into text files if score conversion is not required.

5.3. Interval Calculator

The next constructed module was the interval calculator. It is capable of calculating intervals between two provided notes. Each interval consists of a pair of numbers, where the first number is a diatonic interval, while the second is chromatic. For example, the interval between C and C# is an augmented prime, and can be denoted as [0 1]. It is different from the minor second (as between C and D flat), denoted as [1 1]. Similarly, the distinction between augmented fourth and diminished fifth is evident in the code. The former is denoted as [3 6] while the later as [4 6].

The definition of interval calculator (INTERVAL) works in a fashion that is typical for PROLOG. In particular, it can be used for calculating intervals between notes and calculating notes from the interval provided. For example:

[INTERVAL [C 1] [G 1] *x] will calculate *x = [4 7]
[INTERVAL *x [G 1] [4 7]] will calculate *x = [C 1]
[INTERVAL [C 1] *x [4 7]] will calculate *x = [G 1]

There is also a more generic version, which can calculate parts of notes and parts of intervals. For example:

[interval [C 1] *x [*i 7]] will calculate all three possible combination of values for *x and *i:

*x = [G 1], *i = 4, which is perfect fifth
*x = [Fx 1], *i = 3, which is double augmented fourth
*x = [Abb 1], *i = 5, which is double diminished sixth

5.4. Translation of the Contrapuntal Rules

The bulk of the system code consists of the rules of counterpoint translated directly into PROLOG clauses. They can be divided into the following classes.

5.4.1. Facts

Facts are unconditional truth. They are written as clauses without conditions. In other words, they are statements, which are always true. For example, a step interval (major or minor second up or down) can be coded as the example below. In a similar way, skip and leap intervals can be entered into the system.

[[step_interval [1 2]]]
[[step_interval [1 1]]]
[[step_interval [-1 -2]]]
[[step_interval [-1 -1]]]
[[skip_interval [2 4]]]
…. and so on.

5.4.2. Rules

Rules describe sentences which are true under certain conditions. For example, we may try to say that a melody can only go by step (up or down) or a skip (either direction). One possible way to express this in PROLOG is to use two rules, one saying that melodic interval can be a step, while the other would say it can be a skip.

[[melodic_interval *i] [step_interval *i]]

```
[[melodic_interval *i] [skip_interval *i]]
```

5.4.3. Prohibited constructions

Typically they have several checks, point of no return and a forced failure. A good example of such a rule is prohibited perfect fifths in similar motion. Two rules are necessary. One for prohibited perfect fifths, another for all other cases.

```
[[prohibited_parallel_intervals *i *i] [perfect_interval *i] / fail]

[[prohibited_parallel_intervals *i1 *i2]]
```

The first rule detects the occurrence of two identical perfect intervals. Then, the "cut" operator indicates the point of no return, which is immediately followed by failure. The second definition is used for all other cases.

5.5. Species Selector

The counterpoint generation system should be able to produce all species. As the fifth species counterpoint is a combination of all other species, the system should also be able to generate sequences of bars in different species. However, species can not be mixed completely freely. There are some rules to it as well. In particular, the last bar should always be of the first species, and a fourth species bar cannot be followed by a first species bar.

The construction of a species selector module can be started by translating the rules from English to PROLOG. Initially, we will choose some symbols for bars in different species, such as: bar1, bar2, bar3 and bar4 for the first, second, third and fourth species, respectively. Now we need to tell the PROLOG system that those symbols represent species. It can be expressed as follows:

```
[[species bar1]]
[[species bar2]]
[[species bar3]]
[[species bar4]]
```

Now we will say that the last bar can be of the first species only:

```
[[last_bar bar1]]
```

The next rule will say that the initial bar can be of any species. The rules uses a variable (beginning with star "*"), which has to be a symbol used for a species.

```
[[first_bar *B] [species *B]]
```

Our next step is to define how species can be chained in a sequence.

```
[[species_sequence bar1 *B] [species *B]]
```

This means that a first species bar can be followed by any other species bar.

```
[[species_sequence bar2 *B] [species *B]]
[[species_sequence bar3 *B] [species *B]]
```

This means the same thing for second and third species.

```
[[species_sequence bar4 bar2]]
[[species_sequence bar4 bar3]]
[[species_sequence bar4 bar4]]
```

These three rules mean that a fourth species bar can only be followed by second, third and fourth species.

Now, we can put all this together and create the species selector module. It will take a cantus firmus as the input and create a sequence of bars in different species with the same length as the cantus firmus. It consists of three PROLOG rules. It should be noted that the entire cantus firmus is denoted as *cf. It can be also described as [*cf1 *cf2 : *cfs], where *cf1 is the first note, *cf2 is the second and whatever follows the colon mark is the list consisting of the reminder of the notes. Similar notation is used for bars: such as [*bar1 : *bars], etc.

The first rule says that the first bar must obey the rules regarding the initial bar. When this restriction is satisfied, the species selector can make sure that the rest conform with the rules by calling species_selector_cd rule (which is described later).

```
[[species_selector *cf [*bar1 :*bars]]
        [first_bar *bar1]
        [species_selector_cd *cf [*bar1 : *bars]]
]
```

The second rule deals with the last bar. It should be noted that colon marks are not present, indicating that there are no more remaining notes.

```
[[species_selector_cd [*cf1 *cf2] [*bar1 *bar2]]
        [last_bar *bar2]
]
```

Our last rule takes care of all the notes between the first and the last one. It should be noted how this rule calls itself recursively, and how it strips the notes and bars from the head of the lists. For example, [*cf1 : *cfs] is passed to the recursive call without *cf1, i.e. as *cfs only. The same treatment is applied to the list of bars, i.e. [*bar1 *bar2 : *bars] becomes [*bar2 : *bars].

```
[[species_selector_cd [*cf1 :*cfs] [*bar1 *bar2 :*bars]]
        [species_sequence *bar1 *bar2]
        [species_selector_cd *cfs [*bar2 : *bars]]
]
```

Now the species selector can be applied to the sample cantus firmus, by formulating a PROLOG query:

```
[res
    [species_selector [[C 1] [E 1] [D 1] [C 1]] *bars]
    [show *bars]
]
```

The result will be, of course: [bar1 bar1 bar1 bar1]. All possible combinations can be generated by inducing unconditional failure at the end of the query. In this case, PROLOG will try all available solutions.

```
[res
    [species_selector [[C 1] [E 1] [D 1] [C 1]] *bars]
    [show *bars]
    fail
]
```

These rules illustrate that the species selector can be written in PROLOG and the thinking process behind its construction. Of course, such a module should be expanded to accommodate some extra code for generating skeletal rhythmic patterns for counterpoint. It should be noted that the rules will be applied in the same order as they were written. Therefore it might be useful to modify it slightly to make it possible to apply the rules in random order.

5.6. Scale Selector

A well designed counterpoint generation system should not be specific to any particular scale or mode. It should also be able to use pitch alterations appropriately within the enharmonic context. For this purpose, a mechanism for describing musical scales needs to be implemented.

It has to be noted that a musical scale or mode is not just a collection of pitches. It also carries some rules on how melodies can be constructed. For example, in minor scales, the leading seventh tone is raised whenever melody goes up, etc. (Polglase 2008).

Each musical scale can be constructed using exactly the same methodology as the one described previously about the species selector module. In particular, a scale evaluating module will contain rules on how notes can be combined into melodic sequences. Such a module can function as a kind of plug-in component to the rest of the system. In other words, the counterpoint generating system can be written without worrying about any scale-related intricacies. This functionality can be provided by external modules, specifically designed for all different musical scales.

5.7. Species Solver

Each species of counterpoint will need its own dedicated set of rules. For example, the first species counterpoint needs to worry about consonance, parallel perfect intervals and several others. The second species counterpoint introduces rules regarding the use of dissonances. The third species introduces even more devices, among them nota cambiata. And so on.

The fact that the florid counterpoint is a collection of bars in different species, complicates the architecture of the system. It is therefore practical to design 4 different species solvers, which can deal with just a single bar. The duty of applying correct rules to the entire melodies is now pushed to another module.

5.8. Counterpoint Solver

The duty of the counterpoint solver is just to apply different species solvers to each bar of the counterpoint. It works recursively in a loop in similar way to the species selector. Aside from counterpoints, this solver also returns some more information, such as: species structure, list of vertical intervals (against cantus firmus) and list of horizontal intervals within the melody.

5.9. Main Program

The main program can be constructed by putting all aforementioned modules together. In particular, it can work in the following sequence:

1. Use MIDI keyboard to input cantus firmus.

2. Use species selector module to generate rhythmic pattern of the entire counterpoint.

3. Use counterpoint solver with the cantus firmus and a specific scale (i.e. A-minor) to produce counterpoint.

4. Export generated score and possibly play it on a MIDI enabled instrument.

A surprising functionality of such a system may occur. In practice, PROLOG programs display the ability to perform the reverse transformation. In this case, the counterpoint generating system can, in fact, reconstruct all possible canti firmi if a counterpoint was provided as the input. It also has to be noted that species sequences can be predetermined or generated by the system. In this respect, it is possible to limit the system to generating only one particular species of counterpoint, or florid counterpoints with rhythmic structures provided by the user.

6. CONCLUSIONS

The main benefit of using PROLOG programming language for construction of the counterpoint generating system is the fact that the programmer is not required to design and implement any traditional algorithms. It is sufficient simply to translate contrapuntal rules from English to PROLOG. The language is able to find counterpoints entirely by itself when trying to satisfy logical constraints described as rules. Also, the rules of counterpoint are not written in algorithmic fashion. They do not specify how to construct a counterpoint melody. Instead, they describe what is good and what is prohibited, without giving any recipe. The declarative programming style of PROLOG overcomes this problem, thus making it the ideal tool for building this type of system.

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Artists Presentations

RE-CONTEXTUALISING THE HUMAN BODY AS AN ELECTRONIC PERFORMANCE INTERFACE

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ABSTRACT

An important factor to consider regarding computer based electronic instruments is the de-coupling of the control and feedback of electronic systems in contrast to the traditional instrument/player relationship where the connection is intimate and precise. In exploiting the potential for complex mapping where multiple sonic events can arise from the most minimal of performance gestures, the performer runs the risk of severing the connection between gesture and sonic result. At an extreme, the use of virtual reality systems and the internet has extended the field of performance activity beyond the physical performance space itself. Working within this technological matrix, the traditional conception of the human body as an agent of gesture has come under scrutiny leading some scholars to question the significance of the body in electronic performance. This presentation will advocate a return to the centralisation of the human body in electronic performance drawing on the phenomenological writings of Maurice Merleau-Ponty. Informed in large part by the authors own performance practice, the artist will discuss the benefits of attaching transducers directly onto the performers body. Such configuration allows for the amplification of the inner biological workings of the human organism. Internal body sounds such as heartbeat and breath cycle can then be used as raw compositional material during performance. The author will claim that this approach establishes the material body as a focal point from which additional technology can then be applied. The presentation will conclude by suggesting that the traditional notion of the performer and instrument as being separate entities can no longer be universally assumed and that a re-contextualisation of the 'human body' as an electronic performance interface is worthy of consideration.

WILD WEATHER

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ABSTRACT

My artist talk presents the prototype for the installation of a networked public sound sculpture planned for the Newcastle central business district in late 2011. The project is supported by the Newcastle City Centre Committee who, along with the organisation LiveSites, are seeking to install a range of projects over the coming year and are actively exploring strategies for interactive and innovative projects in the public domain.

THE INSTALLATION

The Wild Weather public sound installation includes five light weight transparent acrylic resonators, (approx 2200mm high X 110mm in outside diameter) suspended side by side and mechanically controlled to oscillate in response to the file transfer of media (predominantly audio & text files) from a remote server to the sculpture's interface (text files will be generated as audio files using text to speech software and the application WireTap Studio). I am currently working with both Arduino and Infusion Systems analog to MIDI digitisers and sensor technologies with standalone (using the Arduino programming language similar to C++) and Max/MSP applications on a MacBook Pro computer to determine the best solution. The options for this installation include using a remote computer streaming media from a server or having all media files transferred (using FTP) to local storage, but this is yet to be determined. The advantage of the local storage option is that all media will continue to be available for playback regardless of Internet connectivity. (See Figure 6).

I plan to use the FTP application Transmit to transfer and cue media files from the wildweather.net.au website to a dedicated server to the sculpture's hard drive. This can be done very effectively through an open source CMS Drupal website where individual or community based users can register and upload content (including a range of media files) for screening and subsequent broadcast, respond to specific calls for works and listen to and see the resonators in action through a webcam installed at the site. If the media uploaded is an audio file it would typically be in the stereo WAV or AIFF format, 16 bit/ 44.1Khz. Its duration would not exceed 4 to 5 minutes and so have an average file size of 40 to 50MB (10MB/ minute). To manage all of the server operations I plan to customise my own (dv) Dedicated-Virtual Server from the company Media Temple. This service is scalable (depending on usage) and would start with 20GB Storage, 512MB RAM and 1TB of Transfer Bandwidth, sufficient to manage and control my initial hosting environment.

The sculpture's sound projection is planned to emanate from each of the five resonators, equipped at their base with a small four inch speaker. The media content transferred via the Internet to the sculpture's computer would playback through any one of these five speakers. I am also experimenting with generating sound from within the five resonators using either prerecorded samples or natural sound. For example, the resonators could contain a quantity of liquid inside their closed end acrylic tubes. When set in motion the natural sound picked up (using very sensitive microphones) by the movement of the liquid inside the oscillating resonators could then be processed using a range of signal processing techniques (such as time stretching and delay) and then amplified back through their speakers. To accomplish this however the movement of liquid inside the resonators would have to be extreme and whether or not this is achievable is yet to be determined. The swing span for example of the resonators would have to be greater than 180 degrees to tip the liquid and the length of the resonators would need to be shorter than I have proposed.

The sculpture is designed as an Internet-orientated real-time interactive system for use by the general public. It is a hyperinstrument designed to be used with an analog to MIDI digitiser and five actuator signals with high resolution to transmit messages through USB to a computer where they are made available at a virtual serial port for analysis and/or control purposes. The research focus of the project is on designing a system (using software, sensors, signal processing and sculptural objects) that manages and broadcasts human response to human ecology, as well as exploring the appropriate protocols and innovative content for interactive art in public environments free of physical, social and/or political constraints. The Wild Weather installation can be installed in any public space with access to electricity, the Internet and an under cover public area with an approximate ceiling height of 4.5 meters, a minimum room width of 3.6 meters and a total swing span of 4.6 meters. The interaction with the sculpture will take place entirely over the Internet and is not constructed for any form of physical human contact.

Project Theme

We make the weather, we take the temperature, the temperature is rising, the weather is wilder. Professor of natural hazards at James Cook University, Cyclone

chaser Jonathan Nott this year warned that, *"town planners have become "very, very complacent" about the dangers of wild weather which is likely to increase during La Nina weather patterns and a shift in the Pacific Decadal Oscillation, which will warm the east Pacific ocean and possibly fuel severe cyclones. Three have hit Queensland's coast already this summer; Yasi, Anthony, and the weakest Tasha, which triggered the flooding rains that left three quarters of the state a disaster zone"* (Bita 2011).

This project addresses this intense and dangerous relationship we have with the weather and focuses on the devastating affects of floods and storms on our landscape and it's population. *"Learning to live with floods rather than fighting the rivers"*, where *"after 10 years of draught the floods have arrived on the Australian landscape"* are now commonly repeated messages in today's media. Central to this project is an investigation into these shared experiences and threatening relationships. Through the use of networked technologies the installation will provide the general population with a means to privately reflect on their experiences and relationships with their local weather environment and where possible share meaningful content that might describe the social and physical impact of weather on their communities. A diversity of stories for broadcast might explore their experiences, observations, memories and perceptions of weather change in different regions and may include flood stories, meteorological stories, cloud stories, rain, storm and hail stories and other observations of change to the weather. An extensive range of form in the content could include sound recordings, anecdotes, recollections, poetry, fragments, old recordings, letters, emails, blogs, lyrics and any other material that can be translated into sound. Prior to the public installation of the work an open call for content will take place and that addressing the theme will be made available for sculptural broadcast.

Project history

This work expands on and is inspired by my previous involvement in a exhibition titled Memory Flows curated by Norie Neumark, Deborah Turnbull and Sophia Kouyoumdjian in May/June 2010. Memory Flows is an ongoing and distributed media art project of the CMAI (Center for Media Arts Innovation) and was exhibited in collaboration with the Sydney Olympic Park Authority and UTS at the Armory Gallery, Newington Armory, Sydney. Norie Neumark explains the exhibition in her article, Memory Flows: Rivers, Creeks & the Great Artesian Basin in that, *"This exhibition works with the figure of memory to suggest that we are in an active and ethical relationship with our waterways. If we think of rivers as having memory - as an affordance we can connect to - we can sense their 'otherness' calling out to us, provoking desire and curiosity about our relations with such natural others. We can sense the flow of memory between ourselves and rivers as a network of complex relations"*. (Neumark 2010). For the Memory Flows exhibition composer/programmer Solange Kershaw and I installed a motion sensitive sound sculpture titled Memory Pendulums. Affected by contact and gravity the gentle swinging movement of the sculpture's pendulums modulated the spoken measure of a poem written for the work by the Australian poet Jill Jones. Her words flowed randomly in time with the playful swinging back and forth of three diamond shaped acrylic water pendulums, each filled to a different capacity with a blue (toxic) liquid, triggering a different memory, dripping into and polluting the muddy waters of the Parramatta River contained in a basin just below the tip of the pendulums. The three pendulums controlled the listening of the poem. If they swung uninterrupted for approximately 90 seconds the listener would hear the entire poem, which in three parts took all three pendulums to swing. The mechanism used to trigger the poem was created using a Max/MSP patch on a MacBook Pro, custom built sensors, mercury switches and the I-CubeX MicroDig.

Memory Pendulums Max/MSP Patch

The central object for this patch was the usb-microdig. The physical USB-microDig from Infusion Systems (Figure 1) is a thumb-sized hardware device that encodes analog signals to MIDI. It includes 8 digital outputs/ inputs that can be used to control actuators or trigger MIDI via sensors. The firmware for the USB-microDig enables it to operate in both stand-alone mode (sensor data is processed before it is transmitted) as well as host mode (raw sensor data is transmitted). The 8 inputs of each 10 bits resolution (1024 steps of each 4.9 mV) can sample at up to 5760 Hz with milliseconds latency to capture even the finest gestures and movements. (Mulder 2011).



Figure 1. I-CubeX USB-microDig

The Max patch was co-written by my project partner Solange Kershaw and myself. It is described as follows: For the Memory Flows exhibition we connected three custom built tilt sensors to the inputs of the USB-microDig. The sensor data was routed to the first three digital outputs and used to control the rest of the patch. The first input in the microdig object was used to determine the port, in this case a virtual serial (usb) port. The next 8 inputs were the sensor inputs in the physical microDig. These sensor inputs had to be turned on, and for the work to self-start, we used the 'loadbang' object. The loadbang object generates a 'bang' when a patcher is loaded, which in turn was connected to a toggle object (toggle object switches on and off, 1 and 0). We actually used a number of loadbangs delayed by 1 second to ensure that the digital outputs were reliably triggered, regardless of how fast or slow the computer, softwares and patch were loading on startup (Figure 2).

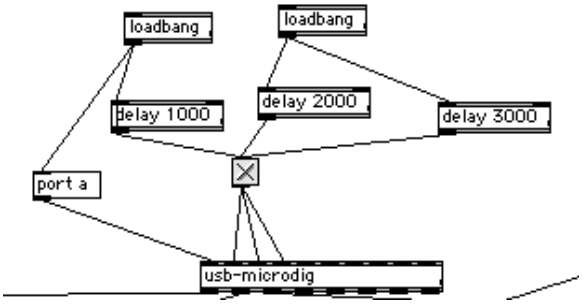


Figure 2. Max Patch loadbangs on startup

The first object connected to the digital output was a number box that displayed the sensor data in the midi range : 0 - 127. To determine whether the tilt switch was in the 'on' position (ie, when the sensor data ouput was 127) the number box was connected to a select object, that would trigger a bang in its output only if its input matched its argument. The select object here triggered two signal paths, one to play the sound file and one to monitor the time elapsed after that initial trigger (in 500ms units). As the pendulums swung through each cycle, the sensor data output changed from 127 to 0 and back again as the tilt switches went through each on/off state. So as to not restart the process and audio file playback on each cycle the right button was connected to a onebang object. The onebang object let a bang in through the left inlet only if a bang had been received in the right inlet (Figure 3).

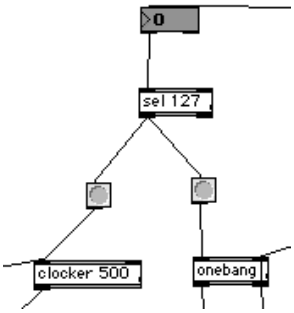


Figure 3. Max Patch onebang object

So each of the onebang objects were connected to a loadbang object so that they were ready at startup to let the first trigger in when the sensor output was equal to 127. Then it did nothing more until it received another bang in its right input inlet, which reset it. The left output of the onebang object was connected to a message box: 1. This message was passed on to the number box below it, that in turn triggered the soundfile playback object (sfplay~)(note: the tilda signifies MSP sound objects for Max/MSP, that is anything to do with sound). The sfplay~ object took a soundfile and a message in its left inlet (the message '1' told it to play the current sound file from beginning to end) and was connected to one or both of the channels of the audio output object (dac~) (the dac~ object is stereo by default).

The sound file must be opened before it's accessible to the sfplay~ object. Connected to the input of the sound file (and its 'open' message) was another loadbang object (Figure 4) so that the file would be accessible when the patch was opened. The same loadbang toggled the dac~ object on, and similarly we added a delay of 500ms to ensure reliability of the sequence.

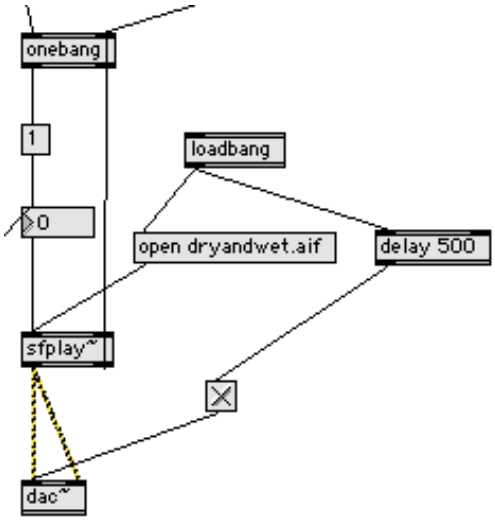


Figure 4. Max Patch onebang, sfplay~ and dac~ objects

As discussed, by using the onebang object we ensured that once the playback was triggered, it continued to play through without being reset each time the tilt switch was being toggled on and off by the swinging pendulum. But we still wanted to stop playback if the pendulum was no longer moving or if someone stopped the pendulum by choice. To do this we monitored the time elapsed between each cycle of the pendulum swing using the clocker object. The clocker object reports the time since it started, with an optional argument to set the output interval in milliseconds, in this case 500 (Figure 5).

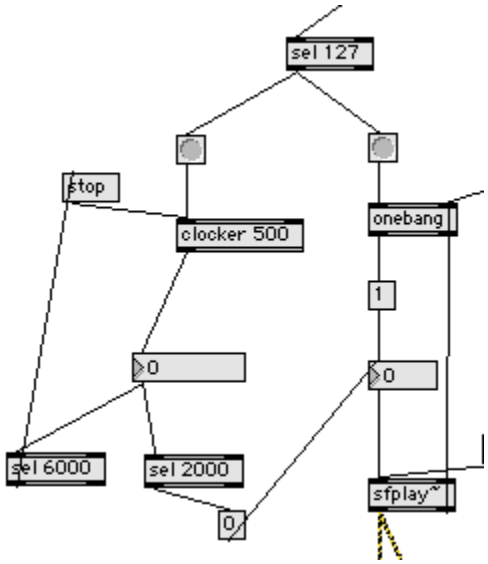


Figure 5. Max Patch clocker object

So from its left output, the select object had triggered the soundfile (onebang) and also started the metronome (clocker). The clocker object was restarted every time the tilt switch toggled on. The number box connected to the clocker's output monitored the time elapsed between each toggle of the switch in units of 500ms, and was in turn connected to two select objects. The first select object output a bang when the time reached 2000ms, the second select object output a bang when the time reached 6000ms (figure 5). If nothing had happened for 2 seconds, ie, the pendulum had stopped swinging for at least that long, the select object sent a bang to the message '0' which was passed on to the input of the sfplay~ object, and stopped playback. The right output of the sfplay~ object triggered a bang when it was done playing, which would happen if it had stopped (0) or if the audiofile had finished playing. That output was connected to the right inlet of the onebang object (figure 5), so it was ready to trigger if the pendulum was moving again. If nothing had happened for another 4 seconds (a total of 6000ms), the next select object sent a message to the clocker to stop monitoring, and the whole patch was idle until a pendulum started swinging. (Kershaw 2011; Castaldi 2011).

The Max/MSP patch we are developing for the Wild Weather installation will work along similar lines to the one we developed for the Memory Flows exhibition but somewhat in reverse where the audio files sent to the patch and cued via FTP will trigger the swinging movement of the resonators. As mentioned, this movement could be triggered by multiple media file upload via FTP and will use 5 actuators located at the top of each of the five resonators to control their back and forth swinging motion. Up to 5 audio files will be cued for playback and previously uploaded files will be looped for random playback. The proposed sound (natural or otherwise) inside the resonators will be fed back into the computer, processed in Max/MSP and amplified back through the resonator's speaker system in real time.

CONCLUSION

This installation sits at the intersection of sonic art, networked media, human ecology and remote user interaction. Through the combined technologies of the Internet, a computer/sensor interface and a public sound sculpture the project will investigate shared experiences and provide a means for the general population to contribute meaningful content that addresses our intense and dangerous relationship with the weather for public and web broadcast. The weather is a contentious and emotionally charged topic for many of us and over time I hope that the work will generate a rich pool of personal response to the global issues of climate change and global warming with the possibility of attracting further attention (from the public and media) and acting as a catalyst for change to the ever increasing dangers of wild weather.

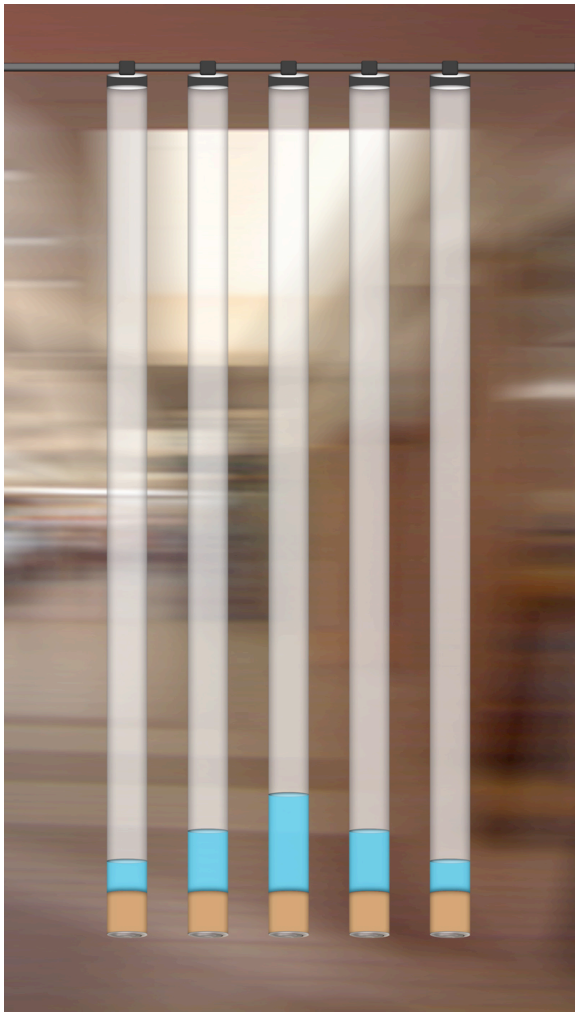


Figure 6. Wild Weather sound installation prototype

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Heritage As Composer Authenticity: A Poietic Approach Towards Sounds With Heritage-Related Quality In Electroacoustic Music Composition.

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ABSTRACT

In seeking an artistic voice and identity, I am interested in applying sounds that relate to my heritage as the foundation of musical composition. The term *heritage* denotes "*valued objects and qualities such as cultural, traditions… that have been passed down from the previous generations; a special or individual possession*" (New Oxford Dictionary, 2010). Heritage encompasses the cultural-inheritance and the life-experience of an individual. It is associated with an individual intrinsically and plays a direct part in authenticity of a creative work. I found in the domain of electroacoustic music, the composer is offered the opportunity to obtain sounds that relate to him or her on a more personal level. For example, the compositional materials can be sounds from the composer's family, childhood, special events, memories, and sounds from the habitual lands that are meaningful to the composer.

First of all, in adapting sounds relating to my heritage as electroacoustic music composition, instruments and sounds that suggest heritage-related quality are considered. From an Asian-Pacific perspective, native instruments such as the New Zealand Maori instruments, and the Asian traditional instruments are mostly made from natural organic substances. For example, the *Koauau* (New Zealand Maori) is made of wood or bone; the *Dizi*, *Xiao* (Chinese), the *Daejeum* (Korea), and the *Shakuhachi* (Japan) are made of bamboo; Tuaroria (New Zealand Maori) and *Muye* (Chinese) are tree leafs used as wind instruments. In the case of the Taonga Puoro, natural organic instruments are produced within the concept of 'individualizing one's own instrument'. For instance, the length of a custom-made Koauau is measured to the length of one's palm. Therefore, the sounds of the instrument are unique to the individual. John Dewey in *Arts as experience* suggests that aesthetic appreciation is inherently connected with the experience of making. From a poietic stand point, the concept of 'making your own unique instrument' provides an opportunity for composers to fully engage with the process of sound making. In addition, the instrument is unique to the instrument producer or performer. The paradigm of making one's own instrument is commonly adapted by composers such as Akio Suzuki, Phil Dadson, and Ajay Kapur et al as the exploration of their artistic voices. The sounds from the custom-made instrument and the performer/producer imply *surrogacy of the self*[1] (composer/performer/instrument maker) within the sounds. Moreover, the use of natural organic substances and the appreciation on natural beauty suggests a harmony between the human and the lands. In ancient Chinese philosophy, the material force and substance of the *five agents* (water, fire, wood, metal and earth) provide beneficial support for all things: "[the five agents] are first used to become an embryo and later for growth and support. … the physical form gets its support everyday, every day the material force enjoys its flourish, and principle attains completion everyday. These things are received as one is born, but as one continues to live for a day, one keeps receiving them for a day…." (Chan, 1963).

A pragmatic approach is adapted within the context of the work. Organic substances from nature such as bamboo, wood, soil, and stones are investigated as the basis of my organic instrument production. Organic sounds from the native landscapes are explored in both concrete and abstract fields with the aid of technology. In response to intuition with adapting sounds from the learned and the primitive schemas, my approach is to explore the inherent quality of organic substances and organic sounds that contains a human inheritance. The findings from the pragmatic

[1] I borrowed the term from Gerardo Dirie's ACMC 2011 presentation *Surrogates of Self: Exteriorising Organic Behaviour in Sound*.

aesthetic approach may support composers, performers, students and teachers who are interested in cultivating the aesthetic individuality through producing musical composition in the similar contexts.

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A PRELIMINARY ENQUIRY INTO SPIRITUAL, RELIGIOUS AND SACRED ELECTROACOUSTIC MUSIC.

Susan Frykberg, MA.

ABSTRACT.

This paper is a consequence of an internet call I put out in January 2011, through the CEC, for electroacoustic music of a sacred/spiritual/religious nature, to be used for my radio show. It is a preliminary response to the surprisingly large number of e-mails I received. It attempts to do the following: To categorise some of the almost one hundred e-mail responses in terms of country and general type of music; to quote six of the composers who e-mailed me; to list future possible areas of enquiry in the area; to let you hear excerpts from the following works: *Requiem* Georges Forget; *Ikaro* Gary Kendall; *Homo Religiosus* Eldad Tsabary.

Composers from the following countries responded: Greece, France, Spain, Mexico, USA, Britain, Ireland, Poland, Canada, Australia, Argentina, New Zealand. Types of works ranged from: soundscape pieces that use sounds from both eastern and western sacred places; compositions In the western christian tradition especially the mass; works that reference buddhist practice; one piece based on an islamic chant; many compositions that use scripture or other religious texts; works that describe themselves generally as transcendent or spiritual; compositions that reference religion in general; pieces that refer to shamanism in both content and practice; those that imply some sort of sacred quality within sound/music itself and those that reference indigenous religious practice.

Below are a six quotes from the many e-mail correspondences.

My composition, *Ikaro*, is my most recent effort at bridging the shamanic world of spirit and ceremony with the world of technological music. This effort emerged for me after some years of study with Peruvian shamans, during which I directly experienced their healing power of their music. **Gary S Kendall**

My whole purpose in dedicating a lot of my work to religion is to find connections and to find the good in all religions (charity, love, spirituality, growth, prayer, faith, and more) **Eldad Tsabary**

I am currently working on a piece for electroacoustic and kata (bo staff) called BO - which is more Asian in its spirituality. I am preparing for my karate black belt test which should be this year and this piece is part of my process to get there. **Elizabeth Hinkle Turner**

In my cycle of four "spirit journeys", *Temple* addresses the Western aspect, *The Way of the Spirit* the Eastern one, and *The Shaman Ascending* the aboriginal. Within each of these pieces, I've used the convolution process to bring out the inner harmonic structure of the source sounds as an aural metaphor for reaching a transcendent or spiritual state. **Barry Truax**

Some of my work uses text, music or ideas that has been used in churches for religious purposes. In my music, I try to bring that sense of God into the concert hall. Other works are more about the process, imitating the sacredness of creation which is one of God's activities. I want to develop a 'sacred' recording process. **Susan Frykberg**
It's religious/spiritual but only in a very abstract, Pythagorean sense. **Bruno Degazio**

Some future possible areas of enquiry may include. developing an accessible discography: collecting papers written on the subject; encouraging further discussion in the area; developing a category scheme that has meaning and relevence.

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Brief biography of Author Susan Frykberg is a composer, poet and radio broadcaster who tries to balance spirituality, creativity and social justice. She has degrees in experimental music and theatre, classics and theology. She taught at Simon Fraser University with Barry Truax for seven years and has also been a guest lecturer at Auckland and Otago universities; the Ontario School of Contemporary Art and the Emily Carr School of Art in Vancouver. She currently resides in Whanganui, New Zealand. She is a member of the Canadian Music Centre and the New Zealand Music Centre(SOUNZ), a past member of The World Forum for Acoustic Ecology, a founding member of the Canadian Electroacoustic Community and has a CD Astonishing Sense..... released under Earsay Editions. www.earsay.com. In 2008, one of her works was performed in Beijing, at the Congress for the International Alliance of Women Musicians and in 2010, at the International Women Composers conference in Pennsylvania. Susan has a 25 year old son Esha. She works part-time as a chaplain for Nazareth Rest Home and has a monthly radio show on electroacoustic music on Radio New Zealand Concert. She has recently begun working with improvising musicians in her Let The Art Sing series in art galleries in New Plymouth and Whanganui, to critical acclaim

THE FUTURE ARCHEOLOGY OF POPULAR MUSIC: MEDIA, AURA, *VANITAS*

Dugal McKinnon
New Zealand School of Music

ABSTRACT.

This talk will examine the core questions posed by my sound installation *Popular Archeology: Cassette, c. 1967-1994* (2010) which interrogates the myth of spatiotemporal totality that characterises the online era, an era in which the entire world is purportedly available in digitised form. But has music's shift to digital disembodiment, via an ongoing series of format changes, left in its wake a trail of superceded objects, devices and media? Some of these live a shadow life as retro-fetish objects: LPs and cassettes continue to circulate where mini-disc and 8-track cassette appear consigned to history. The former have durability greater than digital media, yet the wear of time cannot be effaced from them. Unless digitised they go the way of all flesh, slipping into decrepitude, prey to the imperfections of embodied memory. Does this senescence reinstate a Benjaminian aura to these artifacts of mass (re)production, given that the decay of each object is unique and not reproducible? The installation sonically enacts such a process of techno-cultural decay by simulating accelerated temporality, creating a cloud of sonic dust, out of which emerged privileged remnants: the hooks, licks and phrases crucial to the success of popular music. Is such a process to be read as analog nostalgia for a digital age, or is there something with greater longevity at work here, such as a contemporary iteration of the *vanitas* trope? These questions form the framework for my analysis of the installation, which in turn is directed towards a deeper analysis of the relationship between music, technology and memory, within the contexts of popular music and sound-based art.

Popular Archeology: Cassette, c. 1967-1994 (2010)
35 cassette players and cassette loops, shelving, disused retail space
http://www.lettingspace.org.nz/dugal-mckinnon/



Dugal McKinnon – Biography

Dugal McKinnon is a composer of electronic, instrumental and multimedia work, a sound artist and a writer on contemporary music. His work has received performances in Australasia, Europe, North America and Asia. Current projects include new works for saxophonist Richard Haynes (AUS) and the Lunapark ensemble (NL), both developed during an artist residency at the Centre for Art and Media Technology (ZKM) in Karlsruhe (DE). He is co-director with Sophie Jerram of Now Future, an organisation founded to foster interdisciplinary research and production at the intersection of art and ecology (www.nowfuture.org.nz). Dugal holds a PhD in Composition from the University of Birmingham (UK), where he studied with Dr Jonty Harrison. His doctorate was completed while on a DAAD Stipendium at the Technische Universität Berlin. He teaches sonic art and composition at the New Zealand School of Music (Wellington, NZ) where he is the director of the Lilburn Electroacoustic Music Studios.

ELECTROACOUSTIC MUSIC, ORGANIC SOUNDS AND NETWORKS.

Elizabeth de Vegt
The University of Waikato
Conservatorium of Music

ABSTRACT

This paper outlines my PhD project which draws together current and new music practises by writing music that combines live electronic and acoustic performers with new technology in virtual and real performance spaces. The investigation is inspired by Weinberg's (2005) Interconnected Music Networks and details the aesthetic and creative issues that are considered when collaborating with geographically dispersed musicians. This is an interesting area to research in due to its wide focus and the rapid growing interest in the area of Networked Music. The aim of the project is to develop awareness of new performance spaces while creating music suited to this genre that fulfils the composer's aesthetic and practical considerations while using a wide range of sound and instrument sources.

1. BACKGROUND

1.1. The Composers Aesthetic

"The measure of a work of art is whether one can sense in it the presence of the artist's body…the corporeal presence of the artist emanating from the work." Ostertag (2002)
"When you give a performance that takes your body (or the audience) out of the mundane and into something extraordinary though art, it has a profound appeal." Ostertag (2002)
"Listeners read into the piece as a reflection of the personality of the composer." (Stead 2009)
This paper is part of the author's Creative Phd at The University of Waikato. These are the considerations and aesthetics kept in mind while planning and writing music:

To have a clear sense of intentionality in the style, meaning and purpose of the music.

To appreciate works in which, despite less than perfect performances, the robustness of gesture and intention is clear.

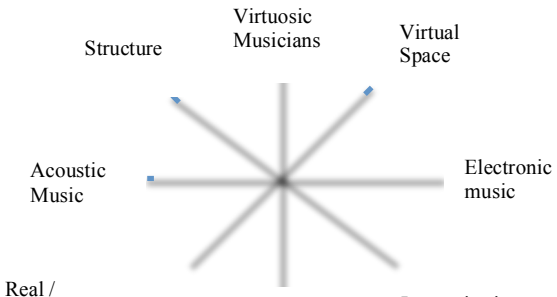
To write music that engages people intellectually and emotionally and provokes a thoughtful response.

To consider at all times the intellectual framework, the idea and its purpose.

To draw from a wide variety of sound sources including natural landscapes and people and utilize a wide range of traditional and non-traditional instruments played by virtuosic and amateur performers.

2. INTRODUCTION

"The world of the acoustic instrument has changed profoundly in the wake of the onslaught of electronic sound and media. How does the instrumentalist (or composer) respond – literally and musically to this." (Emmerson 2009)
This aim of this research is to write live electroacoustic music for the genre of Networked Music. The reason for this is the interest in the growing genre and the ability to apply well researched acousmatic and acoustic composition techniques and practises to this music. There are eight interlocking focus areas involved in this research.



The eight focus areas of the research question.

The research aims to synthesise these eight areas while exploring the new performance spaces offered by advances in high speed internet technology and computers. The exploration will be a combination of notated works written or directed by the composer as well as mixtures of collaborations with geographically dispersed musicians and composers.

The research will tailor writing music for the new performance space of Networked Music and draw effective practises from the tradition of acousmatic and acoustic music performance – practises such as the blending of acoustic music chamber ensembles or the effective combination of live instruments and electronics, the technology and software for the creation of acousmatic and live electroacoustic music. The

difference will be that this will be performed live over online networks. The performance space in this project is either defined as physical - "natural perception of an environment" or telepresence - "mediated perception of an environment" for example a 'distance space viewed through a video camera'. (Steuer 1992)

2.1. Concert Space

Attending live music concerts is a long tradition seeped in history. The architecture of concert halls has been perfected to house classical music, with research and building technique honed to create ideal acoustics.[1] Classical music concerts showcase virtuosity - it is the immediacy of hearing and watching the music performed live that makes it worthwhile to attend.

In the last five years, due to the increase in computer technology, it has become possible to collaborate online with musicians overseas via high speed networks in both real and virtual space.[2] Composers can choose how to use this space and have begun to create a new genre of networked electroacoustic music.

"With no real physical space users have no way to encounter each other as would a chamber ensemble Therefore participants are aware of each other and intuitively recognize opportunities for interaction by simulating the real world patterns of the performance of chamber music and borrow skills learnt while playing in that medium, of awareness and interaction." (Föllmer 2005)

Telepresence is a term coined by Marvin Minsky in 1980; "telepresence is defined as the experience of presence in an environment, by means of a communication medium." (Steuer 1992) The research will draw on this approach during the performance of this music.

"However, when perception is mediated by a communication technology, one is forced to perceive two separate environments simultaneously: the physical environment in which one is actually present, and the environment presented via the medium. The term "telepresence" can be used to describe the precedence of the latter experience in favour of the former; that is, telepresence is the extent to which one feels present in the mediated environment, rather than in the immediate physical environment." (Steuer 1992)

The Syneme Network is a high speed network set up at the University of University of Calgary, Canada in partnership with the University of Waikato. The network "Supports a new program of artistic practice enabled by high speed fibre optic research links that afford multiple artist/participants to engage in real time, high-definition, software supported Telepresence links".[3] The recent work of Whalley (2009, 2010a, 2010b) covers this area and outlines his pioneering work and progress in this field.

2.2. Network Approaches

"There is something magical about gathering together and making sound". (Trueman 2007)
It is exciting to be able to perform music with people of other cultures and nationalities. Prior to the advent of high speed internet networks, musician and composers would have to physically meet to collaborate or jam. The internet is now a "faster and more reliable platform for large scale interconnected musical networks." (Weinberg 2005) The speed and capabilities allow for net music interplay many individuals. (Trueman 2007)

Weinberg in his paper *Interconnected Musical Networks* (Weinberg 2005) describes four approaches to online music making. These approaches consider the "level of connectivity among players and the role of the computer in enhancing the interdependent social relations." These describe the ways in which collaborators can interact with each other over the server, which in turn depends on the technical capabilities of the equipment and people involed. Summarised from Weinberg:
The *Server* Approach allows users to send musical information to a server, although it does not allow for users to interact with one another.
The *Bridge* Approach connects distance users and allows them to perform together as if the same space, "providing a technical solution for imitating traditional group collaboration."
The *Shaper* Approach lets users shape and modify music material generated algorithmically by the networks central system. The players can continuously listen and respond to the music that modified by all participants, however, they cannot directly communicate to each other.
The *Construction Kit* Approach morphs together all three of the above approaches in a centralised network. Participants, most likely highly skilled musicians, can contribute the music, manipulate and shape their own and others' players music, and take part in a collective creation.
For the research in question the aim would be to us the Construction Kit approach as this allows the most flexibility between musicians.

When the geographically displaced music makers are joined together by a high speed network, audiences can then be located anywhere around the world. This includes at the physical performance venue(s) and elsewhere streaming the performance to their computer, thus allowing music to reach wider audiences.

[1] Researched by many scholars - including the Journal of the Acoustical Society of America or research listed on www.acoustics.org/booklist.html.
[2] Real space can be defined as the physical present space in which a concert or chamber ensemble would usually play. Virtual space is defined as the space created by the joining of physical spaces via online networks.
[3] http://syneme.ucalgary.ca/tiki-index.php

3. THE RESEARCH

The research aim will be to write music for the network music genre. The composer aims to write music with flexible compositional length and structures, rather than predetermine these by conventional notation and composition practise. This will allow for freedom within the works structure and will break away from the "fixed–just-press-play" acousmatic works.[1]

The research will define the role of the composer in creating collaborative music for networked music. History accepted the composer as an authoritative music creator however Stuart Sim argues that in the age of Postmodernism the death of a genius composer has happened. (Sim 1998) With the ease of sampling anyone can create music; this has had a major impact on originality, creativity and ownership. Ten years ago composers most often hand wrote their music scores at the piano - nowadays composers work straight onto computers using notation programmes such as Sibelius with the ability to copy and paste others work. The advent of remixes destroy the "notion of authorship", often removing the composers own presence (sense of self) within their work. (Sim 1998) With networked music and cross culture collaboration questions of ownership will arise as to who came up with what idea and who takes final ownership of the work. In all cases the creative process will be documented to aid this.

All composers want to have their music connect to its audience in some format (Landy 2007) - being able to sense the composer's intention and "soul" in the music is one way to judge the authenticity and effectiveness of a composition. (Emmerson 2007) Members of an audience react to different music as the sound is filtered through their life experience and persona – therefore, maybe, the more "composer personality or being" one hears, the more the audience responds or understands to the music. (Emmerson 2007) The music for this project will attempt to adhere to this aesthetic while having purpose and intention.

The sound resources for this music will be chosen carefully – from samples to instrumentation due to cross cultural collaboration. For example consideration as to the combination of various ethnic instrumentalists and what concerns or intrinsic meaning do they hold and if it is complementary or contradictory. Or, performing a work with bird sounds from different cultures with the addition of a panpipe might just be creating crass world music. To give careful thought to these types of considerations will impact the success, sense and intention of the music.

Depending on the artistic concept of the work, samples will be recorded by the composer from a wide variety of places such as instruments, landscape, nature and people. Which sounds will shape the pieces will depend on the countries involved and the context. How these sounds and instruments are then electronically treated is a major concern. Live sound processing will be primarily done in Ableton Live with a range of spectral plugins. The musicians must be able to interact and change in each performance, and the sound must be able to be shaped and evolve over time.[2]

The music will be flexibly scored, able to be played by varying ensembles in more than one context including a mixture of amateur and virtuosic musicians. This will allow for multiple performances in various sized premises [3] or allow different skilled performers to perform a work.[4]

The performance of Classical music has traditionally been exclusively for highly skilled musicians on Western instruments.[5] Contemporary Classical music also requires virtuoso performers, with most having to learn extended techniques on their instruments to stay current in their field.[6] Amateur musicians are usually self-trained, having had no formal lessons and qualifications on their instruments. Between this lies the Intermediate musician – someone with a great love for music and having had formal lessons but not up to a university performance standard. This project wants to utilize musicians of different capabilities.

The research will also investigate the interchangeability of the instrumentation of works, likening back to the Baroque musical tradition where instrumentation was flexible. Written music works were non-instrument specific and could be performed by which ever instrumentalists turned up that day.

The research is also interested in the multiplicity of interpretations that music can have. How will the music differ from performance to performance? And where is the point of difference that it becomes a separate work? (Benson 2003) Dependant on the interchangeability of instrumentation and different characteristics of (for example) two violins performers it would be interesting to see what qualities of the music are different and unique to each performance. Will changing the instrument forces lessen the authenticity of a work and then what constitutes an authentic performance?

The research will allow the music to have an interchangeable purpose dependant on the performance situation allows for further multiple uses. The reality that

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[2] See the work of (Wishart 1996) and Paine (2002)

[3] For example if space is limited one could leave out the contrabassoon.

[4] For example if it is a piece for a university level string orchestra, but transcribed for high school group yet the electronics and multimedia could still be the same – would it work would it still have the same effect or impact?

[5] For example crowd favourites: Violin Virtuoso Nigel Kennedy or tenor Luciano Pavarotti.

[6] For example *Echange* by Iannis Xenakis, for solo bass clarinet and mixed chamber ensemble, this demands phenomenal control of extended techniques on the soloist's behalf.

concert situations may be vastly different, i.e time frame for performance or technical equipment or the artists performing may have different emotional intentions. As the postmodern view is that there is not one right interpretation or truth, the research will focus on writing music that can exist in many forms at once, all of which may be equally valid.

The music will also tryi to enhance live performers' relationships with the score through improvisation and interaction. "Composers refuse to give up on acoustic forces so easily and are happy not just to coexist but to develop and exploit the often difficult relationship with the technology." (Emmerson in Dean 2009) This is a challenge to create well-crafted works that synthesise acoustic and electronic music.

Advances in technology allow realtime control of electroacoustic music, and institutes such as IRCAM [1] specialise in combining computer and acoustic music. Programmes have been written that interact and react to live inputs then create self-generated music. Roger Dannenberg's research at Carnegie Mellon University[2] enables score following programmes that are active and engaged to *"follow the live performance"* as a human would do. (Emmerson 2009)

There are many interactive systems that are "capable of changing and evolving" through their mutual interaction as described in Paine's theories of Interactivity. (Paine 2002) A recent one being Miroslav Spasov ENATCIVE which is a "multi-modal composing system", which creates content and effects in real time to accompany and interact with a performer. (Spasov 2011)

"For improvisers and live electronic performers, composition and performance are so strongly integrated that, for some, they are barely separable. Of course, material may be prepared, but may only be finally formed in the moment of performance. Yet again, new versions of very old processes emerge – recycling of materials, reliance on techniques that are known to work". (Emmerson 2007b)

The research will utilise improvisation to free up the fixed time domain of acousmatic music and to give performers complete control of tempo and expressivity and also, where appropriate, to improvise with the electronic sounds. Emmerson writes how performers can feel "straight-jacketed" by a fixed electronic part and that it surrenders their "precious expressive musical potential" to a machine. (Emmerson 2009)

The research will detail how to perform works at multisite concerts that include physical and virtual spaces.

"Progressively more advanced media technologies will enhance the sense of telepresence in a wide variety of virtual realities. Rapid advances in both multimedia computer technologies and in high-speed data networks hasten the development of a truly global village, in which our ability to interact with friends, family, and others who share interests similar to our own will no longer be limited by physical proximity. Such "virtual communities" represent one of the most exciting aspects of these developing new media, as they offer individuals a method for participation in, rather than mere observation of, the mediated worlds that surround them." (Steuer 2007)

As computer technology advances, the world has become a global village, where people's interaction is no longer limited to the physical but can be mediated via computers and networks. This new virtual space has created an online community and allows the chance for musicians and composer to play music together in new ways.

4. METHOD

The intellectual and creative core of this research will be the creative and collection of original compositions that synthesises the concerns laid out in section 2 and 3. There will be two methodologies used for this research – one creative and the other analytical. The research will be proven by following a logical progression through the writing of the compositions. To begin small compositions will be written that cover one or many of the eight focus areas. For example, a structured piece for a virtuosic musician in virtual space, or an electronic piece for amateur musicians featuring improvisation.

Each compositional process will be reflexive on the methods applied to future work. The skills and ideas learnt through each composition will inform new methods as well as have a retrospective flow on effect to prior work which could be revisited during the process of research.

"For my generation it is considered vulgar to have an audience really, really like a piece on the first hearing. Why would we write music except to move people and to be expressive?" (Benson 2003)

Music will seek to be intentional in its style, meaning and purpose. The music will also seek to engage people intellectually and emotionally while provoking thoughtful responses.

[1] IRCAM = The Institute for Research and Coordination Acoustic/Music. www.ircam.fr.

[2] For more information see www.cs.cmu.edu/~music/ Dannenberg's software is well used – his free audio editor *Audacity* was rated one of the PC World's best 100 products of 2008. http://www.cmu.edu/homepage/innovation/2008/summer/roger-dannenberg.shtml

5. CREATIVE OUTPUT

To begin with work has been written:
Man on the Corner (2010) Multimedia work for soprano, prepared piano, laptop and visuals. Focuses on: Virtuosic performers, improvisation, electronic and acoustic music, real space.

Short spectral work for orchestra expanding acoustic writing techniques.

You'll be in all my Tomorrows (2011) Piano and live electronics work with spectral processing.

After Sengai (2011) Live electroacoustic sound for live film by Bevin Yeatman.

5.1. Proposed Works

Several small acoustic works for virtuosic performers singularly and combined with amateur performers will be written to develop techniques and ideas.
A suite of electroacoustic music works, each 8-12 minute long which feature techniques in exploring spectral music and further refine control of sound processing. (Sethares 1998)
A small series of improvisation sketches to be performed by acoustic and electronic instruments. This will refine notation methods for improvisation and will test out how much music needs to be notated.
A piece for laptop orchestra that will demonstrate performance value without acoustic instruments.
Real Virtual Space /Networked Music – Music written to be performed in physical and virtual space for online festivals.
These works will be written with the final aim to uniquely synthesise the eight focus areas into large Networked Music projects having discovered appropriate methods within each focus area. The work will be with visual artists, musicians, and other artists from around the globe mainly improvised in real time over high speed networks.

The research focuses will be explored first methodically by following a logical progression through the writing of the compositions. To begin small compositions will be written that cover one or many of the following aspects relating to the research question. For example, a structured piece for a virtuosic musician in virtual space, or an electronic piece for amateur musicians featuring improvisation.
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These works will be written to confirm the skills, technology and resources required for each focus. Each compositional process will be reflexive on the methods applied to future work. The skills and ideas learnt through each composition will inform new methods as well as have a retrospective flow on effect to prior work which could be revisited during the process of research. By reflecting on the success of each piece upon completion it will help to make future informed research decisions.

5.2. Performance and Recording

"Even... composers cannot know their work in the whole fullness of its qualification before its performance… [they] sometimes discover only in performance what exactly they have created." (Ingarden quoted in Benson 2003)
"What happens when music-making machines and computers are added to the mix?... The respective roles of concerts and recording have been switched. Recording is the norm and concerts are glorifications of recording." (Lansky 1990)
Each work will be recorded on CD or DVD, notated appropriately and have detailed analysis of its conception to finish process. The aim of writing these works will be to build a coherent answer to the research questions.
All compositions will have a strong focus on technology and live performance. The technology will be a mix of current software such as Ableton Live, Max for Live and ProTools. The scope of this research will not be to programme new live performance software but to find new ways to combine currently created technology. In all cases an essential focus will be the musical outcome and the technology will serve as means with which to make music.
"We tend to think of a recording is a documentation of live performance, and perhaps a less than perfect substitute, an illusion, an incomplete and distorted image of reality". (Lansky 1990)
The created work will be recorded to serve as a documentation of one interpretation of a live event. It will stand as a "self-contained and self-confirming entity", and will be no substitute for witnessing composer/performer music making interactions via live performance. While the studio will be used as a creative tool for final renders of special audio effects, primarily the work will be live performance.

6. CONCLUSION

The intellectual/creative core of this research will be a collection of original compositions that tackle the research aesthetics in a particular way. This work will be documented with the considerations involved with writing this music for physical and virtual space. The initial hypothesis is that the eight aspects of the research will not have equal weighting, some being more important than others and through the unique synthesis and the unusual combination of realtime interactivity, real and virtual space, amateur and professional musicians an original contribution will be made to the genre of Networked Music.

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RE-ASSEMENT OF ELECTROACOUSTIC CHAIN

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ABSTRACT

The focus of my talk will be my work into the re-assessment of the traditional electroacoustic signal chain. This topic comprises the core of my Masters research at the NZSM.

As a basic model this chain can be defined as follows: sound source (within an acoustic space of some kind) to microphone, microphone signal to digital/analog workstation, then through speakers to the human ear via compression and rarefaction of air. It is my intention to reassess this signal chain by instrumentalising and physically manipulating elements in this chain that are generally considered to be static and invariant, such as microphone and loudspeakers as well as the acoustic environment in which sound is produced.
The context of my work emerges from a wider look at signal chains that have been manipulated, bent and broken to create new forms of music and sound. In a popular context, by including the hand *as instrument* into a record playback system (record, stylus, amp, speakers) turntablism and hip-hop were produced. In the context of sound-based arts there are examples ranging from Cage's prepared piano work, to Collins' hardware hacking and circuit bending. The creation of music from non- musical signal chains has also been accomplished (music created from amplified brain waves in Lucier's *'Music for Solo Performer'*).

Some of the techniques I will be employing within my work are; the movement of microphones around a sound source to create spatialisation and filtering within a stereo field and using the resonant bodies of instruments to enhance resonance of a piece of music (pace Lucier's work '*Music for piano with amplified sonorous vessels'* in which vessels are used as chambers to capture and enhance resonance of certain piano notes). Another technique I will be developing is the use of the speaker as an instrument. I will re-evaluate its use in electroacoustic music and exploit the physicalaspects of the speaker that will allow me to use it as a tool for transforming sound, rather than leaving it dormant in the signal chain (mentioned above). This may include covering the speaker to create filtering, moving it to create external spatialisation (Reich; '*Pendulum Music'*, Monahan; '*Speaker Swinging'*) that is not inherent in the aural material, or using it as a resonant body in which the vibrations of the structure itself are used as musical source material.

My talk will outline my reevaluation of models for production and reception of electroacoustic music and seek to develop new techniques to add to the signal chain or break it down to a point where the individual steps either merge or are totally isolated. In doing this, my aim is to move away from relying solely on digital signal processing in the studio to create music, but rather pushing the signal chain towards including tangible objects (such as the microphone, speaker and instruments) that will not only transform, but create and re-contextualise sound. Simply put, I aim to create methods of processing sound that exist on a physical level through approaching loudspeakers and microphones as instruments rather than just a means to capture and reproduce sound.